



SOUVENIR

2D MatTech Global: Fundamentals to Applications

2DMTG 2026

June 24-26, 2026

INDORE, INDIA

Organized By



2D MatTech Global:

Fundamentals to Applications

2D**MTG** 2026

June 24-26, 2026

INDORE, INDIA

Organized By



भारतीय प्रौद्योगिकी संस्थान

Indian Institute of Technology, Indore



विश्वविद्यालय अनुदान आयोग-परमाणु ऊर्जा विभाग वैज्ञानिक अनुसंधान संकुल

UGC-DAE Consortium for Scientific Research, Indore



राजा रामन्ना प्रगत प्रौद्योगिकी केंद्र

Raja Ramanna Centre for Advanced Technology, Indore

Organizing Committee

Patron

Director, IIT Indore

Convener

MEMS/CEVITS IIT Indore

Co-Convener(s)

UGC-DAE CSR, Indore

RRCAT, Indore

Publication Co-convenor

Wiley

Prof. Suhas S. Joshi

Prof. Rupesh S. Devan

Dr. Ram Janay Choudhary

Dr. Ravindra Jangir

Dr. Rohini Kitture

International Advisory Committee

Prof. Kourosh Kalantar-Zadeh

Prof. Umesh V Waghmare

Prof. Robert Weatherup

Prof. T. Venky Venkatesan

Prof. Anirudha V. Sumant

Prof. Sebastien Royer

Prof. Kaustubh R. S. Priolkar

Prof. Amlan J. Pal

Prof. S. B. Ogale

Prof. Yogendra Mishra

Prof. Sanjay Mathur

Mr. Unmesh D. Malshe

Prof. Kalobaran Maiti

Prof. Yuan-Ron Ma

Prof. Jin-Hyeok Kim

Prof. Bharat Jalan

Prof. Motohiko Ezawa

Dr. R Balamuralikrishnan

Prof. Bikramjit Basu

Prof. Babak Anasori

University of Sydney, Australia

President, JNCASR, Bangalore, India

University of Oxford, UK

University of Oklahoma, USA

Argonne National Laboratory, USA

Université du Littoral Côte d'Opale, France

Director UGC-DAE CSR, India

Ex. Director UGC-DAE-CSR, India

Director TCG-CREST, & IISER Pune, India

University of Southern, Denmark

Director, IIMC; University of Cologne,

Germany

Former Director RRCAT, India

Director IACS Kolkata, India

Vice President, Fo Guang University, Taiwan

Chonnam National University, South Korea

University of Minnesota, USA

University of Tokyo, Japan

Director DMRL, India

Director CSIR-CGCRI, India

Purdue University, USA

Local Organizing Committee

Treasurers

Dr. Sunil Kumar	MEMS/CEVITS IIT Indore
Prof. Santosh Hosmani	MEMS IIT Indore

Session Management

Dr. Abhinav Raghuvanshi	Chem IIT Indore
Prof. Santosh Hosmani	MEMS IIT Indore
Dr. Sunil Kumar	MEMS IIT INDORE (C)
Dr. Umesh A. Kshirsagar	Chem IIT Indore
Dr. Venkata Vamsi Koruprolu	MEMS IIT Indore
Dr. Debalya Sarker	UGC-DAE CSR, Indore
Dr. Dinesh Shukla	UGC-DAE CSR, Indore
Dr. Rajeev Batabyal	UGC-DAE CSR, Indore
Dr. Sanjoy Mahatha	UGC-DAE CSR, Indore
Mr. Pranjal Shrimali	MEMS IIT Indore

Award Committee

Dr. Rohini Kitture	Wiley
Dr. Sunil Kumar	MEMS/CEVITS IIT Indore (C)
Prof. Pavan Kankar	ME IIT Indore
Prof. Trapti Jain	EE/CEVITS IIT Indore
Dr. Rajamani Raghunathan	UGC-DAE CSR, Indore
Mr. Mayur Dhake	MEMS IIT Indore
Mr. Pranjal Shrimali	MEMS IIT Indore

Publication & Press

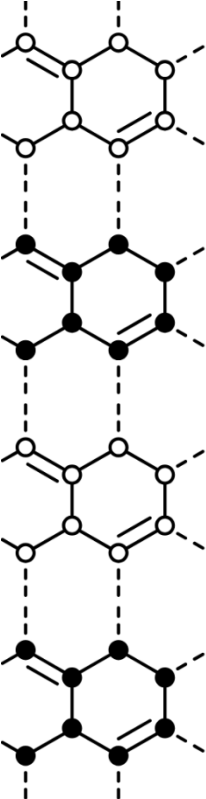
Dr. B. Prathap Reddy	EE/CEVITS IIT Indore
Dr. Nisheeth Kumar Prasad	MEMS IIT Indore
Dr. Onkar S. Game	Phys IIT Indore (C)
Prof. Parasharam M. Shirage	MEMS IIT Indore
Dr. Ranjith Kumar	MEMS IIT Indore (C)
Dr. Sandeep Singh	ME IIT Indore
Dr. Subhadeep Paladhi	EE/CEVITS IIT Indore
Dr. Sumanta Samal	MEMS IIT Indore
Dr. Vinod Kumar	MEMS IIT Indore
Dr. Arvind Yogi	UGC-DAE CSR, Indore
Dr. Devendra Kumar	UGC-DAE CSR, Indore
Dr. Rajamani Raghunathan	UGC-DAE CSR, Indore
Mr. Mayur Dhake	MEMS IIT Indore

Website and Publicity

Dr. Venkata Vamsi Koruprolu	MEMS IIT Indore (C)
Dr. Praveen Velpulla	UGC-DAE CSR, Indore

Registration and Help Desk

Dr. Lokesh Basavrajappa	BSBE IIT Indore
Dr. Mrigendra Dubey	MEMS IIT Indore (C)
Dr. Pravarthana Dhanapal	Chem IIT Indore
Dr. Srashtasrita Das	ChE IIT Indore
Dr. Gunjan Verma	UGC-DAE CSR, Indore
Dr. Praveen Velpulla	UGC-DAE CSR, Indore
Dr. Sudip Pal	UGC-DAE CSR, Indore
Mr. Mahesh Chand Bairwa	MEMS IIT Indore
Mr. Mayur Dhake	MEMS IIT Indore



Session Co-chairs

Prof. Biswarup Pathak
 Prof. I. A. Palani
 Prof. Krushna Mavani
 Prof. Pankaj R. Sagdeo
 Prof. Parasharam M. Shirage
 Prof. Prashant Kodgire
 Prof. Preeti Bhohe
 Prof. Rajesh Kumar
 Dr. Archana Lakhani
 Dr. Mukul Gupta
 Dr. V.R. Reddy

Chem IIT Indore
 ME/CEVITS IIT Indore
 Phys. IIT Indore
 Phys IIT Indore
 MEMS IIT Indore
 BSBE IIT Indore
 Phys IIT Indore
 Phys IIT Indore
 UGC-DAE CSR, Indore
 UGC-DAE CSR, Indore
 UGC-DAE CSR, Indore

Transport and Tour

Dr. Mayur Jain
 Dr. Nisheet Kumar Prasad
 Dr. Onkar S. Game
 Dr. Ranjith Kumar
 Dr. Dilip Kumar
 Dr. Uday Deshpande
 Mr. Mayur Dhake

CE IIT Indore
 MEMS IIT Indore (C)
 Phys IIT Indore
 MEMS IIT Indore
 UGC-DAE CSR, Indore
 UGC-DAE CSR, Indore
 MEMS IIT Indore

Accommodation

Dr. Abhijeet Joshi
 Dr. Dharendra Kumar Rai
 Dr. Jayaprakash Muragesan
 Dr. Saptarshi Ghosh
 Dr. Dilip Kumar
 Dr. R. Venkatesh
 Mr. Shubham Verma

BSBE IIT Indore
 MEMS IIT Indore
 MEMS IIT Indore (C)
 EE IIT Indore
 UGC-DAE CSR, Indore
 UGC-DAE CSR, Indore
 MEMS IIT Indore

Catering

Prof. Amod Umarikar
 Dr. Mayur Jain
 Dr. Pravarthana Dhanapal
 Dr. Sandeep Singh
 Dr. Umesh A. Kshirsagar
 Mr. Mahesh Chand Bairwa
 Mr. Shubham Verma

EE/CEVITS IIT Indore
 CE IIT Indore
 Chem IIT Indore (C)
 ME IIT Indore
 Chem IIT Indore
 MEMS IIT Indore
 MEMS IIT Indore

Global Connect Committee

Website Designing

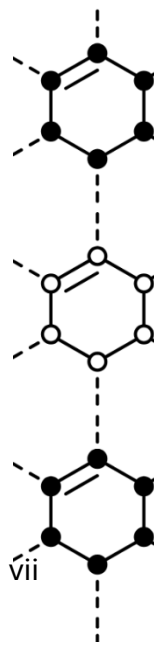
Pranav Manurkar
 Mohit Garhewal
 Rishav Sharma

MEMS, IIT Indore (C)
 MEMS, IIT Indore
 MEMS, IIT Indore

Session Management

Suman Yadav
 Prabhat Kumar Singh
 Bharti Ojha
 Bharthi Palthiya
 Shubham Parihar
 Priti Roy
 Apurva Karahe
 Ankit Yadav
 Kushal Dale

MEMS, IIT Indore (C)
 APSUD, RRCAT, Indore
 UGC-DAE CSR, Indore
 UGC-DAE CSR, Indore
 MEMS, IIT Indore
 UGC-DAE CSR, Indore
 MEMS, IIT Indore
 MEMS, IIT Indore
 Phys, IIT Indore



Registration and Help Desk

Aayushi Miglani	Phys, IIT Indore (C)
Shama Parveen	MEMS, IIT Indore
Manopriya Samtham	MEMS, IIT Indore
Sheetal Yamalakonda	MEMS, IIT Indore
Samraj	UGC-DAE CSR, Indore
Arpita	Phys, IIT Indore
Soumi Mukherji	MEMS, IIT Indore

Awards

Saurav Kumar Dubey	Phys, IIT Indore (C)
Soumi Mukherjee	MEMS, IIT Indore
Shyam Lal Gurjar	Phys, IIT Indore
Rishiraj Awasthi	Phys, IIT Indore
Akhilesh Choudhary	Phys, IIT Indore

Publicity and Press

Rishav Sharma	MEMS, IIT Indore (C)
Ankit Yadav	MEMS, IIT Indore
Ajay Patil	MEMS, IIT Indore
Gaurav Chouhan	UGC-DAE CSR, Indore
Apurva Karahe	MEMS, IIT Indore
Suman Yadav	MEMS, IIT Indore
Ekta Choudhary	Phys, IIT Indore
Shamma Parveen	MEMS, IIT Indore

Transportation

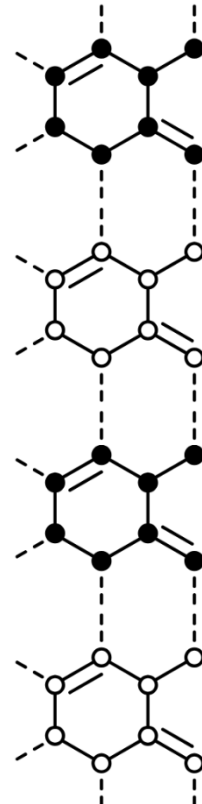
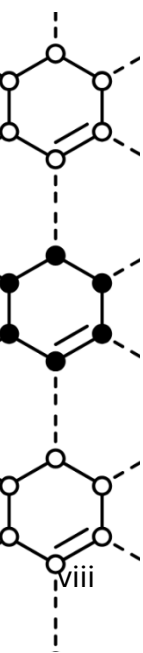
Ajay Patil	MEMS, IIT Indore (C)
Mohit Tiwari	MEMS, IIT Indore
Bhimrao Mohan Hiwarale	Chem, IITI
Yogesh Kuamr Yadav	UGC-DAE CSR, Indore
Rajan Mishra	UGC-DAE CSR, Indore
Shivam Choudhary	UGC-DAE CSR, Indore
Gaurav Chouhan	UGC-DAE CSR, Indore

Accommodation

Ekta Choudhary	Phys, IIT Indore (C)
Aayushi Miglani	Phys, IIT Indore
Apoorv Purohit	MEMS, IIT Indore
Gaurav Tiwari	MEMS, IIT Indore
Saurav Kumar Dubey	Phys, IIT Indore
Jerin K. Roy	Phys, IIT Indore
Dharmendra Kumar	MEMS, IIT Indore

Catering

Manopriya Samtham	MEMS, IIT Indore (C)
Aayushi Miglani	Phys, IIT Indore
Suman Yadav	MEMS, IIT Indore
Saurav Kumar Dubey	Phys, IIT Indore
Rishav Sharma	MEMS, IIT Indore
Akhilesh Choudhary	MEMS, IIT Indore
Rajan Mishra	UGC-DAE CSR, Indore
Yogesh Kumar Yadav	UGC-DAE CSR, Indore
Subho Saha	UGC-DAE CSR, Indore



Dr Anil Kakodkar
Chancellor, Homi Bhabha National Institute,
Chairman, Rajiv Gandhi Science & Technology Commission,
Former Chairman, Atomic Energy Commission

1st June 2026



Message

It gives me immense pleasure to extend my warm greetings and best wishes to all participants, distinguished scientists, researchers, academicians, industry leaders, and students attending the 2D MatTech Global Conference at IIT Indore, being organized by IIT Indore, UGC-DAE CSR Indore, and RRCAT Indore.

The advent of 2D materials has opened up new technological opportunities for breakthroughs across diverse fields such as electronics, energy storage, sensing, quantum technologies, biomedical applications, etc. To leverage these new opportunities, one needs to nurture a full spectrum research translation continuum from basic research to marketable products. IIT Indore, UGC-DAE CSR Indore, and RRCAT Indore, the three Indore based institutions that are organising this Conference, together do bring in significant complementarity in education, research, experimentation and cutting-edge technology development. They also constitute a near holistic ecosystem that can become a full continuum by joining hands with collaborators in gap areas where ever they exist. I must therefore commend the organisers for shaping this conference, in the way they have done.

This conference serves as a vital platform for bringing together experts from across the globe to exchange their respective current pursuits, present their insights and novel findings and explore impact making collaboration opportunities for solving important problems. Such conferences also enable valuable opportunities for research students and young researchers to interact with their counterparts, as well as more experienced and eminent personalities in their field, get noticed in the peer community and widen their research network. I am confident that the deliberations during this conference will stimulate new ideas, strengthen national and international partnerships, and inspire the next generation of researchers working in 2D materials.

I would like to compliment the organizers, advisory committee members, sponsors, and volunteers for their dedicated efforts in planning this important event. I wish the conference every success and hope that all participants have a productive, enriching, and memorable experience.

With best wishes,

A handwritten signature in blue ink, appearing to read 'Anil Kakodkar'.

Anil Kakodkar

7th fl. Central Complex, Bhabha Atomic Research Centre, Trombay, Mumbai 400085
Tel. +912269293793, Email kakodkar@barc.gov.in / kakodkaranil@gmail.com



भारतीय प्रौद्योगिकी संस्थान Indian Institute of Technology Indore



Message

It gives me immense pleasure to welcome all the distinguished guests, plenary and invited speakers, delegates, researchers, and students to the International Conference on **2D MatTech Global: Fundamental to Applications**, proudly hosted by the Indian Institute of Technology Indore.

Material Science has always been the cornerstone of technological evolution, and today, two-dimensional (2D) materials stand at the absolute frontier of scientific discovery. The unique physical, electronic, chemical, and optical properties of these materials have opened up unprecedented avenues for breakthrough innovation. This conference comes at a crucial juncture as the global scientific community works aggressively to transition these remarkable nanomaterials from fundamental laboratory research into scalable, real-world applications.

I am delighted to note that 2D MatTechGlobal 2026 has successfully brought together an exceptional gathering of over 400 participants with over 350 high-quality unpublished paper submissions and a curated lineup of expert-led technical sessions. This forum provides an ideal ecosystem for academia, research institutions, and industry pioneers to deliberate on the future of materials science. The interdisciplinary focus of this event, spanning critical sectors including clean energy storage, healthcare, advanced electronics, sensors, and emerging quantum computing technologies, underscores the transformative potential of the research showcased here.

At, IIT Indore, we have consistently strived to foster an environment of cutting-edge research, innovation, and global collaboration. Hosting an event of this magnitude reinforces our commitment to driving scientific dialogue that addresses both pressing global challenges and future technological needs.

This souvenir serves as a valuable compendium of the collective intellect, creative breakthroughs, and visionary ideas presented during the conference. I would like to extend my heartfelt congratulations to the organizers, various committee members, volunteers, and supporting staff from IIT Indore, UGC-DAE CSR Indore, and RRCAT Indore for their meticulous planning and dedication in bringing this event to fruition. I am also deeply grateful to our esteemed Chief Guest, Dr. Anil Kakodkar, and our distinguished plenary and invited speakers for gracing this platform.

I wish all the participants highly engaging, intellectually stimulating discussions and a memorable experience at IIT Indore. May the collaborations forged during these days lead to transformative breakthroughs in the years to come.

Prof. Suhas S. Joshi,
Director,
Indian Institute of Technology Indore
Patron, 2D MatTechGlobal 2026



विश्वविद्यालय अनुदान आयोग-परमाणु ऊर्जा विभाग वैज्ञानिक अनुसंधान संकुल
UGC-DAE Consortium for Scientific Research, Indore



Message

On behalf of the **UGC-DAE Consortium for Scientific Research (UGC-DAE CSR)**, Indore, it gives me immense pleasure to extend a warm welcome to all delegates, eminent scientists, researchers, and industry participants to the **International Conference on 2D Materials: Fundamentals to Applications**, jointly organized by **IIT Indore, UGC-DAE CSR**, and **RRCAT, Indore**.

In recent years, 2D materials have emerged as a transformative technology, driving innovation across diverse domains such as electronics, energy storage, sensors, photonics, healthcare, and quantum technologies.

UGC-DAE CSR has consistently endeavoured to promote interdisciplinary collaborations, foster cutting-edge scientific research, and build competence within Indian universities and academic institutions. By providing a robust institutional framework, we enable optimal utilization of the mega research facilities established by the DAE, alongside our in-house advanced experimental infrastructure. In this context, the collaboration among UGC-DAE CSR, IIT Indore, and RRCAT holds particular significance, as it brings together expertise in materials synthesis and characterization, synchrotron and laser sciences, device-oriented research, and theoretical modelling.

I am confident that this conference will serve as an exceptional platform for the exchange of ideas, dissemination of recent advances, and the forging of meaningful national and international collaborations. The interactions among participants from academia, research laboratories, and industry will undoubtedly enrich the scientific dialogue and help address the complex challenges associated with 2D materials.

I sincerely congratulate the organizing committee and all partner institutions for their exemplary teamwork in bringing together a distinguished group of experts and young researchers. I hope the conference achieves resounding success and that the scientific discussions contribute significantly to advancing knowledge, fostering collaborations, and strengthening the research landscape in materials science.

With my best wishes for a productive and inspiring conference,

Prof. Kaustubh R. S. Priolkar
Director
UGC-DAE Consortium for Scientific Research



राजा रामन्ना प्रगत प्रौद्योगिकी केंद्र Raja Ramanna Centre for Advanced Technology, Indore



Message

It gives me great pleasure to welcome all the distinguished guests, plenary and invited speakers, delegates, researchers, and students to the **International Conference on 2D MatTech Global: Fundamentals to Applications-2026 (2DMTG-2026)** being organized during **24-26 June 2026** at the **Indian Institute of Technology Indore**. This event is co-organized with the **Raja Ramanna Centre for Advanced Technology (RRCAT)**, and **UGC-DAE Consortium for Scientific Research, Indore Centre**.

The discovery and development of two-dimensional (2D) materials have revolutionized the field of materials science. The theme of the conference, “Fundamentals to Applications” aptly captures the essence of modern scientific research, where a deep understanding of fundamental phenomena serves as the foundation for transformative technologies.

I am delighted to note that 2DMTG-2026 has attracted an exceptional gathering of over 400 participants, along with submission of more than 350 high-quality unpublished research papers. I am confident that the deliberations and interactions during the conference will inspire new research directions, strengthen scientific partnerships, and significantly advance materials science and technology. This conference offers students and young researchers an invaluable opportunity to interact directly with global experts and gain deep insights into this rapidly evolving field.

As a premier institution of Department of Atomic Energy, RRCAT leads in frontier research involving particle accelerators, lasers, synchrotron radiation sources, advanced materials, and related technologies. 2D materials research relies heavily on sophisticated synthesis, characterization & fabrication techniques, many of which are enabled by RRCAT's state-of-the-art facilities and infrastructure. The available expertise and infrastructure at RRCAT plays an important role in bridging fundamental science with industrial innovation.

I extend my sincere congratulations to the organizers, committee members, volunteers, and supporting staff from IIT Indore, UGC-DAE CSR Indore, and RRCAT for their dedicated efforts in organizing this event and bringing esteemed community of researchers and professionals together. I extend my best wishes for the grand success of 2DMTG-2026 and hope that all participants find the conference intellectually stimulating, professionally rewarding, and personally enriching.

With my best wishes,

(Viraj P. Bhanage)

Distinguished Scientist & Director,

Raja Ramanna Centre for Advanced Technology (RRCAT), Indore

Convener and Co-Convener(s)



Message

It is our profound privilege and honor to welcome you all to the International Conference on **2D MatTech Global: Fundamentals to Applications** being hosted at the Indian Institute of Technology Indore from **June 24 to 26, 2026**.

The field of 2D materials has evolved rapidly over the past decade, expanding from basic physical explorations into a driving force behind next-generation transformative technologies. Today, these atomically thin structures are redefining the limits of clean energy storage, medical diagnostics, high-performance sensors, advanced electronics, and quantum technologies. 2D MatTechGlobal 2026 was conceptualized to serve as a vital crucible for these innovations, providing a collaborative environment where cross-disciplinary breakthroughs are shared, scrutinized, and scaled.

Bringing this vision to life has been an immense and rewarding journey, made possible through powerful institutional collaborations. We are deeply grateful to the Directors of the Indian Institute of Technology Indore, UGC-DAE Consortium for Scientific Research (UGC-DAE CSR), and Raja Ramanna Center for Advanced Technology (RRCAT) for their steadfast support, visionary guidance, and for graciously allowing their esteemed organizations to partner in this conference. This synergy between premier national research facilities and academia has immensely elevated the technical depth and research horizon of the event.

We are thrilled to share that the conference has drawn an overwhelming global response, welcoming over 400 participants from over 10 countries. The technical program features an exceptional lineup of expert-led sessions, including insights from renowned global leaders in this field and high-quality unpublished paper submissions. A key highlight of our pre-conference schedule is the specialized, hands-on workshop dedicated to Synchrotron-Based Techniques for Materials Characterization, organized in partnership with RRCAT Indore.

We take great pride in announcing that the highest-quality peer-reviewed submissions from this event will be considered for publication in prestigious Wiley journals, including *Small*, *Small Structures*, *Small Methods*, *Advanced Materials Interfaces*, *Advanced Physics Research*, and *physica status solidi (RRL)*. Furthermore, to champion the next generation of materials scientists, we have instituted competitive accolades, including the Best Thesis Award, Oral Presentation Awards, and Poster Presentation Awards, recognizing novelty, clarity, and industrial relevance.

We express our deepest gratitude to our Patron, the Director of IIT Indore, for his steadfast support. We extend our sincere thanks to our illustrious Chief Guest, Dr. Anil Kakodkar (Chancellor, Homi Bhabha National Institute), our International Advisory Committee Members, plenary and invited speakers, the Organizing Committee Members, and the reviewers. The success of 2DMTG is safeguarded by the financial support from the government funding agencies and private partners. Most importantly, we thank you, the authors and delegates, for choosing this platform to showcase your pioneering discoveries. Last but certainly not least, special thanks to our dedicated student volunteers on a Global Connect Committee, whose relentless hard work made this event possible.

This souvenir serves as a permanent record of the intellect, creative solutions, and forward-looking ideas driving our community forward. We hope that the three days of the conference provide you with stimulating technical discussions, valuable new professional networks, and a memorable stay at the picturesque IIT Indore campus.

We wish you all a highly rewarding and inspiring conference!

Prof. Rupesh S. Devan (IIT Indore)
Dr. Ram J. Choudhary (UGC-DAE CSR Indore)
Dr. Ravindra Jangir (RRCAT Indore)

PROGRAM SCHEDULE

Date and time \ Session	A Maitreyi	B Gargi	C Sandipani	D Kalidas
23/06/2026				
4:00 – 6:00 pm	Registration			
24/06/2026				
8:00 – 9:15 am	Registration			
9:00 – 9:30 am	Breakfast			
9:30 – 11:00 am	Inauguration + Inaugural talk			
11:00 – 11:30 am	High Tea			
11:30 – 12:10 pm	Plenary Talk			
12:10 – 12:15 pm	Break			
12:15 – 01:15 pm	Invited Talks			
01:15 – 02:15 pm	Lunch			
02:15 – 02:55 pm	Plenary Talk			
02:55 – 03:00 pm	Break			
03:00 – 4:00 pm	Invited Talks			
4:00 – 4:15 pm	Tea break			
4:15 – 5:45 pm	Oral			
5:45 – 6:15 pm	Invited Talks			Contrib. Talk
Till Dinner	Thesis Poster presentation (Stage 1)			
7:30pm onwards	Dinner			
25/06/2026				
9:00 – 9:30 am	Breakfast			
9:30 – 10:10 am	Plenary Talk			
10:10 – 10:50 am	Industrial Lecture			
10:50 – 11:05 am	Tea Break			
11:05 – 12:05 pm	Invited Talks			
12:05 – 1:35 pm	Thesis Oral presentation (Stage 2)			
1:35 – 2:30 pm	Lunch			
2:30 – 3:10 pm	Plenary Talk			
3:10 – 3:50 pm	Industrial Lecture			
3:50 – 4:05 pm	Tea break			
4:05 – 5:05 pm	Invited Talks			
5:05 – 5:20 pm	Contrib. Talk			
5:20 – 6:45 pm	Poster Presentation			
6:45 – 7:45 pm	Travel toward Gala dinner			
8:00 pm onwards	Gala Dinner			
26/06/2026				
9:00 – 9:30 am	Breakfast			
9:30 – 10:50 am	Plenary Talk			
10:50 – 11:05 am	Tea Break			
11:05 – 12:05 pm	Invited Talks			
12:05 – 12:20 pm	Contrib. Talk			
12:20 – 01:20 pm	Oral			
1:20 – 2:30 pm	Lunch			
2:30 – 3:40 pm	Poster Presentation			
3:45 onwards	Valedictory			

SPEAKERS

Plenary Speakers



Prof. S. B. Ogale
Director, RISE, TCG-CREST,
Kolkata, India



Prof. Jin-Hyeok Kim
Chonnam National University,
South Korea.



Prof. Bikramjit Basu
CSIR-CGCRI,
India



Dr. K. P. Vijayamohan
IISER Tirupati,
India



Prof. Umesh V Waghmare
JNCASR Bangalore,
India



Prof. Yuan-Ron Ma
National Dong Hwa
University, Hu Taiwan.

Invited Speakers



Prof. Abha Misra
IISc, Bangalore,
India



Prof. Anirudha Sumant
Argonne National Lab.,
USA



Dr. Anjana dogra
National Physical laboratory,
New Delhi, India



Dr. Ashish Mishra
IIT BHU,
India



Dr. Ashutosh Singh
CeNS, Bangalore,
India



Prof. B. Ranjit K. Nanda
IIT Madras,
India



Prof. Babak Anasori
Purdue University,
USA



Prof. Chandra S. Sharma
IIT Hyderabad,
India



Dr. C. Kamal
RRCAT Indore,
India



Prof. Dhanvir S. Rana
IISER Bhopal,
India



Dr. Dhiraj Bhatia
IIT Gandhinagar,
India



Dr. Himlal Bhatt
BARC,
India



Dr. Gopinadhan Kalon
IIT Gandhinagar,
India



Prof. K. D. M. Rao
IACS Kolkata,
India



Dr. Pratap Kumar Sahoo
NISER Bhubaneswar,
India



Dr. Biswarup Pathak
IIT Indore
India



Dr. N. Sriram Gopal
University of Hyderabad,
India



Prof. Rajesh Kumar
IIT Indore,
India



Prof. Nanasahab Thorat
University of Limerick,
Ireland



Dr. Nishad Deshpande
IIIT Surat,
India



Prof. Pankaj Koinkar
Tokushima University,
Japan



Prof. Pavan Nukala
IISc, Bangalore,
India



Prof. Priya Mahadevan
S. N. Bose Kolkata,
India



Dr. Rajamani Raghunathan
UGC-DAE CSR,
India



Prof. Rajendra S. Dayal
Central Univ. of Rajasthan,
India



Prof. Saket Asthana
IIT Hyderabad,
India



Prof. Samaresh Das
CARE, IIT Delhi
India



Prof. Sandip Ghosh
TIFR, Mumbai,
India



Dr. Sanjoy Kumar Mahatha
UGC-DAE CSR, India



Prof. Somnath C. Roy
IIT Madras,
India



Prof. Soon H. Kang
Chonnam National
University, South Korea



Prof. Suman K. Pal
IIT Mandi,
India



Prof. Sumeet Walia
RMIT University,
Australia.



Prof. Tanushree Choudhury
IIT Bombay,
India



Prof. Toshihiro Moriga
Tokushima University,
Japan



Dr. Varun Natu
CSIR-NCL
India



Dr. Vikas Thakare
DRDE, Gwalior
India



Prof. Vilas Pol
Purdue University,
USA



Prof. Surajit Saha
IISER Bhopal, India



Dr. Som Datta Kaushik
UGC-DAE CSR,
India



Dr. Sandip Patil
E-Spin Nanotech Pvt. Ltd,
India



Prof. Rajendra S. Dhaka
IIT Delhi,
India



Dr. Sanjay Kumar Rai
RRCAT,
India

Contributory Speakers



Dr. Sarika Verma
CSIR- AMPRI, Bhopal,
India



Dr. Tanmoy Paul
TCG CREST, Kolkata,
India



Dr. Vidyadhar Singh
JP University Chapra,
India



Dr. Manoj Kumar Gupta
CSIR-AMPRI, Bhopal,
India



Dr. J. P. Yadav
Shivaji University, Kolhapur,
India



Dr. Harishchandra Singh
Amity University,
India



Dr. Anupma Thakur
IISc, Bengaluru,
India



Dr. Kingshuk Roy
TCG CREST, Kolkata,
India



Dr. Manav Saxena
Jain University, Bengaluru,
India



Dr. Varun Harbola
TIFR Mumbai,
India

Advisory Committee



Prof. K. Kalantar-Zadeh
University of Sydney,
Australia



Prof. Umesh Waghmare
President, JNCASR,
India



Prof. S. B. Ogale
RISE, TCG-CREST, Kolkata
India



Prof. Yuan-Ron Ma
Vice President, Fo Guang
University, Taiwan



Prof. Jin-Hyeok Kim
Chonnam National
University, South Korea



Mr. Umesh Malshe
Former Director RRCAT,
India



Prof. Bikramjit Basu
Director CSIR-CGCRI,
India



Prof. Amlan J. Pal
Ex. Director UGC-DAE-CSR,
India



Prof. Kaustubh Priolkar
Director UGC-DAE CSR,
India



Prof. Robert Weatherup
University of Oxford,
United Kingdom



Prof. T. Venky Venkatesan
University of Oklahoma,
USA



Prof. Anirudha Sumant
Argonne National Laboratory,
USA



Prof. Sebastien Royer
Université du Littoral Côte
d'Opale, France



Prof. Yogendra Mishra
University of Southern,
Denmark



Prof. Kalobaran Maiti
Director IACS Kolkata,
India



Prof. Motohiko Ezawa
University of Tokyo,
Japan



Prof. Sanjay Mathur
University of Cologne,
Germany



Prof. Bharat Jalan
University of Minnesota,
USA



Prof. Babak Anasori
Purdue University,
USA



Dr. R. Balamuralikrishnan
Director DMRL,
India

Presenters

Abstract ID	Title and Presenters	Page No.
2DPL01	2D and Mixed Dimensional Materials in Emergent Energy Applications Satishchandra Ogale	23
2DPL02	Engineering earth-abundant catalysts for efficient electrochemical and photoelectrochemical water splitting Jin Hyeok Kim	23
2DPL03	Emerging Innovation ecosystem in Engineering Ceramics and Glass in India Bikramjit Basu	24
2DPL04	Applications of the Two Dimensional Materials for Energy Storage: Battery Electrodes, Electrolytes and Multi-functional Electrocatalysis Vijayamohanan K Pillai	25
2DPL05	Anomalous spintronic dark excitons in 2D ferromagnetic materials Yuan-Ron Ma	26
2DPL06	New Vibrational Spectroscopies based on Quantum Geometry of Electrons: 2D Materials Umesh V. Waghmare	26
2DIT01	Influence of Electric Field on Excitons in Transition Metal Dichalcogenide Semiconductors Ghosh S	27
2DIT02	A Theoretical Perspective of Time Reversal Symmetry Breaking in Low-dimensional Antiferromagnets Rajamani Raghunathan	28
2DIT03	Engineering 2D MoS ₂ via Composite Design, Intercalation, and Defect Modulation for Efficient Hydrogen Evolution Reaction Soon Hyung Kang	29
2DIT04	Ultra-thin materials for next-generation electronics and optoelectronics technologies Sumeet Walia	30
2DIT05	2D Material-Based Membranes for Desalination, Energy Generation, and Ionic Memristor Applications Kalon Gopinadhan	31
2DIT06	Giant piezo shear drives electronic instabilities and phase transitions in layered ferroelectric In ₂ Se ₃ Pavan Nukala	32
2DIT07	Nanofiber Technology & Its Commercial Application: Future Start-Ups Sandip Patil	32
2DIT08	Next Generation Advanced Materials for NBC Mitigation Dr Vikas Baburao Thakare	33
2DIT09	Role of 2D Materials in Energy Storage and Sensing Applications at CARBON Lab Chandra Shekhar Sharma	34
2DIT10	Strain induced polarization in few layer SrTiO ₃ films Priya Mahadevan	35
2DIT11	1D Metal Oxides – 2D Materials Heterojunctions: Design, Fabrication and Photoelectrochemical Properties Somnath C Roy	35
2DIT12	DNA based programmable 2D materials and films for biological applications Dhiraj Bhatia	36
2DIT13	MXene Surface Chemistry: The Interplay Between Synthesis, Colloidal Stability, and Macroscopic Properties Varun Natu	37
2DIT14	Layered Transition Metal Dichalcogenides Materials for High-Performance THz Sources and Detectors Samaresh Das	37
2DIT15	2D Transition Metal Dichalcogenides for Hydrogen Generation and Metal Air Batteries Dr. Ashish Kumar Mishra	38

2DIT16	2D Nanoparticles in Cancer Theranostics: From Atomic Sheets, AI-Integrated to 3D Tumor Models Nanasaheb Thorat	39
2DIT17	Composition-Structure-Property Relationships in Mxenes Babak Anasori	40
2DIT18	2D/3D Heterointegration: A new take on developing diamond-based electronics Anirudha V. Sumant	41
2DIT19	Ultrafast Sustainable Synthesis of Functional Mxenes Vilas G. Pol	41
2DIT20	Competing Disorder, Charge Ordering, and Electronic Correlations Two-Dimensional Mixed Chalcogenides Sanjoy Kr Mahatha	42
2DIT21	Tailoring of BaRuO ₃ /LaNiO ₃ Superlattice to Induce Ferromagnetism Anjana Dogra	43
2DIT22	Development of electrode materials for cost-effective sodium-ion batteries Rajendra S. Dhaka	44
2DIT23	Defect-Driven Tubular Charge Density Wave and Excitonic Luminescence in 2D Quantum Materials Pratap K. Sahoo	45
2DIT24	Strategies for scalable synthesis of layered semiconductors Tanushree H. Choudhury	46
2DIT25	Transverse quantum transport in non-collinear spin textures: Isolated Skyrmions and Crystals B. R. K. Nanda	47
2DIT26	Probing Hidden Carrier Relaxation Pathways in 2D Materials Using Ultrafast Mid-IR Spectroscopy Sri Ram Gopal Naraharisetty	48
2DIT27	Integration of Two-Dimensional Materials and Bulk Ferroelectric Ceramics in Nanogenerators for Enhanced Mechanical Energy Harvesting Saket Asthana	49
2DIT28	Next-Generation Terahertz Functionalities Derived from Emergent Quantum Materials Dhanvir Singh Rana	50
2DIT29	Computational Search on Two-Dimensional Dirac Materials – From Elemental to Compound Systems C. Kamal	51
2DIT30	Enhancing Electrochromic Energy Storage in W ₁₈ O ₄₉ /Ti ₃ C ₂ T _x Composites for Next-Generation Smart Windows Ashutosh K. Singh	52
2DIT31	Research using Indus and possibilities with 4th generation High brightness Synchrotron Radiation source Dr. Sanjay Kumar Rai	52
2DIT32	IR & Raman Spectroscopic Studies to determine Vibrational, Optical, Electronic Properties of 2d Materials under Extreme Thermodynamic Conditions Himal Bhatt	53
2DIT33	Graph Neural Network for High-Throughput Screening of Nanocluster Electrocatalysts Biswarup Pathak	53
2DIT34	Beyond Multimetal Catalysts: Scalable 2D-Integrated β -FeOOH Electrocatalysts for Alkaline Water Splitting Nishad G. Deshpande	54
2DIT35	Role of Neutron Diffraction in Examining Layered Perovskite and Other Emerging Materials S. D. Kaushik	55
2DIT36	Ultrafast Manipulation of Valley Polarization in Two-Dimensional Transition Metal Dichalcogenides Suman Kalyan Pal	55
2DIT37	Novel Phase Transitions in Two-dimensional Ferroelectric In ₂ Se ₃ Surajit Saha	56

2DIT38	Photocatalytic Properties of Three-Dimensionally Ordered Macroporous Metal Oxynitride Inverse Opals Toshihiro Moriga	57
2DIT39	Two-dimensional materials for on-chip micro-supercapacitor Abha Misra	58
2DIT40	Atomically Precise Metal Nanoclusters for Energy and Pharmaceuticals Rajendra Singh Dhayal	59
2DIT41	Electrochromic materials and device: smart windows for multifunctional applications Rajesh Kumar	60
2DIT42	Strategies to Improve the Photocatalytic Performance of Bismuth Halide-Based Nanocomposites Pankaj Koinkar	61
2DIT43	On-Chip Full-UV-Band Photodetectors Enabled by Hot Hole Extraction K. D. M. Rao	62
2DCT01	3D Printing Of Advanced Multifunctional 2D Mxene-Based Nanocomposites Sarika Verma	63
2DCT02	XANES Simulations for Li ₃ InCl ₆ Solid Electrolyte and interfacial reactions using First-Principles Calculations Tanmoy Paul	63
2DCT03	In-Situ TEM Study on the Microstructural Evolution of Size-Selected Tantalum Oxide at High Temperatures via Carbothermic Reaction Vidyadhar Singh	64
2DCT04	Flexible 2D Materials Based Piezoelectric Nanogenerator Dr. Manoj Kumar Gupta	65
2DCT05	Enhanced HER Activity via Binder-Free Co-MOF Thin Films with Optimized Coordination Environment Jyotiprakash Yadav	66
2DCT06	Theory-Guided 2D Semiconductive Materials for Energy Applications: Insights from Ni ₃ TeO ₆ Harishchandra Singh	67
2DCT07	Compositional Engineering of Two-dimensional MXenes for Electrocatalytic Green Hydrogen Production Anupma Thakur	68
2DCT08	Plating Dynamics in Sodium Metal Anodes: Linking Molecular Interactions to Macroscopic Stability Kingshuk Roy	68
2DCT09	Tailoring the Synergy of Transition Metal-Based V ₂ CTX Nanocomposite for Efficient Hydrogen Evolution Reaction Manav Saxena	69
2DCT10	Interface Design Beyond Epitaxy Varun Harbola	70
2DIL01	Ultra-high vacuum systems for scientific research Kevin Yang	71
2DIL02	Accelerating 2D Materials Research Through Thermo Fisher Scientific's End-to-End Characterization Workflow Dr. B. Karthik	71
2DIL03	Crystal structure, complex phase diagram and re-entrant phase transition in NaSrPO ₄ Dr. Mangesh Mahajan	72
2DIL04	Publishing High Impact Research: Insights from the Editor's Desk Dr. Rohini Kitture	73
2DTH01	Sodium p-Nitrophenolate Dihydrate and Sodium 4-Nitrophenoxide 4-Niitrophenol Dihydrate single crystals for NLO and Terahertz applications Sethupathi Dharmalingam	77
2DTH02	Synthesis and Characterization of Transition Metal Carbide (MAX) and MXene Nanoflakes for Energy Storage and Electrocatalysis Srikanth Ragunath	77
2DTH03	Synthesis of Ultrathin 2D Metal Oxide/Hydroxide Nanosheets and Heterostructure for Energy Storage and Conversion Applications	78

	Sayali Patil	
2DTH04	Design and Synthesis of the Highly Porous Solid Electrolyte for Rechargeable Batteries Aparna Potdar	79
2DTH05	Novel approaches to synthesis and characterization of 2D MXene-based refractory carbides Nithin Chandran B S	80
2DTH06	A Study of MXene Monolayers for Multifunctional Applications Using the Density Functional Theory Nidhi Modi	81
2DTH07	Fabrication of large area two-dimensional PtX ₂ (X=S, Se) heterostructures for broadband photodetector studies Gaurav Bassi	82
2DTH08	Development of Photon Upconversion Materials Incorporated Photocatalysts for Hydrogen Generation Using Visible Light Amit Verma	83
2DTH09	Investigation of Multifaceted Applications of BaSnO ₃ : Ceramic and Thin Film Shikha Shrivastava	84
2DTH10	Atomic-Level Doping at Redox-Active Centers in Polyanion Cathodes for Enhanced Sodium-Ion Storage Sharad Pinjari	85
2DTH11	Solar Light Active Nano Ni Decorated TiO ₂ , Nb ₂ O ₅ and SnO ₂ Photocatalysts for Efficient Hydrogen Generation via Water Splitting Priyanka Birla	86
2DTH12	Perovskite Memristors for In-Memory Sensing and Neuromorphic Computing Mubashir Ganaie	87
2DTH13	IoT-Integrated 2D-Material Functionalized AlGaIn/GaN HEMT for Multifunctional Sensing Vikas Pandey	88
2DTH14	Theoretical Framework and Predictive Modeling for Emergent Topological Phases in Quantum Materials Rahul Verma	89
2DTH15	Study of Preparation of Tight Ultrafiltration and Loose Nanofiltration Membranes via Layer-by-Layer Self-Assembly for Selective Separation Urvashi Joshi	90
2DTH16	Unravelling Efficient And Sustainable Next-Generation Solar Cells Via Doping And Interface Engineering Abhishek Srivastava	91
2DTH17	Fe-based Phosphate Materials for Sodium-ion Batteries Archana R. Kanwade	92
2DOA01	Shape-dependent structural, electronic and transport properties of BP nanowires for next-generation nanoscale interconnect Ankita Nemu	97
2DOA02	2D-SnS ₂ Nanoflake-Enabled 2D-0D Heterostructure for High-Detectivity Ultrawideband Photodetection Sanjeev Yadav	97
2DOA03	On-chip Multifunctional Heterostructure Enabled by Hot Hole Extraction Sougata Karmakar	98
2DOA04	Harnessing Compositional Synergy to Accelerated Charge Transfer in NiS/g-C ₃ N ₄ Heterostructures for Oxygen Evolution Reaction Shivprasad Jadhav	99
2DOA05	Exploring Self-Assembled Mask-Assisted Spatial Defect Engineering in 2D TMDs Gayatri Subhalaxmi	100
2DOA06	Integrated Recovery of Rare Earth Elements and Graphenization of Spent Ion-Exchange Resins via Flash Joule Heating Subham Kumar	101
2DOA07	Half-Metallicity in Stable EuBr ₃ and EuI ₃ Single layers : A DFT+U Study Amrendra Kumar	102

2DOA08	Estimation Of Energy Levels Of Platinum Using Classical And Quantum Framework Vishva C, Hari Haran I, Vijay	103
2DOA09	Structural, Thermal, and Electrical Properties of In-Situ Oxygen-Doped Transition Metal Dichalcogenides Diwakar Singh	104
2DOA10	Studies on cation exchange capacity of different clays for Textile effluent treatment Saroj Sepat	105
2DOA11	Engineering Oxygen Vacancy Channels In Bm-Sco Film Via Strain Rajan Mishra	105
2DOA12	Surface acoustic wave-driven dynamic strain engineering in monolayer MoS ₂ Deena Nath	106
2DOA13	Innovation in Topological Insulator-based Photodetector heterojunction for Metastructure Applications: An Effect of Co layer incorporation Puja Dey	107
2DOA14	Tuning the electronic and transport properties of SrNbO ₃ thin films Pampa Sadhukhan	108
2DOA15	Vacancy Defect Effects on Mechanical and Thermal Properties of Janus WS ₂ Monolayer: A Molecular Dynamics Study Ashwani Kushwaha	109
2DOA16	CNC Micro-Milling as a Democratising Subtractive Patterning Platform for Ti ₃ C ₂ T _x MXene-Coated Substrates: A Conceptual Framework for Low-Cost 2D Material Device Fabrication Lalit Bhanwrela	110
2DOA17	Graphene Oxide-Modified Nanofiltration Membranes for Enhanced Separation Performance and Radiation Stability Bitan Ghosh	111
2DOA18	Leakage-Aware Multi-Property Screening of JARVIS-DFT 2D Materials with Cautious Exfoliation-Energy Analysis Md Efatazzaman Efata	112
2DOB01	MoS ₂ Nanoflowers as SERS Substrate for Methyl Orange Detection Sudin De	113
2DOB02	Co-based thin film electrocatalysts with diverse substrates for oxygen evolution reaction Suraj Bulakhe	114
2DOB03	Hot Carrier Dynamics in Bismuth-based 2D Mixed Halide Perovskite Shivani Malvi	114
2DOB04	2D Chiral Metasurface based All-optical Standard Ternary Inverter Jitendra Roy	116
2DOB05	Divergent Optical and Vibrational Evolution in 2D vs. 1D MoS ₂ Monolayers at Cryogenic Temperatures. Akhil Warriar	117
2DOB06	Synergism of graphene and diamond nanoparticles for improved lubrication characteristics of synthetic oil Dr. Ankush Raina	118
2DOB07	Structural Stability and Compressibility Behaviour of Rare-earth Tritellurides under Pressure Sanitha K	118
2DOC01	Data-driven prediction of atomic hydrogen adsorption energies in two-dimensional materials Tejaswini G	119
2DOC02	ZnMn ₂ O ₄ /MXene Composite as a High-Performance Cathode Material for Hybrid Supercapacitors Pulkrit Garg	120
2DOC03	β-AgVO ₃ @MXene Heterostructures with AI-Driven Electrochemical Informatics for High-Performance Asymmetric Supercapacitors Shwetha K P	121
2DOC04	Manganese (Mn) doping in CoFeS ₂ to Enhance the Supercapacitor Performance Poonam Moar	122

2DOC05	Li-Functionalized 2D SiN Monolayer for Hydrogen Storage Application: A DFT Study Trilokchand Kumavat	123
2DOC06	Exploring the possible connection between elastic properties and ionic /electronic transports in solid polymer electrolyte. Krushna Shewale	123
2DOC07	MoS ₂ @Ti ₃ C ₂ @CNT Ternary Hetero-nanostructured functional Anode for Na-Ion Batteries Ashish Yengantiwar	124
2DOC08	Temperature-Controlled Spray Pyrolyzed NiMn ₂ O ₄ Thin Films on Stainless Steel for Enhanced Performance in MXene-Based Hybrid Asymmetric Supercapacitors Sachin Tanwade	125
2DOC09	Palladium-Gold/MXene Nanocomposite as an Efficient Electrocatalyst for Hydrogen Evolution Reaction Mallamma Jinagi	126
2DOC10	Flexible Binder-Free V-Doped FeOOH on Ni Foam as an Efficient Bifunctional Electrocatalyst for Alkaline Water Splitting Haripal Dhayal	127
2DOC11	Scalable Fabrication of Binder-Free Cobalt Boride Phosphide Electrocatalyst for Green Hydrogen Production Susmita Patil	127
2DOC12	Engineering the MXene-BiCuS Interface for Electrocatalytic CO ₂ Reduction to Value-Added Products Chob Singh	128
2DOC13	Phase Engineered 1T-MoS ₂ Electrodes for High-Performance Aqueous Supercapacitors Shama Parveen	129
2DOC14	2D MnTiO ₃ Nanodiscs For Novel Ammonium Ion Solar Supercapacitor Manopriya Samtham	129
2DOC15	Growth of 2D materials by chemical vapour deposition for supercapacitor and electrocatalytic applications Sourabh Barua	130
2DOC16	2D NiMo LDH-MWCNT Heterostructure as a High-Energy-Storage Electrode for Supercapacitors Rajesh Katru	131
2DOC17	Scalable Ti ₂ CT _x MXene for High-Performance HER in Acidic and Saline Media Suman Yadav	131
2DOC18	Screening Hydrogen Adsorption in p-Block-Doped α -CP Monolayers for HER Applications: A First-Principles Study Shardul Vadalkar	132
2DOC19	Atomic-Scale Interfacial Design of Zn ₂ SnO ₄ /2D g-C ₃ N ₄ Heterojunction for Enhanced Pseudocapacitive Energy Storage Dr. Haribhau Gholap	133
2DOD01	Hydrogen-Assisted Selective NO Reduction on Defective Graphene-Stabilized Ni ₄ Single-Cluster Catalysts Sourav Ghoshal	134
2DOD02	Interfacial 2DEG in Oxide Heterostructures: Unlocking Spintronic and Spin-Caloritronic Functionalities Payal Wadhwa	135
2DOD03	Electrocatalytic Detection of Epinephrine using an Engineered 2D Carbon / Metal-Oxide Disposable Sensor in a Point-of-Care setting Ankita Dutta Chowdhury	136
2DOD04	Integration of InAs@ZnSe nanocrystals with 2D Materials for Infrared Optoelectronics Mukesh Kumar Thakur	137
2DOD05	Tribological and Mechanical Behaviour of Spray-Coated MoO ₃ and MoS ₂ -MoO ₃ Coatings on SS321 Steel Satyam Upadhyay	137
2DOD06	Crystal field splittings in rare-earth honeycomb magnets RCl ₃ (R= Yb, Er) Shivam Parasar	138
2DOD07	Scalable Fabrication and Device Architecture Optimization of MXene Gas Sensors for Wearable Applications	139

	Varsha Biswas	
2DOD08	Endodontic file wear assessment using multi-sensor fusion & CRITIC Apoorv Tripathi	140
2DOD09	Bi-Doped g-C ₃ N ₄ as a Sustainable Electrocatalyst for Ammonia Production Aman Bharti	140
2DOD10	Biological Response of ZX50 Mg Alloy: The Role of Surface Microstructure Sheetal Yamalakonda	141
2DOD11	Architecture-Dependent Emergence of Self-Powered Gas Sensing in Junction Engineered WSe ₂ Device Dipanjana Modal	142
2DOD12	WS ₂ vertically aligned sheet like structure for humidity sensing at room temperature Thameemul Ansari M	143
2DOD13	Deterministic Switching of Graphitic Stacking Order via Superlubric Interfaces Nirmal Roy	144
2DOD14	Gate-Tunable Density-of-States Valve for Charge Injection at a Two-Dimensional Superconductor/Semiconductor Interface Abhishek Patil	144
2DOD15	Redox-Active 2D Layers Metal-Organic Framework as Ion Highways Towards Fast and Reversible Ammonium-Ion Supercapacitors Nissar Hussain	145
2DOD16	Capacitive Pressure Sensing Behaviour of Microporous Dielectric Elastomer Composites Dhananjay Sahu	146
2DPA01	True Strain Estimation and Energy Band Gap Modulation in Silicon Nanomembrane-based 2D Structures Soumya Tiwari	151
2DPA02	Impact of Built-in E-field and Covalency on Excitonic Properties of PtXY (X,Y = S, Se, Te) Janus layers Anjana Sudheer	151
2DPA03	Magnetic and transport properties of a Gd/Co multilayer Srijani Haldar	152
2DPA04	Unravelling the lattice architecture of MnCo ₂ O ₄ Chinmay Gavade	153
2DPA05	COUPLED STRUCTURAL, MAGNETIC, AND ELECTRONIC EVOLUTION IN OXYGEN-DEFICIENT BAFeO _{3-x} THIN FILMS Priti Roy	154
2DPA06	Unleashing the Potential of Layered Nb ₂ CTx MXene Inks Towards the Flexible Electronic Applications Calvin Prakash raj	154
2DPA07	Enhanced Valley Polarization and Valley Lifetime in Fe ₃ O ₄ /MoS ₂ Heterostructure via Magnetic Proximity Laxmi Saran	156
2DPA08	Exciton-phonon coupling in few-layer 2D CrCl ₃ Carlo Sta Maria	156
2DPA09	Anisotropic light matter interaction in 2D α -In ₂ Se ₃ Divya Jangra	157
2DPA10	Machine Learning-Accelerated Design of Near-High-Entropy O ₃ -type Low-Strain Cathodes for Sodium-Ion Batteries Novotna Seal	158
2DPA11	Correlation Between Growth Parameters, Morphology, and Defect Formation in CVD-grown MoS ₂ Rohit Kumar Mahto	159
2DPA12	Modulating the optoelectronic properties of MoS ₂ -WS ₂ verticle heterostructure through different strain mechanisms: a DFT study Shakti Singh	160
2DPA13	Self-Powered Photodetection in a 2D MoS ₂ /MoSe ₂ p-p Heterojunction with Enhanced Current Density O L Lenin	160

2DPA14	Interplay of Ripples, Strain, and Grain Boundaries on Thermal Conductivity of Polycrystalline Graphene Abhishek Kumar	161
2DPA15	Impact of Different Metal Precursors on APCVD Growth of α -MoO ₃ Vedant Chavan	162
2DPA16	Synthesis of Layered Two-dimensional Vanadium Disulfide (VS ₂) by Chemical Vapor Deposition for Optoelectronic Applications Ramesh Rajarapu	163
2DPA17	Oxygen-Tuned MoS ₂ Thin Film: From Surface Chemistry to Humidity Sensing Performance Megha Mishra	164
2DPA18	Synthesis and characterization of PDMS based Ru-Si membrane by spin coating method for dissolved oxygen sensor application Gangadhar Hattale	165
2DPA19	Carbon Surface-Engineered CuO/Fe ₂ O ₃ nano-composite as efficient photocatalyst Purnima Patil	166
2DPA20	Valence Band Engineering in Mg-Doped α -GaCrO ₃ for Efficient Electron-Blocking in Photovoltaic Devices Rishav Sharma	167
2DPA21	Harnessing electric-field-driven interfacial charge modulation to control exchange bias in complex oxides Shivam Choudhary	168
2DPA22	Isolation and Purification of Phycobiliprotein from Nostochopsis Lobatus HKAR-21 AYUSHI Kumari	168
2DPA23	Stabilization of Orthorhombic Ferroelectric Phase in HfO ₂ thin film Mangla Nand	169
2DPA24	Ultrafast water flow in graphene oxide and its implications for environmental applications Sangram Behera	170
2DPA25	Study of Dielectric Tunability of Polycrystalline Barium Strontium Titanate Thin Films Priyanshu Chandrakar	171
2DPA26	Structural & electrical studies of epitaxial LaNiO ₃ thin film grown by RF magnetron sputtering Dhananjay Moharana	172
2DPA27	Ni and Mg Doped α -AlCrO ₃ : Wide Band Gap P-type Transparent Conducting Oxides. Prabhat Singh	173
2DPA28	Reactive Glass–Metal Interaction as a Platform for Designing Hybrid Functional Interfaces Sinorul Haque	174
2DPA29	A study of quantum capacitance in defect engineered WS ₂ monolayer Snehal Shinde	174
2DPA30	Superexchange-Induced Positive Exchange Bias and Field-Cooling Reversal In Co/CoO Multilayer Ambrish Dwivedi	175
2DPB01	Spin-Photon Correlation in Few-Layer AgCrP ₂ S ₆ : Unraveling Antiferromagnetic Order Through Helicity Photoluminescence Denny Pratama	176
2DPB02	Circularly Polarized Photoluminescence Helicity in Stacked CrI ₃ van der Waals Layered Crystals Clara Saragih	177
2DPB03	Photo Luminescence features of Ce doped YAG synthesized by modified sol-gel method Nanda Bhatia	178
2DPB04	Anisotropic Phonon Modes in PLD-Grown α - MoO ₃ Thin film probed by Polarization-dependent Raman Spectroscopy Manish Kumar	178
2DPB05	Simultaneous Extraction of Correlated Fermi Surface Parameters in WTe ₂ via Shubnikov-de Haas Oscillations and Profile Likelihood Analysis Jaydeep Lohia	179

2DPB06	Probing Plasmons in MXenes using surface spectroscopy Yugal Rathod	180
2DPB07	Direct Observation of Electric Field-Driven Protein Translocation Through the Single-Walled Carbon Nanotube Embedded in a Lipid Bilayer Membrane Revealed a High Dipole Moment of the Transition State Saurabh Chaubey	181
2DPB08	Curcumin-Derived (3-Aminopropyl) trimethoxysilane-Functionalized Carbon Quantum Dots: A Colorimetric and Fluorometric Nanoprobe for Antioxidant Activity, Liposome Encapsulation, and Picric Acid Detection Abhishek Abhishek	182
2DPB09	High-Resolution Characterization of Å-nm Scale Pores During the Electrical Breakdown; Modulatory Effect of Sucrose on Lipid Bilayer Integrity Km Mamata Patel	183
2DPB10	Probing Strain-Driven Modifications in Electronic Structure of Epitaxial GdVO ₃ Thin Films Bharathi Palthya	183
2DPB11	Resonant Photoemission Study of PLD-Grown Textured Ti ₃ AlC ₂ Thin Films Apurva Karahe	184
2DPB12	In-situ characterization of temperature-dependent thermal conductivity in MXene (Ti ₃ C ₂ Tx) using Raman spectroscopy Priyam Srivastava	185
2DPB13	Evidence of Spin-phonon Coupling in Magnetic Two-dimensional Van Der Waals Single-crystalline Fe _{3-x} CoxGaTe ₂ Rosni Roy	185
2DPB14	Multiplexed Au/Pt/Pd Nanorattles Enable Electronic Tongue Detection of Breast Cancer Biomarkers in Urine Khushal Singh	186
2DPB15	Layer-Controlled Growth of WS ₂ Films on Al ₂ O ₃ and Interface Band Alignment via HXPES Analysis U. K. Goutam	187
2DPB16	Scan-Number Scaling and Structural Convergence in Fluorescence EXAFS: A Reverse Monte Carlo Study of O ₃ -NaFeO ₂ Chiranjeev Rao	188
2DPC01	PEDOT: PSS induced Supramolecular G-Quartet Hydrogel for Diffusion-Controlled High-Performance Flexible Supercapacitor Yoshita Chakravarty	189
2DPC02	Two-Dimensional Molybdenum carbide as an efficient electrocatalyst for Hydrogen Evolution Reaction Shubham Kushwaha	189
2DPC03	Superior 3D Conductive Na ₃ V ₂ O ₂ (PO ₄) ₂ F/rGO Composite Cathode for Advanced Sodium-Ion Batteries Vishal Khetmalis	190
2DPC04	Osmotic Energy Conversion using MoSe ₂ /MoS ₂ Heterostructure Nanopores in Multi-Ion Environments Mukesh Kumar	191
2DPC05	Thermal Stability of 2D PtSSe-SnSSe Janus Lateral Heterostructures – A DFT Study Karthikeyan C	192
2DPC06	A-Site Cu Engineering in Cs ₃ Bi ₂ I ₉ -Derived Perovskite-Inspired Materials for Enhanced Charge Transport and Magneto-Optoelectronic Performance Gopal Sa	193
2DPC07	Screening of Energetically Stable High Entropy Halide Perovskites Using Cluster Expansion Method for Optoelectronic Devices Vinay Bhatt	194
2DPC08	Fluorophosphate: A Promising Cathode for Sodium-Ion Battery Arshiya Pathan	195
2DPC09	Top-Down Synthesis of Ti ₂ CTX MXene from Ti ₂ AlC Precursor for Advanced Electrochemical Energy Storage Suguna Palanimuthu	195

2DPC10	From Jackfruit Waste to Supercapacitor Electrode Neha Joshi	197
2DPC11	Towards defect-free laser-induced graphene coating on copper foil for anode-free Li metal batteries Aarti Gunjal	197
2DPC12	Structural and Thermoelectric Properties of High-Entropy Spinel Oxide Jitendra Kumar	198
2DPC13	Impact of Different Solvents on the Structural, Optical, and Photocatalytic Performance of Pristine Bi ₂ S ₃ Priyanka Jangra	199
2DPC14	Tailoring defects in solution-processed SnO ₂ for enhanced interfacial charge dynamics in perovskite solar cells Yash Taneja	199
2DPC15	Surface-Modified Pt/Al ₂ O ₃ Nanocatalysts for Hydrogen Recombination application in Humid Environments Sushmita Bhartiya	200
2DPC16	Enhancing Mechanical Energy Harvesting with Robust Sponge TENG Parallel Arrays Shubham Thakur	201
2DPC17	Multimodal Computational Insights Into Hydrogen Evolution Reaction Mitigation in Zn Electrolytes: The Role of Anion-Surface Coupling Soumya U K	202
2DPC18	Machine Learning-Driven Multiscale Computational and Experimental Investigation of Graphite-Based Anodes for Lithium-Ion Batteries Antra Mohini	202
2DPC19	Nitrogen Ion Irradiation-Induced Defect Engineering of Ti ₃ C ₂ T _x MXene for Supercapacitor Applications Gauri Salunkhe	203
2DPC20	Graphene Oxide Integrated P (VDF–TrFE) Nanocomposite Films: Studies on Structural, Electrical and Dielectric properties for Energy Harvesting Applications Sampa Sutradhar	204
2DPC21	Type II Based TiSe ₂ /TiO ₂ Heterostructures for Hydrogen Evolution Reactions (HER) Laveena Pursharathi	205
2DPC22	Effect of Epitaxial Strain on the Structural and Magnetic Properties of BiCrO ₃ Thin Films Bharti Ojha	206
2DPC23	Thermo-Mechanical Hybrid Piezo-Pyroelectric Nanogenerator Based on Graphene-Reinforced BCZT-PVDF Composite for Enhanced Energy Harvesting Abhijit Das	207
2DPC24	Layer-Engineered MoS ₂ -Integrated Photo-Triboelectric Nanogenerator for Dual Energy Harvesting and Detection Rudresh Kundugolmath	208
2DPC25	Role of Fe doping on OER activity of Ni ₃ N thin films Shivani Sahu	208
2DPC26	Development and evaluation of transition metal compound-carbon based composites as electrodes for supercapacitor applications Shwetha K P	210
2DPC27	Machine Learning-Guided Electrochemical Behaviour of NiV ₃ O ₈ @MXene Asymmetric Supercapacitors Sudha Kamath	210
2DPC28	Study of Silver-Based Catalyst for Selective Catalytic Reduction of NO _x Using Hydrogen: Water Tolerance Vinod Lohar	211
2DPC29	From Pre-Catalyst to Active Phase: DFT Insights into Surface Evolution of 2D TMPs toward β-NiOOH Interfaces for OER Rohan Arya	212
2DPC30	Ligand-Tuned Co-MOF Thin Films Electrode for Efficient and Scalable Hydrogen Production Reshma Khandekar	213

2DPC31	Photo-Response of Pristine and In-doped ZnO Thin Films prepared via a Cost-effective Spray Pyrolysis using a Medical Nebulizer Mohd Vasim	214
2DPC32	Electrocatalytic OER Activity of Cr Oxyhydroxides, from Molecular Clusters to Surfaces: a DFT Study Shaonsikta Chatterjee	215
2DPC33	Bidirectional Catalytic Functionality of $\text{Ti}_3\text{C}_2\text{-F}$ MXene to Hydrogen Evolution and Storage: Theoretical and Experimental Study Ekta Choudhary	216
2DPC34	Sustainable Hydrogen Generation Through Ammonia-Assisted Water Electrolysis Using Layered Double Hydroxide (LDH) Catalysts Sanjivani Papal	217
2DPC35	Hierarchical Ni-MOF@ $\text{Ti}_3\text{C}_2\text{T}_x$ MXene composite for efficient bifunctional electrocatalytic water splitting Ajay Patil	218
2DPC36	Mechanistic Insights into $\text{CoZn}(\text{VO}_4)_2$ Anodes Using In-Situ and Post-Cycling Analyses Mohd Saqib	218
2DPC37	Supercapacitive Applications of Cobalt and Nickel Derived 3D Graphene Synthesized by Thermal Chemical Vapor Deposition Anshuman Verma	219
2DPC38	Enhancing Cycle Life with solution-processed Laminar Li_3VO_4 Anodes for Li, Na, and aqueous Zn-Ion Batteries Tejveer Anand	220
2DPC39	Unraveling growth dynamics of $\text{Sn}_3\text{O}_4/\text{FTO}$ photoanode for photoelectrochemical water oxidation under directional illumination Aayushi Miglani	221
2DPC40	Elucidating Lithium and Sodium-ion Storage Mechanisms in Layered $\text{Na}_2\text{Ti}_3\text{O}_7$ Akshita Sharma	222
2DPC41	Atomic-Level Engineering of Ni-MOF via Cobalt Incorporation: Structural insights and Enhanced supercapacitor electrode Ankit Yadav	223
2DPC42	3D printed $0.937(\text{Bi } 0.5 \text{ Na } 0.5) \text{TiO}_3 - 0.063\text{BaTiO}_3$ (BNT-BT) for Piezo/photo-catalysis Abhishek Shukla	224
2DPC43	$\text{MoS}_2@\text{Ti}_3\text{C}_2@\text{CNT}$ Ternary Heterostructure as a High-Performance Anode for Sodium-Ion Batteries Trupti Aher	224
2DPD01	Investigation of water retention capacity & swelling kinetics of biocompatible electrospun nanofibers for enhanced wound healing Nisha Kumari	225
2DPD02	Theoretical Investigation on the Magnetic Properties of 2D CrSSe Janus Material Under Tensile Strain Abhinav Gotmare	226
2DPD03	Novel 2D $\text{LiInP}_2\text{Se}_6/\text{WSe}_2$ Heterostructure for Efficient Photocatalytic Water Splitting: First-Principle insights Divyanshu Mishra	227
2DPD04	Molten Salt-Assisted Low-Temperature Synthesis and Characterization of MoAlB MAB Phase and 2D MoB MBene Nanosheets Muhammed Shabeer K V	228
2DPD05	CVD-Grown 2D SnS_2 Memristors for Low-Power Non-Volatile Memory Applications Swaraj Mukherjee	229
2DPD06	Synaptic Performance of 2D Bismuth Oxyiodide (BiOI) Based Memristors for Neuromorphic Computing Saiprasad Sawant	230
2DPD07	Ultrafast Nonlinear Optical Response of Plasmonic-Au Nanoparticles Decorated 2D- V_2O_5 Nanoflakes at Near-IR Wavelengths Shivani Vijay Tiwari	230

2DPD08	Zinc Aluminium Layer double hydroxide- A 2D material for the electrochemical detection of Bacillus Anthracis Diksha Singh	231
2DPD09	Exploring Family of 2D Transition Metal Phosphides Towards Spintronics Application: A DFT Study Hardik Chavadiya	232
2DPD10	IR-Reflective CoAl ₂ O ₄ Blue Pigments: Role of pH and Mineralizers. Samruddhi Mulay	233
2DPD11	Fabrication of 2D TMD-based Electrode for Multianalyte Detection of Neurotransmitters Amit Rana	233
2DPD12	A highly sensitive and efficient room-temperature sensing material for NH ₃ gas: 2D Terraced Microplate MoO ₃ Trupti Patel	234
2DPD13	Defect Engineering in Few-Layer h-BN and MoO ₃ via Oxygen Ion Irradiation for Single-Photon Emission samruddhi kushwaha	235
2DPD14	Selective Detection of Isopropanol Using CuO-Doped SnO ₂ Nanocomposites at Ambient Temperature Sreebalan K	236
2DPD15	Te-Rich MoTe ₂ /MWCNT Heterojunctions with Interface-Induced Carrier Inversion for Ultra-Low-Level Bipolar Detection of NO ₂ and NH ₃ Virendra Choudhary	237
2DPD16	MoS ₂ -Driven Conductive Networks in 3D Porous PDMS for High-Performance Flexible Piezoresistive Detection over a wide pressure range Kiran Kumbhar	238
2DPD17	Performance Benchmarking of CNT and Graphene Nanoplatelet Reinforced PVA Conducting Gel for Self-Powered Artificial Skin Bablesh Gupta	239
2DPD18	All-Carbon Nanostructured Membranes Incorporating Carbon Quantum Dots for Antimicrobial Filtration Technologies Ragini Ahiwale	240
2DPD19	AI-Driven CNT based Sensor Arrays for Selective VOC Detection in Wearable Health Monitoring Platforms Rohan Rohilla	241
2DPD20	Au/ZnO Nanofilm E-Textile Sensor for Respiratory Humidity Monitoring Anupam Kumari	241
2DPD21	Two-Dimensional Halide Perovskite-Based Flexible Memristor Devices for Neuromorphic Applications Koushik Gayen	242
2DPD22	Development of Novel Cost-efficient Environment Friendly High Performing Sulfide Based Thermoelectric Material Pritam Dey	243
2DPD23	Electrochemical Properties of Samarium-Substituted Mn Ferrites Prepared by Sol-Gel Auto-Combustion Technique for Supercapacitor applications Balbhim Maharnavar	244
2DPD24	Ultrasensitive Detection of Mercury Ions Using BaZnV ₂ O ₇ @ Ti ₃ C ₂ Tx MXene Functionalized AlGaIn/GaN HEMT Sensor Pooja Kumari	245
2DPD25	Comprehensive Assessment of GAA, DG, and GAP Vertically Stacked 2-D Nanosheet FETs for Advanced Technology Nodes Aruna Patel	245
2DPD26	Observation of Resonant Tunneling Phenomena through the intrinsic valleys of Monolayer MoS ₂ Akash Sahu	246
2DPD27	Frequency-Selective EMI Shielding with Near-Zero Transmission via Tunable Attenuation in Magnetic-Conductive Layered Structures Prateesh Samal	247

2DPD28	Performance Evaluation and Comparative Analysis of Ternary Circuit based on CNTFET and CMOS Vaibhav Nagrale	248
2DPA31	Sulfur-driven stoichiometric control of tungsten disulfide thin films grown via sulfurization of nanoscale tungsten layers Md Samim Reza	249

SPEAKERS' ABSTRACTS

2D and Mixed Dimensional Materials in Emergent Energy Applications

Satishchandra Ogale

*Research Institute for Sustainable Energy, TCG-CREST, Sector V, Salt Lake, Kolkata-700091,
India*

*Indian Institute of Science Education and Research, Dr. Homi Bhabha Road, Pune-411008,
India*

Email: satishogale@gmail.com

Since the isolation of single layer graphene in 2004 and the discovery of its immensely intriguing properties, research in the field of 2D materials has witnessed a tremendous upsurge during the past two decades. Numerous 2D materials belonging to different families of materials systems have been synthesized, and their science and applications have been intensely pursued. Coupling of these interesting materials with 0D (quantum dots), 1D (nanotubes) and 3D forms of functional materials has led to a host of new materials designs with unlimited range of application-worthy properties. Such mixed dimensional materials have made significant inroads into the arena of clean energy science, which includes energy harvesting (next generation solar cells, green hydrogen, CO₂ and N₂ to value chemicals etc.), storage (batteries and supercapacitors), and conservation (LEDs). In my talk, I will discuss these developments using several examples derived from own research.

References:

1. J. Phys. Energy 8, 015007, 2026,
2. Small e 04995, 2025,
3. Small 21, 2411355, 2025,
4. Ad. Mater. Tech. 10, 2401868, 2025, .
5. ACS Appl. Mater. Interfaces 14, 37076, 2022,
6. Langmuir 38, 3139-3148, 2022,

Engineering earth-abundant catalysts for efficient electrochemical and photoelectrochemical water splitting

Mayur A. Gaikwad, Jin Hyeok Kim*

*Optoelectronics Convergence Research Center and Department of Materials Science and Engineering, Chonnam National University, 300, Yongbong-Dong, Buk-Gu, Gwangju 61186,
South Korea.*

*Corresponding Author E-mail: jinhyeok@chonnam.ac.kr

Water splitting has emerged as a highly promising route for sustainable hydrogen (H₂) production due to its potential for generating high-purity fuel with zero carbon emissions. Currently, noble metal-based catalysts such as IrO₂ and RuO₂ are widely employed for the oxygen evolution reaction (OER), while Pt remains the benchmark for the hydrogen evolution reaction (HER). However, their high cost and scarcity limit large-scale implementation, highlighting the

urgent need for efficient, stable, and cost-effective photo- and electrocatalysts. Recent studies have demonstrated the potential of earth-abundant materials in improving catalytic activity and stability [1]. In this context, my laboratory has developed simple, low-cost, and ultrafast room-temperature deposition techniques for synthesizing various metal oxides, oxyhydroxides, and layered double hydroxides (LDHs) as efficient catalysts for water oxidation. The overall efficiency of water splitting is often hindered by sluggish OER kinetics, which limits HER performance and reduces energy conversion efficiency. To address this challenge, alternative anodic reactions such as methanol, urea, and hydrazine oxidation have been explored, offering lower theoretical overpotentials, improved energy efficiency, and the generation of value-added by-products.

This talk highlights recent advancements in catalyst design and architectural engineering for high-performance photo/electrocatalysts for overall water splitting systems. Particular emphasis will be placed on photoanode development for photoelectrochemical water splitting, focusing on surface, bulk, and interface engineering strategies to enhance charge transport and catalytic activity [2,3]. These approaches significantly improve water oxidation performance and device stability. Furthermore, key challenges and future research directions for the development of next-generation, earth-abundant photo/electrocatalysts are highlighted, providing insights into scalable and efficient hydrogen production technologies.

Keywords: Water splitting; Earth-abundant catalysts; hydrogen evolution reaction (HER); oxygen evolution reaction (OER)

References:

1. Mayur A Gaikwad, Vishal V. Burungale, Dhanaji B. Malavekar, Uma. V. Ghorpade, Umesh P. Suryawanshi, Suyoung Jang, Xinyao Guo, Seung Wook Shin, Jun-Seok Ha, Mahesh P. Suryawanshi, and Jin Hyeok Kim, *Advanced Energy Materials*, 14, (2024), 2303730.
2. Mayur A. Gaikwad, Umesh P Suryawanshi, Uma V Ghorpade, Jun Sung Jang, Mahesh P Suryawanshi, and Jin Hyeok Kim, *Small*, 18, (2022), 2105084.
3. Suyoung Jang, Mayur A Gaikwad, Hongjae Shim, Tae Ei Hong, Maheswari Arunachalam, JunHo Kim, Soon Hyung Kang, Jong-Sook Lee, Jan Seidel, Xiaojing Hao, and Jin Hyeok Kim, *Materials Today Physics*, 57, (2025), 101825.

Emerging Innovation ecosystem in Engineering Ceramics and Glass in India

Bikramjit Basu

*CSIR- Central Glass and Ceramic Research Institute, Kolkata, India
(on deputation from Indian Institute of Science, Bangalore 560012, India)*

India is currently globally considered as a major innovation hub in the global research and development landscape. In the paradigm of advanced materials, the last few decades have witnessed impressive progress in the usage of Glasses and Ceramics from refractories in steel industries, to strategic (space, atomic energy, defense) and healthcare sector. Against this backdrop, I shall present a few illustrative examples of the translational research and technology development in Glasses and Engineering Ceramics, based on the work carried out over last several

decades at CSIR-CGCRI. Those examples are related to the niche technologies based on layered materials (graphene, Mxene), space-grade multifunctional glasses, glass fiber-based lasers and sensors, glass foam bricks, advanced materials for green energy generation and storage. It will be presented as how the translation of these technologies will be benefitted by creating robust translational ecosystem connecting industries, strategic sectors, national labs and academia. Towards the end of my lecture, I shall present the recent efforts in the computational materials and data science domain, with a few examples of the glass design, single atom catalysis or sensor-related applications. This lecture will conclude with a few examples of policy related changes that I have introduced in CSIR-CGCRI.

Applications of the Two Dimensional Materials for Energy Storage: Battery Electrodes, Electrolytes and Multi-functional Electrocatalysis

Vijayamohanan K Pillai

IISER – Tirupati, A P – 517619

vijay@iisertirupati.ac.in

Many daunting challenges in the generation, interconversion and storage of clean energy depends on materials innovation. In this regards, many 2D materials have received considerable attention in recent times blending the synthetic skills of Chemists with atomic molecular level characterization tools of condensed matter physicists in facilitating many applications in electrocatalysis and electrochemical energy storage. My group dabbled up on diverse type of materials including zero dimensional (quantum dots), one dimensional (Carbon nanotubes), and two dimensional (phosphorine) systems and their composites tailored for energy storage applications emphasising on their size/shape dependent properties. For example, most of the Two Dimensional materials have a finite van der Waals gap (v-gap) in between the layers which provides a clear separation between the in-plane and through plane electronic interactions. Although the carrier transport and associated unique features are studied in accurately prepared and precisely characterized monolayers, many applications use multilayers which are conveniently prepared using inexpensive and scalable chemical methods. For example, among various two-dimensional (2D) materials, graphene possesses similar interlayer spacing of multiwalled carbon nanotubes (0.32–0.35 nm) which could be modulated by several strategies such as the intercalation of cations and anions, the nature of the solvent, and surface functionalization. The resultant graphene nanoribbons (~40 nm diameter) formed by the ionic liquid-assisted transformation of multiwalled carbon nanotubes of an average diameter (~21 nm) along with graphene quantum dots having an emission wavelength of 445 nm are more useful for some of the specific applications. The tunability of this spacing depends on the nature of the ionic liquid, the size of the incoming ions, the applied potential, and the time. In this talk several examples of electrochemically prepared 2D materials, their heteroatom doping and surface modification possibility will be discussed using both elemental and compound 2D quantum dots like Graphene, Phosphorene, carbon nitride, Molybdenum Sulphide, Bismuthene and clay based 2D composites for specific applications like electrocatalysis and battery storage.

Keywords: 2D Materials, Quantum Dots, Energy Storage, Na ion Batteries, Composite Electrolytes

References:

1. Domain Size Modulation of Clay–Nanocellulose Composites for Enhanced Na-Ion Transport, Sneha Mandal et al., ACS Appl. Energy Mater., 8/9 (2025) 6071–6086
-

Anomalous spintronic dark excitons in 2D ferromagnetic materialsYuan-Ron Ma^{1,2}¹*Department of Applied Informatics, Fo Guang University, Jiaosi, Yilan, Taiwan*²*Department of Physics, National Dong Hwa University, Shoufen, Hualien, Taiwan,*

*Corresponding Author E-mail: yuanronma@gmail.com

Anomalous spintronic dark excitons, characterized as parallel-spin electron-hole pairs existing at zero magnetic field, exhibit unique physical properties distinct from traditional bright excitons [1-5] and can be directly observed using photomodulated reflectance (PR) spectroscopy. Dark excitons consisting of antiparallel pairs of spin-up (or spin-down) electrons and spin-down (or spin-up) holes have a spintronic character, so there are repulsive forces between spin-up (or spin-down) electrons and spin-down (or spin-up) holes to keep the dark excitons exist much longer than typical free and bound excitons. This is because the free and bound excitons comprised of excited electrons and holes can have the electron-hole recombination to emit photoluminescence (PL), while the spintronic dark exciton have no PL release. The flat conduction band at Fermi level plays an important role in the selection rule for the anomalous spintronic dark excitons, because spin-up (or spin-down) electrons stay there easily.

Keywords: Dark exciton; photomodulated; reflectance; spintronic repulsion;**References:**

1. A. M. Jones, H. Yu, N. J. Ghimire, S. Wu, G. Aivazian, J. S. Ross, B. Zhao, J. Yan, D. G. Mandrus, D. Xiao, W. Yao, and X. Xu, Nat. Nanotechnol., 8 (2013) 634-638
 2. X.-X. Zhang, T. Cao, Z. Lu, Y.-C. Lin, F. Zhang, Y. Wang, Z. Li, J. C. Hone, J. A. Robinson, D. Smirnov, S. G. Louie, T. F. Heinz, Nat. Nanotechnol., 12 (2017) 883-888.
 3. V. Mapara, A. Barua, V. Turkowski, M. T. Trinh, C. Stevens, H. Liu, F. A. Nugera, N. Kapuruge, H. R. Gutierrez, F. Liu, X. Zhu, D. Semenov, S. A. McGill, N. Pradhan, D. J. Hilton, D. Karauskaj, Nano Lett., 22 (2022) 1680-1687.
-

New Vibrational Spectroscopies based on Quantum Geometry of Electrons: 2D Materials

Umesh V Waghmare

*Theoretical Sciences Unit
JNCASR, Bangalore 560 064*

We first present an introduction to quantum geometry of electrons in periodic structures in terms of Berry phases and curvature, which cause bending of electronic trajectories that can be

observed with transport measurements. Time-reversal and inversion symmetries of a periodic system however force the quantum phase space of its electrons to be *flat* with vanishing curvature. With first-principles theoretical analysis, we predict that coupling of dynamical excitations, such as phonons, with electrons can induce oscillations in the quantum geometric curvature with observable signatures in Hall measurements. Illustrating these ideas with examples of 2D materials, we propose a new kind of vibrational spectroscopy technique making system-specific predictions (R Bhuvaneswari et al 2024). This spectroscopy can be realized experimentally with scattering of electromagnetic waves or using circuits. It can be employed in analysis of wider range of systems by interfacing them with a quantum material like graphene, thus opening up new kind of applications of quantum 2D materials wherein they are used as a substrate to sense vibrations of a system in the neighbourhood. We present partial validation with experiments carried out by Deshmukh et al (2026).

Work done in collaboration with R Bhuvaneswari and the group of Mandar M Deshmukh at TIFR Mumbai.

Influence of Electric Field on Excitons in Transition Metal Dichalcogenide Semiconductors

Jindal V¹, Kumar S¹, Jana D¹, Bhuyan S¹, Deilmann T², Ghosh S^{1*}

¹*Dept. of Condensed Matter Phys. and Materials Sci., Tata Institute of Fundamental Research, India*

²*Institut für Festkörpertheorie, Universität Münster, Germany*

*E-mail: sangho10@tifr.res.in

Excitons in transition metal dichalcogenide (TMDC) semiconductors have large binding energies compared to $k_B T \sim 26$ meV at room temperature. Therefore for opto-electronic device applications, it is important to understand how excitons in TMDCs are influenced by an electric field. Optical spectrum of TMDCs can have several exciton-like resonances. How such excitons behave under an electric field also gives us clues about their origins.

Here we first show that in the absorption and reflectance spectrum of bulk 2H-WS₂ there is only a single strong A₁s-exciton resonance around its direct bandgap energy. However when we perturb the sample with a small voltage signal and measure the change in reflectance, i.e. perform electro-reflectance (ER) spectroscopy, we begin to see a second strong resonant feature. This is also seen if the sample is electrically perturbed indirectly through a second pump laser beam, i.e. through photo-reflectance (PR) spectroscopy. We argue that this new resonance arises from the first excited state A₂s exciton of 2H-WS₂ [1]. Thereafter using PR spectroscopy we show that in bulk 2H-MoS₂ there exists an unusual exciton state, referred to as an inter-layer exciton, where the electron and hole reside in adjacent Mo-S layers [2]. Such a state had earlier been identified in bilayer 2H-MoS₂. We then go on to show that the interlayer exciton in bulk 2H-MoS₂ shows large Stark splitting under electric fields, due to their possessing an unusually large electric dipole moment [3]. These results were supported by calculations based on density functional theory, including electron-hole Coulomb interaction via the GW + Bethe-Salpeter equation.

Finally we discuss if large electric fields can actually break up excitons into free electrons and holes. There are reports of observation of Franz-Keldysh oscillations (FKO) in

TMDC semiconductors under large electric fields. We have studied bulk 2H-MoS₂ using ER spectroscopy under increasing dc electric field bias. We find that as the field increases the individual exciton resonances evolve into a regular oscillatory feature. We analyse these in detail to understand if they actually represent FKOs, thereby indicating electric field induced ionization of excitons.

Keywords: Transition Metal Dichalcogenide semiconductors; Excitons; electro-reflectance spectroscopy.

References:

1. V. Jindal, S. Bhuyan, T. Deilmann, and S. Ghosh, Phys. Rev. B 97, (2018), 045211.
 2. V. Jindal, D. Jana, T. Deilmann and S. Ghosh, Phys. Rev. B 102, (2020) 235204.
 3. V. Jindal, T. Deilmann and S. Ghosh, Phys. Rev. B 107, (2023), L241201.
-

A Theoretical Perspective of Time Reversal Symmetry Breaking in Low-dimensional Antiferromagnets

Rajamani Raghunathan

UGC-DAE Consortium for Scientific Research, DAVV Campus, Indore.

E-mail: rajamani@csr.res.in

This talk will examine how spatial symmetries—namely translation, rotation, and reflection—together with their spin counterpart, time-reversal symmetry (TRS), and their combined operations, govern spin-splitting in the electronic band structures of antiferromagnets (AFMs). In conventional AFMs, the combined operation of TRS and a lattice translation is preserved, thereby prohibiting spin-splitting of the energy bands. However, in certain systems this symmetry is broken unless supplemented by an additional rotational operation. Such materials belong to a recently identified class of AFMs known as altermagnets, which exhibit anisotropic spin-splitting between up- and down-spin bands, reminiscent of Rashba-type effects, yet notably arising in the absence of spin-orbit coupling. These concepts will be illustrated through the electronic structure of a two-dimensional bipartite lattice, as well as in hybrid halide perovskites. Furthermore, in systems where the unit cell hosts spin motifs that are not related by translation, rotation, or inversion symmetries, an alternative scenario emerges: isotropic spin-splitting akin to that observed in ferromagnets, but without a net magnetization. Representative results from ladder systems will be discussed. Such spin-split AFMs are expected to exhibit functionalities including spin current generation, the spin Hall effect, and magnetoresistance, positioning them as promising candidates for low-dissipation spintronic applications.

Keywords: Spin-split antiferromagnets; Altermagnets; Hubbard model; 2-d Bipartite lattice; ladder systems.

References:

1. D. Dey, S. Choudhary, S. Ramasesha and R. Raghunathan, Phys. Rev. B, 111, 214439 (2025).
2. S. Jana, D. Dey, M. Kumar, S. Ramasesha and R. Raghunathan, arXiv:2507.09502

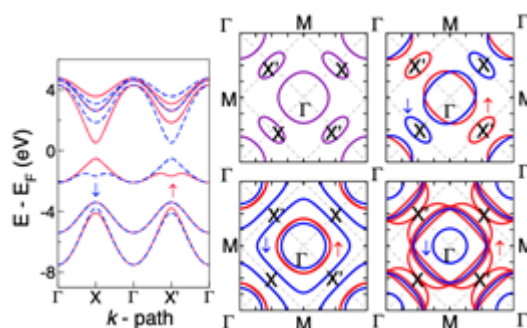


Figure: Band structure and Fermi surface of bipartite lattice in the charge-ordered antiferromagnetic phase exhibiting anisotropic spin splitting.

Engineering 2D MoS₂ via Composite Design, Intercalation, and Defect Modulation for Efficient Hydrogen Evolution Reaction

Jyoti Ganapati Badiger and Soon Hyung Kang*

*Department of Chemistry Education and Optoelectronic Convergence Research Center,
Chonnam National University, Gwangju 61186, Republic of Korea.*

*Corresponding Author E-mail: skang@jnu.ac.kr

Two-dimensional (2D) transition metal dichalcogenides, particularly MoS₂, have emerged as promising electrocatalysts for the hydrogen evolution reaction (HER) due to their tunable electronic structure, layered morphology, and abundant active sites [1]. The semiconducting 2H exhibits limited conductivity, with catalytic activity primarily confined to edge sites, whereas the metallic 1T phase offers enhanced conductivity and abundant active sites across both basal and edge planes [2]. However, the intrinsic limitations of pristine MoS₂—including low conductivity, inactive basal planes, and the tendency of the 1T phase to restack—hinder its overall catalytic efficiency. In this study, three distinct strategies are systematically employed to improve HER performance. First, a MoS₂/MnMoO₄ composite based on the 2H phase is designed, where the synergistic interaction between MoS₂ nanosheets and MnMoO₄ enhances charge transfer and increases the number of accessible active sites without altering the intrinsic phase. Second, electrochemical proton intercalation is utilized to induce and stabilize the metallic 1T phase, accompanied by interlayer expansion, thereby improving conductivity and optimizing hydrogen adsorption kinetics. Third, defect engineering in 1T MoS₂ is achieved through in situ Ni leaching, which introduces lattice distortion and electronic modulation, further boosting intrinsic activity and accelerating reaction kinetics. Collectively, these approaches demonstrate that precise control over phase, interlayer structure, and defect chemistry in 2D MoS₂ enables a significant enhancement in HER activity, as evidenced by the reduced overpotentials of 153 mV_{RHE}, 187 mV_{RHE}, and 8 mV_{RHE} for MoS₂/MnMoO₄, intercalated MoS₂, and Ni-MoS₂, respectively. This highlights a comprehensive strategy for designing efficient and scalable electrocatalysts for sustainable hydrogen production.

Keywords: 2D material, MoS₂, Electrocatalyst, Hydrogen Evolution reaction.

References:

1. X.-L. Fan, Y. Yang, P. Xiao, W.M. Lau, J.Mater.Chem.2(48),2014,20545-20555.
 2. H. Sun, Z. Yan, F. Liu, W. Xu, F. Cheng, J. Chen, Advanced materials.32(3),2020,1806326.
 3. K. Leng, Z. Chen, X. Zhao, W. Tang, B. Tian, C.T. Nai, W. Zhou, K.P. Loh, ACS Nano.10(10),2016, 9208-9215.
-

Ultra-thin materials for next-generation electronics and optoelectronics technologies

Sumeet Walia

*Centre for Opto-electronic materials and sensors, School of Engineering RMIT University,
Melbourne, Australia*

E-mail: sumeet.walia@rmit.edu.au

Atomically thin materials possess unique intrinsic properties and offer exceptional tunability across electronic, optical and mechanical domains. In this talk, I will present our approach to harnessing these properties, guided by application needs and co-designed with industry to accelerate translation into end-user technologies.

We synthesise a diverse range of atomically thin metal oxides, mono- and dichalcogenides, and elemental 2D materials using solid, liquid and vapour phase techniques tailored to targeted applications. Through defect engineering, we have demonstrated neuromorphic vision and processing sensors, while hybridising dissimilar materials has enabled enhanced electronic and optical performance. For example, ultra-thin layers have been leveraged to develop one of the miniaturised photodetectors, capable of sensing across the UV-infrared spectrum.

More recently, we have explored defect formation and passivation strategies in monolayer transition metal dichalcogenides for optoelectronic and sensing applications. Through a cross-disciplinary approach, we deploy the multifunctionality of these material systems to address technological challenges faced by industry partners across multiple sectors.

Translation of these discoveries into real-world sensing solutions is enabled through IC-SenseE, the ARC Industrial Transformation Hub for Intelligent Contaminant Sensing in Complex Environments where we integrate advanced materials, devices and system-level engineering to deliver deployable sensor platforms for environmental, industrial and health monitoring applications.

Keywords: Two-dimensional materials, optoelectronics, defects

References:

1. Ahmed, Taimur, et al. "Fully light-controlled memory and neuromorphic computation in layered black phosphorus." *Advanced Materials* 33.10 (2021): 2004207.
2. Abidi, Irfan H., et al. "Oxygen driven defect engineering of monolayer MoS₂ for tunable electronic, optoelectronic, and electrochemical devices." *Advanced Functional Materials* 34.37 (2024): 2402402.

3. Verma, Ajay Kumar, et al. "Oxygen-passivated sulfur vacancies in monolayer MoS₂ for enhanced piezoelectricity." *ACS nano* 19.3 (2025): 3478-3489.
-

2D Material-Based Membranes for Desalination, Energy Generation, and Ionic Memristor Applications

Kalon Gopinadhan

*Department of Physics & Materials Engineering, Indian Institute of Technology Gandhinagar,
Palaj, Gujarat 382055*

Sub-nanometer protein channels are essential for sustaining biological life. For instance, aquaporins selectively transport water molecules while rejecting ions, including protons. Conversely, potassium and calcium channels selectively transport K⁺ and Ca²⁺ ions, respectively, while excluding others. Mimicking these biological channels is crucial for advanced technologies such as desalination, chemical purification, metal recovery, and reverse electrodialysis.

However, state-of-the-art commercial membranes face severe bottlenecks, including low water flux, poor ion rejection, susceptibility to fouling, and high energy consumption. To address these challenges, we incorporated vermiculite, a 2D material, as an interlayer within a polyamide matrix. This integration induces the formation of Turing patterns, which significantly enhance both water flux and ion rejection by increasing the membrane's effective surface area. Furthermore, these composite membranes exhibit superior antifouling properties compared to pristine polyamide membranes due to an increased surface charge density.

Beyond filtration, we explored the utility of vermiculite membranes for osmotic energy harvesting by exploiting the salinity gradient at the intersection of river water and seawater. Featuring nanochannel dimensions of 0.3–0.5 nm, these membranes demonstrate exceptional cation selectivity and high ionic conductance, yielding a power density that far exceeds the commercial benchmark of 5 W/m². Finally, by engineering these membrane channels into asymmetric geometries, we achieved pronounced ionic memristor behavior with switching times ranging from milliseconds to seconds. This talk will summarise these research findings and discuss future paradigms in 2D material-based membrane research.

References:

1. R. Aparna, S. S. Pranav, K. Gopinadhan, *Desalination*, 616, 119387 (2025).
 2. A. Rath, B. Dhal, S. S. Sarath, K. Gopinadhan, *Cell Reports Physical Science* 4, 101717 (2023).
 3. B. Dhal, A. Puzari, L. H. Yeh, K. Gopinadhan, *ACS Applied Materials & Interfaces* 16, 50, 69556 (2024).
 4. D. Biswabhusan, Y. Noh, S. N. Paltasingh, C. Naman, S. S. Nemala, R. Aparna, K. Suvigya, A. Capasso, S. K. Nayak, L. H. Yeh, K. Gopinadhan, *Nature Communications*, 16, 8542 (2025)
-

Giant piezo shear drives electronic instabilities and phase transitions in layered ferroelectric In_2Se_3

Pavan Nukala

Center for Nanoscience and Engineering, IISc Bengaluru

In_2Se_3 is a prototypical van der Waals ferroelectric material that exhibits a plethora of metastable phases which are energetically within close proximity of each other (<100 meV). Through in-situ microscopy experiments on In_2Se_3 based devices, we show that while the out-of-plane ferroelectricity is robust, the in-plane ferroelectricity is not. Application of in-plane electric field results in inter layer sliding, rotations (Moire lattices), interlayer shear, domain boundaries and their volatile and non-volatile phase-transitions. This is all thanks to the giant piezoelectric shear coefficients ($d_{15}=2800$ pm/V), that transduces field into mechanical effects. In this talk, first I will discuss field induced and temperature induced reversible phase transitions that give rise negative differential resistance in FeSFETs (ferroelectric semiconducting field effect transistors). In the second part, I will discuss in-plane field induced non-volatile phase transitions, including solid-state amorphization that is driven by earthquake like Barkhausen's noise that emanates in these systems due to domain boundary interactions. All-in-all, the sensitivity of In_2Se_3 structure to in-plane electric field, coupled piezo shear stress and diffusionless phase transitions can be exploited to design novel neuromorphic devices.

References:

1. G. Modi, S. Parathe,, P. Nukala*, R. Agarwal* et al., Nature, 2024
2. J. Ghosh, S. Parathe, ..., D. Nath, P. Nukala et al., arXiv:2509.09233

Nanofiber Technology & Its Commercial Application: Future Start-Ups

Sandip Patil

The paradigm of material science research is rapidly shifting towards the realm of nanofiber technology, signalling the arrival of innovative solutions to complex problems. This presentation delves into the pivotal domain of translational research, outlining the path from research findings to business ventures in the vast field of nanofiber technology. The discussion navigates through diverse applications and commercial product developments, spanning critical engineering sectors. Key focal points encompass energy, bio technology and filtration devices. By exploring these specialized fields, this discussion seeks to shed light on the multitude of options inherent in nanofiber technology. Furthermore, the session endeavours to provide actionable insights, guiding future startups on harnessing the unmatched potential of nanotechnology. Through an overview of its applications, this presentation aims to inspire entrepreneurial endeavours that capitalize on the limitless possibilities offered by nanofiber technology.

Next Generation Advanced Materials for NBC Mitigation

Dr Vikas Baburao Thakare

Scientist G & Associate Director

Defence Research & Development Establishment Jhansi Road, Gwalior-474002

Email: vikash.drde@gov.in

The increasing frequency of industrial accidents, environmental contamination, and the persistent threat of chemical warfare and bioterrorism have heightened global concern regarding nuclear, biological, and chemical (NBC) hazards. Effective mitigation of these threats requires advanced protective systems capable of rapidly detecting, adsorbing, and neutralizing a broad spectrum of toxic agents. Over the past several decades, adsorption-based materials have formed the backbone of NBC protection technologies, particularly in personal protective equipment, filtration systems, and collective protection shelters.

Conventional NBC protective systems have relied heavily on carbon-based materials such as granular activated carbon (GAC), activated carbon spheres (ACS), and activated carbon fabrics (ACF). These materials are widely used due to their high specific surface area, well-developed microporous structure, and strong physisorption capacity for a variety of organic vapors and toxic industrial chemicals. Among them, ACF offers additional advantages, such as flexibility, lightweight properties, and faster adsorption kinetics, due to its accessible pore structure. Despite their widespread use, these traditional materials exhibit several limitations, including non-selective adsorption, reduced efficiency against low-concentration or emerging contaminants, limited catalytic degradation capability, and challenges in regeneration and reuse.

To address these shortcomings, recent research has increasingly focused on the development of next-generation materials with enhanced functionality and performance. Nanotechnology has opened new avenues for designing materials with tuneable surface chemistry, high reactivity, and improved adsorption kinetics. Nanomaterials enable precise control over particle size and surface properties, thereby enhancing interaction with hazardous agents. In parallel, metal–organic frameworks (MOFs) and covalent organic frameworks (COFs) have emerged as highly promising candidates due to their ultra-high surface areas, ordered pore structures, and the ability to incorporate functional groups for selective adsorption and catalytic degradation of chemical warfare agents. Emerging materials such as aerogels and hydrogels contribute additional benefits including ultra-low density, high porosity, flexibility, and the ability to capture both particulate and vapor-phase contaminants.

In conclusion, next-generation NBC protective gears represent a transformative evolution in defence technology. By integrating advanced materials with smart functionalities into NBC mitigation systems, these innovations substantially improve soldier protection, enhance comfort, and optimize mission effectiveness across diverse and challenging operational environments.

Keywords: Activated Carbon Fabric(ACF), Covalent organic Frameworks (COF), Metal–organic Frameworks (MOF), Nuclear, Biological and Chemical (NBC) hazards.

Role of 2D Materials in Energy Storage and Sensing Applications at CARBON Lab

Chandra Shekhar Sharma

*Creative & Advanced Research Based On Nanomaterials (CARBON) Laboratory,
Department of Chemical Engineering, Indian Institute of Technology Hyderabad, Kandi,
502285, Sangareddy, Telangana, India
Email id: cssharma@che.iith.ac.in*

The rapid emergence of two-dimensional (2D) materials has significantly advanced the design of high-performance systems for energy storage, and gas sensing applications. At CARBON Lab, we demonstrate how diverse 2D materials—including transition metal dichalcogenides such as MoS₂ and MoSe₂, graphene derivatives (rGO, GQDs), and MXenes—play a central role in improving electrochemical performance through catalytic activity, conductivity enhancement, and interfacial engineering.

In energy storage systems such as Li–CO₂ and Na–S batteries, MoS₂ nanosheets function as efficient electrocatalysts by accelerating redox kinetics and anchoring intermediate species like polysulfides, thereby suppressing shuttle effects and improving cycling stability. Similarly, MoSe₂ nanosheets exhibit high surface reactivity and adsorption capability, enabling sensitive and selective gas sensing at room temperature. Graphene-based 2D materials act predominantly as conductive frameworks; reduced graphene oxide and N-doped graphene enhance charge transport and CO₂ adsorption, while heteroatom-doped graphene quantum dots introduce tunable electronic structures and strong interactions with polysulfide species, reducing capacity fading in metal–sulfur batteries.

MXenes, with their metallic conductivity and surface functional groups, contribute significantly to pseudocapacitive behavior and improve charge storage when integrated into hybrid electrodes. Importantly, these studies highlight that the true potential of 2D materials lies in their incorporation into hybrid and heterostructured systems, where synergistic interactions between components—such as MoS₂ with carbon matrices, MXene with conducting polymers, and graphene with multimetal catalysts—lead to enhanced catalytic efficiency, improved adsorption of reactants like CO₂, and better reversibility of discharge products such as Li₂CO₃. A novel MXene/graphene nanocomposite is explored as a bifunctional cathode to enhance the kinetics of CO₂ reduction and evolution. The synergistic integration of conductive graphene with catalytically active MXene provides a high surface area, improved charge transport, and abundant active sites for electrochemical reactions. These preliminary results indicate enhanced reaction kinetics and reduced polarization compared to conventional carbon-based systems. MXenes are also explored with MOF as composites, to exhibit synergistic performance for Li–CO₂ batteries and supercapacitors.

Overall, this talk will present 2D materials as multifunctional platforms that integrate catalytic activity, electrical conductivity, and structural stability, offering promising pathways for the development of next-generation sustainable energy storage devices and advanced sensing technologies.

Strain induced polarization in few layer SrTiO₃ films

Priya Mahadevan

Department of Condensed Matter and Materials Physics

S.N.Bose National Centre for Basic Sciences,

JD Block, Sector-III, Salt Lake, Kolkata 700106, India

Oxide-based freestanding ferroelectric membranes hold great potential as ideal candidates for next-generation, highly efficient electronic devices, offering unique and advanced functionalities. SrTiO₃ (STO) thin films, nominally quantum paraelectric [1], are known to become ferroelectric under strain [2]. Recently, in-plane ferroelectricity with a transition temperature (T_x) as high as 400 K has been observed in freestanding STO membranes [3], opening up a new platform to investigate the microscopic origins of ferroelectricity in this limit. To date, research has mainly focused on the critical thickness and ferroelectric transition temperature for room-temperature ferroelectricity in STO, with findings suggesting that these properties are strongly influenced by the bulk-to-surface ratio [4]. However, the microscopic origins of ferroelectricity and the thickness dependency of various properties remain largely unexplored. In this study, we focus on the origin of changes in polarization upon strain specifically in few-layer STO. Our results reveal a distinct even–odd layer number effect: even-layered STO exhibits antiferroelectric behavior within a certain strain range, transitioning to ferroelectric beyond that range, whereas odd-layered STO remains ferroelectric throughout. These findings highlight unique thickness- and symmetry-dependent polarization behavior in ultrathin STO, distinguishing it from its bulk counterpart and other conventional ferroelectrics. Our work offers new insights into the mechanisms governing ferroelectricity in low-dimensional systems and contributes valuable perspectives for the design of nanoscale ferroelectric devices.

References:

1. K. A. Müller and H. Bukard, SrTiO₃: Phys. Rev. B 19, 3593 (1979).
2. J. H. Haeni et al., Nature (London) 430, 758 (2004).
3. R. Xu, J. Huang, E. S. Barnard, S. S. Hong, P. Singh, E. K. Wong, T. Jansen, V. Harbola, J. Xiao, B. Y. Wang et al., Nat. Commun. 11, 3141 (2020).
4. H. Sun, J. Gu, Y. Li, T. R. Paudel, D. Liu, J. Wang, Y. Zang, C. Gu, J. Yang, W. Sun, Z. Gu, E. Y. Tsymbal, J. Liu, H. Huang, D. Wu, and Y. Nie, Phys. Rev. Lett. 130, 126801 (2023).

1D Metal Oxides – 2D Materials Heterojunctions: Design, Fabrication and Photoelectrochemical Properties

Somnath C Roy

*Semiconducting Oxide Materials, Nanostructures and Tailored Heterojunctions (SOMNaTH)
Lab, Functional Oxides Research Group and Centre for 2D Materials Research and Innovation,
Department of Physics and School of Sustainability, Indian Institute of Technology Madras,
Chennai 600036, India*

*Corresponding Author E-mail: somnath@iitm.ac.in

Over the past 2 decades, 2D materials have taken over the materials world with several exotic properties that promise to find solutions for many unsolved challenges. Excellent surface to volume ratio, superior mobility, and ease of functionalization make 2D materials highly flexible and versatile for several different applications. Starting from graphene and its derivatives, materials such as TMDCs, MXene, gCN and other related materials are being extensively investigated. On the other hand, metal oxides are ubiquitous materials with excellent chemical and thermal stability and have served us for centuries. Our work aims to explore the best of both classes of materials by design and fabrication of heterojunctions between 1D metal oxides (TiO₂ nanotubes/ CuO nanowires etc) with 2D materials. In particular, we design the heterostructures with superior interfacial properties to enhance light harvesting, charge separation and catalytic activities. In this talk, I shall present some of our recent works in these directions.

Keywords: TiO₂ Nanotubes; CuO Nanowires; Fe₂O₃ Nanorods, Graphene oxide; WS₂

References:

1. Govinda C Behera, Somnath C Roy et al; Small (2024)
 2. Ragulkrishnan V, Somnath C Roy et al, Advanced Sustainable Systems; 9(2) 2400732 (2025);
 3. Garima Gupta, Somnath C Roy et al, Journal of Energy Storage, (2026) 157, 121658
-

DNA based programmable 2D materials and films for biological applications

Dhiraj Bhatia

*Associate Professor, Department of Biological Sciences and Engineering,
Indian Institute of Technology Gandhinagar, IIT Gandhinagar,
Near Palaj Village, Gandhinagar 382355, Gujarat, India
dhiraj.bhatia@iitgn.ac.in*

My laboratory asks how nanometer-sized biomolecules transmit and integrate information across much larger length scales of the orders of cells and tissues. We seek to explore how collections of macromolecules work together to establish a common functional system like cellular pathways, organelles, living cells and further into tissues, organs and entire organisms. Different biomolecules establish long-range orders in living systems by self-assembling into much larger structures, such as molecular complexes, membranes, and cytoskeletal organelles, intra- and inter-cellular contacts, and long range contacts. The main theme of our lab will be to understand the assembly principles of biological systems and the roles they play in living cells, tissues and full organisms...and further developing technologies to modulate the same.

To address these problems, we adopt a multidisciplinary, bottom-up approach using DNA nanotechnology. DNA has immense potential to arrange the matter at nanoscale with extreme robustness and spatial specificity. The compatibility of DNA to interface with other biomolecules like proteins, carbohydrates, lipids, small molecules makes DNA a natural choice of material for bottom-up self-assembly. Thus, we will merge the complex programmability of DNA nanotechnology with the structural and functional diversity of other biomolecules. Our interdisciplinary research, along with national and international collaborations with experts, will

leverage expertise from chemistry, nanotechnology, biophysics, biology, engineering, and medicine.

The overarching goal of the my team would be to translate laboratory findings into the development of new therapeutic strategies.

Keywords: Structural DNA nanotechnology, Biofilms and Hydrogels Biomedical and translational applications.

MXene Surface Chemistry: The Interplay Between Synthesis, Colloidal Stability, and Macroscopic Properties

Varun Natu

The rapid ascent of MXenes in the landscape of 2D inorganic materials is largely attributed to their exceptional conductivity, hydrophilicity, and tunable surface chemistry. However, navigating the transition from as-synthesized powders to functional devices hinges entirely on a rigorous understanding of their colloidal chemistry. This talk explores the critical role that surface terminations play in dictating the colloidal interactions, stability, and ultimate utility of MXene dispersions.

Crucially, these surface functionalities are not mere artifacts; they are fundamentally governed by the chosen synthesis route and selective etching protocols. We will dissect how tailoring the synthesis environment directly controls the terminal group population, which in turn heavily influences the electrostatic and steric stabilization of MXene flakes in suspension. This fundamental colloidal behaviour is the cornerstone for the practical processability of MXene inks, coatings, and films.

Furthermore, the implications of surface chemistry extend far beyond dispersion stability, directly governing the material's macroscopic behaviour. I will discuss how specific surface environments dictate oxidation resistance, effectively extending the functional shelf-life of these sensitive materials. Equally important is the profound impact on electrochemical performance, where surface terminations influence ion intercalation kinetics and interfacial interactions. By examining specific states we demonstrate that mastering the synthesis-surface-property relationship is absolutely essential for unlocking the full potential of MXenes in next-generation energy storage and advanced material engineering.

Layered Transition Metal Dichalcogenides Materials for High-Performance THz Sources and Detectors

Samaresh Das, Suprovat Ghosh, Pinki Yadav and Rahul Mishra

Centre for Applied Research in Electronics, Indian Institute of Technology Delhi, India

*Corresponding Author E-mail: samareshdas@care.iitd.ac.in

The rapid advancement of terahertz (THz) technology for applications in communication, imaging, sensing, and security demands efficient, broadband, and room-

temperature-operable devices. In this context, layered transition metal dichalcogenides (TMDs) have emerged as promising material platforms due to their unique electronic structure, strong spin–orbit coupling, and tunable transport properties. This talk will present recent advances in the development of TMD-based THz sources and detectors, with emphasis on device engineering and underlying physical mechanisms. On the source side, high-efficiency spintronic THz emitters based on Dirac semimetal PtTe₂ [1] and MBE-grown MoTe₂ will be discussed, where enhanced emission arises from strong spin–charge conversion driven by high spin Hall conductivity, and interfacial spin–orbit interactions. Notably, PtTe₂/Co heterostructures demonstrate superior performance compared to conventional metal systems. On the detection side, room-temperature, zero-bias THz detectors based on WSe₂ [2] and PtTe₂ [3] will be presented, exhibiting broadband operation (0.1–1.5 THz), high responsivity, and low noise, eliminating the need for cryogenic cooling and enabling low-power operation. Overall, the talk will highlight how layered TMD materials provide a unified platform for THz emission and detection, paving the way for integrated, energy-efficient THz technologies for next-generation communication and sensing applications.

Keywords: Transition Metal Dichalcogenides; Spintronic THz Emitters; THz detectors

References:

1. P Yadav, C Xinhou, S Bhatt, S Das, H Yang, R Mishra, “Highly Efficient Spintronic Terahertz Emitter Utilizing a Large Spin Hall Conductivity of Type-II Dirac Semimetal PtTe₂,” *Nano Letters*, 24 (7), 2376-2383 (2024)
2. P Yadav, P Kumar, S Ghosh, A Jakhar, R Mishra, S Das, “Room-temperature broadband THz detection at zero bias in layered WSe₂,” *Applied Physics Letters* 127 (13), (2025)
3. A Jakhar, D. Arya, S. Ghosh, S Das, “Zero biased and broadband (0.1-1.5 THz) THz detector using Dirac semimetal platinum telluride”, *IEEE Sensors Letters*. 6, 3501604 (2022)

2D Transition Metal Dichalcogenides for Hydrogen Generation and Metal Air Batteries

Dr. Ashish Kumar Mishra

*Nano Photonics & Energy Materials Laboratory (NPEML)
School of Materials Science and Technology, IIT (BHU), Varanasi*

*Corresponding Author E-mail: akmishra.mst@iitbhu.ac.in

Electrochemical hydrogen generation and metal-air batteries have gained increasing demand as sustainable and efficient energy solutions for clean energy generation and storage. Transition metal dichalcogenides (TMDs) are known to be potential electrocatalysts, offering exceptional performance due to their unique electronic properties, abundant active sites, and high surface area. Here, we discuss the role of TMDs nanostructures like MoS₂ and MoSe₂ with different morphologies in hydrogen production through water splitting and in metal-air batteries like Zn-air, and Al-air batteries. TMDs based electrodes demonstrate efficient hydrogen evolution reaction (HER) and oxygen evolution reaction (OER), suitable for water splitting [1-3]. TMDs-metal oxides based nanocomposites exhibit good electrocatalytic behaviour in oxygen reduction reaction (ORR) and oxygen evolution reaction (OER), suggesting their suitability for metal air

batteries. We have demonstrated the performance of Zn-air and Al-air batteries using these nanocomposites as air cathode [4-3]. The results underscore the potential of TMDs as a versatile and effective catalyst for advancing hydrogen generation and metal-air battery technologies, paving the way for sustainable and high-performance energy systems.

Keywords: Transition metal dichalcogenides, Electrochemical hydrogen production, Metal-air batteries, Oxygen evolution reaction, Hydrogen evolution reaction

Reference:

1. P. K. Maurya, and A. K. Mishra, ACS Appl. Energy Mater. 2023, 7(2), 487-498.
 2. S. Mishra, P. K. Maurya, and A. K. Mishra, Mater. Today Commun. 2020, 25, 101270.
 3. A.K. Mishra and P.K. Maurya, Indian Patent Number 563644.
 4. P. K. Maurya, and A. K. Mishra, J. Phys. Chem. Lett. 2024, 15(5), 1246-1253.
 5. A.K. Mishra and P.K. Maurya, Indian Patent Number 531211.
 6. K. K. Mishra, P. K. Maurya, and A. K. Mishra, Int. J. Hydrogen Energy. 2024, 69, 252-260.
-

2D Nanoparticles in Cancer Theranostics: From Atomic Sheets, AI-Integrated to 3D Tumor Models

Sabya Sacchi Das, Sandeep Somwanshi, Ajay Mali and Nanasaheb Thorat ^{1,2*}

¹*Department of Physics and Bernal Institute, University of Limerick, Castletroy, Limerick, V94T9PX, Ireland*

²*Limerick Digital Cancer Research Centre (LDCRC), University of Limerick, Castletroy, Limerick, V94T9PX, Ireland*

*Corresponding author: nanasaheb.d.thorat@ul.ie

Nanotechnology has emerged as a transformative approach in cancer theranostics, enabling the seamless integration of diagnostic and therapeutic functionalities within multifunctional platforms. Hybrid nanostructures have demonstrated strong potential in targeted drug delivery, multimodal imaging, and controlled therapeutic interventions [1], while the emergence of two-dimensional (2D) nanomaterials presents new opportunities that require robust biological validation and AI-driven analytical approaches for clinical translation. In this work, we present our group's integrated research program in hybrid nanoplatform engineering as a compelling foundation for advancing 2D material-based cancer theranostics. Our established systems encompassing magneto-plasmonic nanoparticles, surface-enhanced Raman scattering (SERS)-active nanomaterials, and exosome-based delivery platforms demonstrate the power of multifunctional nanoarchitectures for simultaneous tumor imaging, targeted drug delivery, and photothermal therapy. A defining contribution of our program is the development of physiologically relevant three-dimensional (3D) biological models, including multicellular spheroids and organoids, that faithfully recapitulate the tumor microenvironment and drug resistance mechanisms observed in vivo providing a critical testbed for next-generation nanomaterial evaluation [2]. Artificial intelligence and deep learning pipelines are further embedded for automated spheroid morphology analysis and predictive therapeutic modelling [3]. We conclude by discussing how these validated platforms and biological models can directly

inform the rational design and preclinical evaluation of emerging 2D nanomaterials including graphene, MXenes, and transition metal dichalcogenides highlighting key challenges and future opportunities at this exciting materials-biology interface.

Keywords: Artificial Intelligence; Cancer Theranostics; Hybrid Nanoplatfroms; Nanomedicine; Nano-Bio Interface

References:

1. Thorat, N.D. et al. Comprehensive approach of hybrid nanoplatfroms in drug delivery and theranostics to combat cancer, *Drug Discovery Today*, 25(7), (2020), 1245-1252.
 2. Phalake, S. et al. Functionalized manganese iron oxide nanoparticles: a dual potential magneto-chemotherapeutic cargo in a 3D breast cancer model, *Nanoscale*, 15, (2023), 15686-15699.
 3. Mali, A.K. et al. A Deep Learning Pipeline for Morphological and Viability Assessment of 3D Cancer Cell Spheroids, *Biology methods & protocols*, 10(1), (2025), bpaf030.
-

Composition-Structure-Property Relationships in MXenes

Babak Anasori

Reilly Rising Star Associate Professor

*School of Materials Engineering and School of Mechanical Engineering
Purdue University, West Lafayette, IN 47907*

MXenes are a large family of 2D transition metal carbides, carbonitrides, and nitrides with over 100 synthesized compositions to date. The rapid growth of the MXene family is driven by designer chemistry control of their composition and structure, including the transition metal and surface functional groups, the non-metal X sublattice, and atomic-layer configurations. This compositional diversity governs chemical ordering, atomic-level defects, and surface chemistry, ultimately shaping MXenes properties. MXenes offer chemical and structural tunability, high electrical conductivity, hydrophilicity, functionalized surfaces, and mechanical robustness.

In this talk, I will start with the composition–structure relationships in MXenes and then focus on our recent efforts on compositionally complex MXenes and defect engineering to control stability, phase transformation, and functional properties. I will discuss our work on the transition from order to disorder at sub-nanometer thicknesses across 40 different MXenes with up to nine transition metals due to entropy effects. I will also present how controlling defects, including metal and non-metal vacancies and substitution, enables tuning of oxidation resistance, electrocatalytic behavior, and high-temperature stability.

2D/3D Heterointegration: A new take on developing diamond-based electronics

Anirudha V. Sumant

Center for Nanoscale Materials, Argonne National Laboratory 9700 S. Cass Ave, Lemont, IL 60439 USA

*Corresponding Author E-mail: sumant@anl.gov.in

Diamond is a highly promising material for next-generation power electronics due to its exceptional thermal conductivity, wide bandgap, high electric field strength, radiation hardness, and superior carrier mobilities. However, the advancement of diamond-based devices has been constrained by critical challenges, including the lack of reliable n-type transport at room temperature, high synthesis temperatures, and limited scalability.

In this talk, I will present our recent efforts at Argonne to overcome these limitations through heterointegration with two-dimensional (2D) and other semiconductor materials [1-2]. In particular, we address the long-standing challenge of n-type doping in diamond by integrating n-type 2D molybdenum disulfide (MoS_2) monolayers with boron-doped p-type single-crystal diamond to form 2D/3D heterostructure PN junction diodes [3]. These devices demonstrate excellent room-temperature rectification, with a forward current density of $\sim 4000 \text{ A/cm}^2$, an ideality factor of ~ 3.7 , and a rectification ratio approaching 10^6 . We further show that the rectification behavior arises from interlayer recombination of majority carriers, governed by direct tunneling and Fowler–Nordheim tunneling mechanisms. Beyond device demonstration, I will discuss strategies for wafer-scale integration and the potential for operation under extreme temperature conditions. These results highlight a scalable pathway toward high-performance diamond-based electronics enabled by strategic heterointegration with 2D materials.

Keywords: Heterointegration, Diamond, 2D materials, n-type doping, microelectronics

References:

1. D. Berman, S. A. Deshmukh, B. Narayanan, S. KRS Sankaranarayanan, Z. Yan, A. A. Balandin, A. Zinovev, D. Rosenmann, A. V. Sumant, *Nature Communications*, 7 (1), (2016), 12099
2. J. Yu, G Liu, A.V. Sumant, V. Goyal, A.A Balandin, *Nano letters* 12 (3), (2012), 160
3. A. Wali, R. Padhan, R. Divan, L. Stan, N. Pradhan, A. V. Sumant, *Nano Letters*, 25 (42), (2025), 15353.

Ultrafast Sustainable Synthesis of Functional MXenes

Vilas G. Pol^{1,*}, M. Sai Bhargava Reddy¹

¹*Davidson School of Chemical Engineering, Purdue University, West Lafayette, 47907, IN, USA*

*Corresponding Author E-mail: vpol@purdue.edu

MXenes, a rapidly emerging class of two-dimensional transition metal carbides and nitrides[1], are conventionally synthesized using hydrofluoric acid (HF) etching[2] or high-

temperature molten salt routes[3], both of which suffer from safety concerns, high energy consumption, and slow reaction kinetics. In this work, we report an ultrafast, fluorine-free microwave synthesis strategy for the efficient production of MXenes under ambient conditions. This approach integrates microwave-driven dielectric heating with Lewis acidic molten salts (LAMS) and carbon-based microwave susceptors to achieve rapid and selective etching of MAX phases. By irradiating the dry mixture in a conventional microwave (900 W), complete removal of the A-layer is achieved within 5 minutes, reducing synthesis time by several orders of magnitude compared to traditional methods. Density functional theory (DFT) calculations reveal that elevated temperatures and solid–liquid phase transitions reduce diffusion barriers and promote favorable reaction thermodynamics, enabling ultrafast etching. The synthesized $\text{Ti}_3\text{C}_2\text{Cl}_2$ MXene exhibits expanded interlayer spacing, high structural integrity, and tunable surface chemistry. Electrochemical studies demonstrate excellent Li-ion storage performance, delivering a high specific capacity ($\sim 220 \text{ mAh g}^{-1}$ at 0.1 A g^{-1}), strong rate capability, and outstanding cycling stability, attributed to a dominant pseudocapacitive mechanism. This work establishes our method as a scalable, energy-efficient, and environmentally benign route for MXene production, offering significant potential for advanced energy storage applications and beyond.

Keywords: MXene; Microwave synthesis; Molten salt; Energy storage; Sustainable.

References:

1. A. VahidMohammadi, J. Rosen, Y. Gogotsi, Science 2021, 372, DOI 10.1126/science.abf1581.
2. K. R. G. Lim, M. Shekhirev, B. C. Wyatt, B. Anasori, Y. Gogotsi, Z. W. Seh, Nat. Synth 2022, 1, 601.
3. V. Kamysbayev, A. S. Filatov, H. Hu, X. Rui, F. Lagunas, D. Wang, R. F. Klie, D. V. Talapin, Science 2020, 369, 979.

Competing Disorder, Charge Ordering, and Electronic Correlations Two-Dimensional Mixed Chalcogenides

Sanjoy Kr Mahatha*

UGC-DAE Consortium for Scientific Research, University Campus, Khandwa Road, Indore 452001

*Email: skmahatha@csr.res.in, sanjoymahatha@gmail.com

Transition metal mixed chalcogenides, where two different chalcogen species occupy ligand sites, offer a versatile platform to study the interplay between charge order, lattice distortion, and electronic correlations. In this talk, I will present our recent investigations of two such systems, rhombohedral 6R-NbSeTe and ligand-disordered 1T-TaSeTe , using a combination of diffraction, spectroscopy, and transport techniques. In 6R-NbSeTe , perfect self-stacking of alternating 1T and 1H layers lead to temperature-independent short-range charge correlations with a unique propagation vector, quenched long-range charge density wave (CDW) order, and the emergence of inhomogeneous short-range magnetic order, both driven by Se/Te ligand disorder. Angle resolved photoemission spectroscopy (ARPES) reveals mixed trigonal and

octahedral Nb band contributions at the Fermi level without gap opening. In contrast, 1T-TaSeTe retains a robust incommensurate CDW down to low temperatures, albeit with a reduced transition temperature compared to 1T-TaSe 2. ARPES shows a pronounced pseudogap arising from Fermi surface nesting, while transport measurements reveal semiconductor-like behavior, likely linked to disorder-induced scattering and in-gap states. Together, these results illustrate how ligand disorder can both suppress and preserve charge order, tune competing ground states, and create fertile conditions for unconventional electronic behavior in layered chalcogenides.

References:

1. S. K. Mahatha, J. Phillips, J. Corral-Sertal, D. Subires, A. Korshunov, A. Kar, J. Buck, F. Diekmann, G. Garbarino, Y. P. Ivanov, A. Chuvilin, D. Mondal, I. Vobornik, A. Bosak, K. Rossnagel, V. Pardo, A. O. Fumega, S. Blanco-Canosa, Self-Stacked 1T-1H Layers in 6R-NbSeTe and the Emergence of Charge and Magnetic Correlations Due to Ligand Disorder, *ACS Nano* 18, 32, 21052 (2024).
2. J. Sharma, S. Choudhury, M.-J. Huang, J. Buck, S. Blanco-Canosa, K. Rossnagel, S. K. Mahatha, Persistent incommensurate charge density wave in chalcogen-disordered 1T-TaSeTe, *Appl. Phys. Lett.* 127, 243101 (2025).

Tailoring of BaRuO₃/LaNiO₃ Superlattice to Induce Ferromagnetism

Annu^{1,2}, Geetanjali Sehgal¹, Charan Singh³, Vivek Malik³, B. Satpati⁴, Anjana Dogra^{1,2,*}

¹CSIR-National Physical Laboratory, New Delhi-110012, INDIA

²Academy of Scientific and Innovative research (AcSIR), Ghaziabad-201002, INDIA

³ Department of Physics, Indian Institute of Technology, Roorkee, INDIA

⁴ Surface Physics & Material Science Division, Saha Institute of Nuclear Physics, 1/AF, Bidhannagar, Kolkata 700 064 INDIA

*Corresponding Author E-mail: anjana.nplindia@csir.res.in

The oxide heterostructures have shown exotic properties due to the strong charge transfer and hybridization at their interface, which are not shown by their pristine form. These perovskite oxides can be modified at the nanoscale by making layered structure and further their properties can be tuned for spintronics applications. Our research of inducing ferromagnetism at the interface of BaRuO₃/LaNiO₃ superlattice on SrTiO₃(STO)- a property not present in either material alone, is based on the same idea of charge transfer at the interface of these two materials. For this, we have prepared a BaRuO₃(BRO) and LaNiO₃(LNO) targets using solid state method and thin films were grown by Pulsed Laser deposition (PLD) under different oxygen pressures and laser frequencies to optimize structural and magnetic properties. Heterostructure samples with different thickness of BRO and LNO were prepared under optimized conditions. The number of consecutive layers were kept constant i.e., 11(6 BRO layers and 5 LNO layers). The thickness of LNO layers was kept constant (5nm) with varying thickness of BRO (2,3,4,5nm). The structural characterization of samples like HRXRD, AFM, HRTEM and XRR shows well defined layered structure of the samples and support the epitaxial growth of the materials. The magnetic measurements like M-H and M-T proves the induced ferromagnetism at the interface. M(H) at different low temperatures show hysteresis which is persistent till room temperature. The graphs

show two-step reversal magnetization because of different coercive field values for both the compositions. However, the saturation magnetization (M_s) was found to vary with temperature, indicating temperature-dependent spin alignment. In future, spectroscopic techniques like XAS/XLD can confirm the charge transfer at the interface through the change in the oxidation states of materials. Our findings give insight into tuning the magnetic properties of the oxide materials which can be used for spintronics applications.

Keywords: Superlattices, Interfacial ferromagnetism Magnetization, Charge transfer

References:

1. Nakao H, Sudayama T, Kubota M, et al. Magnetic and electronic states in $(\text{LaMnO}_3)_2(\text{SrMnO}_3)_2$ superlattice exhibiting a large negative magnetoresistance. *Phys Rev B*. 92 (2015) 245104.
2. A Liu, P., & Ning, X. (2020). High-spin state and magnetic coupling induced through interfacial orbital reconstruction observed in $\text{SrRuO}_3/\text{LaNiO}_3$ superlattice. *Materials Research Letters*, 8(5) (2020), 173–179.

Development of electrode materials for cost-effective sodium-ion batteries

Rajendra S. Dhaka

Department of Physics, Indian Institute of Technology Delhi, Hauz Khas-110016, New Delhi

*Corresponding Author E-mail: rsdhaka@physics.iitd.ac.in

In recent years, energy storage devices especially rechargeable batteries have become part of our daily life in portable electronic devices like mobiles, laptops as well as being used at large scale in electrical vehicles (EV), etc. Among various types of batteries, the only commercially established and feasible technology is based on Li-ion intercalation, mainly due to the advantage of its high energy density, good rate kinetics and light weight. However, the main concern comes from the limited availability and uneven distribution of Li in the earth crust. This motivates researchers to search for alternative avenue, and the Na-ion batteries are one of the potential candidates particularly for grid level storage due to its low cost and abundance on the globe, and therefore extensive research has been started in this field over past few years [1]. In this direction, it is vital to find suitable and high-performance electrode materials. However, there are many challenges due the larger size of the Na-ion as compared to the Li. Therefore, constant efforts are going on to search new electrode materials (negative as well as positive) for improving the electrochemistry. Recently, we have developed/tested few cathode and anode materials, indigenous separators and studied their electrochemical performance to understand the diffusion kinetics, stability, rate capability and coulombic efficiency in sodium ion batteries [2-7]. In this talk, I will present some of our recent results towards development of cost-effective/high-performance sodium-ion batteries for energy storage applications.

Keywords: Cathode Materials, Sodium-ion batteries

References

1. Madhav Sharma, R. Gulati, and R. S. Dhaka*, *Coord. Chem. Rev.* 543, 216912 (2025).
2. Jayashree Pati, and R. S. Dhaka*, *Journal of Power Sources* 609, 234646 (2024).
3. S. K. Sapra, J.-K. Chang, and R. S. Dhaka*, *ACS Appl. Mater. Interfaces* 16, 43535 (2024).

4. S. K. Sapra, M. Das, M.W. Raja, J.-K. Chang, R. S. Dhaka*, J. Mater. Chem. A 13, 1996 (2025).
 5. S. K. Sapra, J.-K. Chang*, and R. S. Dhaka*, Energy Storage Materials 81, 104478 (2025).
 6. Madhav Sharma and R. S. Dhaka*, Small 21, 2505200 (2025).
 7. M. K. Singh, J. Pati, D. Seth, J. Prasad, M. Agrawal, M. Ali Haider, J.-K. Chang, & R. S. Dhaka* Chemical Engineering Journal 454, 140140 (2023).
-

Defect-Driven Tubular Charge Density Wave and Excitonic Luminescence in 2D Quantum Materials

Pratap K. Sahoo

Ion Beam and Nanomaterials Laboratory (IBNL), School of Physical Sciences, National Institute of Science Education and Research, Bhubaneswar, An OCC of Homi Bhabha National Institute, Jatni Odisha, 752050, India

*Corresponding Author E-mail: pratap.sahoo@niser.ac.in

Understanding the excitonic properties by controlling defects in the 2D-nanomaterials is of great interest to the large community of advanced quantum materials and application groups. Depending on the doping parameters, various defects can be created and annihilated in the host matrix either by ion implantation or chemical doping that can modulate the optical, magnetic, and electronic properties of the materials. This talk emphasizes the excitonic properties of two-dimensional Titanium Diselenide (TiSe₂) that exhibit tunable charge density wave (CDW) order below a thermal transition scale (T_{CDW}) $\approx$ 200 K and defect-induced ferromagnetism, as well as bandgap tunability in TiSe₂ and Bi₂Se₃ 2D materials. Low-energy ion implantation was employed to introduce Au as a controlled disorder potential in TiSe₂ and Bi₂Se₃ by the ion beam fluence. We demonstrate a notable effect of in-plane vibrational modes via excitons, where disorder enhances the temperature windows of short-range CDW correlations to 2.5 times the T_{CDW} of the Au-free sample [1-3]. Along with that, the excitons are responsible for the ferromagnetic order that is tuned with Au impurities. A qualitative model has been proposed for the experimentally observed magnetization as a function of the ion fluence, corroborated with high-resolution transmission electron microscopy. Depending on the Au nanoparticle size, magnetization saturates faster at a much lower applied magnetic field. The co-existence of CDW, excitons, and magnetism holds the potential to expand the range of 2D ferromagnetic materials for spintronic and magnetic-sensing device applications.

Keywords: 2D materials, TiSe₂, Defect, Luminescence, excitons

References:

1. Utkalika P. Sahoo, Anamitra Mukherjee, and Pratap K. Sahoo; ACS Applied Electron. Materials; (2022), 4, 7, 3428.
 2. Utkalika P. Sahoo, Spandan Anupam, Bidyadhar Das, Mrinal K. Sikdar, Laxmipriya Nanda, and Pratap K. Sahoo; ACS Applied Nano Materials, (2022) 5, 3, 4072.
 3. Soumendra K. Das, Utkalika P. Sahoo, Bidyadhar Das, Gurupada Ghorai, Prasanjit Samal, Pratap K. Sahoo; Advanced Quantum Technologies; (2023) 202300169.
-

Strategies for scalable synthesis of layered semiconductors

Ayushi Tyagi¹, Diwakar Singh¹, Ramesh Rajarapu¹, Gayatri Subhalaxmi¹, Ray Zhu², Mikhail Chubarov², Nadire Nayir^{2,3}, Joan M. Redwing² and Tanushree H. Choudhury^{1,*}

¹*MEMS Department, Indian Institute of Technology Bombay, Mumbai, India*

²*2D Crystal Consortium, Pennsylvania State University*

³*Paul Drud Institut Fur Festkorperelektronik, Berlin, Germany*

*Corresponding Author E-mail: tanuhc@iitb.ac.in

Layered semiconductors like transition metal dichalcogenides (MX₂, M=Mo, W and X=S, Se) exhibit novel layer-dependent properties. These atomically thin materials have been successfully integrated in various lab-scale optoelectronic and quantum applications. A key hurdle in the industrial integration of these materials is scalable techniques for synthesizing large-scale layered semiconductors.

This talk will discuss two contrasting techniques to achieve large-scale synthesis. Metalorganic chemical vapor deposition (MOCVD) of MX₂ will be presented with emphasis on epitaxial growth using surface chemistry modulation. A multi-step precursor modulation growth method was developed to independently control nucleation density and the lateral growth rate of monolayer domains on the substrate [1]. Using this approach, uniform, coalesced monolayer and few-layer TMD films were obtained on 2" sapphire substrates at growth rates on the order of ~1 monolayer/30 min. Controlling the growth temperature and chalcogen flux was crucial in establishing an epitaxial relation. Chemical engineering of the step results in control of the orientation of growth [2,3].

In addition to the MOCVD process, a low-cost solution synthesis of layered transition-metal oxides will be discussed, with an emphasis on solvent and substrate engineering. Using specific temperature-time exposure, phase-pure coalesced films of layered transition metal oxides, like α-MoO₃ were synthesized. Doping of these films was also explored to modulate their electrical properties. The impact of processing conditions on the structural and electrical properties in both approaches will be demonstrated.

Keywords: MOCVD; Layered semiconductors; solution synthesis

References:

1. Mikhail Chubarov*, Tanushree H Choudhury*, Danielle Reifsnnyder Hickey, Saiphaneendra Bachu, Tianyi Zhang, Amritanand Sebastian, Anushka Bansal, Haoyue Zhu, Nicholas Trainor, Saptarshi Das, Mauricio Terrones, Nasim Alem, Joan M Redwing, ACS Nano, 15, 2532 (2021)
2. Haoyue Zhu*, Nadire Nayir*, Tanushree H Choudhury*, Anushka Bansal, Benjamin Huet, Kunyan Zhang, Alexander A Poretzky, Saiphaneendra Bachu, Krystal York, Thomas V Mc Knight, Nicholas Trainor, Aaryan Oberoi, Ke Wang, Saptarshi Das, Robert A Makin, Steven M Durbin, Shengxi Huang, Nasim Alem, Vincent H Crespi, Adri CT van Duin, Joan M Redwing, Nature Nanotechnology, 18, 1295 (2023)
3. Tanushree H Choudhury, Nicholas Trainor, Chen Chen, Mikhail Chubarov, Saiphaneendra Bachu, Kasra Momeni, James Spencer Lundh, Danielle Reifsnnyder Hickey, Tianyi Zhang, Amritanand Sebastian, Haoyue Zhu, Baokun Song, Yueli Chen, Benjamin Huet, Anushka Bansal, Sukwon Choi, Nasim Alem, Mauricio Terrones, Deep Jariwala, Saptarshi Das, Joan M Redwing, 2D Materials, 12, 045009 (2025)

Transverse quantum transport in non-collinear spin textures: Isolated Skyrmions and Crystals

B. R. K. Nanda^{1,2}, Arijit Mandal^{1,2}, Hareram Swain^{1,2}, Sashi Satpathy^{1,2,3}

¹ *Condensed Matter Theory and Computational Lab, Department of Physics, IIT Madras, Chennai, 600036, India*

² *Center for Atomistic Modelling and Materials Design, IIT Madras, Chennai, 600036, India*

³ *Department of Physics and Astronomy, University of Missouri, Columbia, MO, 65211, USA*

E-mail: nandab@iitm.ac.in

Propagation of electrons in the field created by chiral spin textures is intriguing and it offers richness in topological quantum transport involving charge, spin, and orbital. In this talk, we will first explore the skyrmion crystals of various kinds of crystals to establish topological spin, charge and orbital Hall effects by employing a tight-binding model Hamiltonian. In the second part of the talk, we will discuss the dynamical process of scattering of the spin- half particles in the form of Gaussian wavepackets from isolated skyrmions with the aid of the non-relativistic time-dependent Schrödinger equation. The scattering cross section shows a rich angular dependence and is deterministically influenced by the iterative flipping of the spin state inside the skyrmion. The latter leads to a set of non-trivial outcomes which include finite transmission and reflection probabilities irrespective of interaction strength, formation of secondary wavefronts associated with back-converted spin components, and a long-lived quasi-bound state at the scattering center. Our work provides key foundation for understanding of the transport properties of collective chiral objects, such as the skyrmion crystals and the like. With these studies, we have also developed a new numerical recipe, applicable for any non-collinear spin textures.

Keywords: Skyrmions, Topological Hall Effect, Scattering, Quantum Dynamics

References:

1. Tuning the band topology and topological Hall effect in skyrmion crystals via the spin-orbit coupling, A. Mandal, S. Satpathy, and B. R. K. Nanda, Phys. Rev. B, 112, 064428 (2025)
2. Quantum Dynamics of Electron Scattering from Skyrmions, H. K. Swain, A. Mandal, S. Satpathy, and B. R. K. Nanda, Phys. Rev. B, 113, 104446 (2026)

Probing Hidden Carrier Relaxation Pathways in 2D Materials Using Ultrafast Mid-IR Spectroscopy

Sri Ram Gopal Naraharisetty*

School of Physics, University of Hyderabad, Hyderabad – 500046, India

*Corresponding Author E-mail: nsrgsp@uohyd.ac.in

Two-dimensional transition metal dichalcogenides (TMDCs) have emerged as promising materials for next-generation optoelectronic and valleytronic technologies owing to their strong light–matter interaction, tunable excitonic properties, and atomically thin nature. However, their performance is critically governed by defect states, many-body interactions, and ultrafast carrier relaxation pathways that are often inaccessible through conventional optical techniques. In this talk, I will present our recent advances in the development of ultrafast visible-pump/mid-infrared probe spectroscopy and its application to investigate carrier, exciton, and defect dynamics in layered TMDC systems. Using a home-built ultrafast mid-IR spectroscopy platform, we directly probe intraband and intra-excitonic transitions to reveal defect-assisted nonradiative recombination in few-layer and monolayer MoS₂. Our studies uncover polarization-dependent carrier relaxation, defect-state saturation effects, and the role of vanadium doping in modifying nonradiative decay channels. Mid-IR probing further reveals how sulfur-vacancy defects alter excitonic interactions and many-body phenomena across the exciton Mott transition, leading to significant changes in carrier mobility and recombination dynamics. Our recent investigations on MoS₂–MoSe₂ heterostructures demonstrate the formation and defect-mediated relaxation of interlayer excitons, providing insights into charge-transfer processes relevant for energy conversion and valleytronic applications.

In addition to scientific discoveries, I will discuss the development of compact ultrafast mid-IR pulse-generation schemes and instrumentation advances, including microscope-integrated transient absorption measurements for micron-scale samples. These studies establish ultrafast mid-IR spectroscopy as a powerful tool for directly probing hidden electronic states and carrier relaxation pathways in low-dimensional materials, enabling rational defect engineering for future quantum, photonic, and optoelectronic technologies.

Keywords: Ultrafast spectroscopy; Mid-IR transient absorption; TMDCs; Exciton dynamics; Defect engineering.

References:

1. Shilpa Mangalassery, Dipak Maity, Bedant K. Deka, Tharangattu N. Narayanan, Sri Ram G. Naraharisetty, npj 2D Mater Appl 10, 14, 2026.
2. Vinod K. Rajput, Dipak Maity, Bedanta K. Deka, Tharangattu N. Narayanan, Sri Ram G. Naraharisetty, J. Phys. Chem. C, 2023 127, 27, 13120–13129.
3. Vinod k Rajput, Sajin Ponna, Sri Ram G Naraharisetty, Pramana – J. Phys. (2022) 96:198.

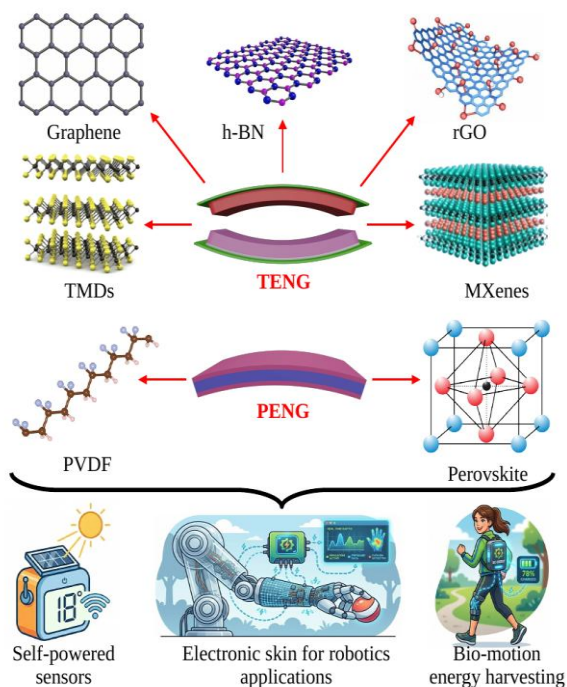
Integration of Two-Dimensional Materials and Bulk Ferroelectric Ceramics in Nanogenerators for Enhanced Mechanical Energy Harvesting

Saket Asthana*

Department of Physics, IIT Hyderabad, 502285, India

*Corresponding Author E-mail: asthanas@phy.iith.ac.in

Triboelectric and Piezoelectric nanogenerators (TENGs and PENGs) have become an attractive option for harvesting mechanical energy and enabling self-powered sensing. However, their practical use is limited by low surface charge density, quick charge loss, and unstable output. In this study, we introduce a design strategy that effectively combines two-dimensional (2D) layered materials, such as transition metal dichalcogenides (TMDs) like MoS₂ and WS₂, along with graphene derivatives¹. We pair these with bulk ferroelectric ceramics (BaTiO₃, PVDF composites)² as complementary layers in a single PENG design while employing them individually in TENG devices. The 2D materials provide a very high specific surface area, adjustable tribopolarity through phase engineering (1T/2H polymorphism) and surface modifications, as well as inherent photoresponsive charge generation. The ferroelectric bulk component offers a steady, poling-induced internal electric field, which helps reduce charge drift, improve electrostatic induction efficiency, and stabilize the V–Q–x relationship during cyclic mechanical stress. Our systematic evaluation shows that these modified structures achieve higher surface charge density in TENG and PENG devices. We observe improvements in open-circuit voltage and short-circuit current density, thanks to the enhanced coupling between the triboelectric and piezoelectric fields at the material interface. Additionally, the device shows the ability to self-power photodetection over a wide UV–NIR spectral range, resulting from the modulation of the tribosurface potential by photo-induced carriers. This work offers a new design approach for future high-performance TENGs and PENGs and has immediate applications in wearable electronics, structural health monitoring, and ambient energy harvesting systems.



Keywords: Triboelectric nanogenerators, Piezoelectric nanogenerator, 2D materials, Ferroelectrics, Energy harvesting

References:

1. Chahal, Sumit, et al. ACS Applied Energy Materials 8.23 (2025): 17431-17441.

2. Jena, Lalit Kumar, et al. ACS Applied Electronic Materials 7.3 (2025): 1314-1328.
-

Next-Generation Terahertz Functionalities Derived from Emergent Quantum Materials

Dhanvir Singh Rana

Department of Physics, Indian Institute of Science Education and Research (IISER) Bhopal, India

The terahertz (THz) spectrum (0.1-10 THz) offers promising opportunities in the development of next-generation data processing, quantum memory and communication technologies. These aspirations are driven by simultaneous developments in the THz spectrum and, the area of spintronics, magnonics, caloritronic, and some emergent quantum materials. In ordered magnets, magnons - the quanta of spin waves - envisage the prospects of non-Boolean-based spin wave computation, magnon logic gates, and so on. Magnonics has been widely explored in ferromagnets resulting in gigahertz magnons. However, the area of magnonics is being extended to ultra-low dissipation and ultrafast THz region exploiting the antiferromagnet magnons in this region. This offers THz radiation-based technological utility of antiferromagnets which, so far, have played only a passive role in conventional magnetic devices. THz radiation couples mainly with the magnetic-dipole moment of the spins unveiling not only the low-energy antiferromagnetic magnons but an interplay of magnons with other quasiparticles such as phonons, polaritons, and many more. A control over such magnon can be contemplated in magnetoelectric/multiferroic systems, wherein the spin and electric orders are entangled, resulting in electric as well as magnetic control of magnons. These facets of magnon along with a control of the dielectric medium via magnetic phase transition offer several new THz functions which are potentially relevant in futuristic computational and communication technologies. In this talk, I'll present our recent work in exploration of a variety of non-collinear magnets for magnon and magnon-phonon excitation modes in the THz spectral region, considering some examples; i) the $A_4B_2O_9$ ($A=\text{Co}$ and $B=\text{Nb, Ta}$) family of non-collinear magnetoelectric antiferromagnetic exhibit a multitude of low-energy antiferromagnetic resonances comprised of magnons, phonons, and hybridized spin-phonon coupled modes and the notion of beyond conventional magnon, that is, electromagnon, ii) the second example is based on RCrO_3 ($R=\text{rare-earth}$) orthochromates in which weak magnetic moment is accompanied by sub-THz magnon modes along with complex crystal-field excitations, and iii) using magneto-elastic and magnetoelectric transition induced drastic changes in dielectric medium to modulate THz phase in sub-THz region – a much desired attribute for wave propagation relevant for 6G and beyond technologies.

References:

1. Jayaprakash Sahoo et al, Advanced Functional Materials (accepted, 2026)
 2. B. S. Mehra et al, Advanced Science e75555 (2026)
 3. Santhosh Kumar et al, Advanced Optical Materials 8, 1900958 (2020)
 4. B. S. Mehra and D. S. Rana, Nature Physics 20, 687 (2024) [News and Views]
 5. A Shyam et al, Applied Physics Reviews 12, 021420 (2025)
 6. Sanjeev Kumar et al, Applied Physics Reviews 12, 031423 (2025)
 7. B Mehra et al, Physical Review Applied 23, 054081 (2025)
-

Computational Search on Two-Dimensional Dirac Materials – From Elemental to Compound Systems

C. Kamal

*Theoretical and Computational Physics Laboratory, Semiconductor Lasers Device and Devices Division,
Raja Ramanna Centre for Advanced technology, Indore-452013, India
Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai-400094, India
e-mail: ckamal@rrcat.gov.in*

Dirac material is a class of novel materials in which the behavior of charge carriers lying close to the Fermi level is governed by the Dirac-like equation. This is in contrast to the ordinary materials whose properties are described by the Schrodinger equation. The Dirac materials show many exciting physical properties which are important from both fundamental as well as application perspectives. Graphene is a well-known example for two-dimensional (2D) Dirac material. In this 2D monolayer, the charge carriers behave as massless Dirac-Fermions due to the presence of peculiar linear dispersion which leads to formation of Dirac cone (DC) in energy versus momentum space $[E \text{ vs } (k_x, k_y)]$ around the highly symmetric K and K' points in the Brillouin zone (BZ) at the Fermi level. Many interesting properties of graphene, such as Klein tunneling, anomalous half-integer quantum hall effect, etc. are associated with the DC. There has been growing interest in searching for new graphene-like honeycomb 2D monolayers made up of other p-block elements which may show similar properties and also provide better tunability of their properties as compared to graphene. Over the years, several theoretical and experimental investigations on p-block atom-based 2D materials have been reported. In our group, we have been performing extensive theoretical studies on physical properties of 2D materials such as silicene, arsenene, aluminene, borophene and binary/ternary monolayers. In this talk, I will discuss the interesting physical properties of 2D Dirac materials, both elemental and compound (binary and ternary), made up of p-block elements. This includes silicene, BN/silicene, CAs₃[1], CPAs₂ [2] and C₂As₆(1-x)P_{6x} [3] monolayers. I will also discuss the role of symmetry and valence orbitals of constituent elements on the formation Dirac cones and suggest ways to maneuver the Dirac cones.

Keywords: 2D Dirac materials, Electronic structure, Dirac cone and density functional theory

References:

1. C. Kamal, Physical Review B, 100, 205404 (2019)
 2. A. Kumar, C. Kamal, Journal of Physical Chemistry C, 127, 23370 (2023)
 3. A. Kumar, C. Kamal, Journal of Physical Chemistry C, 128, 18351 (2024)
-

Enhancing Electrochromic Energy Storage in $W_{18}O_{49}/Ti_3C_2T_x$ Composites for Next-Generation Smart Windows

Pritha Dutta, Sukanya Goswami, Rahuldeb Roy, Ashutosh K. Singh*

Centre for Nano and Soft Matter Sciences (CeNS), Bengaluru 562162, India

*Corresponding Author E-mail: aksingh@cens.res.in

Smart windows capable of dynamically modulating near-infrared and visible light have emerged as a transformative technology for improving indoor comfort and privacy while integrating energy storage functionalities. However, conventional transition-metal oxides, such as tungsten oxides, despite their widespread use in electrochromic applications, suffer from inherent limitations, including low coloration efficiency and slow response times. This talk presents a novel approach to overcoming these challenges strategically by incorporating $Ti_3C_2T_x$ MXene into oxygen-deficient tungsten oxide ($W_{18}O_{49}$). By optimizing solvothermal synthesis parameters and solvent selection, we developed a 5 wt.% $Ti_3C_2T_x$ - $W_{18}O_{49}$ composite (5MX-WO) that demonstrates exceptional electrochromic properties. The composite achieves 72% optical modulation at 700 nm, rapid switching speeds (6.5 s coloration, 5.6 s bleaching), high coloration efficiency of approximately $182 \text{ cm}^2 \text{ C}^{-1}$, and an areal capacitance of 25 mF cm^{-2} . The enhanced performance stems from the MXene's facilitation of superior electronic and ionic transport pathways. Furthermore, we fabricated a $5 \times 5 \text{ cm}^2$ complementary electrochromic device integrating 5MX-WO as the working electrode with spray-coated NiO as the counter electrode. This device exhibits 61% optical modulation at 850 nm, fast response times, and excellent cyclic stability over 1000 cycles. These findings establish $W_{18}O_{49}/Ti_3C_2T_x$ composites as promising candidates for next-generation electrochromic energy storage systems, offering a pathway toward multifunctional smart windows that simultaneously address energy efficiency and environmental sustainability.

Keywords: MXene; Tungsten oxide; oxygen-deficient; Electrochromic; Smart Windows.

References:

1. Pritha Dutta, Sukanya Goswami, Rahuldeb Roy, and Ashutosh K Singh, *SMALL*, 34 202503529 (2025).
2. Pritha Dutta, Kavya V P, Ramya Prabhu and Ashutosh K Singh, *Journal of Materials Chemistry A* (2026), DOI: <https://doi.org/10.1039/D6TA01049K>.

Research using Indus and possibilities with 4th generation High brightness Synchrotron Radiation source

Dr. Sanjay Kumar Rai

*Head, Indus Synchrotron Utilization Division
Raja Ramanna Centre for Advanced Technology*

Indus 1 and Indus 2 are the only two operating synchrotron radiation sources in India. These sources have been operating in round the clock mode since February 2010 and have been

in extensive use by the research community of India. Currently, there are 20 operational beamlines on Indus 2 and 7 on Indus 1 and a few more are expected to be available for users shortly. The talk will give a flavor of the kind of research being done at Indus-1 and 2. The requirement for a High Brightness Synchrotron Radiation Source (HBSRS) for the Indian scientific community will be explained. How HBSRS will support cutting-edge research in diverse fields like materials science, biology, medicine, and nanotechnology, to foster innovation in indigenous industries. Some details of Indian plan for developing our own HBSRS will be discussed.

IR & Raman Spectroscopic Studies to determine Vibrational, Optical, Electronic Properties of 2d Materials under Extreme Thermodynamic Conditions

Himal Bhatt

Bhabha Atomic Research Centre, Trombay, Mumbai 400085

Homi Bhabha National Institute, Mumbai 400094

hbhatt@barc.gov.in

Infrared and Raman spectroscopic techniques are mostly used to determine local molecular structure by studying the bond vibrations in materials research. However, in 2d/ layered materials, these techniques have evolved as powerful methods to study complex optical parameters, spin-phonon coupling, defects, metal to insulator transition, charge carrier concentration and inter- and intra-band electronic transitions, in addition to the vibrational properties. Our recent high pressure – low temperature studies on one of the most fundamental 2d material, graphene, have shown that the charge carrier concentrations may reduce under external pressures of upto 4 GPa in few layers graphene. This challenges the contemporary idea of very high conductivity of graphene due to dirac fermions in conical band structure, thereby paving the way to find band gap opening possibilities for sensing applications. In another layered system of Mg₃Sb₂ derivatives, we find that the optical phonon anharmonicity influences a high temperature phase transition and defect structure in the system, leading to novel thermoelectric properties, for battery applications. The details of our high pressure, variable temperature studies on these two systems will be discussed to corroborate the recent findings in the layered 2d system MXene.

Graph Neural Network for High-Throughput Screening of Nanocluster Electrocatalysts

Biswarup Pathak,¹ Rahul K Sharma,¹ Harpriya Minhas¹, Ayushi Pareek¹

¹*Department of Chemistry, IIT Indore, Indore*

*Corresponding Author E-mail: biswarup@iiti.ac.in

Nanoclusters represent an emerging class of heterogeneous electrocatalysts with properties tunable at the atomic scale, enabling reactivity beyond traditional structure–activity relationships. However, their catalytic behavior is highly complex due to diverse morphologies

and structural fluxionality, leading to multiple coexisting isomers. In this work, we employ advanced graph neural network (GNN) models trained on a density functional theory (DFT)-curated dataset of bimetallic nanocluster alloys for the oxygen reduction reaction (ORR). A systematic benchmarking of state-of-the-art GNN architectures reveals that attention-based models provide superior predictive accuracy for catalytic activity. The proposed framework is scalable, interpretable, and capable of capturing morphology-dependent reactivity in nanoclusters. This approach significantly accelerates the discovery of high-performance nanocluster catalysts and can be extended to a broad range of catalytic processes beyond.¹⁻³

Keywords: GNN; DFT; CHGNet; Nanocluster; ORR

References

1. R. K. Sharma, H. Minhas, and B. Pathak, *J. Phys. Chem. Lett.*, 17, (2026), 2689–2701.
2. S. Maity, R. K. Sharma, H. Minhas, and B. Pathak, *Small*, (2026), in press.
3. A. S. Nair and B. Pathak, *WIREs Comput. Mol. Sci.*, 11, (2021), e1508.

Beyond Multimetal Catalysts: Scalable 2D-Integrated β -FeOOH Electro catalysts for Alkaline Water Splitting

Nishad G. Deshpande

Department of Physical and Applied Sciences, Indian Institute of Information Technology, Surat
 nishad.deshpande@iiitsurat.ac.in

As an alternative to conventional oxygen evolution reaction (OER) electro catalysts based on complex bi- or multimetallic Layered double hydroxides (LDHs), we report the design of a simple, self-supported, and single-metal-ion OER electro catalyst integrated with 2D material functionality. In this work, β -like FeOOH nanosword architectures are encapsulated within conductive Reduced graphene oxide (rGO) nanosheets and directly grown on formable Ni foam substrates via a cost-effective repeated soaking and quenching process. The incorporation of the 2D rGO network provides a high surface area, enhanced electrical conductivity, and efficient charge transport pathways. Structural analysis reveals that ultrathin rGO sheets uniformly wrap the β -like FeOOH nanoswords, forming a porous and interconnected hybrid framework. This 2D/1D integrated architecture delivers an exceptionally low overpotential of 210 mV at 10 mA cm^{-2} and maintains stable operation for over 40 hours in alkaline media. Furthermore, a high current density of $\sim 300 \text{ mA cm}^{-2}$ is achieved at cell voltages below 1.8 V, demonstrating its suitability for practical alkaline water electrolysis. Detailed electrochemical and physical investigations indicate that the synergistic interaction between the activated Ni foam substrate, β -phase FeOOH nanosword morphology, and the conductive 2D rGO matrix significantly enhances intrinsic charge transfer and catalytic activity. The presence of the 2D material not only improves electron mobility but also reinforces structural integrity, leading to superior long-term durability. This work highlights a scalable strategy for integrating low-dimensional nanostructures with 2D materials to develop efficient, cost-effective, and industrially viable OER electro catalysts.

Keywords: 2D/1D Hybrid Nanostructures; Layered double hydroxides; Reduced graphene oxide; Oxygen Evolution Reaction (OER).

Role of Neutron Diffraction in Examining Layered Perovskite and Other Emerging Materials

S. D. Kaushik

¹UGC-DAE Consortium for Scientific Research, Mumbai Centre, 246-C CFB,
BARC Campus, Mumbai 400085

*Corresponding author: sdkaushik@csr.res.in / somdattakaushik@gmail.com

In pursuit of materials for futuristic industrial/device applications, perovskite materials have shown enormous versatility. They can be designed in various form such as layered (Ruddleson-popper, Aurivillius), cubic perovskite, double perovskite, triple perovskite while exhibiting unique variation in structural and physical properties viz. ferroelectricity, electrical conductivity, magnetodielectric, magnetoresistance etc. thus, intensively being explored by research group across the world [1,3].

Among several perovskites, layered perovskites consist of infinite sized 2-dimensional (2 D) slab of ABO₃ type structure separated by some motif. Aurivillius compounds are one class of compound exhibiting layered perovskite structure where (A_{n-1}B_nO_{3n+1})₂- is interleaved between (Bi₂O₂)₂⁺ where n belongs to number of perovskite type layers where ions at A (Ca²⁺, Ba²⁺, Sr²⁺, Bi³⁺, La³⁺) and B (Fe³⁺, Ti⁴⁺, Nb⁵⁺, V⁵⁺, W⁶⁺) site can be chosen appropriately. Here, we will discuss the role of temperature dependent neutron diffraction (ND), recorded at PD-III diffractometer at Dhruva, India in examining structural parameter on layered perovskite Bi₅FeTi₃O₁₅ aurivillius compound and their correlation with physical properties such as magnetic, dielectric etc. [1-3].

Recent results of temperature dependent neutron diffraction studies on other emerging materials will also be discussed to further elucidate the importance of structural (crystal and magnetic) characterization and establishing the link in explaining physical properties viz. dielectric, specific heat, AC Susceptibility.

References:

1. D. Prajapat, A. Surampalli, I. Schiesaro, S. D. Kaushik, C. Meneghini, A. Sagdeo, V. G. Sathe, V. Siruguri, E. Welter, V. R. Reddy, J. Phys. Condens. Matter 22 (2021) 355803.
2. V. Siruguri, P. D. Babu, M. Gupta, A. V. Pimpale, P. S. Goyal, Pramana 71 (2008) 1197.
3. R. Jena, K. Chandrakanta, P. Pal, M. F. Abdullah, S. D. Kaushik, A. K. Singh, International Journal of Minerals, Metallurgy and Materials 28 (2021) 1063.

Ultrafast Manipulation of Valley Polarization in Two-Dimensional Transition Metal Dichalcogenides

Suman Kalyan Pal^{1,2}

¹School of Physical Sciences, Indian Institute of Technology Mandi, Kamand, Mandi 175005,
Himachal Pradesh, India

²Advanced Materials Research Centre, Indian Institute of Technology Mandi, Kamand, Mandi
175005, Himachal Pradesh, India

*Corresponding Author E-mail: suman@iitmandi.ac.in

Two-dimensional transition metal dichalcogenides (TMDs) host momentum-inequivalent but energetically degenerate valleys at the K and K' points of the Brillouin zone, arising from broken inversion symmetry. This enables a valley degree of freedom that can be selectively addressed with circularly polarized light, positioning TMDs as promising platforms for valleytronics and emerging quantum technologies. A central challenge, however, lies in the rapid loss of valley polarization and the limited control over intervalley scattering. In this talk, I will present our recent efforts to achieve ultrafast control and enhancement of valley polarization in monolayer TMDs, with an emphasis on MoS₂. Using helicity-resolved transient absorption spectroscopy, we directly track valley-selective carrier dynamics with femtosecond resolution by independently exciting and probing specific valleys.

We demonstrate that alloying and defect engineering provide effective routes to tailor intervalley interactions. In particular, vanadium doping in monolayer MoS₂ significantly alters depolarization pathways, leading to extended valley lifetimes [1]. We further show that substrate-induced strain and controlled defect densities offer additional knobs to modulate scattering processes [2]. Notably, under B-exciton excitation, we uncover spin- and valley-selective coupling between vanadium-induced defect states and higher-energy excitons, giving rise to unconventional intervalley scattering channels that depend sensitively on dopant concentration [3]. These results reveal the pivotal role of spin-selective defect-exciton interactions in governing valley dynamics and establish practical strategies for engineering robust, long-lived valley polarization. This work advances the fundamental understanding of valley physics and opens new directions for designing 2D materials for valleytronic and quantum device applications.

Keywords: 2D TMDs; Valley degree of freedom; Valley polarization; Helicity-resolved transient absorption spectroscopy; Valley dynamics.

References:

1. D. Maity, B. Upadhyay, J. J. Panda, K. R. Sahoo, A. Sreekant, G. Rajalakshmi, S. K. Pal, T. N. Narayanan, *Phys. Rev. Mater.*, 8, (2024), 064004.
2. A. Soni, N. S. Kamath, IS Fatima, Y-Y Shen, W. H. Chang, A. De Sarkar, S. K. Pal, *Nanoscale*, 17, (2025), 25727–25739.
3. N. S. Kamath, A. Soni, D. Maity, T. N. Narayanan, S. K. Pal, *J. Phys. Chem. C*, 130, (2026), 4504–4513.

Novel Phase Transitions in Two-dimensional Ferroelectric In₂Se₃

Saswata Talukdar, Mitanshi Gupta, Devesh Negi, Suvodeep Paul, Shalini Badola,
Ravi Shankar Singh, and Surajit Saha*

*Department of Physics, Indian Institute of Science Education and Research Bhopal, 462066,
India*

*Corresponding Author E-mail: surajit@iiserb.ac.in

A phase transition represents a collective reorganization of the internal degrees of freedom of a material in response to external stimuli, leading to the emergence of distinct phases separated by well-defined phase boundaries.[1] Among various phase transitions, structural transition plays a pivotal role in determining the materials' properties, especially, in two-dimensional (2D) layered materials, as they involve changes in crystal structure, symmetry, and

the emergent properties.[2] In two-dimensional (2D) layered materials, the weak van der Waals coupling between adjacent layers makes their crystal structures highly susceptible to external perturbations such as temperature, pressure, strain, and thickness. These perturbations can induce changes in stacking order, lattice distortions, and atomic configurations, which are reflected in its various physical properties and the evolution of phonon modes.[3] Temperature-dependent Raman spectroscopy is a powerful tool for probing structural phase transitions in 2D materials, providing insights into variations of phonons associated with these transitions. However, beyond identifying distinct structural phases, understanding the origin of structural phase transitions requires exploring transition pathways and energy barriers. This knowledge is crucial for understanding the kinetics, reversibility, and the potential existence of intermediate states during phase evolution. In recent years, the complex characteristics and polymorphic nature of indium selenide (In_2Se_3) have distinguished it as a prominent candidate among 2D semiconductors for ferroelectric memory, nanoelectronic, and wearable flexible electronic applications. Despite comprehensive research on its thermal, electrical, and optical properties, the structural dynamics, especially, the phase transitions over temperature and its correlation with layer thickness, remain elusive. In this work, temperature-dependent Raman spectroscopy of bulk $\alpha\text{-In}_2\text{Se}_3$ reveals well-defined spectral signatures corresponding to its multiple polymorphic phases, providing direct evidence of its reversible as well as irreversible structural transitions. Our results also show that the phase transition is governed not solely by temperature, but it is further influenced by the local strain, which forms wrinkles on the surface of the material, playing a crucial role in directing the transition pathway. Moreover, comparative Raman measurements on exfoliated flakes highlight a pronounced thickness-dependent modulation of the transition behaviour, underscoring the critical role of dimensionality in governing phase stability and evolution. Altogether, this work presents a comprehensive picture of the phase transitions in In_2Se_3 , offering valuable insights for its integration into future high-performance phase-change memory and flexible electronics applications.

Keywords: Phase transition, 2D materials, Raman spectroscopy, In_2Se_3 , Thickness.

References:

1. H. Yang, S. W. Kim, M. Chhowalla, and Y. H. Lee, *Nature Physics*, 13, (2017), 931-937.
2. W. Li, X. Qian, and J. Li, *Nature Reviews Materials*, 6, (2021), 829-846.
3. M. S. Choi, B. K. Cheong, C. H. Ra, S. Lee, J. H. Bae, S. Lee, G. D. Lee, C. W. Yang, J. Hone, and W. J. Yoo, *Advanced Materials*, 29, (2017), 1703568.

Photocatalytic Properties of Three-Dimensionally Ordered Macroporous Metal Oxynitride Inverse Opals

Taiki MAEKAWA^{1,2}, Geoffrey I. N. WATERHOUSE³, Toshihiro MORIGA^{1,*}

¹*Department of Applied Chemistry, Graduate School of Science and Technology for Innovation, Tokushima University, 2-1 Minami-Josanjima, Tokushima 770-8506, Japan*

²*Department of Life Science and Applied Chemistry, Graduate School of Engineering, Nagoya Institute of Technology, Gokiso-Cho, Showa-Ku, Nagoya 466-8555, Japan*

³*School of Chemical Sciences, The University of Auckland, Private Bag 92019, Auckland, New Zealand*

*Corresponding Author E-mail: moriga@tokushima-u.ac.jp

We synthesized TaON inverse opal (IO) photocatalysts with a well-defined three-dimensionally ordered macroporous (3DOM) structure using a combination of colloidal crystal templating and thermal nitridation[1]. These materials exhibited vivid structural color and tunable photonic band gaps (PBGs), allowing precise alignment with the optical absorption edge. Compared to conventional fine powders, the 3DOM TaON IOs showed significantly enhanced hydrogen evolution, owing to improved light harvesting, efficient charge carrier separation, and enhanced mass transport[2]. In particular, the slow photon effect played a key role: aligning the red edge of the PBG with the absorption edge enhanced light absorption and electron–hole pair generation, while aligning the blue edge suppressed carrier recombination without sacrificing absorption, both resulting in increased H₂ production.

In addition, we developed a synergistic strategy to further improve oxynitride photocatalysts by combining 3DOM photonic crystal structures with defect control through cation stoichiometry in LaTiO₂N[3]. The formation of a stable 3DOM structure in this multi-cation system was achieved via a sacrificial template method, aided by citric acid chelation of metal precursors. By slightly adjusting the La/Ti ratio, we successfully suppressed defect formation during nitridation. In lanthanum-rich samples, a La₂O₃ phase formed on the surface, acting as a protective layer that inhibits excessive reduction of titanium and reduces anion vacancy formation. As a result, these samples exhibited more than fourfold higher photocatalytic activity compared to bulk fine powders. Overall, this work demonstrates that integrating photonic crystal engineering with precise compositional control is a powerful strategy for designing highly efficient oxynitride photocatalysts.

Keywords: inverse opal photonic structure; photonic band gap; photocatalytic enhancement; oxynitride

References:

1. T. Maekawa, H. Maekawa, Y. Ikeda, T. Onoe, G.I.N. Waterhouse, K. Murai, T. Moriga, *Open Ceram.*, 20, (2024) 100688.
2. T. Maekawa, Y-S. Huang, N. Tateishi, A. Nakanishi, T. Onoe, Y. Dong, G.I.N. Waterhouse, K. Murai, T. Moriga, *J. Solid State Chem.*, 329, (2024) 124404.
3. T. Maekawa, N. Tateishi, Y. Ikeda, H. Maekawa, G.I.N. Waterhouse, K. Murai, T. Moriga, *Chem. Mater.*, 37, (2025) 7439-7449.

Two-dimensional materials for on-chip micro-supercapacitor

Santilata Sahoo, Namrata Harbola, Prem Sagar Shukla, Abha Misra*

Department of Instrumentation and Applied Physics, Indian Institute of Science, Bangalore, Karnataka 560012

*Corresponding Author E-mail: abha@iisc.ac.in

Revolutionary two-dimensional (2D) materials have opened a wide avenue of new applications, including energy storage. Nanomaterials have so far played a vital role in the development of micro-supercapacitors; however, available fabrication methods have limited the size of planar micro-supercapacitors to the millimetre scale only. In this talk, I will highlight the

research in my group on exploiting the properties of 2D materials to develop an on-chip supercapacitor with unprecedented energy storage capability. The field effect transistors (FET) are used as novel electrodes for a gate-tunable density of states in 2D semiconductors. Molybdenum disulfide (MoS₂) and graphene interfaces are used to fabricate a symmetric micro-supercapacitor on Si/SiO₂ substrate for an on-chip device. Back-gate bias-controlled electric double-layer enhances ion interaction at both edges and interlayers of MoS₂, thereby increasing the electrochemical capacitance by three orders of magnitude. We also used carbon nanotubes (CNTs) to investigate the effect of alignment on the formation of an efficient network in a planar supercapacitor. A notable difference is observed in the optimal alignment of the CNTs between the electrodes.

Keywords: Micro-supercapacitor, MoS₂, Micro-electrodes

References:

1. On-chip micro-supercapacitor of aligned CNT/MnO₂, Materials Today Communications, 50, 114350, 2026.
2. Gate Field Induced Extraordinary Energy Storage in MoS₂-Graphene-Based Ultramicro-Electrochemical Capacitor, ACS Energy Lett. 2023, 8, 3, 1510–1519.

Atomically Precise Metal Nanoclusters for Energy and Pharmaceuticals

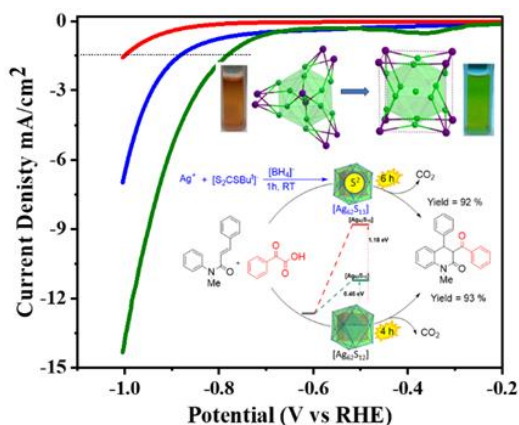
Rajendra Singh Dhayal

*Department of Chemistry, School of Chemical Sciences and Pharmacy,
Central University of Rajasthan, Ajmer-305817, Rajasthan*

*Corresponding Author E-mail: rajendra.dhayal@curaj.ac.in

Dichalcogenolates effectively stabilise air- and moisture-resistant metal (Cu/Ag) nanoclusters, including well-defined Cu₂₀–Cu₃₂ species, which release hydrogen and electrocatalytically reduce CO₂ at low overpotentials. On the other hand, asymmetric dichalcogenolates stabilised [Ag₂₁{S₂P(OR)(p-C₆H₄OCH₃)₂}₁₂]⁺ NCs facilitate being an eight-electron superatom (1S₂₁P₆), in which eight capping Ag atoms exhibit structural isomerism with documented isoelectronic [Ag₂₁{S₂P(OiPr)₂}₁₂]PF₆, with tuning of luminescence properties. Ferrocenyl dithiophosphonates stabilized silver clusters demonstrate a μ 1-coordinate S site acting as the reaction center for H₂ evolution in water splitting. The synthesised Ag-S type quantum dots versus superatom as nano-catalyst reveal a significant role of a single sulfur atom in modulating the decarboxylative radical cascade reaction. Mixed-Valent Cu(I)/Cu(II) nanocluster Cu₅₅ reveals dual (HER vs OER) activity in water splitting and catalytic in carboboration reactions with activation of B-halide bond.

Keywords: Metal Hydrides; Nanoclusters; Dichalcogenides; Energy Harvesting.



References:

1. D. K. Jangid, A. Prakash, S. K. Bose, and R. S. Dhayal, *Angew. Chem. Int. Ed.* 65 (2026), e23716.
 2. A. Prakash, S. Basappa, R. R. Urkude, R. Jangir, R. S. Dhayal, and S. K. Bose, *Angew. Chem. Int. Ed.* 64(2025), e202418901.
 3. D. K. Jangid, P. Kumari, S. K. Bose, K. K. Haldar, K. Mondal, and R. S. Dhayal *Inorg. Chem.* 65(2026), 2599.
-

Electrochromic materials and device: smart windows for multifunctional applications

Rajesh Kumar

Department of Physics, Institute of Technology Indore Khandwa Rd, Simrol, Indore-453552, India

The modern era is an era of science and technology that affects all aspects of life. The simplest physical manifestation of the term 'technology' takes place in the form of a simple and user-friendly device. The design and development of a smart device, capable of performing at its best, depends on the level of scientific understanding of the basic principles that make the actual backbone on which the technology relies. This leads to progressive development and research in electronics, materials science and related fields. An electrochromic device (ECD) is a perfect example of a smart yet very simple form of a device that can be useful for a range of applications, including electrochromism: the phenomenon of bias-induced optical modulation, shown primarily by at least one of the components in an ECD. This talk attempts to provide an update on different materials' regimes and highlights the basic requirements that a material must have to be suitable for use as the active component of an ECD. This gives the details of several electrochromic materials, and their electrochemical, electrochromic and other functional properties in brief. An attempt has been made to cover a subset of the vast superset of known electrochromic materials, which can be extended as a detailed comprehensive review in the future. Based on the type(s) of materials used in an ECD, device paradigms are defined, which have also been summarized through different device regimes, namely 'all-organic', 'all-inorganic' and 'hybrid'. To provide a glimpse, ECDs that are multifunctional and that show other functionalities like memory, and supercapacitive and sensing characteristics will also be discussed.

Strategies to Improve the Photocatalytic Performance of Bismuth Halide-Based Nanocomposites

Yasuyuki Maeda¹, Niloy Paul¹, Chang-Mou Wu³, Akihiro Furube⁴, and Pankaj Koinkar^{2,4,*}

¹*Graduate School of Science and Technology for Innovation, Tokushima University, Tokushima 770-8501, Japan*

²*Graduate School of Technology, Industrial and Social Science, Tokushima University, Tokushima 770 8506, Japan*

³*Department of Materials Science and Engineering, National Taiwan University of Science and Technology, Taipei 10607, Taiwan, ROC*

⁴*Institute of Post-LED Photonics, Tokushima University, Tokushima 770 8506, Japan*

*Corresponding Author E-mail: koinkar@tokushima-u.ac.jp

Bismuth halide-based materials (BiOX, where X = Cl, Br, or I) have emerged as promising photocatalysts owing to their distinctive layered crystal structures, suitable bandgap energies, internal electric fields, and excellent chemical stability. These characteristics enable efficient charge generation and transport, making BiOX materials attractive for applications in environmental remediation and solar energy conversion. However, their widespread practical implementation is still limited by several intrinsic drawbacks, particularly the rapid recombination of photogenerated electron-hole pairs and insufficient absorption of visible light. Addressing these challenges is therefore critical for realizing their full photocatalytic potential.

This talk provides a comprehensive overview of current strategies for enhancing the photocatalytic performance of BiOX-based nanocomposites. First, morphological engineering approaches are discussed, with particular emphasis on the fabrication of two-dimensional nanosheets and three-dimensional hierarchical structures. These architectures significantly increase specific surface area and expose more active sites, thereby improving photocatalytic reaction kinetics. Next, recent progress in heterojunction construction is summarized, with a focus on the design of effective interfaces that promote directional charge transfer and inhibit carrier recombination via favorable band alignment. In addition, defect engineering strategies, such as the deliberate introduction of oxygen vacancies, are examined for their ability to tune the electronic structure and extend light absorption. Noble metal modification, which exploits surface plasmon resonance, is also reviewed as an effective method for enhancing visible-light utilization.

The synergistic effects of these modification strategies are evaluated through representative applications, including the degradation of organic pollutants. Drawing on recent advances, this talk highlights that the rational integration of interfacial engineering, defect control, and crystal facet regulation offers a highly effective pathway to developing high-performance bismuth halide photocatalysts, contributing to sustainable environmental purification and solar-driven energy production.

Keywords: Bismuth Halide; Nanocomposites; Photocatalysis; Heterojunction; Charge Separation

References:

1. D. Nakayama, C. M. Wu, K. G. Matora, P. Koinkar and A. Furube, *New Journal of Chemistry*, 47, 22078-22089, 2023.
2. Y. Maeda, C. M. Wu, K. G. Matora, P. Koinkar, and A. Furube, *Journal of Materials Science: Materials in Electronics*, 36, (2025) 1407.

3. K. S. Lin, A. Furube, T. Katayama, P. Koinkar, and C. M. Wu, *Modern Physics Letters. B*, 39, (2025) 2441007.

On-Chip Full-UV-Band Photodetectors Enabled by Hot Hole Extraction

Sougata Karmakar¹, Kuntal Singh¹ et al., K. D. M. Rao^{1*}

¹*School of Applied and Interdisciplinary Sciences, Indian Association for the Cultivation of Science (IACS), Kolkata 700032, India ¹*

*Corresponding Author E-mail: saiskdmrao@iacs.res.in

Achieving on-chip full-UV-band photodetection across UV-A (315–400 nm), UV-B (280–315 nm), and UV-C (100–280 nm) regions remains challenging due to the limitations of conventional materials, which often exhibit narrow spectral response and require high operating voltages. In this work, we present a self-driven, on-chip photodetector based on a micro–nano engineered heterostructure comprising hybrid gold nanoislands (Au NIs) embedded in H-glass and cesium bismuth iodide (Cs₃Bi₂I₉). The Au nanoislands act as catalytic nucleation centers, enhancing crystallinity and promoting the vertical growth of interconnected petal-like Cs₃Bi₂I₉ microstructures. This micro–nanostructured interface plays a crucial role in improving light absorption and charge transport. Under UV illumination, energetic hot carriers are generated within the Au nanoislands. A built-in electric field at the heterojunction facilitates efficient separation of these carriers, enabling direct transfer of hot holes into the valence band of Cs₃Bi₂I₉ while minimizing recombination losses. The device exhibits an ultrahigh open-circuit voltage of 0.6 V, high responsivity of 0.88 A/W, and an ultralow detection limit of 90 nW/cm² under self-driven operation. These performance metrics surpass those of conventional thin-film UV photodetectors. Furthermore, the device demonstrates excellent operational stability, maintaining consistent performance under ambient conditions for up to eight months. This work highlights the importance of micro–nano structural engineering and hot-carrier dynamics in achieving high-performance, self-powered photodetectors. The proposed architecture provides a scalable pathway for next-generation full-UV-band optoelectronic devices with potential applications in environmental monitoring, flame detection, biomedical diagnostics, and secure communication systems.

Keywords: Micro–nanostructures; Optoelectronic devices; Photodetectors; Halide perovskites; Broadband detection

References:

1. K. Singh, M. Hossain, P. K. Nayak, S. Karmakar, A. Tripathi, P. Sarkar, P. Rudra, A. V. Muhammed Ali, G. K. Grandhi, P. Vivo, S. Mondal, K. V. Adarsh, D. Ghosh, and K. D. M. Rao, *Mater. Horiz.*, 12 (2025), 3865–3877.
2. S. Karmakar, S. Ash, S. Haque, N. K. Murugasenapathi, M. Sridevi, I. Mandal, G. Ghorai, A. V. Muhammed Ali, N. N. Gosvami, N. M. A. Krishnan, S. Kanungo, M. Chirumamilla, T. Palanisamy, R. K. Singh, A. R. Allu, and K. D. M. Rao, *ACS Nano*, 19 (2025), 6309–6319.
3. D. Nasipuri, S. Karmakar, A. H. Sarkar, N. Goyal, S. Banerjee, R. Das, N. Ravishankar, K. D. M. Rao, and N. Pradhan, *Adv. Mater.*, (2025), doi:10.1002/adma.202510226.

3D Printing Of Advanced Multifunctional 2D Mxene-Based Nanocomposites

Sarika Verma^{1,2,*}

¹Council of Scientific and Industrial Research- Advanced Materials and Processes Research Institute, Hoshangabad Road, Bhopal (M.P.), 462026, India.

²Academy of Council Scientific and Innovative Research, Ghaziabad, U.P., India.

*Correspondence: Sarika Verma Email: sarika.ampri@csir.res.in

This work focuses on the development of advanced multifunctional nanocomposite materials based on two-dimensional (2D) MXenes for additive manufacturing applications. Titanium carbide MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) and molybdenum carbide MXene (Mo_2CT_x) were incorporated with functional metal oxide nanofillers such as copper oxide (CuO), neodymium oxide (Nd_2O_3), and other ceramic reinforcements to formulate highly stable and printable nanocomposite feedstocks printable material. The prepared formulations were optimized for rheological behavior and processed through paste extrusion/direct ink writing (DIW) and fused deposition modeling (FDM)-based 3D printing techniques to fabricate customized geometries with controlled microstructures.

The integration of two-dimensional (2D) MXenes with metal oxide nanoparticles significantly enhances the electrical conductivity, thermal stability, mechanical strength, and multifunctional performance of the 3D printed structures. The developed nanocomposites demonstrate promising characteristics for a wide range of applications including electromagnetic interference (EMI) shielding, energy storage systems, flexible electronics, sensing devices, antibacterial coatings, catalytic systems, and thermal management materials. The study highlights the potential of combining emerging 2D materials with advanced additive manufacturing technologies to produce next-generation smart and high-performance engineered components. The proposed approach offers a scalable and versatile pathway toward the fabrication of multifunctional materials for aerospace, biomedical, environmental, and electronic applications.

Keywords: Additive Manufacturing; MXene; 3D printed Ink

XANES Simulations for Li_3InCl_6 Solid Electrolyte and interfacial reactions using First-Principles Calculations

Tanmoy Paul^{1,2,*}

¹Research Institute for Sustainable Energy (RISE), TCG Centres for Research and Education in Science and Technology, Sector V, Kolkata, 700091, India

²School of Natural Sciences, TCG CREST, Sector V, Kolkata, 700091, India

*tanmoy.paul@tcgcrest.org

A detailed analysis of the theoretical X-ray absorption near-edge structures (XANES) for the Cl and In K-edge in Schottky (LiCl), Frenkel (Cl and In) induced Li_3InCl_6 solid electrolyte employing density-functional theory calculations is presented. The supercell core-hole method is used for the description of electron-hole interactions even for $\text{Li}_3\text{InCl}_6|\text{Li}$ solid and reactive interface. I have found close agreement between the spectra obtained from experimental and that obtained from full-potential linear augmented-plane-wave method. The theoretical spectra of Cl K-edge originated from both 8j and 4i sites of Li_3InCl_6 compound show six features: a double peak at 2825 eV, two weak peaks around 2831 eV, two around 2832 eV and 2834 eV and a hump around 2836 eV. The spectral features of the Cl K-edge match with the Cl s , p_x , p_y , and p_z PDOS since Cl K-edge spectrum should come from $1s \rightarrow 4p$ transition. A shoulder peak at 2820 eV is caused by Cl bonding to oxidized metal centers ($p \rightarrow d$ orbital mixing). The peak at 2825 eV mainly originates from Cl p_z while the shoulder of this peak is caused by Cl p_{xy} hybridization. Peak positions in the computed spectra appear to be correlated with changes in the Li-Cl-Li, In-Cl-Li angles as well as Cl coordination numbers in Cl Frenkel defect containing structure. The theoretical spectra of In K-edge calculated for both 2a and 4g sites do not show any significant shift but a splitting of peak around 27997 eV for the spectrum at 2a site. I have found 3 peaks of which most intense peak at 27955 eV which corresponds to $3+$ state with that obtained for In_2O_3 compound. The peak at 27955 eV and 27981 eV come due to In $1s \rightarrow 5p$ transition although partial p - d hybridization is observed. The Cl K-edge spectrum looks similar with that obtained for bulk for $\text{Li}_3\text{InCl}_6|\text{Li}$ interface whereas significant peak shift is noted for Cl at the interface.

Keywords: XANES; K-edge; Core hole; DFT; Solid electrolytes;

References:

1. D. Cabaret, F. Mauri, and G. S. Henderson, Phys Rev B, 75, (2007), 184205.
2. T. Tamura, M. Kohyama, and S. Ogata, Phys Rev B, 96, (2017), 035107.
3. X. Li, J. Liang, J. Luo, M. N. Banis, C. Wang, W. Li, S. Deng, C. Yu et al., Energy Environ. Sci., 12 (2019), 2665-2671.

In-Situ TEM Study on the Microstructural Evolution of Size-Selected Tantalum Oxide at High Temperatures via Carbothermic Reaction

Vidyadhar Singh^{1*}

¹Department of Physics, Jai Prakash University, Chapra, Bihar-841302, India.

*E-mail: vsraj47@gmail.com

The formation process of refractory carbides by carbothermal reduction has long been a focus of intense theoretical and experimental efforts [1-2]. Among the existing refractory-metal carbides, tantalum carbide has numerous specific applications due to its very high hardness and the highest melting temperature of 4,000 °C [3]. In this study, we use in situ transmission electron microscopy (TEM) at high temperatures to investigate the carbothermic reaction of size-selected amorphous tantalum oxide nanoparticles. Size-selected amorphous tantalum oxide nanoparticles were synthesized by magnetron-sputtering inert-gas condensation method [4]. We found that amorphous tantalum oxide converts to pure tantalum carbide when heated to 1,200 °C in situ TEM on a carbon support under vacuum. This is remarkably low given the bulk attributes of this

refractory material system ($\sim 2,000$ °C), and previous reports for this process. In situ atomic-resolution TEM observations combined with electron energy-loss spectroscopy (EELS) and selected-area electron diffraction (SAED), as a function of temperature, confirm the reduction of oxygen from the tantalum oxide nanoparticles, consumption of carbon from the support by the reacting nanoparticles, and complete formation of stoichiometric TaC. Ultimately, these findings offer broad and exciting prospects for understanding the carbothermal reduction mechanism of other carbides.

Keywords: Carbides; Gas-phase Synthesis; High Resolution Transmission Electron Microscopy; Carbothermic Reaction

References:

1. V. G. Pol, S. V. & Pol, A. Gedanken Adv. Mater. 23 (2011) 1179–1190.
2. L., Zhang, J. Hu, A. A. & Voevodin, H. Fong, Nanoscale, 2 (2010) 1670–1673.
3. A. Gubernat, “Pressure less sintering of single-phase tantalum carbide and niobium carbide.” J. Eur. Ceram. Soc. 33 (2013) 2391–2398.
4. V. Singh, P. Grammatikopoulos, C. Cassidy, M. Benelmekki, M. Bohra, Z. Hawash, K.W. Baughman, M. Sowwan “Assembly of tantalum porous films with graded oxidation profile from size-selected nanoparticles.” J. Nanopart. Res. 16 (2014) 2373.

Flexible 2D Materials Based Piezoelectric Nanogenerator

Manoj Kumar Gupta*

*CSIR-Advanced Materials and Processes Research Institute Bhopal, Madhya Pradesh 462026
India*

*Academy of Scientific and Innovative Research (AcSIR), Ghaziabad, Uttar Pradesh, India
E-mail: mkgupt.ampri@csir.res.in**

Flexible two-dimensional (2D) piezoelectric nanogenerators have emerged as promising energy-harvesting devices capable of converting mechanical energy into electrical energy for powering nanosystems and low-power portable electronics. Among various 2D materials, monoelemental borophene and transition-metal dichalcogenides (TMDs), particularly MoS₂, have attracted significant attention due to their unique structural, electronic, and piezoelectric properties.

In this work, we have successfully synthesized several novel 2D materials, including piezoelectric semiconducting ZnO nanosheets, few-layer crystalline MoS₂ nanosheets, and chemically exfoliated borophene nanosheets through cost-effective, low-temperature processes. Flexible piezoelectric nanogenerators based on vertically aligned ZnO nanosheets and borophene-MoS₂ nanosheet heterostructures were fabricated and device performance were evaluated for mechanical energy harvesting. The ZnO nanosheet-based nanogenerator generated a DC-type output current density of $1.2 \mu\text{A cm}^{-2}$ under vertical mechanical pressure. In contrast, the borophene-MoS₂ heterostructure-based nanogenerator exhibited a significantly enhanced performance, delivering a high output voltage of 22 V and an ultrahigh current density of $34 \mu\text{A cm}^{-2}$ under identical loading conditions. Furthermore, the output characteristics of the flexible borophene-MoS₂ nanogenerator were investigated under UV illumination, demonstrating its capability as a self-powered UV photodetector. These results highlight the potential of 2D borophene-MoS₂ heterostructures for next-generation flexible energy-harvesting and self-powered sensing applications.

Keywords: Nanogenerator ; 2D Materials ; Energy Harvesting ; Dielectric

Reference:

1. Charu Sharma, Avnish Kumar Srivastava Manoj Kumar Gupta* 2D borophene–MoS₂ nanosheet heterostructure as a flexible piezoelectric nanogenerator for self-powered piezotronic sensing, *Communications Materials (Nature Portfolio)* 6 (1), 1-15, 2025.
2. Priyanka Arya, Manoj K Gupta*, Umesh Tiwari, High performance flexible surface functional mediated piezoelectric MXene Ti₃C₂Tx nanosheets based nanogenerator, , *Journal of Alloys and Compounds*, 1043, 2025, 184283.
3. D.K. Kashyap, A.K. Chaturvedi, A. Pappu, A.K. Srivastava, M.K. Gupta, Hydrophobic Fully Wearable Piezoelectric WS₂ Quantum Dot-Polyaniline-Based Nanogenerator, , *ACS Applied Electronic Materials*, 7, 2025, 3090–3103.

Enhanced HER Activity via Binder-Free Co-MOF Thin Films with Optimized Coordination Environment

Reshma Khandekara, Susmita Patila, Rohit Sutara, Rahul Patilb, Ashok Chougalec, Jyotiprakash Yadav ^{a*}

^a *Energy Research and Thin Film Laboratory, USIC, Shivaji University, Kolhapur, 416004, Maharashtra, India.*

Department of Physics, Shri Yashwantrao Patil Science College, Solankur, Shivaji University, Kolhapur, 416212 Maharashtra, India.

^c *Department of Chemistry, The New College, Kolhapur, Shivaji University, Kolhapur, 416012, Maharashtra, India.*

*Corresponding author: Dr. Jyotiprakash B. Yadav

E-mail: jbyadavphy@gmail.com

The development of cost-effective electrocatalysts for the hydrogen evolution reaction (HER) is essential to minimize dependence on scarce and expensive noble metals while advancing sustainable hydrogen production technologies. In this work, we report a facile and binder-free approach for the fabrication of cobalt-based metal–organic framework (Co-MOF) thin films directly grown on a stainless-steel substrate via a chemical method. The metal-to-ligand ratio was systematically tuned (1:3, 1:4, 1:5, and 1:6) to optimize the electrocatalytic performance. Among the prepared samples, the optimized electrode, denoted as (1:5) CoM-3, exhibits the best HER activity, requiring an overpotential of 258 mV to achieve a current density of 10 mA cm⁻², along with a Tafel slope of 76 mV dec⁻¹, indicating favourable reaction kinetics. The electrode also demonstrates a high electrochemically active surface area (ECSA) of 11.97 cm², suggesting an increased density of accessible active sites. Furthermore, evaluation in a prototype water electrolysis system reveals that the optimized catalyst delivers a maximum hydrogen generation rate of 625.33 mL h⁻¹, outperforming other compositions. The binder-free growth of Co-MOF thin films directly on the conductive substrate enhances interfacial contact, leading to improved electrical conductivity and more efficient charge transfer during the catalytic process. In addition, the electrode shows excellent operational stability, maintaining consistent performance over 100 hours of continuous electrolysis. Overall, these findings demonstrate that Co-MOF thin films

represent a promising, low-cost, and scalable class of electrocatalysts for HER, with significant potential for application in sustainable hydrogen production and clean energy technologies.

Theory-Guided 2D Semiconductive Materials for Energy Applications: Insights from Ni_3TeO_6

Javier Fernández-Catalá¹, Andrey A. Kistanov¹, Yang Bai¹, Wei Cao¹, Harishchandra Singh^{1,2,*}

¹*Nano and Molecular Systems Research Unit (NANOMO), University of Oulu, Oulu, FIN-90014, Finland*

²*Amity Institute of Applied Sciences, Amity University Uttar Pradesh, Noida – 201313, India*

*Corresponding Author E-mail: singh85harish@gmail.com

The synergy of theory and experiment is pivotal for unveiling the potential of new energy materials. Guided by first-principles calculations predicting a stable 2D monolayer of Ni_3TeO_6 (NTO) with a reduced bandgap relative to its bulk counterpart, we synthesised two-dimensional NTO nanosheets via a one-pot shape-controlled hydrothermal route. The theoretical design involved coordination of metalloid chalcogen, slicing the bulk trigonal R3 structure, geometrical optimisations, and stability verification. Experimental results confirm the theoretically predicted enhanced semiconducting properties, demonstrating superior photoconductivity and wavelength-selective photodetection. Most significantly, the 2D nanosheets achieve visible-light-driven hydrogen evolution without any co-catalyst or sacrificial scavenger — a direct consequence of the low dimensionality enabling efficient photoinduced charge migration to surface-active sites. The photocatalytic time-course measurements confirm both activity and aqueous stability of the nanosheets over extended reaction periods. The unique polar space group (R3) of the NTO structure, combined with the increased surface-to-volume ratio inherent to 2D morphology, underpins both the enhanced optoelectronic response and the emerging photocatalytic functionality absent in the bulk phase. This work establishes 2D Ni_3TeO_6 as a promising candidate for photocatalysis and optoelectronic applications, and demonstrates a robust theory-to-synthesis-to-application pipeline applicable to broader families of 2D metal tellurate semiconductors [1,2,3].

Keywords: 2D materials; Ni_3TeO_6 ; photocatalytic hydrogen evolution; first-principles calculations; optoelectronics

References:

1. J. Fernández-Catalá, A. A. Kistanov, Y. Bai, H. Singh, and W. Cao, npj 2D Materials and Applications, 7, (2023), 48.
 2. H. Singh, M. Veijola-Kara, E. Rani, L. S. Bezerra, P. Talebi, H. L. S. dos Santos, A. K. Patel, M. Huttula, and P. H. C. Camargo, ACS Applied Nano Materials, 7, (2024), 12480–12489.
 3. J. Fernández-Catalá, L. Jussila, M. Daboczi, F. Temerov, S. Eslava, R. Greco, and W. Cao, Crystal Growth & Design, 23, (2023), 8828–8837.
-

Compositional Engineering of Two-dimensional MXenes for Electrocatalytic Green Hydrogen Production

Anupma Thakur

Department of Materials Engineering, Indian Institute of Science, Bangalore, KA 560012, India

*Corresponding Author E-mail: anupmathakur@iisc.ac.in

MXenes, with their $Mn+1XnTx$ chemical formula, have one of the most diverse chemical compositions and structures among the family of two-dimensional (2D) materials [1]. This notable compositional diversity, spanning across the transition metal carbides, nitrides, and carbonitrides, positions MXenes as potential material candidates for applications ranging from energy conversion and storage to biomedicine. The rapid growth of the 2D MXenes family is driven by their tailorable surface chemistry, diverse compositions, structures, and architectures, including the transition metal and surface functional groups, non-metal X sublattice, and atomic-layer configurations [2]. This compositional diversity controls MXenes' structure-property relationships, which tailor their functional performance in various application domains. In this talk, we will explore how variations in compositional diversity, atomic arrangement, atomistic design, and material engineering strategies enable the systematic fine-tuning of the properties of 2D MXenes and their applications in clean energy [3]. The discussion will center around the six key factors that play a critical role in tunability and controlling MXenes' performance for electrocatalysis, including composition (M, X, and Tx), the thickness of the 2D flake (number of atomic layers), lateral flake dimensions, surface adatoms, intercalating ions, atomic defects (e.g., vacancies), and heteroatom doping [3]. The discussion will highlight recent advances in the growing research on MXenes and offer forward-looking perspectives on leveraging MXene chemistry to develop high-performance 2D catalysts critical for achieving a carbon-neutral economy.

Keywords: MXenes; 2D materials; Electrocatalysis; Hydrogen; Clean energy

References:

1. Thakur, A., Kim, J., Wyatt, B. C., Gogotsi, Y., & Anasori, B. *Nature Reviews Materials*, (2026), 1-16.
2. Thakur, A., & Anasori, B. *Science*, (2024), 383(6688), 1182-1183.
3. Thakur, A., Chandran, N., Wyatt, B. C., Bedford, A., Firouzjaei, M. D., Kamath, K., Muhammad S. & Anasori, B. *Materials Today* (2025) 88, 1-11.

Plating Dynamics in Sodium Metal Anodes: Linking Molecular Interactions to Macroscopic Stability

Kingshuk Roy^{1,*}

¹ *aResearch Institute for Sustainable Energy, TCG Centres for Research and Education in Science and Technology, Salt Lake, Kolkata 700091, West Bengal, India*

Email address: kingshuk.roy@tcgcrest.org

The promise of sodium metal batteries, particularly anode-less configurations, rests on mastering one deceptively simple event: the first nucleation of sodium on a bare substrate. Yet, this event is governed by a complex web of interfacial kinetics, solvation chemistry, nucleation energetics, and evolving interphases. This talk will present a ground-up exploration of sodium metal anode stability, integrating molecular-level electrolyte behavior, current collector surface chemistry, and spatially modulated ion flux. This talk will begin with probing the interfacial charge transfer kinetics and solvation-desolvation dynamics using fast-scan voltammetry, capturing transient plating behavior in real time. By combining variable-temperature nuclear magnetic resonance, Raman spectroscopy, and several other fundamental material and electrochemical characterization techniques across length and time scales, this talk will help uncover how solvation asymmetry and mobility dictate desolvation energetics and SEI evolution. These insights are distilled into mechanistic descriptors that quantify electrolyte efficacy. Building on this, the talk will also demonstrate how spatially engineered sodiophilic domains engineer sodium ion flux and stabilize deposition morphology. Various degrees of post-cycling in-situ measurements will reveal that edge-localized electric fields and chemical defects can guide uniform plating and suppress dendrites. Together, these findings reframe the problem of Na metal stability; not as a materials limitation, but as an interfacial design challenge, solvable through a unified, multiscale strategy linking molecules to morphology.

Keywords: Sodium Metal Anode, Interfacial Kinetics, Desolvation Dynamics, Spatial Ion Flux Engineering, Multiscale Characterization

Tailoring the Synergy of Transition Metal-Based V₂CT_x Nanocomposite for Efficient Hydrogen Evolution Reaction

Mallamma Jinagi ¹, Seiji Yamazoe ², Sébastien Royer ³, Manav Saxena ^{1*}

¹ *Centre for Nano and Material Sciences, Jain (Deemed-to-be University)
Jain Global Campus, Kanakapura, Ramanagara, Bangalore, 562112, India.*

² *Department of Chemistry, School of Science, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo, 113-0033, Japan.*

³ *Unité de Chimie Environnementale et Interactions sur le Vivant, Dunkerque, France*

*Corresponding author E-mail: s.manav@jainuniversity.ac.in

Electrocatalytic water splitting is a powerful technology that produces clean hydrogen without greenhouse gas emissions, enabling the mass production of hydrogen via a renewable-energy route [1] [2]. Highly active, low-overpotential electrocatalysts are essential for achieving fast kinetics in the hydrogen evolution reaction (HER) [3]. MXene, an emerging family of 2D layered materials consisting of transition-metal carbides, nitrides, and carbonitrides, has attracted significant attention for energy conversion applications [4]. To create an efficient electrocatalyst for HER, the high binding energy of MXene must be reduced through electronic modification of active sites [4].

The excellent catalytic activity and corrosion resistance in acidic media make noble metal nanoparticles (NPs) the best HER catalysts. Tuning the MXene surface properties and forming composites with noble metals is necessary to improve the electrocatalytic properties of V-based MXene. In this context, attention may shift toward the use of alloyed NPs on an MXene

support, which could alter the MXene substrate's electronic and geometric properties [1]. We propose the optimized PdAuV₂CT_x composite to modify the electronic structure and enhance catalytic activity toward HER. Herein, we report a highly efficient MXene nanocomposite electrocatalyst for the hydrogen evolution reaction in acidic media, with an overpotential of 23 mV, a Tafel slope of 40 mV dec⁻¹, and stability for 100 hours.

Keywords: MXenes; Hydrogen evolution reaction, PdAuV₂CT_x Nanocomposite.

References:

1. Darabdhara, G., Amin, M. A., Mersal, G. A., Ahmed, E. M., Das, M. R., Zakaria, M. B., & Boukherroub, R. (2015). Journal of Materials Chemistry A, 3(40), 20254-20266.
2. Kwak, K., Choi, W., Tang, Q., Kim, M., Lee, Y., Jiang, D. E., & Lee, D. (2017). Nature communications, 8(1), 14723.
3. Sun, Y., Lee, J., Kwon, N. H., Lim, J., Jin, X., Gogotsi, Y., & Hwang, S. J. (2024). ACS nano, 18(8), 6243-6255.
4. Park, S., Lee, Y. L., Yoon, Y., Park, S. Y., Yim, S., Song, W. & An, K. S. (2022). Applied Catalysis B: Environmental, 304, 120989.

Interface Design Beyond Epitaxy

Varun Harbola^{1*}

¹Department of Condensed Matter Physics and Materials Science, Tata Institute of Fundamental Research, Mumbai 400005, India

**v.harbola@tifr.res.in*

The study of thin films, their interfaces and thin film heterostructures has been a cornerstone of experimental research at reduced dimensions. Advancements in epitaxial techniques have enabled the engineering of atomically precise and well-defined interfaces. However, the parameter space of interfaces enabled by epitaxy is determined by the lattice symmetry of the substrate and the plane on which epitaxy takes place. This for such interfaces is taken as a given. I will take this opportunity to focus on recent developments in oxide growth enabling the separation of the grown thin films from the growth substrate, resulting in freestanding oxide membranes [1]. These membranes have allowed for unprecedented access to avenues in oxides, with novel symmetry disallowed interfaces, which go beyond the epitaxially possible atomically sharp interfaces. I will show striking phenomena we have observed with an exemplary SrTiO₃ (001) on a sapphire (0001) interface [2], through which I hope to convey how these developments in oxides promise a fertile ground for remarkable discoveries in material science and physics.

Keywords: Freestanding Membranes; Heterostructures; Oxides; Interfaces

References:

1. Di Lu, David J Baek, Seung Sae Hong, Lena F Kourkoutis, Yasuyuki Hikita, Harold Y Hwang, Nature Materials, 15, (2016), 1255-1260.
2. Hongguang Wang*, Varun Harbola*, Yu-Jung Wu, Peter A van Aken, Jochen Mannhart, Advanced Materials, 36, (2024), 2405065.

Ultra-high vacuum systems for scientific research

Kevin Yang

AdNaNo-tek Corp, Taiwan

Recent advancements in quantum computing, advanced electronics, and surface science demand unprecedented control over thin-film purity, interface sharpness, and material stoichiometry. For achieving these properties requires ultra-high vacuum (UHV) conditions and highly specialized deposition and etching systems. This talk introduces AdNaNo-Tek Corporation's solutions in customizable, research-grade UHV cluster and tunnel systems, engineered to bridge the gap between complex material design and reliable experimental execution.

We will highlight systems designed for complex heterostructure growth, including Molecular Beam Epitaxy (MBE), Electron-Beam Evaporation, and Magnetron Sputtering, as well as specialized platforms like the Josephson Junction E-Beam Evaporator, which facilitates sequential shadow evaporation, etching, and oxidation without breaking UHV. Furthermore, we address a notorious bottleneck in high-temperature oxide research: the rapid failure of conventional resistive heaters in reactive environments. We present a 5-Axis Laser Heating Manipulator capable of heating substrates in oxygen-rich or atmospheric conditions by eliminating the in-chamber filament entirely, thereby maximizing system uptime and process stability for Pulsed Laser Deposition (PLD) and oxide MBE.

Finally, the discussion will cover interconnected UHV multi chamber systems that seamlessly combine thin-film synthesis modules (such as Atomic Layer Deposition) with advanced, in-situ analytical instrumentation. By examining real-world material applications and system integrations, this presentation demonstrates how application-tailored UHV engineering empowers researchers to accelerate discoveries in nanoscale film materials, advanced functional oxides, and quantum device fabrication.

Accelerating 2D Materials Research Through Thermo Fisher Scientific's End-to-End Characterization Workflow

Dr. B. Karthik, Senior Product Manager

*Materials and Structural Analysis Division
Thermo Fisher Scientific India Pvt. Ltd.*

Two-dimensional (2D) materials, including graphene, transition metal dichalcogenides (TMDs), hexagonal boron nitride (h-BN), and van der Waals heterostructures, continue to drive advances in electronics, photonics, energy storage, catalysis, and quantum technologies.¹⁻⁴ Realizing the full potential of these atomically thin materials requires comprehensive characterization across multiple length scales to correlate morphology, structure, composition, and surface chemistry.

Thermo Fisher Scientific provides an end-to-end characterization workflow for 2D materials research. The Apreo SEM enables rapid assessment of surface morphology, layer uniformity, and defect distributions, while Scios and Helios DualBeam platforms support site-specific sample preparation and cross-sectional analysis. The Talos TEM delivers high-

throughput structural and compositional characterization, and the Spectra (S)TEM provides atomic-resolution imaging and spectroscopy to reveal interfaces, lattice defects, dopants, and strain. Complementing electron microscopy, K-Alpha, Nexsa, and ESCALAB QXi+ XPS systems provide quantitative elemental and chemical-state analysis of surfaces and interfaces. Together, these integrated solutions accelerate the discovery, understanding, and development of next-generation 2D materials.

References:

1. Novoselov, K. S. et al. Electric Field Effect in Atomically Thin Carbon Films. *Science* 306, 666–669 (2004).
2. Geim, A. K. & Grigorieva, I. V. Van der Waals Heterostructures. *Nature* 499, 419–425 (2013).
3. Chhowalla, M. et al. The Chemistry of Two-Dimensional Layered Transition Metal Dichalcogenide Nanosheets. *Nature Chemistry* 5, 263–275 (2013).
4. Mak, K. F. & Shan, J. Photonics and Optoelectronics of 2D Semiconductor Transition Metal Dichalcogenides. *Nature Photonics* 10, 216–226 (2016).

Crystal structure, complex phase diagram and re-entrant phase transition in NaSrPO₄

Mangesh Mahajan*, M. Schreyer¹, G. Nénert¹

Malvern Panalytical, India & Almelo¹, The Netherlands

The crystal chemistry of AIBIXO₄ (AI = Alkali ion, BII = alkali-earth ion, X = P, V, As) is very rich and has been widely investigated, particularly the phosphate family [1]. In recent years, we have been investigating the crystal structures [2,3] and magnetic properties of some compositions within the AIBIXO₄ series [4]. However, despite its simple chemistry NaSrPO₄ has never been reported so far. Here, we present the synthesis, crystal structure and phase transitions of this phosphate. Surprisingly, this material exhibits a complex structure (31 atoms in the asymmetric unit-cell, Z = 10) at room temperature characterized by a strongly under bonded Na atom. This under-bonded atom is responsible for the complex and rich phase diagram as function of temperature as illustrated in Figure 1. NaSrPO₄ exhibits 4 phase transitions between room temperature and 750°C. Besides its rich phase diagram, NaSrPO₄ exhibits a re-entrant phase transition slightly below 600°C before to reach a hexagonal paraelastic phase at high temperature. In addition, we show that the sequence of phase transitions is strongly driven by the history of the sample and several phases can be quenched at room temperature.

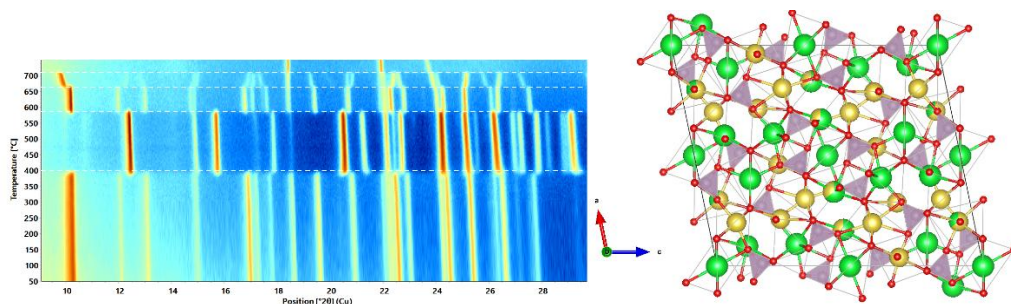


Figure 1. (Left) Temperature phase diagram of NaSrPO₄ and (right) the crystal structure of the a re-entrant polymorph.

Reference:

1. V. A. Isupov, *Ferroelectrics*, 274(1), 203–283 (2002)
 2. G. Nénert, P. O'Meara, T. Degen, *Phys Chem Minerals* 44, 455–463 (2017)
 3. G. Nénert, *Z. Kristallogr.*; 232(10), 669–674 (2017)
 4. G. Nénert, et al., *Inorg. Chem.*, 52, 9627–9635 (2013).
-

Publishing High Impact Research: Insights from the Editor's Desk

Dr. Rohini D. Kitture

*Deputy Editor, Small, Small
Methods, Small Structures, Small Science*
Physical Sciences Portfolio, Wiley*

Publishing high impact research in materials science requires not only scientific excellence but also strategic positioning, clear communication, and an understanding of editorial expectations. This talk offers insights from the editor's desk on navigating the publishing landscape, with a focus on materials science and 2D materials. It will outline how journals evaluate manuscripts, from scope fit and initial screening to peer review and final decisions, and highlight common reasons for desk rejection, such as limited novelty, unclear narrative, or weak presentation.

The session will emphasize crafting compelling manuscripts by targeting the right audience, selecting suitable journals, and presenting data effectively. Key aspects such as writing strong abstracts, designing impactful figures, and preparing effective cover letters will be discussed, along with the role of peer review in improving manuscript quality.

Finally, the talk will touch on evolving trends in scholarly publishing, including open access, research visibility, and ethical considerations such as AI use. Overall, it aims to equip researchers with practical strategies to enhance the visibility and impact of their work.

THESIS' ABSTRACTS

Sodium *p*-Nitrophenolate Dihydrate and Sodium 4-Nitrophenoxide 4-Nitrophenol Dihydrate single crystals for NLO and Terahertz applications

D.Sethupathi^{1,*} and Prasanta Kumar Datta ²

¹ Department of Medical Physics, Bharathiar University, Coimbatore – 641 046, Tamilnadu, India.

² Department of Physics, Indian Institute of Technology, Kharagpur – 721 302, Westbengal, India.

*Corresponding Author E-mail: sethupathidharmalingam@gmail.com

The sodium *p*-Nitrophenolate dihydrate (NPNa.2H₂O) and sodium 4-Nitrophenoxide 4-Nitrophenol dihydrate Na(4-np).(4-npH).2H₂O single crystals were grown from liquid solution growth method at room temperature. The symmetrical behavior of the grown crystal is analyzed by crystallographic structure using single crystal x-ray diffraction. The Attenuated Total Reflectance (ATR) and Potassium Bromide (KBr) pellet Fourier transform infra-red (FT-IR) spectroscopic measurement methods are adopted for the compound functional vibrational behavior analysis and their resultant vibrational recorded information's are compared. The nonlinear optical frequency conversion behavior of the grown crystal compound is studied by Kurtz–Perry powder second harmonic generation (SHG) measurement method and compared with standard KDP crystal reference. The NPNa.2H₂O single crystal performs **4.2** times and Na (4-np).(4-npH).2H₂O crystal performs **5.0** times that of standard potassium dihydrogen phosphate **KDP** crystal performance. Presence of hydrates phase of water component plays the significant role in crystal phase in ultrafast dynamics. The THz-Time domain spectroscopic behavior of the sodium *p*-Nitrophenolate dihydrate (**010**) and the Sodium 4-Nitrophenoxide 4-Nitrophenol dihydrate (**001**) single crystals were performed at nitrogen atmosphere. **Ti:Sa** laser radiation **808 nm** used to excite the material using the source as ZnTe (**110**) orientation. The THz-TDS response of the crystals are compared with standard reference upto **0.2 to 3.0 THz** and also we could found to confirm beyond the standard reference response up to **5.5 THz** among the maximum **10 THz** range.

Keywords: Crystal Growth; FT-IR spectroscopy; SHG; THz-TDS spectroscopy; Refractive index.

References:

1. Sethupathi D. Thesis Reservoir Shodhganga.

Synthesis and Characterization of Transition Metal Carbide (MAX) and MXene Nanoflakes for Energy Storage and Electrocatalysis

Srikanth Ragunath B*

SASTRA Deemed to be University, Thirumalaisamudram, Thanjavur 613401, Tamil Nadu, India

*Corresponding Author E-mail: srikanthragunath96@gmail.com

Rapid growth in population and industrialization created a huge demand for superior energy storage devices. Thus, the researchers are in constant search of developing exotic materials towards energy harvesting and storage devices. Batteries, supercapacitors, and fuel cells received

a great attention to meet the global demand for energy storage. Among them, supercapacitors are widely preferred as energy storage devices due to higher power density, longer cycle life, and fast charging. Recently, “MXenes” – a new class of layered two-dimensional (2D) nanomaterials have been widely employed as electrode material in energy storage applications due to their exciting properties. $\text{Ti}_3\text{C}_2\text{T}_x$ MXene is synthesized from the Ti_3AlC_2 MAX phase a class of hexagonal layered transition metal carbides, nitrides and carbonitrides. MAX phases are synthesized by various methods at a relatively high temperature ($> 1000^\circ\text{C}$). Despite various synthesis approaches, homogeneous mixing of elemental precursor powders plays vital role in uniform distribution of particles and control over the final composition. Traditionally, a high energy ball milling unit is extensively used for mixing the elemental powders although it is time consuming (10 – 72 hr). Further, frictional effect between the powder and balls causes undesired oxidation and contamination of the precursor powders. Therefore, it is necessary to look for an alternative method of mixing the elemental precursor powders without compromising the mixing efficiency.

In this thesis, Ti_3AlC_2 MAX phase is synthesized by using a probe sonication (PS) assisted mixing of elemental precursor powders followed by pressureless sintering under inert atmosphere. A detailed study reveals that a sonication time of 60 min followed by pressure-less sintering at 1500°C for 2 hr in inert condition produces $>99\%$ phase pure Ti_3AlC_2 MAX. Further, pristine Ti_3AlC_2 MAX phase is used as the main precursor to synthesis multilayered ($\text{ML-Ti}_3\text{C}_2\text{T}_x$) and delaminated ($\text{d-Ti}_3\text{C}_2\text{T}_x$) MXene. The electrochemical performance of the delaminated $\text{d-Ti}_3\text{C}_2\text{T}_x$ MXene exhibits a specific capacitance of 255 F g^{-1} at a specific current of 0.5 A g^{-1} . A symmetric supercapacitor fabricated using $\text{d-Ti}_3\text{C}_2\text{T}_x$ MXene delivered a specific energy of 1.75 Wh kg^{-1} at a specific power of 194.32 W kg^{-1} and retains 68.5% capacitance over 5,000 cycles. The electrocatalytic activity of multilayered ($\text{ML-Ti}_3\text{C}_2\text{T}_x$) MXene towards hydrogen evolution reaction (HER) in acidic medium shows a better HER activity with a low overpotential of 147 mV at a current density of 10 mA cm^{-2} . Further, $\text{ML-Ti}_3\text{C}_2\text{T}_x$ showed a two-fold increase in the HER activity under continuous operation (50 hr, 5,000 cycles) when compared with Pt/C, suggesting good stability and durability in hydrogen production at a high rate in an acidic medium. Therefore, the present thesis work suggests the effective synthesis of phase pure Ti_3AlC_2 MAX by a facile PS approach.

Keywords: MAX Phase; MXene Nanoflake; Probe Sonication; Supercapacitor; Electrocatalyst

Synthesis of Ultrathin 2D Metal Oxide/Hydroxide Nanosheets and Heterostructure for Energy Storage and Conversion Applications

Sayali Patil ¹, Manav Saxena ¹, *

¹Centre for Nano and Material Sciences, Jain Global Campus, Jakkasandra, Ramanagaram, Bangalore- 562112, India.

*Corresponding Author E-mail: ap.sayali@jainuniversity.ac.in, s.manav@jainuniversity.ac.in

The growing demand for sustainable energy solutions necessitates the development of advanced materials that can efficiently store and convert energy. In this regard, two-dimensional (2D) metal oxide/hydroxide nanosheets have emerged as promising candidates due to their unique physicochemical properties, high surface area, and tunable electronic structures. This thesis focuses on the synthesis, structural engineering, and electrochemical evaluation of ultrathin 2D metal oxide/hydroxide nanosheets and their heterostructures for energy storage and conversion applications, particularly in supercapacitors and oxygen evolution reaction (OER) catalysis. The study begins with the synthesis of ultrathin, large-area crystalline $\beta\text{-Ni(OH)}_2$ nanosheets, which

exhibit excellent bifunctional performance in both supercapacitors and OER. Their superior electrochemical behavior is attributed to the high crystallinity, abundant active sites, and layered morphology. Building upon this, a scalable approach is employed to fabricate flexible micro-supercapacitors using $\text{Ni}(\text{OH})_2$ nanosheets, achieving enhanced volumetric capacitance and mechanical stability, thereby demonstrating practical applicability in flexible electronics. Further, vertical heterostructures of $\text{Ni}(\text{OH})_2$ and $\text{Co}(\text{OH})_2$ are designed to unlock synergistic effects between the two components. The resulting heterostructures significantly boost the energy and power density of symmetric solid-state micro-supercapacitors, along with excellent flexibility and cycling stability. Lastly, bimetallic NiMoO_4 nanosheets are engineered through precursor ratio tuning to optimize the structural and electronic properties. The resulting material shows remarkable performance in both supercapacitors and OER, underscoring the advantage of compositional modulation. Overall, this work highlights the potential of rationally designed 2D metal hydroxide/oxide nanosheets and their heterostructures as efficient multifunctional materials for next-generation energy storage and conversion devices.

Keywords: 2D nanosheets, Large-area, Heterostructures, Electrocatalysis, Energy storage devices.

References:

1. S. A. Patil, P. B. Jagdale, N. Barman, A. Iqbal, A. Sfeir, S. Royer, R. Thapa, A. K. Samal, and M. Saxena*, *J. Colloid Interface Sci.*, 674, (2024), 587.
2. S. A. Patil, P. B. Jagdale, N. Barman, A. Sfeir, M. Pathak, S. Royer, R. Thapa, A. K. Samal, and M. Saxena*, *ChemSusChem*, 18, (2025), e202500850.
3. S. A. Patil*, P. B. Jagdale, A. Iqbal, S. Reza, M. Jinagi, P. Rajput, A. Sfeir, S. Royer, R. Thapa, A. K. Samal, and M. Saxena*, *J. Mater. Chem. A*, 13, (2025), 15782.

Design and Synthesis of the Highly Porous Solid Electrolyte for Rechargeable Batteries

Name: Mrs. Aparna Suyog Potdar Guide Name: Dr. Milind V. Kulkarni Co-Guide Name: Dr. Bharat B. Kale

*CENTRE FOR MATERIALS FOR ELECTRONICS TECHNOLOGY (C-MET), PANCHWATI,
OFF PASHAN ROAD, PUNE-411008, INDIA*

Growing concerns over energy security and climate change have made efficient, safe energy storage a global priority. Among available technologies, lithium-ion batteries stand out for their versatility and performance yet their reliance on liquid electrolytes continues to raise serious concerns around safety, thermal runaway, and operational limitations. Solid-state electrolytes present a compelling path forward, but achieving adequate ionic conductivity and mechanical durability at room temperature remains an ongoing challenge that this thesis directly confronts.

To address this, three novel polymer-based electrolyte systems were designed, synthesized, and rigorously evaluated. The first involved a PVDF-HFP matrix enriched with UIO-66, a zirconium-based metal-organic framework, alongside a pyrrolidinium ionic liquid. This combination yielded exceptional ionic conductivity, strong thermal resilience, and reliable electrochemical behaviour when assembled with a LiFePO_4 cathode. The second system explored a 2D Ni-BDC metal-organic framework as a composite filler, which naturally facilitated continuous ion-transport channels, translating into impressive rate performance and stable

capacity retention under demanding current conditions. The third electrolyte, built around a polymeric ionic liquid system comprising PDADMA-TFSI, PVDF-HFP, and LiTFSI, achieved a favourable balance of high conductivity, enhanced lithium-ion transference, and durable cycling behaviour suited to high-voltage applications.

Across all three systems, the deliberate incorporation of metal–organic frameworks, ionic liquids, and polymeric ionic liquids consistently improved flexibility, suppressed lithium dendrite formation, and widened electrochemical stability windows shortcomings that have long held back solid-state battery technology.

Taken together, these findings make a strong case for synergistic material design as a practical and scalable route to next-generation solid-state lithium-ion batteries ones that are safer, longer-lasting, and genuinely ready for real-world energy demands.

Novel approaches to synthesis and characterization of 2D MXene-based refractory carbides

Nithin Chandran B.S.^{1,2}, Anupma Thakur², Babak Anasori^{2,3*}, Ravi Kumar^{1*}

¹*Research Center on Ceramic Technologies for Futuristic Mobility and Laboratory for High-Performance Ceramics, Department of Metallurgical and Materials Engineering, Indian Institute of Technology-Madras (IIT Madras), Chennai 600036, India*

²*School of Materials Engineering, Purdue University, West Lafayette, IN 47907, United States*

³*School of Mechanical Engineering, Purdue University, West Lafayette, IN 47907, United States*

*Corresponding Author E-mail: banasori@purdue.edu, nvrk.iitm@gmail.com

Two-dimensional (2D) MXenes are nanometer-thick sheets of transition metal carbides, nitrides, or carbonitrides, with high-temperature stability [1]. MXenes can be used as nanofillers and functional additives in ceramic hybrids, enhancing sintering and mechanical and electrical properties. In this study, we systematically investigate the incorporation of titanium carbide ($\text{Ti}_3\text{C}_2\text{T}_x$) MXene into boron carbide (B_4C) and other refractory carbides using a one-step electrostatic self-assembly method. We tuned the zeta potential of B_4C and $\text{Ti}_3\text{C}_2\text{T}_x$ MXene solutions, gradually added B_4C into MXene solution, and prepared green bodies with 1 to 10 vol% $\text{Ti}_3\text{C}_2\text{T}_x$ [2]. We examined MXene phase stability, reactions, and phase transformations in B_4C via direct-current spark plasma sintering at 1925 °C under 70 MPa vacuum. To understand the reaction pathways, we conducted stepwise sintering at 800 °C to 1800 °C in 200 °C increments. X-ray diffraction and scanning electron microscopy results revealed that $\text{Ti}_3\text{C}_2\text{T}_x$ reacts with B_4C at ~ 1000 °C to form TiB_2 , with complete conversion at 1800 °C, resulting in a TiB_2 - B_4C structure. For samples with 4 vol% MXene or higher, B_2O_3 formed above 1200 °C due to MXene oxygen terminations and disappeared above 1800 °C. The overall oxygen content in MXene-containing samples was lower than in B_4C sintered without MXene, indicating its reducing nature as a 2D carbide. Incorporating 10 vol% MXene and the formation of TiB_2 doubled the hardness while achieving a relative density of ~ 99 %. Compared to traditional bulk TiC , 2D MXene sheets provide a larger surface area and uniform coverage of B_4C particles, lowering diffusion energy and enhancing sintering rates. This study demonstrates a new approach to using MXene as a nanometer-thick additive to uniformly distribute TiB_2 in B_4C and enhance its structural properties.

Keywords: MXene, high-temperature, refractory carbide, self-assembly, nanoceramics

References:

1. Anupma Thakur, Nithin Chandran, Brian C. Wyatt, Annabelle Bedford, Mostafa Dadashi Firouzjaei, Krutarth Kamath, Muhammad Sharif Uddin, and Babak Anasori. *Materials Today* (2025).
2. Nithin Chandran BS, Anupma Thakur, Srinivasa Kartik Nemani, Niraja Moharana, Christian Paul Hardy Scott, KC Hari Kumar, Ravi Kumar, and Babak Anasori. *Advanced Composites and Hybrid Materials* 8, no. 4 (2025): 308.

A Study of MXene Monolayers for Multifunctional Applications Using the Density Functional Theory

Nidhi Modi¹

¹*Department of Physics, Sir P. T. Sarvajani College of Science (Autonomous), Surat- 395001.*

*Corresponding author: nidhimodi2000@gmail.com

The thesis entitled “*A Study of MXene Monolayers for Multifunctional Applications Using Density Functional Theory*” focuses on the investigation of two-dimensional (2D) MXene materials using Density Functional Theory (DFT) for multifunctional applications in advanced energy and optoelectronic devices. The work explores the structural stability, electronic, optical, hydrogen storage, and metal-ion battery properties of carbide and nitride MXene monolayers, highlighting the role of surface functionalization, pressure, and strain engineering in tuning their performance. The first chapter presents an introduction to 2D materials, MAX phases, and MXenes, including their synthesis, surface functionalization, battery applications, and hydrogen storage potential, along with the motivation and outline of the thesis. The second chapter discusses the theoretical framework and computational methodology based on DFT, including Kohn–Sham equations, pseudopotentials, exchange-correlation functionals, and plane-wave basis sets. The third chapter investigates Janus surface functionalized yttrium-based MXene Y₂C and examines its structural, electronic, and optical properties. Various optical parameters such as absorption coefficient, refractive index, extinction coefficient, reflectivity, and energy loss spectrum are analyzed. The fourth chapter studies the effect of pressure on the Sc₂CBr₂ monolayer, revealing pressure-induced bandgap reduction and semiconductor-to-metal transition along with tunable optical behavior. The fifth chapter demonstrates the suitability of strain-engineered ScYCCl₂ monolayer as an anode material for Na, K, Li, and Mg-ion batteries due to its high theoretical storage capacity and low open-circuit voltage. The sixth chapter explores yttrium-based nitride MXene Y₂N as an efficient anode material for metal-ion batteries by evaluating adsorption energy, gravimetric capacity, and open-circuit voltage. The seventh chapter investigates the hydrogen storage capability of the Y₂N monolayer. Stability is confirmed through phonon dispersion and molecular dynamics studies, while hydrogen adsorption behavior is analyzed for both atomic and molecular hydrogen. The final chapter summarizes the major findings and discusses future research directions. Overall, the thesis presents a systematic and original computational study of MXene monolayers using first-principles DFT calculations. The work provides valuable insights into the structural, optoelectronic, and energy storage properties of MXenes, demonstrating their potential for next-generation sustainable energy applications.

Keywords: Density Functional Theory; 2D MXenes; Optoelectronics Applications; Metal-ion Battery applications; Hydrogen Storage Applications.

References:

1. M. Naguib, Curr. Opin. Solid State Mater. Sci. 24 (2020) 100809.
2. K. Xiong, Z. Cheng, J. Liu, P.-F. Liu, Z. Zi, RSC Adv. 13 (2023) 7972–7979.
3. Y.Z. Chu, K.C. Lau, Int. J. Hydrogen Energy 57 (2024) 1144–1151.

Fabrication of large area two-dimensional PtX₂ (X=S, Se) heterostructures for broadband photodetector studies

Gaurav Bassi^{* 1,2}, Mukesh Kumar¹, Tanushree H. Choudhury².

¹Functional and Renewable Energy Materials Laboratory, Department of Physics, Indian Institute of Technology Ropar, Rupnagar, Punjab-140001, India

²Department of Metallurgical and Materials Science, Indian Institute of Technology Bombay, Powai, Mumbai-400076, Maharashtra, India.

Corresponding author email: gaurav.bassi@iitb.ac.in

In recent years, transition metal dichalcogenides (TMDCs) such as PtS₂ and PtSe₂ have attracted significant attention for photodetector applications due to their narrow band gaps (0–1.65 eV), high carrier mobility, and excellent ambient stability. However, controlled large-area growth of PtS₂ and the development of high-performance PtX₂ (X = S, Se) based broadband photodetectors, particularly in the near-infrared (NIR) region, remain challenging.

My thesis addresses these challenges by controlling the synthesis of PtX₂ (X = S, Se) films and their heterostructures for broadband photodetection applications. In the first part, large-area PtS₂ films were successfully grown on multiple substrates, including SiO₂/Si, sapphire, Si, and mica using the thermally assisted conversion (TAC) method [1]. To achieve controlled large area growth, various growth parameters such as pre-coated Pt film thickness, sulfur amount (used during sulfurization), and carrier gas flow rate were optimized to get the best quality PtS₂. The results were further confirmed by Raman and X-ray photoelectron spectroscopy (XPS) characterizations [1].

In the second part, PtS₂/MoS₂ heterostructure fabrication and its photodetector performance were studied. The fabricated PtS₂/MoS₂ heterostructure demonstrated broadband response from 400–1200 nm with a maximum responsivity of 30.2 A/W under 900 nm illumination. A detailed band alignment study (type-I) and photodetector working mechanism were also explained using XPS results. [2].

Finally, PtSe₂/MoS₂ heterostructure was developed to overcome the high dark current issue in bare PtSe₂. The PtSe₂/MoS₂ heterostructure achieved a ~10³ order of reduction in dark current (compared to PtSe₂ device) and enhanced detectivity to 9.8 × 10¹² Jones. Moreover, the heterostructure exhibited broadband visible-to-NIR detection and ultrafast response times of 103/117 μs [3].

In conclusion, my thesis work demonstrates the large-area fabrication of PtX₂ (X = S, Se) and their heterostructures for next-generation photodetector optoelectronic applications.

Keywords: 2D materials, PtS₂, PtSe₂, Photodetector, Heterostructure

References:

1. Bassi G, Wadhwa R, Deswal S, Kumar P and Kumar M 2023 Journal of Alloys and Compounds 955 170233
2. Bassi G, Wadhwa R and Kumar M 2024 Advanced Optical Materials 12 2301899
3. Bassi G, Dahiya R, Chakkar A G, Kumar P and Kumar M 2025 r Applied Surface Science 681 161598.

Development of Photon Upconversion Materials Incorporated Photocatalysts for Hydrogen Generation Using Visible Light

Amit Kumar Verma¹, A.S.K. Sinha¹, Shikha Singh¹

¹*Department of Energy and Human Sciences, Rajiv Gandhi Institute of Petroleum Technology, Jais, Amethi, India-229304*

*Corresponding Author E-mail: shikhas@rgipt.ac.in, asksinha@rgipt.ac.in

The Sun continuously delivers nearly 1.74×10^{17} W of solar energy to the Earth; however, conventional semiconductor photocatalysts utilize only a small portion of solar spectrum because of their fixed band structures and limited band gap energy. Photon upconversion (UC)-assisted photocatalysis offers a promising route for harvesting sub-bandgap photons, yet the efficiency of UC-based systems is often limited by poor charge transport and weak interfacial interactions. This thesis presents a comprehensive materials-engineering strategy for developing rare-earth-based UC-integrated semiconductor photocatalysts for efficient visible-light-driven hydrogen generation.

A series of engineered UC-semiconductor hybrid systems, including CeF₃:Ho³⁺-integrated TiO₂ nanosystems, YAlO₃:Ho³⁺/Er³⁺-coupled- polydopamine functionalized β -SiC heterostructures, and nitrogen-doped graphene oxide nanospheres, were synthesized through in-situ polyol reduction, sol-gel, polymerization and hydrothermal approaches. Beyond material synthesis, systematic optimization of synthesis parameters, and morphology evolution revealed strong correlations between structural morphology, and upconversion efficiency. Morphology-controlled spherical and nanotubular architectures exhibited distinct charge transport and light-conversion characteristics, while dielectric spectroscopy demonstrated that reduced dielectric loss and enhanced electrical conductivity significantly improve charge separation and carrier mobility. The optimized CeF₃:Ho³⁺/TiO₂ nanosystem achieved a visible-light hydrogen evolution rate of 79.85 $\mu\text{mol h}^{-1}$ with an apparent quantum efficiency of 4.00% and solar-to-hydrogen conversion efficiency of 1.37%. Wavelength-dependent studies confirmed broad-spectrum photocatalytic activity originating from Ho³⁺-mediated upconversion pathways. Furthermore, polydopamine-functionalized β -SiC@UC systems exhibited nearly 4.6-fold enhanced hydrogen generation and ~2.9-fold higher transient photocurrent response compared to pristine β -SiC, highlighting the synergistic interplay between photon upconversion, interfacial charge mediation, and semiconductor band alignment. Complementary studies on nitrogen-doped graphene oxide nanospheres further revealed oxygen reduction and selective oxygen evolution (ORR&OER) along with the role of heteroatom-induced band engineering in photocatalysis.

This research exhibits that integrating upconversion materials with scalable semiconductors provides a powerful route for accessing the full solar spectrum, enabling efficient

hydrogen production. The outcomes contribute fundamental understanding and practical pathways for developing next-generation multifunctional catalysts for sustainable solar-chemical energy technologies.

Keywords: Photon upconversion; Photocatalysis; Hydrogen evolution; Structural engineering; Sustainable energy.

References:

1. A.K. Verma, P. Tripathi, A.S.K. Sinha, S. Singh, *ChemSusChem*, 19, (2026), e202501580.
2. A.K. Verma, P. Tripathi, H. Kumar, P. Singh, A.S.K. Sinha, S. Singh, *ACS Applied Energy Materials*, 16, (2025), 11977-11987.
3. A.K. Verma, P. Tripathi, A. Dubey, A.S.K. Sinha, S. Singh, *ACS Applied Energy Materials*, 6, (2023), 5739-5752.

Investigation of Multifaceted Applications of BaSnO₃: Ceramic and Thin Film

Shikha Shrivastava^{1,2,*}

¹*CSIR-National Physical Laboratory, Dr. K.S. Krishnan Marg, New Delhi, India 110012*

²*Academy of Scientific and Innovative Research, (AcSIR), Ghaziabad, India 201002*

*Corresponding Author E-mail: shikhashri2416@gmail.com

BaSnO₃ (BSO) is a cubic perovskite oxide that has attracted considerable attention because of its wide bandgap (~3.4 eV), high electron mobility, and remarkable thermal and chemical stability. These unique properties make BSO a promising candidate for advanced electronic and optoelectronic applications such as transparent conductive oxides, ultraviolet photodetectors, gas sensors, and high-temperature dielectric materials. Owing to its structural stability over a broad temperature range, BSO is also considered suitable for oxide-based semiconductor technologies. This thesis investigates the structural, dielectric, gas sensing, and optoelectronic properties of BSO in both ceramic and thin-film forms.

The first part of the thesis examines the influence of temperature on the structural and dielectric behaviour of BSO. Temperature-dependent X-ray diffraction, Raman spectroscopy, and room-temperature neutron diffraction were employed to analyse the structural characteristics of the material. Surface morphology, homogeneity, and elemental composition were confirmed using scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDX). The results reveal that BSO maintains its cubic crystal symmetry even at elevated temperatures, while the unit cell volume gradually increases due to thermal expansion. High-temperature dielectric studies indicate a high dielectric constant at lower frequencies with increasing temperature, which may arise from oxygen vacancies and lone-pair effects. The second part focuses on the carbon monoxide (CO) gas sensing performance of BSO thin films with different thicknesses. Improved electrical conductivity and enhanced gas sensing response were observed with decreasing film thickness, mainly due to a higher concentration of Sn⁴⁺ ions and an increased density of grains and grain boundaries. AFM and XPS analyses confirmed efficient adsorption and desorption of oxygen species on the film surface. Finally, detailed optical and photoconductive studies, including ultrafast transient spectroscopy, revealed thickness-dependent carrier dynamics such as

trapping, relaxation, and recombination, highlighting the potential of BSO for next-generation optoelectronic and sensing devices.

Keywords: PLD, BaSnO₃ Thin films, Oxide Heterostructures, Gas Sensors, Photo Sensors.

References:

1. Shikha Shrivastava et al, J. Alloy. Compd. 957 (2023), 170458.
2. Shikha Shrivastava et al., Sensors and Actuators: B.Chemical 400 (2024), 134882.
3. Shikha Shrivastava et al., Journal of Physics D: Applied Physics, (58) 105110 (2025).

Atomic-Level Doping at Redox-Active Centers in Polyanion Cathodes for Enhanced Sodium-Ion Storage

Sharad Dnyanu Pinjari¹, Rohit Ranganathan Gaddam^{2,*}

¹Department of Chemical Engineering, Indian Institute of Science Education and Research Bhopal, India.

*Corresponding Author E-mail: rohitg@iiserb.ac.in

Sodium-ion batteries (NIBs) are promising alternatives to lithium-ion systems due to the abundance and low cost of sodium resources; however, their widespread application is hindered by the limited performance of cathode materials. This work presents a comprehensive strategy to overcome intrinsic limitations of Fe-based polyanion cathodes through atomic-level doping, polyanion engineering, and entropy-driven multi-element substitution. In the phosphosulfate system NaFe₂PO₄(SO₄)₂, Mg²⁺ substitution (x = 0.05) induces lattice expansion and enhances Na⁺ diffusion, delivering 111 mAh g⁻¹ at C/20 with 86.45% capacity retention over 100 cycles [1]. To engineer wider Na⁺ diffusion channels, a monoclinic phosphomolybdate framework, NaFe₂(PO₄)(MoO₄)₂, was developed. Indium substitution (x = 0.1) reduces polarization and migration barriers, achieving ~112 mAh g⁻¹ with 75% retention after 650 cycles [2]. In-situ X-ray diffraction confirms a highly reversible single-phase solid-solution mechanism. Furthermore, an advanced entropy engineering strategy was employed by introducing multiple cations (Mn, Cr, Al, Zn, and In) into the phosphomolybdate framework to form a high-entropy NASICON-type cathode [3], NaFe_{1.8}(MnCrAlZnIn)_{0.2}(PO₄)(MoO₄)₂. The multi-ion cation substitution significantly increased configurational entropy, leading to enhanced structural stability, suppressed phase transitions, and improved ionic/electronic transport properties. Electrochemically, NFM'PM20 exhibited a high discharge capacity of 125.21 mAh g⁻¹ at 0.1C, excellent rate capability (82 mAh g⁻¹ at 1C), and outstanding long-term cycling stability, with 84.3% retention after 400 cycles, 83.5% after 500 cycles at 2C, and 65.8% after 2000 cycles at 5C. In-situ XRD analysis revealed negligible lattice strain (~ 0.0091), confirming highly reversible solid-solution behavior. The formation of a stable and uniform cathode-electrolyte interphase (CEI) was also observed, contributing to enhanced cycling stability. This work establishes a rational design framework linking atomic-scale modifications to macroscopic electrochemical performance, offering new insights for the development of high-energy and long-life sodium-ion battery cathodes.

Keywords: Sodium-ion batteries; Polyanion cathodes; NASICON; Atomic-level doping; High-entropy materials.

References:

1. S. D. Pinjari et al., Chemical Engineering Journal, vol. 493, p. 152485, 2024.
2. S. D. Pinjari et al., Small, vol. 21, no. 18, p. 2501004, 2025.
3. S. D. Pinjari et al., Advanced Functional Materials, vol. 36, no. 14, p. e17539, 2026.

Solar Light Active Nano Ni Decorated TiO₂, Nb₂O₅ and SnO₂ Photocatalysts for Efficient Hydrogen Generation via Water Splitting

Priyanka N. Birla^{1,*}, Manish D. Shinde¹, Sudhir S. Arbuj¹, Sunit B. Rane¹, Bharat B. Kale^{1,2}

¹*Centre for Materials for Electronics Technology, Off Pashan Road, Panchwati, Pune-411008, India.*

²*Pimpri Chinchwad University, Pune*

*Corresponding Author E-mail: priyankanikhilbirla@gmail.com

The urgent need for sustainable and carbon-neutral energy solutions has intensified research on hydrogen (H₂) as a next-generation fuel, owing to its high energy density and clean combustion. Among available approaches, photocatalytic water splitting offers a direct and environmentally benign route for solar-to-hydrogen conversion; however, its practical deployment is limited by rapid charge recombination and reliance on noble-metal co-catalysts. In this work, a facile, industry adoptable and scalable electroless deposition strategy is demonstrated for the controlled integration of nickel (Ni) as a low-cost, earth-abundant co-catalyst on nanostructured TiO₂, SnO₂, and Nb₂O₅ photocatalysts. The engineered Ni–semiconductor interfaces induce efficient charge separation via Schottky barrier formation, significantly enhancing photocatalytic performance. Comprehensive structural, morphological, surface area and spectroscopic analyses (XRD, FESEM/TEM, BET, XPS, UV-DRS, Raman, FTIR, ICP-OES, and PL) confirm intermittent nanoscale Ni film deposition (thickness ~1–5 nm) and its pronounced effect on electronic structure and light absorption behaviour. The Ni-modified semiconductor systems exhibit markedly improved charge carrier dynamics. Photocatalytic hydrogen evolution results reveal a clear performance hierarchy across the systems. Ni@TiO₂ delivers a ~3.5-fold enhancement (1600 μmol g⁻¹ h⁻¹) compared to pristine TiO₂ (470 μmol g⁻¹ h⁻¹) with excellent stability. Ni@SnO₂ demonstrates visible-light activity with an optimized rate of ~50 μmol g⁻¹ h⁻¹ at 0.3 wt.% Ni. Strikingly, Ni@Nb₂O₅ achieves the highest hydrogen evolution rate of ~1900 μmol g⁻¹ h⁻¹ at 0.3 wt.% Ni, nearly six times higher than its pristine counterpart. The superior performance is attributed to efficient electron trapping by Ni and accelerated interfacial charge transfer kinetics. This work establishes a robust, noble-metal-free design paradigm for high-efficiency photocatalysts and provides a scalable pathway toward practical solar hydrogen generation.

Keywords: Photocatalytic water splitting; Hydrogen generation; Nickel co-catalyst; Semiconductor nanomaterials; Solar energy conversion.

References:

1. P. N. Birla, S. S. Arbuj, R. Chauhan, M. D. Shinde, S. B. Rane, S. W. Gosawi, B. B. Kale, *International Journal of Hydrogen Energy*, 172 (2025) 151210.
2. P. N. Birla, S. Arbuj, R. Chauhan, M. Shinde, S. Rane, S. Gosavi, B. Kale, *Nanoscale*, 16 (2024) 17838–17851.
3. P. N. Birla, S. Arbuj, M. D. Shinde, S. Joseph, S. Rane, S. Kulkarni, B. Kale, *RSC Advances*, 13 (2023) 20068–20080.

Perovskite Memristors for In-Memory Sensing and Neuromorphic Computing

Mubashir Mushtaq Ganaie^{1,2} *, Satyajit Sahu¹, Mahesh Kumar²

¹Department of Physics, Indian Institute of Technology Jodhpur, Rajasthan, 342030, India

²Department of Physics, Indian Institute of Technology Jodhpur, Rajasthan, 342030, India

*Corresponding Author E-mail: ganaie.1@iitj.ac.in

The rapid proliferation of smart sensors and edge devices in the Internet of Things era demands breakthroughs in energy-efficient electronic architectures. Conventional systems based on the von Neumann architecture face performance bottlenecks due to the physical separation of sensors, memory arrays and processing units.¹ This leads to high energy consumption, data redundancy and limited speed, while sensory neurons in the retina of eye can simultaneously sense and perform preliminary processing, before transmitting the data to brain. In-sensor computing, a paradigm where sensing, memory, and computation functionalities are co-integrated into a single device, offers a promising solution to overcome these limitations.¹ Memristors have attracted significant attention for enabling such functionality. The core objectives of my PhD work have been (i) in-memory sensing, integrating sensing and memory within a single memristor device and (ii) self-powered operation, aimed at reducing external power requirements for emerging neuromorphic systems. To address these objectives, my work has focused on perovskite-based memristor devices. I started with AlFeO₃ based perovskite oxide memristors.¹ I used γ -ray responsive resistive switching (RS) to demonstrate that these devices can be used as radiation sensitive memory that senses and stores at the same time. While AlFeO₃ demonstrated in-memory sensing capability, they showed limited optical absorption, restricting their applicability for optoelectronic neuromorphic systems. This limitation motivated the transition toward halide perovskite materials, which possess strong light absorption, with efficient coupling between optical stimuli and memristive behavior. Following up from those studies, I studied 3D lead halide perovskite-based solaristor devices that were able to combine visible light sensing, photovoltaic effect and synaptic functions. However, it showed some disadvantages like absence of RS capabilities. I moved to 2D lead-based layered halide perovskites.² Those systems offer better structural stability than 3D systems and also possess RS characteristics. However, because of their relatively high band gap, they were less effective for light-induced in-memory sensing. Finally, I studied lead-free 2D layered halide perovskite systems.³ Those devices displayed UV-sensitive in-memory sensing function, as well as co-existing electrical and optical synaptic plasticity. I explored both memristor and solar-cell-like device architectures, motivated by the fact that they share similar layer stacks and electrode configurations. This structural similarity allows to leverage the photovoltaic behavior of solar-cell architecture to drive synaptic behaviour, enabling self-powered operation without external bias. Overall, this thesis has provided a framework regarding materials/devices that can facilitate in-memory sensing across a wide spectral range, synaptic functionality, and self-powered operation.

Keywords: Memristor; In-memory sensing; Neuromorphic computing; Solar cell; Perovskite.

References:

1. M. M. Ganaie, A. Kumar, A. K. Shringi, S. Sahu, M. Saliba and M. Kumar, Adv. Funct. Mater., 34, (2024), 2405080.
2. M. M. Ganaie, G. Bravetti, S. Sahu, M. Kumar and J. V. Milić, Mater. Adv., 5, (2024), 1880.
3. M.M.Ganaie, M. Mohammadi, M. Loizos, K. Rogdakis, R. M. Ansari, G. Bravetti, M. Ghasemi, M. R. Golobostanfard, K. Kumar, S. Ahmad, S. Sahu, E. Kymakis, W. Tress, J.V. Milić, M. Kumar, Mater. Horiz. 2026.

IoT-Integrated 2D-Material Functionalized AlGaIn/GaN HEMT for Multifunctional Sensing

Vikas Pandey¹, Prof Mahesh Kumar² and Dr. Ankur Gupta³

¹*Interdisciplinary Research Program, Indian Institute of Technology Jodhpur, Jodhpur, Rajasthan 342030 India*

²*Department of Electrical Engineering, Indian Institute of Technology Jodhpur, Jodhpur, Rajasthan 342030 India*

³*Department of Mechanical Engineering, Indian Institute of Technology Jodhpur, Jodhpur, Rajasthan 342030 India*

*Corresponding Author Email: pandey.21@iitj.ac.in

There is a growing deployment of hydrogen energy systems, increasing concerns over environmentally sensitive toxic gases, and stringent radiation safety requirements in nuclear and aerospace applications. All these factors have intensified the demand for sensing technologies that operate reliably at room temperature, exhibit high selectivity, provide fast response times, and maintain long-term stability under harsh conditions. Conventional sensors are hindered by high power consumption, slow kinetics, limited selectivity, and poor environmental robustness. Moreover, they remain largely incompatible with modern IoT architectures and rarely exploit data analytics for signal interpretation or predictive monitoring. This research addressed these gaps by working towards a unified, multifunctional sensing platform based on AlGaIn/GaN High Electron Mobility Transistors (HEMTs) functionalized with 2D nano materials and catalytic nanostructures for H₂, NH₃, and ionizing (γ) radiation detection. The inherent sensitivity of the two-dimensional electron gas (2DEG) at the AlGaIn/GaN interface to surface charge changes is analytically utilized to realize room-temperature sensing across multiple domains.

The work begins with the fabrication and optimization of AlGaIn/GaN HEMTs. It acts as the sensing backbone. Building on this, a Pd nanoparticle(np) functionalized HEMT H₂ sensor was developed. The sensor exploited the catalytic properties of Pd, demonstrating a response time of 29 s and a recovery time of 52 s, good selectivity, and a sub-ppm detection of up to 0.5 ppm. To enhance the performance, a Pd/g-C₃N₄ hybrid-functionalized HEMT was developed. It exhibited an extended linear detection range of 0.5-1000 ppm and a higher sensitivity of 0.67 μ A/ppm. It demonstrated robust performance across a range of humidity and temperature conditions, as well as long-term operational stability. This work presents the performance enhancement achieved through the synergistic combination of a 2D nanomaterial with a catalytic nanostructure. Extending to toxic gas detection, an NH₃ (ammonia) sensor was designed. A Co₃O₄ thin film, which exhibits NH₃ chemisorption and redox activity, was sputtered onto the HEMT active region. Cu-np, which is converted to CuO and enhances catalytic dissociation and interfacial charge transfer, was deposited on the film. The sensor showed a response of ~17% to 10 ppm NH₃ at room temperature. Its detection range was 1-200 ppm. The AlGaIn/GaN HEMT was tested under ionizing γ -radiation and exhibited radiation hardness. Functionalized with BST, the device served as a γ -ray dosimeter, exhibiting a monotonic drain-current response to doses up to 10 Gy.

All the developed sensors were integrated onto NodeMCU-based IoT platforms for remote monitoring and data acquisition. To further enhance the system's performance, a machine-learning (ML) framework was validated on the Pd/g-C₃N₄ H₂ sensor. For the ML implementation, features were extracted from the sensor data. The features were subjected to Principal Component

Analysis (PCA), Gaussian Process Regression (GPR), and Neural Network Models to estimate concentrations and provide real-time predictions. The implementation is generalizable to all sensors on this platform. Overall, this work demonstrates a scalable, robust, and intelligent sensing system, with potential for reliable deployment across a wide range of sensing applications.

Keywords: AlGaIn/GaN HEMT, 2D Nanomaterial Functionalization, Multifunctional Gas Sensing, IoT-Integrated Sensing Platform, Machine Learning for Sensor Analytics

References:

1. Vikas Pandey, A. Kumar, A. S. Razeen, A. Gupta, S. Tripathy, M. Kumar (2024), IEEE Sensors Journal, Vol. 24, No. 24, pp. 40409-40416.
2. Vikas Pandey, A. Kumar, S. P. Bharath, C. B. Auti, P. Kumar, A. Gupta, S. Tripathy, M. Kumar (2025), Sensors and Actuators B: Chemical, 139217.
3. Vikas Pandey, N. Jingar, S. S. Barala, S. Sahu, M. Kumar (2025), India Patent, Application No. INA202511021069.

Theoretical Framework and Predictive Modeling for Emergent Topological Phases in Quantum Materials

Rahul Verma

Department of Condensed Matter Physics and Materials Science, Tata Institute of Fundamental Research, Mumbai 400005, India
E-mail: rahul.verma@tifr.res.in

Topology has emerged as a fundamental organizing principle in condensed-matter physics, expanding the classification of quantum phases beyond conventional symmetry-breaking states and revealing materials with symmetry-protected excitations and quantized responses. First-principles electronic-structure methods have been essential for predicting these phases and connecting theoretical concepts to experimental observables[1]. However, widely used symmetry-based classification schemes including symmetry indicators and topological quantum chemistry, rely primarily on irreducible representations at high-symmetry points and often overlook material-specific features such as effective masses, band anisotropy, spin texture, and exchange-correlation effects that influence observable responses[1]. Consequently, diagnosing topology solely from symmetry data may overlook critical physics or leave an incomplete understanding of how topological phases manifest in real materials.

In my thesis, we develop a unified framework that integrates symmetry analysis with detailed electronic-structure calculations to reveal how material-dependent properties shape band topology and quantum responses. Using this framework, we show that puckered group-Va monolayers host obstructed atomic insulator and high-spin-Chern insulating phases with an emergent spin-U(1) quasi-symmetry, leading to a nearly quantized spin Hall conductivity of $\sim 4e^2/h$, the largest reported to date[2]. In transition-metal tetraphosphides MP_4 , pronounced direction-dependent effective masses in crossing bands generate hybrid nodal-line fermions featuring both type-I and type-II dispersions[2]. Building on these insights, we introduce a bottom-up design strategy that employs controlled symmetry lowering to create $1T' - MA_2Z_4$ monolayers and square-octagonal lattices, realizing quantum spin Hall phases with nearly quantized spin Hall conductivity and high-order van Hove singularities[3]. Finally, by examining KZnBi with multiple exchange-correlation functionals, we demonstrate that an accurate treatment of electronic interactions is crucial for reproducing the experimentally observed Dirac semimetal state, highlighting how XC-driven orbital reordering determines the topological character[3].

Together, these results establish a materials-focused, symmetry-guided framework for discovering, understanding, and engineering topological phases, providing guiding principles for designing quantum materials with robust and technologically relevant responses [2-3].

Keywords: Electronic structure methods; Topological materials; Quantum spin Hall insulators; Bottom-up material design; Topological semimetals

References:

1. B. Singh, H. Lin, and A. Bansil, *Advanced Materials*, 35, (2023), 2201058; B. Bradlyn, L. Elcoro, J. Cano et al. *Nature*, 547, (2017), 298-305.
2. R. Verma, S.-M. Huang and B. Singh, *Physical Review B*, 110, (2024), 165122; B. Patra, R. Verma, S.-M. Huang and B. Singh, *Physical Review B*, 108, (2023), 235136;
3. R. Verma, Y. Vardhan, H. Lin and B. Singh, *Advanced Physics Research*, (2026); R. Verma, B. Patra and B. Singh, *Electronic Structure*, 5(4), (2023) 045011; C. Patra, T. Agarwal, R. Verma et al., *Physical Review Letters*, 135, (2025), 216002.

Study of Preparation of Tight Ultrafiltration and Loose Nanofiltration Membranes via Layer-by-Layer Self-Assembly for Selective Separation

Urvashi S. Joshi¹

¹*Department of Chemical Engineering, Visvesvaraya National Institute of Technology, Nagpur*

*Corresponding author: urvashijoshi121@gmail.com

The development of energy-efficient and fouling-resistant membrane systems for selective molecular separation remains a major challenge in sustainable wastewater treatment and resource recovery. This thesis presents the design and engineering of tight ultrafiltration (TUF) and loose nanofiltration (LNF) membranes through controlled layer-by-layer (LbL) self-assembly of functional polyelectrolytes for high-performance dye/salt fractionation and molecular separation under low-pressure operation.

The work demonstrates the fabrication of selective separation barriers via sequential and concentration polarization-driven rapid self-assembly approaches, enabling tunable membrane nano-interfaces and transport pathways. The influence of assembly sequence, polyelectrolyte composition, and non-stoichiometric polyelectrolyte complexation on membrane morphology, surface charge, hydrophilicity, fouling resistance, and separation performance was systematically investigated. The developed membranes exhibited high dye rejection, enhanced salt permeation, low fouling characteristics, and stable long-term separation performance suitable for industrial wastewater treatment applications.

A major outcome of this thesis is the establishment of rapid and scalable strategies for engineering selective and low-fouling membrane interfaces using advanced polyelectrolyte assemblies. The findings provide significant insights into interfacial transport modulation and selective separation mechanisms in self-assembled membrane systems. The research contributes toward sustainable separation technologies with potential applications in textile wastewater treatment, resource recovery, and energy-efficient molecular fractionation.

The thesis resulted in multiple high-impact SCI publications in leading journals including *Journal of Membrane Science*, *ACS Applied Materials & Interfaces*, and *Journal of Materials Chemistry A*.

Keywords: Layer-by-layer self-assembly; Polyelectrolytes; Tight ultrafiltration; Loose nanofiltration; Dye/salt fractionation

UNRAVELLING EFFICIENT AND SUSTAINABLE NEXT-GENERATION SOLAR CELLS *via* DOPING AND INTERFACE ENGINEERING

Abhishek Srivastava¹ and Parasharam M. Shirage^{*1}

¹*Advanced Functional Materials Research Group (AFMRG)-Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore-453552, India*

^{*} Corresponding Author: Prof. Parasharam M. Shirage

Email: pms Shirage@iiti.ac.in, paras.shirage@gmail.com

The core theme of my thesis is the development of economical, environmentally benign, and stable photovoltaic technologies through rational material selection and interface engineering. The research journey began with synthetic N719 dye-based DSSCs with passivated TiO₂ photoanodes, followed by a transition to naturally derived beetroot dye with Fe-doped TiO₂ photoanodes to promote sustainability and reduce cost. Although natural dyes offer ecological advantages due to their strong absorption in the visible region, their limited efficiency and stability have prompted the exploration of alternative photo absorbers. This led to the implementation of lead-free Cs₂AgBiBr₆ double perovskite (CABB) in an HTL-free architecture, achieving a PCE of 1.63% under open-air fabrication with >95% PCE retention after 200 h, highlighting its excellent environmental stability.

To understand the efficiency limitations, systematic numerical simulations using SCAPS-1D were conducted for HTL-based and HTL-free device architectures with TiO₂, SnO₂, WO₃, and ZnO as ETLs. The optimized HTL-based configuration yielded PCEs exceeding 27%, while HTL-free devices showed only ~4% lower efficiency, providing critical insights into defect tolerance, optimal thickness, doping density, interfacial recombination limits, and resistive parameters. Guided by these findings, dual interfacial passivation using 2D Ti₃C₂T_x MXene at the FTO/TiO₂ and TiO₂/CABB interfaces was implemented to optimize the band alignment and suppress recombination. This strategy enhanced the experimental device's performance from 3.74% to a champion PCE of 6.92%, among the highest reported for CABB-based double perovskite solar cells.

Hence, special attention has been given to the role of doping, nano-structuring, interfacial passivation, and device engineering in optimizing charge transport, defect tolerance, and band alignment, making these materials highly promising for next-generation sustainable photovoltaic and optoelectronic applications.

Keywords: TiO₂, Cs₂AgBiBr₆, 2D Ti₃C₂T_x-MXene, dye-sensitized solar cell, double perovskite solar cell

References:

1. Srivastava, A., Satrughna, J. A. K., Tiwari, M. K., Kanwade, A., Yadav, S. C., Bala, K., & Shirage, P. M. *Materials Today Communications*, 35 (2023), 105686.
2. Srivastava, A., Satrughna, J. A. K., Tiwari, M. K., Kanwade, A., Yadav, S. C., Bala, K., & Shirage, P. M. *Energy Advances*, 2 (2023), 1, 148-160.
3. Srivastava, A., Ito, Y., Lee, H., Ohshita, Y., Ogura, A., Dhonde, M., & Shirage, P. M. *ACS Applied Electronic Materials*, 7 (2025), 4, 1531-1549.
4. Srivastava, A., Kasliwal, M., & Shirage, P. M. *Solar Energy*, 298 (2025), 113733.
5. Srivastava, A., & Shirage, P. M. *ACS Appl. Energy Mater.* 9 (2026), 4, 2422-2433.

Fe-based Phosphate Materials for Sodium-ion Batteries

Archana R. Kanwade, Prof. Parasharam M. Shirage

Thesis Supervisor: Prof. Parasharam M. Shirage pms Shirage@iiti.ac.in

*Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology
Indore, Khandwa Road, Simrol, Madhya Pradesh, India – 453 552*

The growing global demand for sustainable and cost-effective energy storage technologies has accelerated research beyond conventional lithium-ion batteries. Among emerging alternatives, sodium-ion batteries (SIBs) have attracted considerable attention due to the natural abundance, low cost, and wide availability of sodium resources. However, the development of high-performance cathode materials remains a key challenge because the larger ionic radius and higher molar mass of Na^+ ions often result in sluggish diffusion kinetics, structural instability, and relatively lower energy density. In this regard, polyanionic phosphate materials have emerged as promising cathodes owing to their strong structural stability, high operating voltage, and thermal safety. This thesis focuses on the rational design and optimization of NASICON-type polyanionic phosphate cathodes for sodium-ion batteries through compositional tuning and interfacial engineering strategies. Initially, vanadium content was reduced in $\text{Na}_3\text{VFe}(\text{PO}_4)_3$ by partial substitution with Fe to develop a cost-effective cathode while maintaining structural stability and electrochemical performance. Furthermore, electrolyte-dependent investigations were carried out to understand the electrode–electrolyte interfacial chemistry and its influence on electrochemical behavior in high-voltage systems. Subsequently, further reduction of vanadium content was achieved through multi-transition-metal substitution by incorporating Fe and Mn, leading to improved structural stability and enhanced sodium-ion diffusion kinetics. To further enhance electrochemical performance, heteroatom doping and conductive carbon network engineering were employed to improve electronic conductivity and facilitate faster charge transport. The resulting materials exhibited improved specific capacity, enhanced rate capability, and excellent long-term cycling stability. Comprehensive physicochemical and electrochemical characterization techniques were used to establish correlations between structural features and electrochemical properties. In addition, theoretical studies were carried out to investigate sodium-ion migration pathways, binding energies, and electronic structures within the cathode framework, providing deeper insight into the sodium-ion storage mechanism. Post-electrochemical analyses were also performed to examine structural evolution and interfacial stability after prolonged cycling. Overall, this thesis provides important insights into the structure–property performance relationships of polyanionic phosphate cathodes and offers effective strategies for designing high-performance and sustainable sodium ion batteries suitable for next-generation large-scale energy storage applications.

Keywords: NASICON compound; Electrolyte compatibility; Phosphate compounds; Sodium-ion battery; Density functional theory.

References:

1. A. Kanwade, S. Gupta, A. Kankane, A. Srivastava, S. C. Yadav and P. M. Shirage, *Sustain. Energy Fuels*, (2022), 6, 3114–3147.
2. A. R. Kanwade, J. A. K. Satrughna, S. M. Rajore, P. Pawar, S. S. Mali, J. V. Patil, C. K. Hong, R. C. Dutta and P. M. Shirage, *Small*, (2025), 21(34), 2502979.

3. A. R. Kanwade, A. K. S. Jena, S. M. Rajore, S. S. Mali, J. V. Patil, C. K. Hong and P. M. Shirage, *J. Mater. Sci.*, (2024), 59, 8956–8972.
-

ORAL ABSTRACTS

Shape-dependent structural, electronic and transport properties of BP nanowires for next-generation nanoscale interconnect

Ankita Nemu, Pankaj Srivastava, Neeraj K. Jaiswal

Nano Materials Research Laboratory, Department of Engineering Sciences, ABV- Indian Institute of Information Technology and Management, Gwalior, M.P., 474015, India

*Corresponding Author E-mail: ankita.nemu@gmail.com

One-dimensional (1-D) nanowires have unique structural, electronic, and transport properties that come from being quantum confined and having fewer dimensions. This makes them good candidates for next-generation nanoelectronic devices. Among various low-dimensional systems, III-V compound nanowires have attracted considerable interest owing to their tunable electronic properties and applicability in device fabrication. In this research, we aim to systematically examine the impact of geometry on the structural stability, electronic band structure, and transport properties of boron-phosphide nanowires (BPNWs). Utilizing first-principles density functional theory, linear, zigzag, and triangular configurations are analyzed to elucidate geometry-induced transitions in stability and electronic behavior, yielding insights for the design of nanoscale electronic materials. We performed the geometry optimization using the generalized gradient approximation (GGA) functional with the Perdew- Burke-Ernzerhof (PBE) parameterization. We used plane wave functions with a cutoff energy of 70 Ry, and we did Brillouin zone integration with a k-point sampling of $1 \times 1 \times 50$.

Depending upon the binding energy analysis, linear BPNWs are the most energetically stable configuration. Zigzag and triangular nanowires are still possible to make, even though they are not as stable. Calculations of electronic structure show that linear BPNWs are semi-metallic because the valence and conduction bands overlap at the Fermi level. On the other hand, zigzag and triangular geometries are metallic because the bands cross the Fermi level. An analysis of the density of states reinforces these conclusions further. The modulation of electronic behavior caused by geometry shows how important structural design is for customizing the properties of nanowires. These results give important information for the smart design of nanoscale electronic and transport devices that use boron and phosphide.

Keywords: BPNW, band gap, semi-metallic, I-V properties, Nano-interconnect

References:

1. K.S. Novoselov, A. Geim, Nat. Mater., 6, (2007), 183–191.
2. E. Gadret, G.O. Dias, L.C.O. Dacal, M.M. de Lima, C.V.R.S. Ruffo, F. Iikawa, M.J.S.P. Brasil, T. Chiamonte, M.A. Cotta, L.H.G. Tizei, D. Ugarte, A. Cantarero, Phys. Rev. B, 82, (2010), 125327.
3. J. Mallet, M. Molinari, F. Martineau, F. Delavoie, P. Fricoteaux, M. Troyon, Nano Lett., 8, (2008), 3468–3474.

2D-SnS₂ Nanoflake-Enabled 2D–0D Heterostructure for High-Detectivity Ultrawideband Photodetection

Sanjeev Mani Yadav ^{1*}, Amritanshu Pandey ²

¹Department of Electronics and Communication Engineering IIIT Vadodara Gujarat-India

²Department of Electronics Engineering, IIT (BHU) Varanasi-India

*Corresponding Author E-mail: sanjeev_yadav@iiitvadodara.ac.in

Two-dimensional (2D) materials have emerged as a transformative class of nanomaterials owing to their high surface-to-volume ratio, tunable electronic properties, and superior charge transport characteristics. These features enable strong light–matter interaction and efficient carrier dynamics, making 2D systems highly promising for next-generation optoelectronic applications, particularly broadband photodetection. However, standalone 2D materials often suffer from limited spectral coverage, restricting their performance across wide wavelength ranges. To overcome this challenge, we report a 2D–0D hybrid heterostructure comprising SnS₂ nanoflakes and Cu₂SnS₃ (CTS) quantum dots for ultrawideband photodetection spanning 300–1100 nm (UV–Vis–NIR) [1]. The device design exploits the strong UV absorption and excellent carrier transport properties of SnS₂, combined with the extended NIR sensitivity of CTS quantum dots, resulting in a synergistic enhancement in photodetection performance. Integration of quantum dots with 2D materials has been shown to significantly improve interfacial charge transfer and spectral response [2]. The materials were synthesized via a cost-effective solvothermal method, yielding SnS₂ nanoflakes (~146 nm) and CTS quantum dots (~3.2 nm) [1–2]. In the hybrid architecture, SnS₂ serves both as an active UV-absorbing layer and as an efficient carrier transport pathway for photogenerated carriers from CTS QDs, minimizing recombination losses.

The fabricated device exhibits high sensitivity and responsivity across UV–Vis–NIR regions, with peak detectivity on the order of 10¹³ Jones, demonstrating its potential for advanced optoelectronic applications.

Keywords: 2D–0D Hybrid Heterostructure; Ultrawideband Photodetector; SnS₂ Nanoflakes; Cu₂SnS₃ Quantum Dots; UV–Vis–NIR Photodetection.

References:

1. Sanjeev Mani Yadav and Amritanshu Pandey. IEEE Transactions on Electron Devices 67, no. 5 (2020): 2008–2015.
2. Sanjeev Mani Yadav and Amritanshu Pandey. IEEE Sensors Journal 21, no. 18 (2021): 19978–19983.

On-chip Multifunctional Heterostructure Enabled by Hot Hole Extraction

Sougata Karmakar¹, Soham Ash¹, Sinorul Haque², Amarnath R. Allu^{2*}, K.D.M. Rao^{1*}

¹*School of Applied and Interdisciplinary Sciences, Indian Association for the Cultivation of Science, Kolkata 700032, India*

²*CSIR-Central Glass and Ceramic Research Institute, Kolkata 700032, India*

Corresponding Author e-mail: mallik2arjun@gmail.com

Achieving on-chip photodetection across the entire ultraviolet spectrum, spanning UV-A (315–400 nm), UV-B (280–315 nm) and UV-C (100–280 nm), remains a significant challenge due to the intrinsic limitations of conventional materials, which typically exhibit narrow spectral sensitivity and high operating voltages. In this work, we report a self-powered, on-chip photodetector based on a heterostructure of hybrid gold nanoislands (Au NIs) embedded in H-glass and cesium bismuth iodide (Cs₃Bi₂I₉). The Au NIs act as catalytic nucleation centers, promoting improved crystallinity and enabling the vertical growth of interconnected petal-like Cs₃Bi₂I₉ thin films. A built-in electric field at the heterojunction facilitates efficient separation of

hot carriers, where photogenerated hot holes in the Au NIs under UV illumination are injected into the valence band of $\text{Cs}_3\text{Bi}_2\text{I}_9$, thereby suppressing recombination losses. The device exhibits a high open-circuit voltage of 0.6 V, a notable responsivity of 0.88 A/W, and a low detection limit of 90 nW/cm², surpassing the performance of previously reported thin-film UV photodetectors operating in self-driven mode.

In addition to photodetection, the heterostructure demonstrates enhanced NO_2 sensing under UV irradiation and efficient hydrogen evolution via photocatalytic H_2 splitting. The petal-like morphology provides a high density of active sites, contributing to improved gas interaction and catalytic activity. Furthermore, the device maintains stable performance under ambient conditions for up to eight months, indicating excellent long-term reliability. Overall, this heterostructure, governed by efficient hot carrier dynamics, presents a promising platform for broadband UV photo detection along with enhanced NO_2 gas detector and self-biased hydrogen generation via H_2 splitting with potential applications in environmental monitoring, flame sensing, biomedical diagnostics, secure communication technologies and green energy generation.^{1,2}

Keywords: full-UV-band photodetectors, hot hole extraction, Au nanoislands, self-powered devices

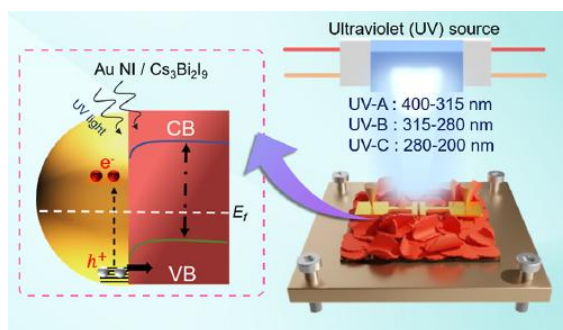


Figure 1. Schematic of the hybrid Au NIs/ $\text{Cs}_3\text{Bi}_2\text{I}_9$ heterostructure for multifunctional applications.

References:

1. J. Gangareddy, P. Rudra, M. Chirumamilla, S. Ganiseti, S. Kasimuthumaniyan, S. Sahoo, K. Jayanthi, J. Rathod, V. R. Soma, S. Das, N. N. Gosvami, N. M. A. Krishnan, K. Pedersen, S. Mondal, S. Ghosh and A. R. Allu, *Small*, 2024, 20, 1–18.
2. S. Karmakar, S. Ash, S. Haque, N. K. Murugasenapathi, M. Sridevi, I. Mandal, G. Ghorai, A. V. Muhammed Ali, N. N. Gosvami, N. M. A. Krishnan, S. Kanungo, M. Chirumamilla, T. Palanisamy, R. K. Singh, A. R. Allu and K. D. M. Rao, *ACS Nano*, 2025, 19, 6309–6319.

Harnessing Compositional Synergy to Accelerated Charge Transfer in $\text{NiS/g-C}_3\text{N}_4$ Heterostructures for Oxygen Evolution Reaction

Shivprasad K. Jadhav¹, Sharad B. Patil^{1*}

¹Centre for Interdisciplinary Research, D. Y. Patil Education Society (Deemed to be university), Kolhapur, Maharashtra.

*Corresponding Author Email: sharadpatil.cir@dypgroup.edu.in

To make water electrolysis commercially viable, it is essential to design a robust, sustainable, and efficient electrocatalyst to boost the sluggish electrocatalytic oxygen evolution (OER) reaction. This study emphasises a binder-free, controlled synthesis of nickel sulfide-graphitic carbon nitride film using a successive ionic layer adsorption and reaction (SILAR) approach. This work demonstrates that exfoliated graphitic carbon nitride (g-C₃N₄) nanosheets alter the material's physicochemical properties and significantly influence the electrochemical performance. The optimised content of g-C₃N₄ nanosheets (0.1 mg mL⁻¹) requires the lowest overpotential, 262 mV to split water at a current density of 25 mA cm⁻², with a Tafel slope of 78.45 mV dec⁻¹. The prepared electrocatalyst shows excellent OER stability for 96 h at a current density of 100 mA cm⁻². Additionally, the assembled electrolyser required a cell voltage of 1.60 V to achieve a current density of 10 mA cm⁻². Thus, the present work demonstrates a feasible approach to improve the OER performance and stability of non-noble electrocatalysts.

Keywords: 2D Graphitic carbon nitride; Electrocatalytic OER; Nickel sulfide; SILAR.

References:

1. Y. Luo, J. Qin, G. Yang, S. Luo, Z. Zhao, M. Chen, J. Ma, Chemical Engineering Journal, 410, (2021), 128394.
2. W. Zhang, S. Kong, M. Nie, J. Wen, C. He, International Journal of Hydrogen Energy, 159, (2025), 150500.
3. T. Singh, D. Deepak, A. Panghal, SK. Saxena, B. Mahato, AS. Shekhawat, SS. Roy, Journal of Materials Chemistry A, 13, (2025), 37614.

Exploring Self-Assembled Mask-Assisted Spatial Defect Engineering in 2D TMDs

Gayatri Subhalaxmi¹, Tanushree H Choudhury^{1*}

¹*Department of Metallurgy and Material Sciences, Indian Institute of Technology, Bombay, India*

*Corresponding Author Email: tanuhc@iitb.ac.in

Defect engineering in two-dimensional (2D) transition metal dichalcogenides (TMDs) provides a powerful pathway to tailor their optoelectronic properties. Achieving local control over defects is critical for integrating these materials into next-generation electronic, photonic, and quantum devices. However, existing approaches, including electron-beam lithography, plasma irradiation, and chemical treatments, are often limited by poor scalability, lack of site selectivity, or material specificity, restricting their broader applicability.

While self-assembled colloidal masks have been previously utilized with 2D TMDs primarily for structural patterning and strain engineering, their application for spatially controlled defect engineering remains largely unexplored. Here, we explore a lithography-free and material-independent strategy for defect patterning using a self-assembled colloidal mask [1]. Polystyrene beads are spin-coated on a MoS₂ film to form hexagonally close-packed monolayer regions separated by boundaries. The polystyrene beads were used as a shadow mask on monolayer MoS₂, followed by ultraviolet-ozone (UV-O₃) exposure to induce localized defect formation [2]. Photoluminescence (PL) and Raman mapping confirm that these defect-induced modifications

are spatially confined and highly reproducible, enabling the formation of large area patterns. PL mapping reveals a clear periodic modulation of the optical response, characterized by systematic quenching of excitonic emission, spectral broadening, and peak shifts, indicative of defect-assisted non-radiative recombination and local electronic perturbations [3]. Raman mapping shows a spatially periodic variation in the A_{1g} – E_{2g}^1 peak separation, reflecting localized changes in doping and defect density induced by UV– O_3 exposure. It is important to note that the impact of UV-ozone treatment is most prominent at the boundaries of the polystyrene mask. Control experiments were conducted with and without the mask to determine the impact of our process.

This method offers a low-cost, scalable, and versatile platform for engineering spatially programmable functionalities in 2D materials, with potential applications in tunable optoelectronic devices, sensor arrays, and integrated photonic systems.

Keywords: Defect engineering, 2D TMDs, Self-assembled mask, UV–ozone treatment

References:

1. J. C. Hulthen, D. A. Treichel, M. T. Smith, M. L. Duval, T. R. Jensen, and R. P. V. Duyne, *The Journal of Physical Chemistry B*, 103 (1999), 3854–3863.
2. C. Jung, H. I. Yang, and W. Choi, *Nanoscale Research Letters*, 14 (2019), 278.
3. K. F. Mak, K. He, C. Lee, G. H. Lee, J. Hone, T. F. Heinz, and J. Shan, *Nature Materials*, 12 (2013), 207–211.

Integrated Recovery of Rare Earth Elements and Graphenization of Spent Ion-Exchange Resins via Flash Joule Heating

Subham Kumar, Rajath Alexander, Jyoti Prakash, Kinshuk Dasgupta*

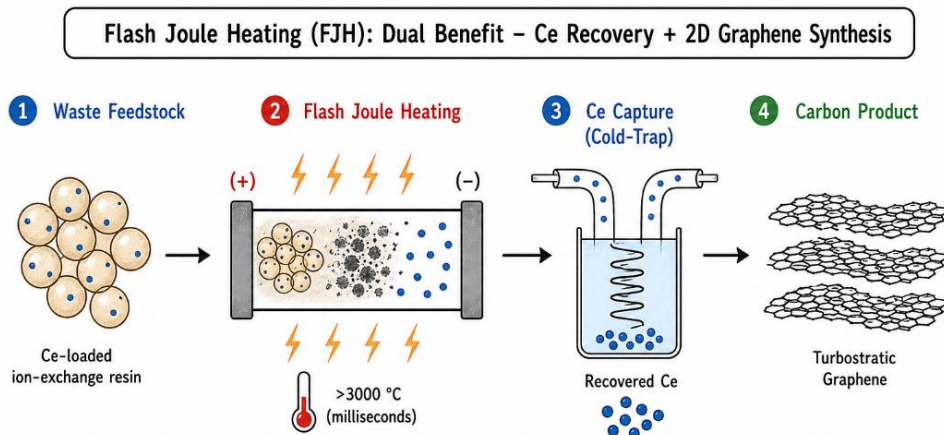
¹ *Homi Bhabha National Institute, Anushakti Nagar, Mumbai, India – 400094*

² *Materials Group, Bhabha Atomic Research Center, Mumbai, India – 400085*

* Corresponding author. E-mail address: kdg@barc.gov.in, Tel no: +91 22 25594951

The surging demand for rare earth element (REE) alongside the growing environmental burden of spent polymeric ion-exchange resins presents a dual sustainability challenge. In this work, we report a rapid and scalable strategy that integrates REE recovery with the synthesis of high-quality 2D carbon materials via flash Joule heating (FJH). Exhausted cerium (Ce)-loaded ion-exchange resin beads are subjected to ultrahigh temperatures ($>3000^\circ\text{C}$) within milliseconds, leading to complete resin pyrolysis and simultaneous volatilization of embedded REE. The volatilized Ce is efficiently captured using a cold-trap system, while the residual carbon reorganizes into turbostratic graphene. ICP-AES analysis confirms complete removal of Ce from the resulting graphene framework. Structural characterization reveals high-quality graphene with a low I_D/I_G ratio (0.12 ± 0.04), high I_{2D}/I_G ratio (3.0 ± 0.20), and expanded interlayer spacing ($3.41 \pm 0.03 \text{ \AA}$), along with characteristic TS1 and TS2 Raman features indicative of turbostratic stacking and weak interlayer interactions. Notably, the process achieves Ce recovery efficiency beyond 70%, enabling resource reutilization. This solvent-free, single-step approach offers a compelling route for converting waste ion-exchange resins into valuable 2D graphene while recovering critical elements. By coupling waste remediation with advanced material synthesis,

the method aligns with circular economy principles and highlights the potential of flash Joule heating in next-generation 2D materials manufacturing.



Keywords: Flash Joule Heating; Graphene; Decontamination; Waste Upcycling; Rare Earth Element

Reference:

1. S. Kumar, R. Alexander, A. Kaushal, J. Prakash, U. K. Goutam, and K. Dasgupta, *Diamond and Related Materials*, (2025), 113109.

Half-Metallicity in Stable EuBr_3 and EuI_3 Single layers: A DFT+ U Study

Amrendra Kumar^{1,2*}, G. Tejaswini³, D. Murali³, C. Kamal¹²

¹*Theoretical and Computational Physics Lab, Semiconductor Lasers and Devices Division, Raja Ramanna Centre for Advanced Technology, Indore - 452013, India*

²*Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai-400094, India* ³*Department of Sciences, Indian Institute of Information Technology Design and Manufacturing (IIITDM), Kurnool - 518008, Andhra Pradesh, India*

*Corresponding author: amrendra623k@gmail.com

Intrinsically magnetic 2D materials have attracted significant attention following the experimental discovery of 2D quantum magnets such as CrI_3 and CrGeTe_3 in 2017, owing to their importance for both spintronics and the fundamental understanding of magnetism in reduced dimensions. In this work, we employ density functional theory (DFT) to investigate EuX_3 ($\text{X}=\text{Br}, \text{I}$) single layers, which are found to be energetically, dynamically, and mechanically stable. The calculated binding energies are -3.165 and -2.748 eV/atom for EuBr_3 and EuI_3 , respectively, and their corresponding Young's modulus values are 5.72 N/m and 7.17 N/m. The ferromagnetic configuration is found to be energetically favorable, and the results are presented accordingly. The ferromagnetic-anti ferromagnetic energy difference is 174 meV (EuBr_3) and 135 meV (EuI_3), corresponding to a Curie temperature of about 580 K and 673 K, respectively, within the Ising model. Electronic structure calculations reveal that the spin-up bands cross the Fermi level, while a large band gap of ~4 eV is observed in the spin-down channel, indicating half-metallic behavior

in both systems. The calculated total magnetic moment of the EuI_3 (EuBr_3) single layer is 12 (12) μ_B per unit cell, primarily originating from the localized f -electrons of Eu atoms, each contributing 6.7 (6.4) μ_B . The I (Br) atoms exhibit small magnetic moments -0.2 (-0.1) μ_B , indicating weak hybridization between Eu- f and I (Br)- p states. Incorporating the Hubbard U results in enhanced band splitting and a larger gap in the spin-down channel, while maintaining the half-metallic character and magnetic moment. A detailed analysis of the density of states reveals that the Eu $4f$ electrons predominantly occupy the spin-up orbitals, contributing to a large magnetic moment, while the spin-down orbitals remain largely unoccupied and contribute mainly to the conduction states.

Keywords: 2D materials; quantum magnetism; half-metallicity; monolayers

References

1. B. Huang, G. Clark, E. Navarro-Moratalla, et al., *Nature*, 546, (2017), 270--273.
2. C. Gong and X. Zhang, *Science*, 363, (2019), eaav4450.
3. W. Ji, *ACS Applied Electronic Materials*, 4, (2022), 3166--3167.

Estimation Of Energy Levels Of Platinum Using Classical And Quantum Framework

Hariharan I¹, Vijay A¹, Vishva C¹, Anuradha R¹, Moorthi P²

*Computer Science and Engineering
Nanoscience and Technology
Sri Ramakrishna Engineering College
Coimbatore, Tamil Nadu
anuradha.r@srec.ac.in

This project is about estimating the energy levels of platinum using quantum chemistry and quantum computing algorithms in the PySCF (Python-based Simulations of Chemistry Framework) software. We took platinum because it is one of the most important metals to be noted as it plays a major role in the fields of fuel cells, automobile industries, medicines due to its unique characteristics and features. Here we found energy levels like ground state multiplicity, electron configurations, HOMO LUMO energies, binding energies, bond length, structure, fermi level and etc. We used PySCF software to perform Hartree-Fock and DFT calculations on single platinum atom, Pt_2 dimer, Pt_4 tetrahedral clusters and Pt_{10} surface models with def2-SVP basis sets and effective core potentials (ECP). The results shows that the platinum atom's ground state is triplet with energy of -118.39 Hartree. The dimer (Pt-Pt) has the optimal bond length around like 2.3-2.7 Å with binding energy of approximately 43-50 kcal/mol and this indicates that this (Pt-Pt) has moderate covalent bonding. The Pt_4 tetrahedral cluster shows cohesive energy consistent with metallic bonding while the Pt_{10} surface model gives the electronic structure that is suitable for catalytic applications. Then we developed an interface between these classical calculations and quantum computing by mapping the molecular Hamiltonian to a 2-qubit system. Using the Variational Quantum Eigensolver (VQE) algorithm with COBYLA optimization, we successfully obtained the ground state energy with high accuracy. The quantum system produced final states with the probabilities associated with the covalent and ionic bond characters. This work shows that quantum algorithms can effectively simulate chemical bonding in transition metal systems and hence it provides a foundation for the future quantum enhanced materials discovery and catalyst design.

Keywords: Quantum Chemistry; Platinum Energy levels; PySCF; VQE; COBYLA optimization;

References:

1. M. A. G. De Souza, G. M. A. Almeida, and A. C. P. Bitencourt, *Journal of Computational Chemistry*, Vol. 47, No. 1, (2026), Article Number: 10.1002/jcc.70289.
2. S. Watson, J. Robledo-Moreno, and W. Kirby, arXiv preprint, arXiv:2508.02578, (2025).
3. S. McArdle et al., *Journal of Physical Chemistry A*, Vol. 129, No. 27, (2025), pp. 5923–5930.

Structural, Thermal, and Electrical Properties of In-Situ Oxygen-Doped Transition Metal Dichalcogenides

Diwakar Singh, Gayatri Subhalaxmi, Ramesh Rajarapu, Tanushree H. Choudhury

¹Metallurgical Engineering and Materials Science Department, Indian Institute of Technology Bombay, Mumbai, India

*Corresponding Author E-mail: tanuhc@iitb.ac.in

Transition metal dichalcogenides (TMDCs) are two-dimensional semiconductors with promising optical and electronic properties for optoelectronic and nanoelectronic applications.¹ Doping and alloying provide effective routes to tune these properties; however, oxygen incorporation in MoS₂ has predominantly been explored through ex-situ treatments, often resulting in contradictory reports of photoluminescence² and charge-transport behavior.³ In-situ oxygen doping during growth remains comparatively underexplored, particularly with respect to its influence on growth kinetics and multifunctional material properties.

This ongoing study aims to investigate the effect of in-situ oxygen incorporation on the structural, optical, electrical, and thermal properties of MoS₂ and WS₂ grown via powder-source chemical vapor deposition using MoO₃ or WO₃ and sulfur powder precursors. Controlled amounts of oxygen are introduced during growth along with Ar or Ar/H₂ carrier gases to systematically modulate oxygen incorporation. Structural and chemical characterization is being performed using Raman spectroscopy and X-ray photoelectron spectroscopy, while optical properties are being examined using photoluminescence spectroscopy. Electrical properties are planned to be characterized using conductive AFM and Kelvin Probe microscopy. Thermal properties are being explored through optothermal Raman techniques (using temperature and power dependant Raman measurements).

Preliminary observations indicate that the presence of a small amount of oxygen (~5%) during growth significantly enhances TMDC nucleation and lateral growth, resulting in increased crystallite size and film coverage compared to oxygen-free conditions. Oxygen incorporation in WS₂ shows redshift in PL spectra and a new peak around 1.57 eV is observed. These early results suggest that oxygen plays a critical role in governing growth dynamics and its optical properties. The outcomes of this work are expected to provide systematic insight into oxygen-mediated growth control and property modulation in two-dimensional TMDCs.

Keywords: *in-situ* oxygen doping; Transition metal dichalcogenides (TMDCs); Nucleation and growth; MoS₂ and WS₂;

References:

1. Ding, J., et al.; *Journal of Physics D: Applied Physics*, 54 (17) (2021), 173002.
2. Nan, H., et al.; *ACS nano*, 8(6), (2014), 5738.
3. Islam, et al.; *Nanoscale*, 6(17), (2014), 10033.

Studies on cation exchange capacity of different clays for Textile effluent treatment

Saroj R. Sepat, Pratibha Pandey, Subham Srivastava, A. R. Shrivastava

Electron Microscopy Division, Defence Research & Development Establishment, Gwalior

Corresponding Author Email: rkpratibha.drde@gov.in

Fuller's Earth is a montmorillonite clay with nano dimensional aluminosilicate layered structure. The cation exchange capacity (CEC) of six different clays, namely Red clay-1 (Local Market), Fuller's earth-1 (HIMEDIA), Red clay -2 (Local Market), Vermiculite (Local Market), Fullers earth -2/FE-2 (Local Market), and Potato field clay (Etawah, UP) was determined in this study to select the best clay for cationic dye removal from effluent. Detailed morphology and composition of the clays were characterized using FESEM and EDX. The FE clay is constituted of 2D nanoflakes and nanorods having Al, Si, O, Fe, Mg, Na, K, C and N as structural element. Methylene blue has been selected as a model cationic dye. The methylene blue (MB) removal capacity from its aqueous solution was assessed for all 6 clays. MB is used in large scale as a cationic textile dye mainly to colour jeans and is commonly present in textile effluent. It is a potential genotoxic material which contaminates water bodies upon discharge of textile effluent. Fullers earth-2 sieved, procured from local market of Gwalior, has shown remarkable adsorption capacity (11.26%, w/w) for methylene blue out of six selected clay's. Wide variation was observed in MB adsorption capacity of various clay's at room temperature i.e. Red clay-1 2%, Fuller's earth-1 0.157%, Red clay-2 2.16%, Vermiculite 0.7%, Fuller's earth-2 11.26% and potato field clay 3.75%. The FE-2 clay is primarily a nanoclay (FESEM) having highest methylene blue adsorption capacity. It is sustainable and environment friendly, abundant in nature, cheaply available and can be used as a cost effective effluent treatment solution by the industries in place of directly disposing coloured effluent i.e. cationic dyes into water bodies.

Key words: Clay, Methylene Blue, Sustainability, Environmental pollutants, Cation exchange capacity

ENGINEERING OXYGEN VACANCY CHANNELS IN BM-SCO FILM VIA STRAIN

Rajan Mishra,¹ Sourav Chowdhury,² Shivam Choudhary,¹ Ram Janay Choudhary^{1*}

*1UGC-DAE Consortium for Scientific Research, Khandwa
Road, Indore 452001, India*

*2Deutsches Elektronen-Synchrotron DESY, Notkestraße 85, 22607
Hamburg, Germany.*

Ram Janay Choudhary*: ram@csr.res.in

Oxygen vacancies are not merely defects in materials but play a crucial role in determining their mechanical, optical, electrical, and catalytic properties.[1] Understanding and controlling these vacancies is essential for the development of advanced energy conversion and storage technologies.[2] SrCoO_{2.5} exhibits a brownmillerite structure with alternating layers of

metal–oxygen octahedra and tetrahedra. When grown on SrTiO_3 (STO) substrates, the oxygen vacancy channels (OVCs) are typically aligned horizontally.[2] Tuning these OVCs strongly influences magnetic anisotropy and resistive switching behavior, which is important for memristor applications. In this work, BM-SCO thin films were deposited on $\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3$ – PbTiO_3 (PMN-PT) and BaTiO_3 (BTO) substrates using pulsed laser deposition. The lattice parameters of PMN-PT (4.0 Å) and BTO (3.994 Å) induce sufficient strain to transform horizontal OVCs into vertical ones without ionic liquid gating. The structural and magnetic properties of films grown on STO, PMN-PT, and BTO were systematically compared. Additionally in SRO/SCO/Nb-STO heterostructures, XRD measurements reveal a bias-induced topotactic phase transition from brownmillerite $\text{SrCoO}_{2.5}$ to perovskite SrCoO_3 (Figure 2), driven by electric-field-controlled oxygen migration. XRD and magnetization (M–T and M–H) measurements confirm the orientation and evolution of OVCs. These tunable vacancy channels enable fast ionic transport, offering promising opportunities for energy storage, neuromorphic devices, and oxide electronics.[3]

Keywords- oxygen vacancy channels , topotactic phase transition, electric-field-driven oxygen migration

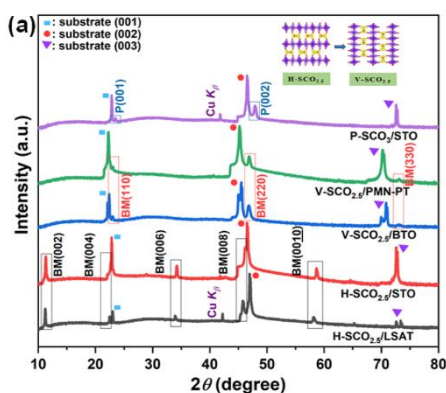


Fig 1. XRD data, reveals H-SCO_{2.5} converted into V-SCO_{2.5} when PNMPT or BTO substrates are utilized.

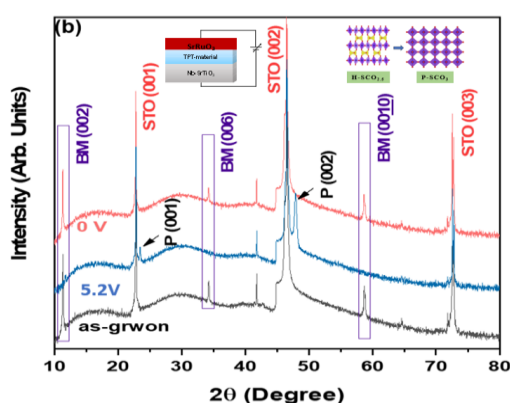


Fig 2. XRD shows Topotactic Phase transition from $\text{SrCoO}_{2.5}$ to SrCoO_3 using small bias of 5.2 volt.

References:

1. Hyeon Han and Stuart S. P. Parkin ACS Nano 2022 16 (4), 6206-6214
2. Chowdhury, S. ACS Appl. Electron. Mater. 2021, 3, 3060– 3071

Surface acoustic wave-driven dynamic strain engineering in monolayer MoS_2

Deena Nath^{1‡}, Rohit Dhawa^{1‡}, Lucky Donald Kunshi¹, Madhu Thalakulam^{1*}

¹School of Physics, Indian Institute of Science Education and Research, Thiruvananthapuram, 695551, Kerala.

*Corresponding Author E-mail: madhu@iisertvm.ac.in

[‡]Equal Contribution.

Strain engineering is a powerful approach for tuning the physical properties of materials and has already demonstrated success in the semiconductor industry through strained silicon technology, which enhances both electron and hole mobility [1]. Various methods have been explored to introduce strain in 2D materials, including mechanical bending and stretching, thermal expansion, epitaxial growth and so on [2].

In this work, surface acoustic waves (SAWs) are utilised to induce dynamic strain in monolayer MoS₂. Prior to device fabrication, a COMSOL Multiphysics simulation was performed to optimise the structural variables and evaluate the device response, including S-parameters and resonant frequency [3]. SAWs are generated using interdigital transducers (IDTs) fabricated on Y-cut 128° LiNbO₃ substrate by standard photolithography, followed by aluminium deposition. Exfoliated MoS₂ flakes are transferred onto the IDT-patterned LiNbO₃ substrate using a dry transfer technique.

Raman spectroscopy reveals two characteristic vibrational modes, one is the E_{12g} mode at ~381.75 cm⁻¹ (in-plane vibration), and the other is the A_{1g} mode (out-of-plane vibration) at ~401.04 cm⁻¹. With increasing RF power (0 to 16 dBm in steps of 4 dBm), both modes exhibit a systematic red shift, indicating a reduction in vibrational frequencies. Photoluminescence spectra show a single emission peak at 1.824 eV, which also red-shifts with increasing RF power, depicting a reduced emission energy. From the COMSOL simulation, the estimated stress value generated by SAW has been calculated. As the RF power increases from 0 to 16 dBm, the stress value is systematically increasing from 1.2×10^{-8} N/μm² to 7×10^{-8} N/μm². The behavior of stress vs RF power plot is consistent with Raman and PL data. Furthermore, Two-probe electrical transport measurements demonstrate intrinsic semiconductor behaviour, with a significant change in the slope of the I-V characteristics under RF power increment. Overall, our work provides a novel route to tune the properties of MoS₂ using SAW-induced dynamic strain.

Keywords: Dynamic strain engineering, MoS₂; Surface acoustic wave; Raman spectroscopy; COMSOL.

References:

1. S.H. Huang; T.M. Lu; S.C. Lu; C.H. Lee; C. W. Liu; D. C. Tsui, Applied Physics Letters, 101, 2012, 042111.
2. A. P. John, A. Thenapparambil, M. Thalukulam, Nanotechnology, 31, 2020, 275703.
COMSOL Multiphysics Users Guide, Ver 3.2 (2005).

Innovation in Topological Insulator-based Photodetector heterojunction for Metastructure Applications: An Effect of Co layer incorporation

Puja Dey^{1,2*}, J. N. Roy^{1,2}, Subhadip Paul¹, Nitish Ghosh¹, and Nibir K Dhar³

¹ Department of Physics, Kazi Nazrul University, Asansol, 713340, India,

²Centre for Organic Spintronics and Optoelectronics Devices (COSOD), Kazi Nazrul University, Asansol, 713340, India

³VCU College of Engineering, Virginia Commonwealth University, Richmond, VA, USA

*Corresponding Author E-mail: puja.dey@knu.ac.in

Innovation in Topological Insulator (TI) based Low Noise, Broadband Infrared photodetectors at elevated temperatures is aimed for IR detection technology and next-generation high-performance focal plane arrays. As per general consensus, in case of TI/ferromagnet heterostructures, there is magnetic proximity effect, which induces high-temperature magnetic ordering in TI. Such ordering might result in the breaking of time reversal symmetry at topological surface states and thereby open a gap in the topologically protected spin momentum locked

surface state [1]. Therefore, inclusion of a magnetic layer adjacent to TI layer in a heterostructure, is supposed to enhance its topological surface conduction.

In this attempt, fabrication of ITO/Bi₂Se₃(10 nm)/Au (S1) and ITO/Bi₂Se₃(10 nm)/Co(10 nm)/Au (S2) heterostructures have been done through e-beam evaporation technique. I-V characteristic study on these heterostructures, in the range from ± 0.000001 V to 2 V, exhibit transition from “admixture of bulk non-conducting and surface conducting state” to “dominating surface conducting state” mediated by bias voltage. In S1, the transition has been obtained at 0.25 V, whereas in S2, similar transition occurs at much less voltage of 0.15 V. Furthermore, pretty less current in S1 has been detected at conducting state than S2. Much larger retention time of conducting topological state has been obtained in S2 than S1. Our experimental results provide evidences that Co incorporation in the heterojunction yielding enhanced electrical conductivity as well as larger retention time. A proof of checking of topological conducting surface state has been performed through I-V measurement under the application of in-plane and out of plane magnetic field. Finally, a positive photocurrent generation has been monitored with optimum photocurrent to dark current ratio of 9×10^3 % under broad band infrared light illumination [2]. This is in conformity with the prominent absorption peak at 1035 nm in absorption spectrum of Bi₂Se₃ film.

Keywords: Topological Insulators; I-V characteristics; Infrared detectors; Heterostructures; Broad band detectors.

References

1. J. Henk et al., Phys. Rev. Lett. 109 076801 (2012). Md Minhaj Ali, Nitish Ghosh, B. Mukhopadhyay, P. Dey, J. N. Roy and Nibir K Dhar, Journal of Alloys and Compounds 1040, 183397 (2025).

Tuning the electronic and transport properties of SrNbO₃ thin films

Pampa Sadhukhan^{1*}, Zahra Minabi¹, Merit Spring¹, Deepnarayan Biswas², Tien-Lin Lee², Louis Veyrat³, Manuel Meyer¹, Fabian Hartmann¹, Martin Kamp¹, Michael Sing¹, and Ralph Claessen¹

¹ Physikalisches Institut and Würzburg-Dresden Cluster of Excellence ct.qmat, Universität Würzburg, 97074 Würzburg, Germany

² Diamond Light Source Ltd., Didcot, United Kingdom

³ CNRS, Laboratoire National des Champs Magnétiques Intenses, Université Grenoble-Alpes, Université Toulouse 3, INSA-Toulouse, EMFL, Toulouse, France

*Corresponding Author E-mail: pampasadhukhan89@gmail.com

The realization of quantum materials (QMs) with both electron correlations and topological properties is currently under active investigation. In this context, 4d transition metal oxides offer a good balance between e-e correlations and spin-orbit coupling. Recently, a topological band structure was predicted for tetragonal SrNbO₃ (SNO) [1]. Strained SrNbO₃/SrTiO₃(001) (SNO/STO) films show an extremely high magnetoresistance (150,000 % at 14T) and mobilities (80,000 cm²/V·s) [2, 3]. The understanding of these unique properties not yet clear. Here, we report a systematic investigation of SNO/STO electronic structure, combined with the transport property. We have grown epitaxial SrNbO₃ thin films on SrTiO₃ (001) substrates using pulsed laser deposition and performed in-situ x-ray photoemission spectroscopy. Ultra-thin film is dominated by Nb⁵⁺, whereas the intensity of Nb⁴⁺ and Nb³⁺ increases and Nb⁵⁺ decreases with increasing film thickness and vice versa with increasing oxygen pressure. Concomitantly, valence band spectra exhibit a reduced density of states at the Fermi level for the thinner films as well as with increasing oxygen pressure. The resistivity data show thickness-

dependent and oxygen pressure-induced semimetallic to metallic behavior, which is consistent with our XPS results. We obtained extremely high magnetoresistance (MR) of 53000% at 2K and 9T for 18uc SNO/STO, which is in agreement with the literature. Our study demonstrates that the control over the oxygen pressure or oxygen stoichiometry or growth temperature as well as film thickness, provides a unique way to enhance MR values in SNO/STO. Moreover, the STO substrate also plays an important role in high MR in SNO, whereas SNO grown on KTaO_3 substrate under similar conditions does not show high MR. In addition, a charge transfer is observed from Nb 4d states to Ti 3d states at the SNO/STO interface, which suggests 2D-electron gas formation at the interface.

Keywords: Magnetoresistance; thin film; electronic structure

References:

1. N. Mohanta, J. M. Ok, J. Zhang, H. Miao, E. Dagotto, H. N. Lee, and S. Okamoto, *Phys. Rev. B* 104, (2021), 235121.
2. J. M. Ok, N. Mohanta, J. Zhang, S. Yoon, S. Okamoto, E. S. Choi, H. Zhou, M. Briggeman, P. Irvin, A. R. Lupini, Y.-Y. Pai, E. Skoropata, C. Sohn, H. Li, H. Miao, B. Lawrie, W. S. Choi, G. Eres, J. Levy, H. N. Lee, *Sci. Adv.* 7, (2021), eabf9631.
3. J. Zhang, J. M. Ok, Y.-Y. Pai, J. Lapano, E. Skoropata, A. R. Mazza, H. Li, A. Huon, S. Yoon, B. Lawrie, M. Brahlek, T. Z. Ward, G. Eres, H. Miao, and H. N. Lee, *Phys. Rev. B* 104, (2021), L161404.

Vacancy Defect Effects on Mechanical and Thermal Properties of Janus WSSe Monolayer: A Molecular Dynamics Study

Ashwani Kushwah¹, Amit Singh^{1*}

¹*Department of Mechanical Engineering, Indian Institute of Technology Bombay, Mumbai 400076, India*

*Corresponding Author E-mail: ashwani12@iitb.ac.in

Janus WSSe, with sulfur (S) on top and selenium (Se) on the bottom chalcogen layer, breaks the out-of-plane mirror symmetry and exhibits unique physical properties absent in symmetric counterparts such as WS₂ and WSe₂. While the mechanical behavior of pristine Janus WSSe has been reported [1], the effect of atomic vacancy defects—unavoidable during chemical vapor deposition synthesis—on its mechanical properties remains unexplored.

Molecular dynamics (MD) simulations using the Stillinger–Weber (SW) potential [3] are performed to investigate the effects of W, S, and Se monovacancy defects on the fracture strength and Young's modulus of Janus WSSe monolayer. A rectangular supercell of 10,080 atoms is subjected to uniaxial tensile loading along the zigzag (X) and armchair (Y) crystallographic directions at 300 K. Vacancy concentrations of 1%, 2%, 4%, and 6% are considered for each defect type.

The pristine WSSe exhibits fracture strengths of 15.44 GPa (zigzag) and 15.85 GPa (armchair), confirming mechanical anisotropy consistent with the literature [1]. Among the three vacancy types, W-vacancy (VW) causes the most severe mechanical degradation, reducing fracture strength by up to 43.7% and 47.4% in the zigzag and armchair directions, respectively, at 6% concentration. A key novel finding arises from the Janus structural asymmetry: Se-vacancy (VSe) is more damaging than S-vacancy (VS) under zigzag loading, whereas VS causes greater

reduction under armchair loading. This direction-dependent reversal of chalcogen vacancy damage—not reported previously—provides direct evidence that the out-of-plane Janus asymmetry interacts with crystallographic anisotropy to govern fracture behavior. Young's modulus shows comparatively lower sensitivity to vacancies, decreasing by only 5–19% across all cases. These findings offer new insights into defect–property relationships in Janus 2D materials with implications for WSSe-based nanodevice design [2].

Keywords: Janus WSSe; Vacancy defects; Molecular dynamics; Mechanical properties; Fracture strength.

References

1. F. Yang, J. Shang, L. Kou, C. Li, and Z. Deng, *Nanomaterials*, 12, (2022), 1910.
2. H. Qin, K. Ren, G. Zhang, Y. Dai, and G. Zhang, *Phys. Chem. Chem. Phys.*, 24, (2022), 20437–20444.
3. J.-W. Jiang, *Acta Mech. Solida Sin.*, 32, (2019), 17–28.

CNC Micro-Milling as a Democratising Subtractive Patterning Platform for $\text{Ti}_3\text{C}_2\text{T}_x$ MXene-Coated Substrates: A Conceptual Framework for Low-Cost 2D Material Device Fabrication

Lalit Bhanwrela¹

*1Department of Electrical & Electronics Engineering, Shri Vaishnav Vidyapeeth
Vishwavidyalaya (SVVV), Indore 453111, Madhya Pradesh, India
lalitbhanwrela@gmail.com*

Two-dimensional (2D) materials, principally $\text{Ti}_3\text{C}_2\text{T}_x$ MXenes, have emerged as a compelling frontier in case of functional materials science based on the characteristics including metallic conductivity ($\sim 15,000 \text{ S cm}^{-1}$), mechanical flexibility, and electromagnetic interference (EMI) shielding effectiveness exceeding 50 dB at sub-micrometre thicknesses — positioning them as critical enablers for next-generation electronics, wearable sensors, and energy storage [1,2]. However, converting these characteristics into high-precision-patterned functional devices remains formidable. Available subtractive patterning techniques such as photolithography, electron-beam lithography, focused ion beam milling, and laser ablation impose prohibitive barriers: resist-mediated contamination that structurally degrades atomically thin films [3], capital equipment costs spanning tens to hundreds of crore rupees, and cleanroom dependencies that systematically exclude academic institutions and MSMEs in emerging economies from 2D material device research.

This paper presents the utilization of a precision CNC micro-milling platform as a democratising subtractive patterning tool for 2D material-coated substrates a low-cost solution. Basically a CNC platform developed by implementing hardware components such as Mitsubishi CNC M80 controller with three-axis CNC H80 servo-driven motion, and developed machine achieves validated positioning accuracy of $\pm 0.042 \text{ mm}$ along with 1σ repeatability of $\pm 0.019 \text{ mm}$ at a total system cost of approximately ₹5 lakh — a reduction of one to two orders of magnitude relative to photolithographic and laser ablation alternatives. The proposed machine can be used in terms of solution-processable $\text{Ti}_3\text{C}_2\text{T}_x$ MXene deposition by spray coating onto FR-4 or

polyimide substrates with CNC-defined isolation channel patterning, targeting EMI shielding trace geometries, micro-supercapacitor interdigitated electrodes, and IoT antenna structures, without photoresists, chemical developers, or inert atmosphere requirements.

A detailed literature review carried out to perform comparative analysis that quantifies the proposed approach against established methods across feature resolution, equipment cost, chemical footprint, substrate compatibility, and throughput. This work represents the first conceptual articulation of CNC micro-milling as a 2D material patterning modality, providing the methodological foundation to democratise precision device fabrication for resource-constrained research environments.

Keywords: $\text{Ti}_3\text{C}_2\text{T}_x$ MXene patterning; CNC micro-milling; EMI shielding; 2D material fabrication; democratising fabrication.

References:

1. Lulu Li, Cheng-Zhang Qi, Mengjie Chen, ACS Applied Materials & Interfaces 2024 16 (10), 13082-13090
 2. F. M.Oliveira, J.Azadmanjiri, X.Wang, M.Yu, Z.Sofer, Small Methods 2023, 7, 2300112
 3. Alessandro Enrico, Oliver Hartwig, Nikolas Dominik, ACS Nano 2023 17 (9), 8041-8052
-

Graphene Oxide-Modified Nanofiltration Membranes for Enhanced Separation Performance and Radiation Stability

Bitan Ghosh*, A. K. Ghosh and K. P. Bhattacharyya

*Membrane Development Section, Desalination and Membrane Technology Division
Bhabha Atomic Research Centre, Trombay, Mumbai – 400 085*

*Corresponding Author E-mail: bitanghosh@barc.gov.in

The increasing demand for efficient and sustainable water purification, chemical separation, and industrial filtration technologies has driven urge for significant advancements in nanofiltration (NF) membranes. Advanced nanofiltration membrane development is focused on enhancing permeability and selectivity, fouling resistance, sustainability in harsh environments including radiation stability, etc. through innovations in materials and fabrication techniques such as development of thin-film nanocomposites and surface modification [1]. The unique 2D structure and properties of graphene oxide nanoparticles make them promising candidates targeting multidimensional performance enhancement of NF membranes.

In this study, graphene oxide (GO) nanoparticles are incorporated onto thin film composite membrane surfaces during interfacial polycondensation reaction between a bivalent amine and a trivalent acid chloride. The interlayer channels prevailing in graphene oxide nanoparticles provide extra pathways for passage of water, when incorporated in polymeric membranes [2]. In addition to this, the surface negative charge of these nanoparticles contributes to the Donnan exclusion properties of the membranes, favouring an enhanced salt rejection by the membranes without any compromise in rejection [3]. Under irradiation, the radicals generated by homolytic cleavage of C-H bond of the polymer, gets delocalised in the aromatic ring of the graphene oxide, the reduces availability to initiate further reaction leading to crosslinking or degradation of the membrane. Thus, the nanocomposite membranes have shown radiation resistant properties and retained performances upto 100 kGy irradiation dose, while radiation effect became prominent at 75 kGy for control membranes (Table 1). Tensile strengths of TFC and TFC-GO membranes further supports the higher radiation stability of TFC-GO membranes (Table 1). These advance nanocomposite NF membranes are thereby finding expanding

applications across water softening at low/high pressure, treatment of radioactive wastes generated in nuclear power plants, offering energy-efficient alternatives to conventional separation processes while addressing global challenges in water scarcity and environmental sustainability.

Table 1: Membrane Performances and Characteristics: Pre and Post Irradiation

Membrane	Dose (kGy)	Tensile strength (MPa)	Water Flux (LMD)	% Rejection of Na ₂ SO ₄
TFC	0	8.09	910	92
	50	8.18	890	94.1
	75	10.24	535	97
	100	6.84	1527	85.27
TFC-GO	0	10.55	951	98.7
	50	11.63	948	98.1
	75	11.63	951	98.0
	100	11.62	958	98.9

Keywords: Graphene Oxide; Nanoparticle; Membrane; Nanofiltration; Radiation resistant.

References:

1. Z. Yang, Y. Zhou, Z. Feng, X. Rui, T. Zhang, Z. Zhang, *Polymers*, 11 (2019) 1252
2. D.R. Dreyer, S. Park, C.W. Bielawski, R.S. Ruoff, *Chemical society reviews*, 39 (2010) 228-240
3. H.M. Hegab, L. Zou, *Journal of Membrane Science*, 484 (2015) 95-106

Leakage-Aware Multi-Property Screening of JARVIS-DFT 2D Materials with Cautious Exfoliation-Energy Analysis

Md. Efatuazzaman Efata^{1*}, Md. Saiful Islam¹, Sayed Md. Fazle Rabbi², Md. Saymon Hosen Polash³

¹*Institute of Information and Communication Technology (IICT), Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh*

²*Department of Electrical and Electronic Engineering (EEE), Ahsanullah University of Science and Technology (AUST), Dhaka, Bangladesh*

³*Department of Computer Science and Engineering (CSE), East West University (EWU), Dhaka, Bangladesh*

*Corresponding Author E-mail: efatuazzaman@gmail.com

Two-dimensional (2D) materials offer a broad design space for electronic, optoelectronic, and emerging device applications, yet reliable computational screening depends on disciplined treatment of target provenance, descriptor leakage, and validation design. We present a leakage-controlled materials-informatics workflow for multi-property analysis of JARVIS-DFT 2D materials data [1]. Three complementary computed targets are examined: band gap, formation energy per atom, and exfoliation energy, interpreted respectively as a DFT-level electronic-structure descriptor, a soft thermodynamic-accessibility descriptor, and a computational proxy for 2D-layer accessibility [2]. No claim is made regarding energy-above-hull stability, experimental synthesizability, or direct exfoliability.

Property-specific datasets are constructed with explicit target-alias control and strict exclusion of target columns, duplicate target descriptors, and target-derived labels from the feature space. Three nested descriptor protocols are compared: composition-only features; composition

combined with transparent structural descriptors; and the latter augmented with non-target DFT-assisted descriptors. Models are evaluated under grouped, leakage-aware validation to reduce overoptimistic estimates when chemically similar materials co-occur across training and test folds [3], with baseline comparison and residual-based error analysis.

Preliminary results indicate that band-gap and formation-energy regression are comparatively more robust under grouped validation, whereas exfoliation-energy regression remains more challenging owing to limited target coverage, skewed distribution, and long-tail behavior. Exfoliation energy is therefore reported with explicit caution as a screening proxy, emphasizing residual analysis, threshold-based prioritization, and uncertainty-aware interpretation. The workflow provides a reproducible, scientifically cautious framework for prioritizing 2D materials across electronic, energetic, and exfoliation-energy-related computed descriptors.

Keywords: 2D materials; JARVIS-DFT; Materials informatics; Leakage-aware validation; Multi-property screening

References:

- [1] K. Choudhary et al., “The Joint Automated Repository for Various Integrated Simulations (JARVIS) for data-driven materials design,” *npj Computational Materials*, 6, 173 (2020).
- [2] M. Ashton, J. Paul, S. B. Sinnott, and R. G. Hennig, “Topology-Scaling Identification of Layered Solids and Stable Exfoliated 2D Materials,” *Physical Review Letters*, 118, 106101 (2017).
- [3] B. Meredig et al., “Can machine learning identify the next high-temperature superconductor? Examining extrapolation performance for materials discovery,” *Molecular Systems Design & Engineering*, 3, 819–825 (2018).

MoS₂ Nanoflowers as SERS Substrate for Methyl Orange Detection

Sudin Kumar De^a, Pintu Kumar Yadav^b, Rohit Kumar Gupta^a, S. K. Dwivedi^b and Ashish Kumar Mishra^a

^aNano Photonics & Energy Materials Laboratory (NPEML), School of Materials Science and Technology, Indian Institute of Technology (BHU), Varanasi, India

^bDepartment of Physics, Udai Pratap (Autonomous) College, Varanasi

E-mail: sudinkumarde.rs.mst24@itbhu.ac.in

Two-dimensional transition metal dichalcogenides (TMDs) have emerged as promising materials due to their exceptional physicochemical properties, including high chemical stability and semiconducting nature. Among them, molybdenum disulfide (MoS₂) has drawn a lot of interest because of its high carrier mobility, ability for charge transfer, tunable band gap, and compelling optical responses, which make it appropriate for use in photonics, optoelectronics, energy storage, and sensing applications. One prominent optical sensing method, involving the use of TMDs nanostructures, is surface enhanced Raman spectroscopy (SERS) technique [1-3]. In this work, hydrothermally synthesized MoS₂ nanoflowers have been examined as SERS substrate for detecting dye. Different characterization techniques like SEM, Raman, and XRD have been used to analyze the morphology and phase of the synthesized MoS₂ nanoflowers. The MoS₂ nanoflowers coated over stainless-steel substrates were subsequently employed as SERS platforms for the detection of methyl orange (MO) at low concentrations. The SERS substrate shows a detection limit of 10⁻⁷ M for methyl orange, indicating good sensitivity. This performance arises from the high surface area and efficient charge transfer of the MoS₂ nanoflowers, suggesting its strong potential for low-cost and highly sensitive chemical sensing applications.

Keywords: MoS₂ nanoflowers; Hydrothermal synthesis; Surface-enhanced Raman spectroscopy (SERS); Methyl orange

References:

1. B. Sharma, R. R. Frontiera, A. I. Henry, E. Ringe, and R. P. Van Duyne, *Materials Today*, 15, 2012, 16-25.
 2. A. H. Puichaud, C. Flament, A. Chniouel, F. Lomello, E. Rouesne, P. F. Giroux, H. Maskrot, F. Schuster, and J. L. Béchade, *EPJ Nuclear Sciences & Technologies*, 5, 2019, 23.
 3. R. K. Gupta, A. Singh, and A. K. Mishra, *Optical Materials*, 155, 2024, 115866.
-

Co-based thin film electrocatalysts with diverse substrates for oxygen evolution reaction

Suraj C. Bulakhe^{1,2}, R. J. Deokate^{1,*}

¹*Vidya Pratishthan's Arts, Science and Commerce College, Baramati, Pune, Maharashtra, 413 133, India*

²*Loknete Gopinathji Munde Arts, Commerce and Science College, Mandangad, Maharashtra, 415 203, India*

*Corresponding Author E-mail: deokate2000@yahoo.co.in

The development of cost-effective, high-efficiency cobalt-based electrocatalysts remains a critical hurdle in sustainable water splitting. This study details a facile electrodeposition method for synthesizing cobalt oxide (Co₃O₄) thin films on diverse conducting supports, including nickel foam (NF), copper foil (CF), and stainless steel (SS). Our findings indicate that the Ni-foam substrate promotes the formation of highly crystalline Co₃O₄ structures and provides an intricate 3D network that optimizes mass transport and electron-transfer kinetics. Consequently, the Co₃O₄@NF electrode outperforms both CF and SS counterparts, requiring an overpotential of only 305 mV to reach a current density of 10 mA/cm² in an alkaline media. Furthermore, the catalyst maintains robust stability over 48 hours with a Faradaic efficiency of 65%, offering a scalable route for industrial-grade water splitting applications.

Keywords: Electrodeposition, Co₃O₄, Electrocatalyst, OER

References:

1. X. Zou, Yu. Zhang, *Chem. Soc. Rev.* 44 (15) (2015), 5148–5180.
 2. X. Hu, X. Tian, Y. W. Lin, Z. Wang, , *RSC advances*, 9 (54) (2019), 31563-31571
 3. A. V. Ravindra, B. C. Behera, P. Padhan, *Journal of Nanoscience and Nanotechnology*, 14 (7) (2014), 5591-5595.
-

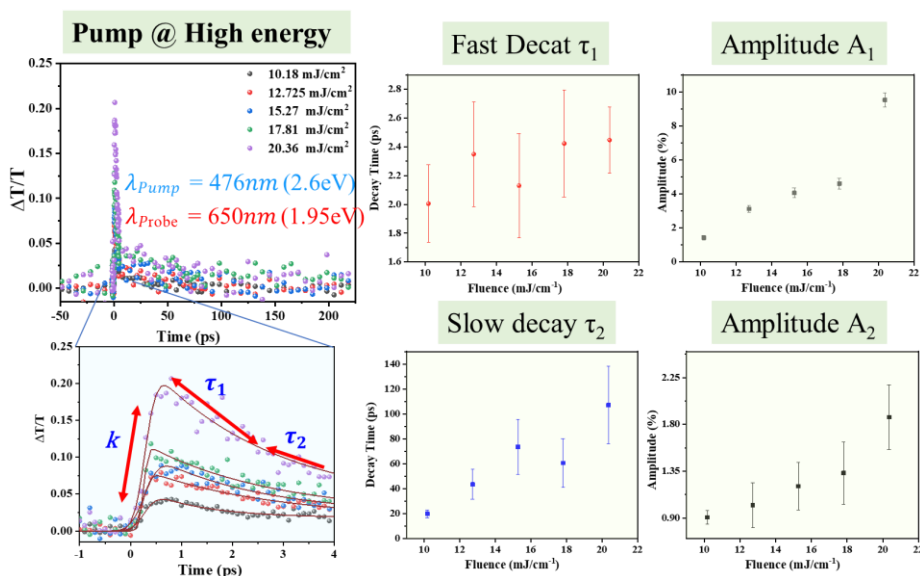
Hot Carrier Dynamics in Bismuth-based 2D Mixed Halide Perovskite

Shivani Malvi¹, Salahuddin Khan^{2,3}, Saleem Khan^{2,3}, Kuntal singh⁴, K.D.M Rao⁴, Tarun Kumar Sharma^{2,3}, Vasant G. Sathe¹, and Praveen K. Velpula¹

¹ UGC-DAE Consortium for Scientific Research Indore 452001, India²Raja Ramanna Centre for Advanced Technology, Indore, India³Homi Bhabha National Institute, Mumbai⁴School of Applied and Interdisciplinary Sciences (SAIS), Indian Association for the Cultivation of Science, Kolkata, 700032 India*Corresponding Author E-mail: velpulacsr@gmail.com

Metal-halide perovskites have emerged as an important class of semiconductors for next-generation optoelectronic and photovoltaic applications due to their exceptional optical absorption, long carrier diffusion lengths, structural versatility, tunable band gaps, and solution-processable synthesis. Among them, low-dimensional and lead-free bismuth-based halide perovskites are attracting growing interest because of their enhanced environmental stability, reduced toxicity, and strong excitonic effects. In particular, understanding hot-carrier dynamics in these materials is crucial for developing high-efficiency optoelectronic and hot-carrier photovoltaic devices, where suppression of rapid carrier cooling can enable efficient extraction of excess carrier energy before thermalization. Here, we investigate the hot carrier relaxation processes and optical properties of bismuth-based 2D mixed-halide perovskite $\text{Cs}_3\text{Bi}_2\text{Br}_3\text{I}_6$ (CBBI) bulk single crystal using X-ray diffraction (XRD), UV–visible absorption spectroscopy, photoluminescence (PL), Raman spectroscopy, and ultrafast transient absorption spectroscopy techniques. Carrier-density-dependent hot-carrier relaxation reveals two distinct decay components: a fast decay of ~ 2.5 ps, attributed to longitudinal optical (LO) phonon emission [1], and a slower decay component is ~ 30 ps at lower carrier densities that is getting slower up to ~ 100 ps with increasing carrier density. This slower decay process could be associated with carrier relaxation mediated by the hot phonon bottleneck effect [2]. The observed prolonged hot-carrier lifetime highlights the potential of CBBI for hot-carrier photovoltaic and advanced optoelectronic applications.

Figure 1: Femtosecond carrier dynamics of CBBI single crystal using a pump of 2.6 eV and probe of 1.95 eV.



Keywords: 2D Metal-Halide Perovskites, Hot carriers, Carrier dynamics, Phonon-Bottleneck.

References:

1. D. Samanta et al, Phys. Rev. B 108, 1–10 (2023).
2. Yang J, Wen X, Xia H, et al, Nature communication, 14120, (2017)

2D Chiral Metasurface based All-optical Standard Ternary Inverter

Sudam Saha¹, Puja Dey^{1,2} and Jitendra Nath Roy^{1,2*}

¹Department of Physics, Kazi Nazrul University, Asansol, West Bengal, India 713340

²Centre for Organic Spintronic & Optoelectronic Devices, Kazi Nazrul University, W.B., 713340

*Corresponding Author E-mail: jn.roy@knu.ac.in

The super chiral exceptional points (SCEP) in plasmonic non-Hermitian metasurface exhibits ideal circular dichroism (CD) with enhanced circular polarization conversion efficiency. All-optical polarization encoded standard ternary inverter (STI) has been theoretically designed realizing anti-chiral SCEPs.

An all-optical ternary inverter processes optical signals of three valued logic states (0,1,2) instead of binary logic (0,1). The earlier reported inverters are bulky in size and required high intense laser beam [1]. H. Li et al. have designed two enantiomers with opposite chirality of plasmonic non-Hermitian chiral metasurface composed of silver nanorods [2]. The enantiomers support twin-SCEP and exhibit ideal CD response up to 0.996 at 1550 nm. At 1550 nm, one type of Enantiomer (A) passes and converts right circularly polarized (RCP) light to left circularly polarized (LCP) light and other type of Enantiomer (B) behaves rightly opposite of Enantiomer A. The unit cells of the enantiomers are shown in Fig.1. A standard ternary inverter has been realized with these enantiomers and can be verified by Jones matrix. The three logical states are defined as “0” = LCP light, “1” = no light (NL), and “2” = RCP light respectively. The inverter circuit design and corresponding truth table are shown in Fig.2. (BE is beam expander and BC is beam combiner). The designed inverter will operate at 1550 nm.

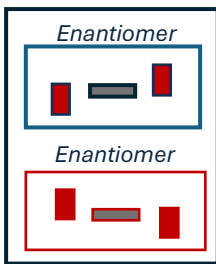


Fig. 1 Unit Cells of the Enantiomers

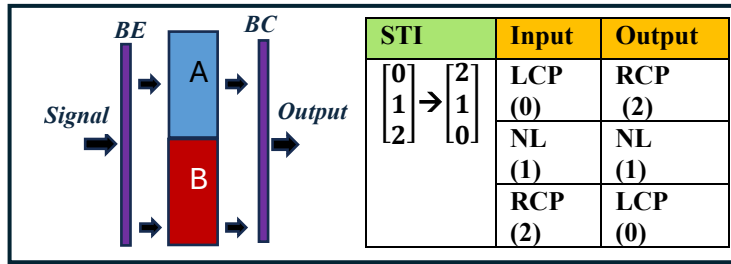


Fig. 2 Standard Ternary Inverter & Truth Table

Keywords: Non-Hermitian metasurface, Super chiral exceptional points, Standard ternary inverter, Optical logic.

References:

1. Tanay Chattopadhyay, Panchatapa Bhowmik and Jitendra Nath Roy, Journal of Optical Society of America B (OSA), 29 (2012) 2852-2860.
2. Haojie Li et. al., Optics Express 34, (2026), 6276-6286.

Acknowledgement: This material is based upon work supported by the Air Force Office of Scientific Research under award number FA2386-25-1-4052.

Divergent Optical and Vibrational Evolution in 2D vs. 1D MoS₂ Monolayers at Cryogenic Temperatures

Akhil R Warriar^{1,2}, Kishore K. Madapu^{1,2,*}

¹Indira Gandhi Centre for Atomic Research, Kalpakkam, Tamil Nadu, India.

²Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai, 400094, India.

* arw23@igcar.gov.in

Although monolayer MoS₂ (1L-MoS₂) exhibits a high absorption coefficient, its photoluminescence (PL) quantum yield (QY) falls short of theoretical expectations [1]. While the optical absorption can be enhanced by increasing the material thickness, multilayer MoS₂ possesses an indirect band gap, which significantly reduces its PL efficiency. Therefore, achieving multilayer structures with electronically non-interacting layers is essential for improving the PL QY. We have fabricated non-interacting MoS₂ nanoscrolls from monolayers using the evaporation of a low-surface-tension solvent [2]. The optical properties of nanoscrolls were evaluated at cryogenic temperatures.

Figure 1a shows optical micrographs of nano-scrolls and remnant 1L-MoS₂. The nanoscrolls exhibit enhanced PL intensity (Fig. 1b). At low temperature, defect-bound exciton emission (DBE) (1.4–1.7 eV) dominates the PL emission of nanoscrolls (Fig. 1d). The enhanced DBE is attributed to efficient heat dissipation facilitated by the nanoscrolls [3]. Interestingly, the intensity ratio of A_{1g} and E_{2g} modes of nanoscrolls is found to depend on sample temperature (Fig. 1e & 1f). These findings suggest that nanoscrolls exhibit a unique excitonic landscape, thereby providing a pathway to tune the optoelectronic performance of TMDCs through morphological engineering.

Keywords: Monolayer MoS₂; Nano scrolls; TMDCs; Cryogenic PL.

References:

1. Madapu, K. K., Bhuyan, C. A., Srivastava, S. K., & Dhara, S. J. Mater. Chem. C, 9(2021), 3578-3588/10.
2. Bhuyan, C. A., Madapu, K. K., Prabakar, K., Ganesan, K., Amirthapandian, S., & Dhara, S. ACS Appl. Nano Mater., 8(2025), 8993-9003/17
3. Madapu, K. K., Bhuyan, C. A., & Dhara, S. Phy. Rev. B, 108(2023), 085429/8.

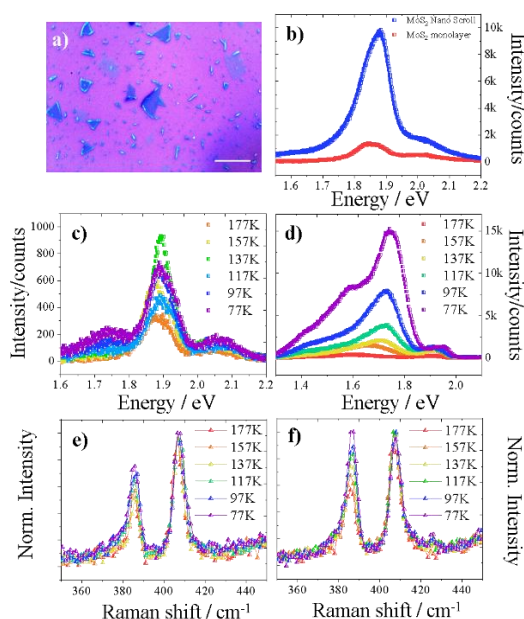


Fig.1 a) Optical microscopy image of the MoS₂ nanoscrolls (scale bar: 10 μ m), b) PL spectra of nanoscroll and monolayer MoS₂. c) and d) PL spectra at various temperatures for monolayer and nanoscroll MoS₂, respectively. e) and f) Raman spectra at different temperatures for monolayer and nanoscroll MoS₂, respectively.

Synergism of graphene and diamond nanoparticles for improved lubrication characteristics of synthetic oil

Jagdeep Chail, Mir Irfan Ul Haq, Ankush Anand, Ankush Raina* Jagdeep Chail, Mir Irfan Ul Haq, Ankush Anand, Ankush Raina*

School of Mechanical Engineering, Shri Mata Vaishno Devi University, Katra, JK, India, 182320

*Corresponding Author E-mail: ankush.smvd@gmail.com

Addition of nanoparticles has led to a significant improvement in the tribological characteristics of the lubricating oils. The use of anti-wear and anti friction nano additives owing to their higher surface area to volume ratio results in the improved characteristics of lubricating oils. In this study, graphene and diamond nanoparticles were investigated in the PAO oil for steel/steel pair. The results were obtained for different loading conditions (20-80 N) for varied concentration of nano particles. Experiments were carried out to evaluate the frictional behavior of different lubricating oils under varied parameters. The result obtained therein demonstrated the strong effect of different nano particles owing to the individual or combined effect of following effects that includes mending, rolling and film formation. The hybrid lubricants prepared by the use of both diamond and graphene nanoparticles resulted in the better frictional and surface properties owing to their combined effect. Surface analysis revealed that the damage to the surface was less for the cases where nanoparticles were used as additives in lubricating oil.

Keywords: Nanoparticles; Lubricating oil; Tribology.

Structural Stability and Compressibility Behaviour of Rare-earth Tritellurides under Pressure

Sanitha K¹, Anand Kumar P¹, Shyam Kumar S², Anuj Upadhyay³, Sanjay Kumar N R^{1,*} and Rajesh Ganesan²

Affiliation: ¹High Pressure Studies Section, Materials Science Group, Indira Gandhi Centre for Atomic Research, A CI of Homi Bhabha National Institute, Kalpakkam, 603 102, Tamil Nadu, India.

²Materials Chemistry & Metal Fuel Cycle Group, Indira Gandhi Centre for Atomic Research, A CI of Homi Bhabha National Institute, Kalpakkam, 603 102, Tamil Nadu, India.

³Accelerator Physics and Synchrotrons Utilization Division, Raja Ramanna Center for Advanced Technology, Indore 452013, India.

*Corresponding Author Email: sanju@igcar.gov.in

Low dimensional systems have gained recent attention in condensed matter physics due to their advanced physical properties which are not attainable in their three-dimensional counterparts. These materials exhibit unique quantum phenomena such as charge density wave (CDW) and superconductivity which are arising due to electron-phonon and electron-electron interactions. Recently, layered rare-earth tritelluride compounds (RTe₃) have gained much attention and are added to the list of existing low-dimensional systems. RTe₃ compounds

generally crystallize in orthorhombic structure with Cmc₂m space symmetry. The crystal structure of RTe₃ compounds consists of double layers of square-planar Te sheets, separated by corrugated R-Te slabs along the b axis [1]. The CDW phase in RTe₃ compounds are found to be suppressed with pressure [2]. The formation of CDW often results in a periodic lattice distortion, which in turn leads to structural modification in materials. In this study, we report high pressure structural stability of three RTe₃ compounds (CeTe₃, PrTe₃ and NdTe₃) investigated by in-situ X-ray diffraction technique. RTe₃ samples are synthesized by non-isothermal isopiestic method. Ambient Cmc₂m phase of RTe₃ compounds is preserved up to the highest pressure studied. Bulk modulus of CeTe₃, PrTe₃ and NdTe₃ are estimated to be 37(2) GPa, 38(5) GPa and 33(5) GPa respectively. RTe₃ compounds exhibited anisotropic compression along different crystallographic axes, with b axis being more compressible. This is understood in terms of weak bonding along b axis. The bond lengths of R-Te (in plane) and R-Te (out of plane) shows a monotonic decrease and a crossover is observed at ~ 4 GPa. This crossover may be due to the suppression of CDW phase in these compounds.

Keywords: High pressure; Rare-earth tritellurides; Structural stability; Bulk modulus

References:

1. E. D. DiMasi, M. C. Aronson, J. F. Mansfield, B. Foran, and S. Lee, Phys. Rev. B, 52, (1995), 14516.
2. Jan Kopaczek, Han Li, Kentaro Yumigeta, Renee Sailus, Muhammed Y. Sayyad, Seyed Tohid Moosavy, Robert Kudrawiec, and Sefaattin Tongay, J. Mater. Chem. C, 10, (2022), 11995-12000.

Data-driven prediction of atomic hydrogen adsorption energies in two-dimensional materials

G. Tejaswini¹, Anjana E Sudheer¹, Amrendra Kumar^{2,3}, M Vallinayagam^{4,5}, Mani Lokamani⁶, Attila Cangi^{6,7}, M. Zschornak^{4,5}, C. Kamal^{2,3}, D. Murali^{1, *}

1Department of Sciences, Indian Institute of Information Technology Design and Manufacturing Kurnool, Andhra Pradesh 518008, India

2Theory and Simulations Laboratory, Theoretical and Computational Physics Section, Raja Ramanna Centre for Advanced Technology, Indore 452013, India

3Homi Bhabha National Institute, Training School Complex, Mumbai 400094, India

4Institute of Experimental Physics, TU Bergakademie Freiberg, Leipziger Str. 23, 09596 Freiberg, Germany

5Technical Physics, University of Applied Sciences, Friedrich-List-Platz 1, 01069 Dresden, Germany und Wirtschaft, Friedrich-List-Platz 1, 01069 Dresden, Germany

6Helmholtz-Zentrum Dresden-Rossendorf, D-01328 Dresden, Germany

7Center for Advanced Systems Understanding (CASUS), D-02826 Görlitz, Germany

*Corresponding Author E-mail: dmurali@iiitk.ac.in

The growing use of artificial intelligence and machine learning in materials science is accelerating the discovery of efficient materials for diverse applications [1]. In this work, we develop a machine learning model to predict atomic hydrogen adsorption energies, which characterize the efficiency of the hydrogen evolution reaction in the overall water-splitting

process. Two-dimensional transition-metal and post-transition-metal dichalcogenides and halides have emerged as promising candidates for photocatalytic applications [2]. Herein, we design two-dimensional monolayers consisting of a post-transition-metal atom sandwiched between two chalcogen or halide layers, or between one chalcogen and one halide layer, which possess 1T structural symmetry. We consider both symmetric regular and asymmetric Janus layers in this study. The study involves performing large-scale Density Functional Theory-based electronic structure calculations to understand the energetics of the intermediate state, i.e., the H-atom adsorption step in the hydrogen evolution reaction. We analyze trends in H adsorption energy data across a wide range of monolayers. The positive and negative adsorption energies indicate the repulsive and attractive interaction of H with the material. H coverage concentration is accounted for by considering supercells of different sizes. The calculated adsorption energies across a diverse range of monolayers are useful parameters for photocatalytic hydrogen production and hydrogen storage applications. Further, we design and engineer features that influence adsorption energies and develop a supervised machine learning model that accurately predicts H adsorption energy in post-transition-metal-based monolayers. Finally, we assess the performance and robustness of the developed model by calculating various metrics, including the root mean squared error, coefficient of determination (R^2), and training and validation loss curves.

Keywords: Artificial intelligence; Machine learning; Overall Water Splitting; Hydrogen evolution reaction; Density Functional Theory;

References:

1. G Tejaswini, Anjana E Sudheer, Amrendra Kumar, Pavan Kumar Perepu, M Vallinayagam, C Kamal, S Mani Prakash, M Posselt, M Zschornak, D Murali, J. Mater. Chem. A, 13, (2025), 19896-19910.
2. Anjana E Sudheer, Golla Tejaswini, Matthias Posselt, D Murali, J. Mater. Chem. A, 13, (2025), 34873-34895.

ZnMn₂O₄/MXene Composite as a High-Performance Cathode Material for Hybrid Supercapacitors

Pulkit Garg, Ankit Tyagi*

Department of Chemical Engineering, Indian Institute of Technology

Jammu, Jammu, Jammu & Kashmir, 181221, India

*Corresponding Author. Email: ankit.tyagi@iitjammu.ac.in

In the growing demand for sustainable and high-performance energy storage systems, aqueous zinc-ion hybrid supercapacitors (ZHSCs) have gained significant attention owing to their inherent safety, environmental friendliness, and cost-effectiveness [1]. However, there is a lack of cathode materials that exhibit high specific capacity, high rate capability, and long life cycle stability. In this work, we have synthesized a novel metal-doped ZnMn₂O₄/MXene composite. Incorporation of metal ions into the ZnMn₂O₄ spinel structure tunes the electronic structure for enhanced electrical conductivity and faster diffusion kinetics through providing extra redox-active sites and oxygen vacancies [2].

MXenes, a class of 2D transition metal carbides and nitrides, have been considered to be promising substrates by virtue of their high conductivity, large surface area, and excellent hydrophilicity [3]. MXene can act not only as a fast electron transport pathway but also as a support to accommodate volume changes during repeated Zn²⁺ intercalation/deintercalation. Physical characterizations were done using XRD, FE-SEM, and XPS analyses. The results verify the successful incorporation of metal without disturbing the spinel phase, with a uniform

distribution of metal-doped ZnMn₂O₄ nanoparticles across the MXene sheets. Electrochemical performance was evaluated using cyclic voltammetry, galvanostatic charge-discharge, and electrochemical impedance spectroscopy. The composite cathode exhibited high specific capacitance, improved rate capability, and excellent cycling stability in a 2 M ZnSO₄ + 0.1 M MnSO₄ electrolyte. The synergistic effect between MXene and metal-doped ZnMn₂O₄ facilitates rapid ion diffusion and efficient electron transport, making it a promising cathode material for ZHSC.

This study introduces a different view on engineering composite materials for hybrid supercapacitors using multivalent metal doping and 2D conductive frameworks.

Keywords: MXene; hybrid supercapacitor; energy storage;

References:

1. Y. Wang, S. Sun, X. Wu, H. Liang, W. Zhang, Status and Opportunities of Zinc Ion Hybrid Capacitors: Focus on Carbon Materials, Current Collectors, and Separators, *Nano-Micro Lett.* 15 (2023) 78
2. Q. Cheng, K. Wang, Z. Yang, Y. Li, G. Zhang, Surface oxygen vacancies induced by copper doping in hexagonal ZnMn₂O₄ nanoplates for high efficiency photothermocatalytic oxidation of toluene, *Sep. Purif. Technol.* 354 (2025) 128743.
3. X. Li, Z. Huang, C.E. Shuck, G. Liang, Y. Gogotsi, C. Zhi, MXene chemistry, electrochemistry and energy storage applications, *Nat. Rev. Chem.* 6 (2022) .

β -AgVO₃@MXene Heterostructures with AI-Driven Electrochemical Informatics for High-Performance Asymmetric Supercapacitors

Shwetha K P^{1,2,†}, Nagendra Babu A P³, Sudha Kamath M K^{1,2,†}.

¹Centre for Nanomaterials and Devices, RV college of Engineering, Bengaluru 560056, India

²Department of Physics, RV College of Engineering, Bengaluru, 560059, India.

³Department of Physics, Jnana Bharathi Campus, Bangalore University, Bengaluru, 560056, India

[†]Corresponding authors: sudhakamath@rvce.edu.in , shwethaprabhakararao@gmail.com

The advancement of high-performance asymmetric supercapacitors (ASCs) requires a comprehensive understanding of interfacial charge-transfer kinetics and predictive performance modelling. This study presents a comprehensive experimental-computational framework for the hydrothermal synthesis of β -AgVO₃@MXene (AGV-MX) heterostructures. Structural and spectroscopic analyses confirm the presence of distinct heterointerfaces between monoclinic β -AgVO₃ and MXene nanosheets, leading to enhanced electronic modulation and crystallinity. According to Dunn's electrochemical analysis method, charge storage is shifting from diffusion-controlled to capacitive-dominant. The AGV-MX electrode has a specific capacitance of 827 Fg⁻¹ at 1 Ag⁻¹ and a retention rate of 91.04% over 10,000 cycles. To optimize the ASC architecture, a comparative analysis of negative electrode materials (MXene, rGO, and AC) was performed, with activated carbon (AC) emerging as the superior option for charge balance and potential window enhancement. The AGV-MX//AC asymmetric device has a stable 2.0 V operating window, an energy density of 68 Wh kg⁻¹ at 1058 W kg⁻¹, and 89.04% capacitance during long-term cycling. Beyond conventional testing, a novel data-driven analytical layer is introduced. By integrating Quantum Kernel Principal Component Analysis (QK-PCA) for nonlinear signature representation

and Ising-model-based segmentation for kinetic behaviour classification, we demonstrate the ability of Physics-Informed Machine Learning (PIML) to accurately forecast electrochemical responses and fluctuating interfacial resistances. Finally, the practical utility of the AGV-MX//AC system is validated through powering infrared-sensor highlighting a cohesive methodology for intelligent, high-performance energy storage technologies.

Keywords: β -AgVO₃@MXene; asymmetric supercapacitors; interfacial charge kinetics; electrochemical informatics; physics-informed modelling.

References:

1. B. R. C. de Menezes, R. G. Ribas, V. M. Schatkoski, T. L. do Amaral Montanheiro, C. Y. Koga-Ito, and G. P. Thim, SN Applied Sciences, 1 (2019) 1327.
2. M. Pudukudy, Q. Jia, H. Wang, S. Shan, and R. Rajendran, Materials Science in Semiconductor Processing, 107 (2020) 104824.

Manganese (Mn) doping in CoFeS₂ to Enhance the Supercapacitor Performance

Poonam Moar¹, Sourabh Barua^{1,*}

*Affiliation:*¹Department of Physics, Central University of Punjab, Ghudda, Bathinda, 151401

*Corresponding Author E-mail: (sourabh.barua@cup.edu.in)

Mn-doped CoFeS₂ have attracted attention as an electrode material for high-performance supercapacitors improved electrical conductivity and multiple redox-active sites. In this study, we reported Mn-doped CoFeS₂ nanostructures were prepared through a hydrothermal route. The introduction of Mn increased the number of electroactive sites, accelerated charge transfer, and improved ion transport¹²³. Structural characterisation such as X-ray diffraction (XRD), scanning electron microscopy (SEM), EDX confirmed the formation of porous Mn doped CoFeS₂ nanostructures. Electrochemical measurement Cyclic voltammetry, galvanostatic charging discharging (GCD), we calculated the specific capacitance, energy density, power density. Electrochemical results showed a specific capacitance of 429.61 F g⁻¹ at 0.5 A g⁻¹, an energy density of 59.28 Wh/kg, and a maximum power density of 10645.71 W/kg. Mn doped CoFeS₂ delivered high energy and power density as compared to CoFeS₂. These findings indicate that Mn doping is an effective strategy to enhance the electrochemical performance of CoFeS₂ for next-generation energy storage devices.

Keywords: 2D materials; Supercapacitor; Cyclic Voltammetry, Energy Storage

References:

1. Song K, Chen X, Yang R, et al. Chem Eng J. 2020;390(January):124175
2. Wang X, Wang Z, Cao Y, et al. Journal of Solid State Chemistry 2024;331(January).
3. Wang M, Wang J, Wu G, et al. Sep Purif Technol. 2025;363(P1):131972.

Li-Functionalized 2D SiN Monolayer for Hydrogen Storage Application: A DFT Study

Trilokchand L. Kumavat^{1,*}, Jai Kishan Sahu¹, Yogesh Sonvane¹

¹*Advanced Materials Lab, Department of Physics, Sardar Vallabhbhai National Institute of Technology, Surat 395007, India*

*Corresponding Author E-mail: kumavattrilok95@gmail.com

Herein, we studied the pristine and Li-functionalized 2D SiN monolayer for hydrogen storage applications using density functional theory (DFT) calculations. The SiN monolayer is an indirect band gap semiconductor with a band gap of 1.72 eV using PBE functional. The pristine SiN monolayer shows a very weak interaction with the H₂ molecule leading to a lower binding energy and charge transfer. However, the Li-functionalization significantly enhances the binding strength of H₂ molecule with the SiN nanosheet. The Li-functionalized SiN nanosheet adsorbs a maximum of 24 H₂ molecules with an average adsorption energy of -0.11 eV/H₂ and a maximum gravimetric storage capacity of 5.9 wt%. The hydrogen desorption temperature is found to be 116 K, which indicates that H₂ can be released from the nanosheet efficiently at low temperature and pressure. This work provides useful insights for designing the 2D materials towards an improved hydrogen storage mechanism.

Keywords: SiN monolayer; Hydrogen Storage; Density functional theory.

References:

1. X. L. Liu, S. P. Shi, L. L. Tang, K. Diao, J. B. Hu, and Y. L. Liu, *Fuel*, 407, (2026), 137264.
2. P. Kerdsoombut, P. Latthiwan, N. Thatsami, P. Reunchan, A. Boonchun, T. Hussain, S. Pinitsoontorn, and T. Kaewmaraya, *Journal of energy storage*, 143, (2026), 119608.
3. J. Wang, D. Chen, S. Xiao, X. Zhang, and B. Xiao, *Journal of energy storage*, 145, (2026), 119942.

Exploring the possible connection between elastic properties and ionic /electronic transports in solid polymer electrolyte.

Krushna Shewale¹, Amruta Killedar¹, Vaibhav Kathvate², Swati Jadhav¹ and Satishchandra Ogale^{*1,3}

¹ *Center for Energy Science, Indian Institute of Science Education and Research, Pune-411008, India*

² *Dr. Vishwanath Karad MIT World Peace University (MIT-WPU), Pune-411038, India*

³ *Research Institute for Sustainable Energy (TCG-CREST) Kolkata-700091, India*

*Corresponding Author E-mail: satishogale@gmail.com

Solid polymer electrolytes (SPEs) have attracted significant interest in solid-state battery technology due to their enhanced safety and superior mechanical stability compared to conventional liquid electrolytes. However, their practical implementation is hindered by relatively

low ionic conductivity and capacity fading during prolonged cycling. In this study, we systematically investigate the influence of salt concentration, mechanical stress, and inorganic filler incorporation on the ionic and electronic conductivity of SPEs. Polyacrylonitrile (PAN) based electrolytes were synthesized using dimethylformamide (DMF) as a solvent, with varying concentrations of lithium perchlorate (LiClO_4) relative to polymer weight. Additionally, SiO_2 nanoparticles were introduced as a filler to enhance the mechanical strength and modulate ion transport behavior. The pressure-dependent ionic conductivity of SPEs, both with and without filler, was evaluated over a range of applied pressures up to 3 MPa. The results reveal a strong dependence of ionic conductivity on salt concentration, governed by ion dissociation, while filler incorporation enhances mechanical robustness and influences ion transport pathways. Furthermore, applied pressure significantly impacts conductivity, highlighting the role of free volume and structural rearrangement in ion conduction. A remarkable hysteresis of the properties is also noted and studied. These results establish a direct correlation between composition, mechanical properties and ionic transport in SPE systems.

Keywords: Solid polymer electrolyte, LiClO_4 , PAN, SiO_2

References:

1. S. K. Fullerton-Shirey and J. K. Maranas, *Journal of Physical Chemistry C*, 114, (2010), 9196–9206.
2. Y. W. Chen-Yang, Y. T. Chen, H. C. Chen, W. T. Lin, and C. H. Tsai, *Polymer*, 50, (2009), 2856–2862.
3. W. Jia, Z. Li, Z. Wu, L. Wang, B. Wu, Y. Wang, Y. Cao, and J. Li, *Solid State Ionics*, 315, (2018), 7–13.

MoS₂@Ti₃C₂@CNT Ternary Hetero-nanostructured functional Anode for Na-Ion Batteries

Trupti Aher^{1,2}, Ajay Yadav² and Ashish Yengantiwar^{1*}

¹*Department of Physics, Fergusson College (Autonomous), Pune affiliated to Savitribai Phule Pune University, 411004, Maharashtra, India*

²*Modern College of Arts, Science and Commerce, Shivajinagar, Pune 411005, India*

*Corresponding Author Email: ashish.yengantiwar@fergusson.edu

A MoS₂@Ti₃C₂@CNT ternary hetero-nanostructure was synthesized via a hydrothermal method through in-situ growth of MoS₂ on Ti₃C₂ MXene in the presence of carbon nanotubes (CNTs), enabling effective integration of layered and conductive components. The Ti₃C₂ MXene was derived from the Ti₃AlC₂ MAX phases via selective etching of Al layers using a chemical etching process, followed by delamination to obtain few layer Ti₃C₂ nanosheets. The composite was characterized to verify the formation of the hetero-nanostructure and the presence of constituent phases. Electrochemical performance was evaluated using cyclic voltammetry (CV) and galvanostatic charge–discharge measurements in the voltage range of 0.01–3.0 V vs. Na⁺/Na. The CV profile exhibits a distinct cathodic peak in the first cycle attributed to Na⁺ intercalation and solid electrolyte interphase (SEI) formation. From the 2nd to 5th cycles, the CV curves show good overlap, indicating stable and reversible sodium storage behavior. Charge–discharge

measurements display an initial irreversible capacity loss in the first cycle, followed by stable voltage profiles in subsequent cycles. The final cycles demonstrate consistent reversible capacity with reduced polarization. The electrode delivers a reversible specific capacity of $\sim 510 \text{ mAh g}^{-1}$ at a current density of 0.2 A g^{-1} over 100 cycles. The improved performance is associated with enhanced charge transport and synergistic interaction among the components within the hetero-nanostructure.

Keywords: Sodium-ion batteries; $\text{MoS}_2@\text{Ti}_3\text{C}_2@\text{CNT}$; Hydrothermal synthesis; MXene; Anode materials.

References:

1. C. Wu, X. Wang, Y. Zhang, and J. Liu, *Journal of Materials Science & Technology*, 100, (2022), 101–109.
2. B. Wen, H. Li, X. Zhao, and Y. Chen, *Energy & Fuels*, 36, (2022), 11290–11299.
3. D. Er, J. Li, M. Naguib, Y. Gogotsi, and V. B. Shenoy, *ACS Applied Materials & Interfaces*, 6, (2014), 11173–11179.

Temperature-Engineered Spray Pyrolyzed NiMn_2O_4 Thin Films on Stainless Steel for High-Performance MXene-Based Hybrid Asymmetric Supercapacitors

Sachin B. Tanwade^{1,2} Ramesh J. Deokate^{1*}

¹*Vidya Pratishthan's Arts, Science and Commerce College, Baramati 413 133, (MS), India.*

²*Army Institute of Technology, Pune 411 015, (MS), India.*

*Corresponding author: rjdeokate@gmail.com

The development of low-cost, scalable, and high-performance electrode materials is crucial for next-generation supercapacitor devices. Spinel NiMn_2O_4 is a promising pseudocapacitive material; however, its electrochemical performance strongly depends on microstructural tuning and electronic conductivity. In this work, NiMn_2O_4 thin films were synthesized on stainless steel substrates via spray pyrolysis at 350, 375, 400, and 425 °C to optimize their electrochemical performance. X-ray diffraction (XRD) confirmed the spinel NiMn_2O_4 phase, while X-ray Photoelectron Spectroscopy (XPS) revealed mixed Ni and Mn oxidation states responsible for redox activity. Scanning Electron Microscopy (SEM) analysis showed temperature-dependent morphological variations influencing porosity and electrolyte accessibility. Electrochemical studies in 2 M KOH (three-electrode system with Hg/HgO reference) demonstrated temperature-dependent pseudocapacitive behavior, with variations in specific capacitance, rate capability, and charge-transfer resistance. An asymmetric supercapacitor was assembled using NiMn_2O_4 as the positive electrode and MXene as the negative electrode, both on SS substrates. The device exhibited enhanced electrochemical performance and good stability due to the synergistic combination of redox-active NiMn_2O_4 and highly conductive MXene. This work highlights a simple and scalable approach for engineering spinel oxide thin films for hybrid supercapacitor applications.

Keywords: NiMn₂O₄ thin films, Spray pyrolysis, MXene electrodes, Asymmetric supercapacitor, Pseudocapacitive energy storage.

Palladium-Gold/MXene Nanocomposite as an Efficient Electrocatalyst for Hydrogen Evolution Reaction

Mallamma Jinagi¹, Manav Saxena²

¹Centre for Nano and Material Sciences, Jain (Deemed-to-be University)

²Jain Global Campus, Kanakapura, Ramanagara, Bangalore, 562112, India.

*Corresponding author E-mail: s.manav@jainuniversity.ac.in

The ever-increasing demand for energy accompanied by global warming and environmental deterioration has triggered the way towards the utilization of hydrogen, referred to as the “fuel of the future”. [1] [2] Recently, Pd (Palladium) was investigated as an alternative to Pt (Platinum) due to its high abundance and enhanced catalytic activity. However, as compared to Pt, Pd displayed less electrocatalytic performance towards the hydrogen evolution reaction (HER). Thus, modifying the surface properties, coupled with metal alloying, is necessary to improve the electrocatalytic properties of Pd NPs.

On the other hand, efficient catalysts with minimal content of catalytically active noble metals are essential for transitioning to a clean hydrogen economy. Catalyst supports that can immobilize and stabilize catalytic nanoparticles and facilitate the supply of electrons and reactants to the catalysts are therefore needed. Being hydrophilic and more conductive compared with carbons, MXenes have shown promise as catalyst supports. [3] In this context, attention may be shifted toward the use of alloyed NPs of Pd on Ti₃C₂T_x support, which could bring changes in the electronic and geometrical properties of pure Pd. [1] Herein, we report a highly efficient Palladium-Gold/MXene electrocatalyst with an overpotential of 41 mV at a current density of -10 mA cm⁻², a Tafel slope of 49 mV dec⁻¹, and a stability of 57 hours for hydrogen evolution reaction in acidic media. The catalyst displays the Volmer-Heyrovsky mechanism. The underlying principles of optimization of hydrogen binding energy, kinetics of atomic redistribution, high atom economy, and the metal deposition process on MXene nanosheets are also revealed in this work.

Keywords: Hydrogen evolution reaction; MXenes; Interfacial electronic coupling; Transition Metal-Based MXene nanocomposite.

References:

1. Darabdhara, G., Amin, M. A., Mersal, G. A., Ahmed, E. M., Das, M. R., Zakaria, M. B., & Boukherroub, R. (2015). Journal of Materials Chemistry A, 3(40), 20254-20266.
 2. Kwak, K., Choi, W., Tang, Q., Kim, M., Lee, Y., Jiang, D. E., & Lee, D. (2017). Nature communications, 8(1), 14723.
 3. Sun, Y., Lee, J., Kwon, N. H., Lim, J., Jin, X., Gogotsi, Y., & Hwang, S. J. (2024). ACS nano, 18(8), 6243-6255.
-

Flexible Binder-Free V-Doped FeOOH on Ni Foam as an Efficient Bifunctional Electrocatalyst for Alkaline Water Splitting

Haripal Singh Dhayal, Nishad G. Deshpande*

*Department of Physical and Applied Sciences, Indian Institute of Information Technology Surat,
Kholvad campus, Surat-394190, Gujarat, India*

*E-mail address: nishad.deshpande@iiitsurat.ac.in

The development of efficient and low-cost electrocatalysts for water splitting is essential for sustainable hydrogen production. Herein, we report a facile hydrothermal method for fabricating vanadium-doped iron oxyhydroxide (V:FeOOH) on flexible nickel foam (Ni foam) as a binder-free bifunctional electrocatalyst for alkaline water-splitting. Vanadium incorporation effectively modulates the electronic structure of FeOOH, enhances electrical conductivity, and improves the adsorption/desorption behavior of reaction intermediates, leading to superior electrocatalytic activity.

The porous three-dimensional Ni foam acts as a conductive and flexible scaffold, providing enhanced surface area, rapid electron transport, and efficient electrolyte diffusion. Electrochemical studies reveal that pristine FeOOH and V:FeOOH electrodes exhibit overpotentials of 280 and 190 mV, respectively, at 10 mA cm^{-2} , confirming the significant enhancement after vanadium doping. The catalyst also demonstrates favorable reaction kinetics and excellent durability in alkaline media.

Furthermore, the bifunctional V:FeOOH/Ni foam electrode achieves efficient overall water splitting with a low cell voltage of 1.54 V at 10 mA cm^{-2} along with long-term operational stability. This study demonstrates a simple and scalable strategy for designing flexible transition-metal oxyhydroxide electrocatalysts for high-performance hydrogen generation applications.

Keywords: Water Splitting; Electrocatalyst; Transition metal oxyhydroxide; Doping; Bifunctional activity.

Scalable Fabrication of Binder-Free Cobalt Boride Phosphide Electrocatalyst for Green Hydrogen Production

Susmita S. Patil^a, Reshma V. Khandekar^a, Jyotiprakash B. Yadav^{a*}

*^aEnergy Research and Thin Film Laboratory, USIC, Shivaji University, Kolhapur 416004,
Maharashtra, India*

*Corresponding Author: jbyadavphy@gmail.com (Dr J. B. Yadav)

Water has emerged as a promising green hydrogen source, with significant attention for its potential applications in the energy sector. However, developing efficient electrocatalysts for hydrogen production remains challenging, particularly with transition metal-based compounds. In the present work, Cobalt-Boride-Phosphide (Co-B-P) electrocatalysts were successfully synthesized using a simple, scalable, and cost-effective Successive Ionic Layer Adsorption and Reaction (SILAR) technique by systematically varying the phosphorus-to-boron (P/B) precursor molar ratios (0.25, 0.66, 1.5, and 4). The effect of P/B precursor molar ratio on the structural,

morphological, and compositional analysis of the synthesized electrocatalysts was comprehensively investigated, which confirmed the formation of a Co-B-P/Co₃O₄ heterointerface with a porous nanosphere morphology at a P/B ratio of 1.5. Electrochemical analysis confirmed that CoBP3 electrode (P/B = 1.5) exhibits superior HER performance with a low overpotential of 100 mV at 10 mA cm⁻² current density. Furthermore, the catalyst achieved a hydrogen production rate of 1125 mL h⁻¹ in a scale-up prototype water electrolyzer, indicating its practical applicability for large-scale hydrogen generation. The enhanced catalytic activity is attributed to the synergistic effect of the Co-B-P/Co₃O₄ heterointerface, which facilitates rapid charge transfer and promotes the hydrogen spillover phenomena, thereby improving the intrinsic catalytic activity.

Keywords: cobalt boride phosphide; SILAR; Hydrogen spillover; electrocatalysis;

Engineering the MXene-BiCuS Interface for Electrocatalytic CO₂ Reduction to Value-Added Products

Chob Singh¹, Arvind H. Jadhav^{1*}

¹*Centre for Nano and Material Sciences (CNMS), JAIN University,
Jain Global Campus, Bangalore 562112, Karnataka, India*

* j.arvind@jainuniversity.ac.in

Electrochemical reduction of carbon dioxide to value-added products is a promising strategy for achieving a sustainable carbon cycle [1, 2]. In this study, a series of Bi-Cu bimetallic sulfide loaded on Ti₃C₂ MXene (BCS@MXC) composites catalysts were synthesized via a facile self-impregnation deposition approach. The as-synthesized catalysts were characterized via various analytical and spectroscopic techniques and followed for eCO₂RR. The 2-BCS@MXC catalyst exhibited excellent electrochemical activity with a high current density of -34.91 mA/cm² and a lower potential of -0.59 V vs. RHE in 0.1 M KHCO₃ electrolyte for eCO₂RR. The Electrocatalytic measurements were carried out in H-type electrolysis cell with the catalyst loading of 0.64 mg/cm². The 2-BCS@MXC catalyst exhibited the excellent selectivity for HCOOH with a Faradaic efficiency of 85% at an optimal potential -0.59 V vs. RHE. The open flakes of MXene and intercalation with BiCuS on MXene provided better CO₂ adsorption for the eCO₂RR activity. The 2-BCS@MXC showed a low charge transfer resistance (R_{ct}) of 16.37 Ω at an applied potential of -0.59 V vs. RHE and a higher Cdl value of 5.27 mF cm⁻², exhibiting the better reaction kinetics and higher electrochemically active surface area. The 2-BCS@MXC could operate upto 12 h without loss in current density, showing better stability of the catalyst. Sulfur incorporation into the BCS@MXC lattice played a pivotal role in promoting H* formation, stabilizing the HCOO* intermediate for the formation of HCOOH, and enhancing catalytic activity. This study underscores the potential of structural engineering in MXene-based catalysts and highlights BCS@MXC as a promising material for efficient CO₂RR to value-added products.

Keywords: Ti₃C₂ MXene; bimetallic effect; BiCuS; eCO₂RR; value-added products

References:

1. Li, L., A. Ozden, S. Guo, F.P. García de Arquer, C. Wang, M. Zhang, J. Zhang, H. Jiang, W. Wang, and H. Dong, Nature Communications, 12 (2021) 5223.
 2. Singh, C., H. Manjunath, S.C. Alla, N.K. Chaudhari, A.K. Samal, and A. H. Jadhav, ACS Applied Nano Materials, 8, (2025) 18662-18679.
-

Phase Engineered 1T-MoS₂ Electrodes for High-Performance Aqueous Supercapacitors

Shama Parveen¹, Rishav Sharma¹, Apurva Karahe¹, Manopriya Samatham¹, Harshada Jadhav¹, Rupesh S. Devan¹,

¹*Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology*

Indore, Simrol, Khandwa Road, Indore, 453552, India.

*Corresponding Author E-mail: rupesh@iiti.ac.in

The pursuit of green energy storage requires electrode materials with superior conductivity and rapid charge kinetics. Metallic 1T-phase MoS₂ outperforms semiconducting 2H MoS₂ due to its high conductivity, abundant basal-plane active sites, and expanded interlayers in multidirectional 2D nanosheets. These properties accelerate charge transfer and ion diffusion kinetics. Here, we synthesized phase 1T MoS₂ via a facile hydrothermal method and evaluated its supercapacitor performance in 1 M (NH₄)₂SO₄ aqueous electrolyte (potential window: -0.6 to 0.3). The electrode delivered specific capacitance of 589 mF cm⁻² at 3 mA cm⁻². This improved performance resulted from efficient NH₄⁺ intercalation enabled by its small hydrated radius (3.3 Å) and higher mobility within the expanded 1T interlayers, coupled with reversible proton-assisted pseudocapacitance, as confirmed by XRD and CV analysis. The neutral pH of 1 M (NH₄)₂SO₄ further mitigates Faradaic side reactions. These results reveal critical electrolyte-electrode synergies in TMDs and establish phase engineering of MoS₂ as a scalable pathway for advanced, high-performance supercapacitors.

Keywords: 1 T MoS₂, Supercapacitor, Electrochemical measurement, nanosheets

References:

1. J. Dai, Y. Chunying, Y. Xu, X. Wang, S. Yang, D. Li, L. Luo, L. Xia, J. Li, X. Qi Adv. Mater., 39 (2023) 2303732.
2. A. Muharrem, D. Voiry, M. Chhowalla, , Nat. Nanotechnol., 4 (2015), 313-318.

2D MnTiO₃ Nanodiscs For Novel Ammonium Ion Solar Supercapacitor

Manopriya Samtham,^a Ekta Choudhary,^a Shama Parveen,^a Ankit Yadav,^a Ram J. Choudhary,^{b*} Rupesh S. Devan^{a,c*}

^a *Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.*

^b *UGC-DAE Consortium for Scientific Research, Khandwa Road, Indore, 452001, India.*

^c *Centre for Electric Vehicle & Intelligent Transport System, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.*

*Corresponding Author E-mail: rupesh@iiti.ac.in, ram@csr.res.in

Defect engineering acts as an efficient method to modulate the microstructure and electronic structure of electrode for rechargeable photo-assisted supercapacitor serving as a promising means to improve the efficiency of photogenerated carrier separation. In this study, we have synthesized defects induced 2D MnTiO₃ (MTO) nanodiscs via hydrothermal method at different temperatures (200, 225, 250, 275 °C) to enhance the intrinsic conductivity and ion migration pathways. XRD and TEM studies revealed that pristine MTO have ordered structure

and good crystallinity, whereas MTO-225 sample exhibits periodic point defects, and in MTO-275 sample, phase segregation started with MnO_2 formation at the nanodiscs edges. Defects (metal or oxygen vacancies) promoted the generation and effective separation of photogenerated electron-hole pairs under light which facilitates ammonium (NH_4^+) ion on the surface. As a result, MTO-225 has shown better electrochemical performance i.e. 760 mF/cm^2 @ 2 mA/cm^2 current density under illumination, 10% higher than that of dark. The fabricated MTO-225|| $\text{Ti}_3\text{C}_2\text{-F}$ MXene asymmetric device demonstrated its promising practical potential. This work establishes a novel pathway for optimizing photoactive electrode materials by leveraging the interplay between light-responsive charge carrier engineering and adaptive structural reconstruction, offering critical insights for next-generation high-performance supercapacitors.

Keywords: MnTiO_3 ; defects; ammonium ion; solar supercapacitor

References:

1. M. Samtham, A. Miglani, A. Patil, V. Dharavath, S. Bimli, H. Srivastava, R. Jangir, Y. R. Ma, Ram J. Choudhary, and R.S. Devan, *J. Materials Chemistry A*, 14, (2026), 12699.
2. M. Samtham, A. Patil, E. Choudhary, H. Srivastava, R. Jangir, Ram J. Choudhary,* and R.S. Devan, *Small Methods*, 9,(2025), 2401441.
3. H.S. Jadhav, E. Choudhary, M. Samtham, A. Patil, S. Yadav, S.R. Mulani, R. Urkude, R. Jangir, P.A Shaikh, R.S. Devan, *J. Materials Chemistry A*, 14, (2026), 4024-4035.

Growth of 2D materials by chemical vapour deposition for supercapacitor and electrocatalytic applications

Hadin Rahman V P¹, Poonam Moar¹, Pooja, Chaithra M¹, Sourabh Barua^{1,*}

Affil¹Department of Physics, Central University of Punjab, Ghudda, Bathinda 151401.

*Corresponding Author E-mail: (sourabh.barua@cup.edu.in)

Chemical vapour deposition (CVD) is a proven technique of growing nanomaterials of 2D materials like Bi_2Se_3 , CoS_2 , NbSe_3 , and others. For example, niobium selenides, particularly NbSe_2 and Nb_2Se_3 , have attracted significant attention due to their low dimensional electronic behavior. Nb_2Se_3 is a quasi-one-dimensional system exhibiting anisotropic transport properties and unconventional electronic phases, including non-Fermi liquid behavior and complex electronic transitions reported in recent studies [1]. However, controlled synthesis of Nb_2Se_3 using scalable techniques such as chemical vapour deposition (CVD) remains largely unexplored. Many of these materials find application in renewable and carbon friendly energy solutions such as supercapacitors and electrocatalysis. In this work we will describe our attempts at growing novel 2D materials using CVD for their application in electrode materials for supercapacitors and water electrocatalysis. The structural, morphological, and compositional properties of the synthesised materials are systematically characterised using techniques such as X-ray diffraction (XRD), scanning electron microscopy (SEM), and energy-dispersive X-ray spectroscopy (EDX). The electrochemical performance of these materials is evaluated for supercapacitor applications through cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS).

Keywords: 2D materials; CVD; Supercapacitor; Electrocatalyst.

References:

1. Hu, R., Lauritch-Kullas, K., O'Brian, J., Mitrovic, V. F., & Petrovic, C. (2007). Anisotropy of electrical transport and superconductivity in metal chains of Nb_2Se_3 . *Physical Review B*, 75(6).

2. Jaber, S. Y. S., Ghaffarinejad, A., Khajehsaeidi, Z., & Sadeghi, A. (2023). The synthesis, properties, and potential applications of CoS₂ as a transition metal dichalcogenide (TMD). *International Journal of Hydrogen Energy*, 48(42), 15831–15878.
3. Li, S., Wu, J., Yao, J., & Yang, Y. (2025). NbS₂ Nanoflakes grown on carbon fiber paper for efficient and stable hydrogen evolution reaction. *Chemical Physics Letters*, 876, 142261.

2D NiMo LDH-MWCNT Heterostructure as a High-Energy-Storage Electrode for Supercapacitors

Rajesh Katru¹, K. Uday Kumar^{1*}

¹*Energy Materials and Devices (EMD) Laboratory, Department of Physics, National Institute of Technology, Warangal 506004, Telangana, INDIA.*

*Corresponding Author E-mail: kanapuram.udaykumar@nitw.ac.in

This study presents a conductive matrix design to overcome the intrinsic limitations of nickel molybdenum (NiMo) layered double hydroxide (LDH). The pristine NiMo-LDH and its composites with multi-walled carbon nanotubes (MWCNTs) at varying concentrations of (5, 10, 15 mg) were synthesized via a solvothermal method. The phase, morphology, microstructure, and porosity were characterized using XRD, SEM, TEM, XPS, and BET techniques. For the electrochemical characterization in a three-electrode configuration (3M KOH) vs. Ag/AgCl as reference electrode and platinum spring as the counter electrode, the optimal 10 mg MWCNT in NiMo-LDH (L+C (10mg)) yielded a specific capacitance of 1128.8 Fg⁻¹ at 1 Ag⁻¹ (GCD) with distinct redox peaks in CV and retained a capacitance of 86% after 10000 cycles. The fabricated asymmetric supercapacitor (ASC) with rGO anode delivered 151 F/g at 1.5 Ag, and the energy density (E) and Power density (P) were measured as 21 Wh/kg and 750 W/kg, respectively. The cyclic stability for 8000 cycles was maintained to be 78% at 2 A/g current density, with a coulombic efficiency of nearly 100% even after 8000 cycles. Practical demonstrations included powering blue/red LEDs and a DC toy motor for several minutes. Therefore, collectively, these results highlight the composites as promising electrode materials for durable, high-energy-density storage in portable devices.

Keywords: NiMo LDH, MWCNT, sustainable energy storage, Supercapacitors, ASC device

References:

1. Katru, R. et al. VS 4 –NiAl LDH composite electrodes for next-generation high-performance supercapacitors. *Journal of Materials Chemistry C* 13, 16024–16039 (2025).
2. Khanapuram, U. K. et al. Harvesting energy from friction: the revolutionary decade of triboelectric nanogenerators. *Advanced Powder Materials* 5, 100373 (2026).
3. Liang, R. et al. A high mass loading flexible electrode with a sheet-like Mn₃O₄/NiMoO₄@NiCo LDH on a carbon cloth for supercapacitors. *RSC Advances* 13, 33463–33470 (2023).

Scalable Ti₂CT_x MXene for High-Performance HER in Acidic and Saline Media

S. Yadav,^a A. Patil,^a R. Sharma,^a A. Karahe,^a R. Jangir,^{b,c,*} R. S. Devan^{a,d,e,*}

^aDepartment of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India

^bAccelerator Physics and Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Indore 452013, India

^cHomi Bhabha National Institute, Anushakti Nagar, Mumbai, 400094, India

^dCentre for Electric Vehicle & Intelligent Transport System, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India

^eMultidisciplinary Academic Research Centre, National Dong Hwa University, Hualien 974301, Taiwan

*Corresponding author: email: rjangir.cat@gmail.com & rupesh@iiti.ac.in

MXenes have come out to be versatile class of 2D electrocatalyst for hydrogen production yet targeting high activity and stability in diverse (acidic & saline) electrolytes and at industrial current densities remain a big bottleneck. Herein, we present Ti₂C the thinnest and lightest MXene, possess a combination of structural and electronic attributes often not observed simultaneously with other higher order MXene. Its minimal layer framework provides maximum accessible basal plane active Ti sites, highest surface to bulk atom ratio, ideal H adsorption energetics enabling rapid proton activation and efficient charge transfer. The intact Ti-C bonding along with surface functionalities impart further resistance to proton and chloride induced degradation, allowing multilayer Ti₂CT_x MXene to deliver an HER overpotential of 95mV in acidic medium and 567 mV in saline medium (3.5wt% NaCl) at 10 mA/cm² with long term cyclic stability of 120h and 70 h respectively. For broad scale applicability, Ti₂CT_x was integrated onto a three-dimensional stainless-steel mesh, achieving a current density of 50mA/cm² with an overpotential of 250 mV in acidic medium. These findings position Ti₂CT_x as a robust, chloride-tolerant, and industrially relevant MXene catalyst for efficient hydrogen production in both acidic and saline electrolytes.

Keywords: Ti₂CT_x MXene, electrocatalysis, substrate adaptability, medium versatility, saline, HER, OER, cyclic stability

References:

1. Zehtab Salmasi, M., et al., Energy Reviews, 2024. 3(3): p. 100070.
2. Faridi, S., et al., Journal of Materials Chemistry A, 2025. 13(22): p. 16481-16490.
3. Lim, K.R.G., et al., Nature Synthesis, 2022. 1(8): p. 601-614.

Screening Hydrogen Adsorption in p-Block-Doped α -CP Monolayers for HER Applications: A First-Principles Study

Shardul Vadalkar^{1*}, Heli Mistry², Zarna D. Ponkiya², Narayan N. Som³, Prafulla K. Jha²

Affiliation: ¹Department of Applied Science & Humanities, Sardar Vallabhbhai Patel Institute of Technology, Vasad, Gujarat – 388306, India

²Department of Physics, Faculty of Science, The Maharaja Sayajirao University of Baroda, Vadodara, Gujarat – 390002, India

³Centre for Advanced Materials and Technologies (CEZAMAT), Warsaw University of Technology, 19 Poleczki St, Warsaw 02-822, Poland

*Corresponding Author E-mail: shardulvadalkar@gmail.com

Efficient hydrogen adsorption is a key descriptor in the screening of catalysts for sustainable hydrogen production via the hydrogen evolution reaction (HER). Among emerging two-dimensional (2D) materials, α -CP has attracted theoretical attention due to its distinctive

electronic properties. Although not yet experimentally realized, it was proposed by Özdamar *et al.* in 2018 as a member of the stable A_2B_2 -type monolayer family, motivating further exploration of its potential applications^[1]. In this work, we investigate the hydrogen adsorption behavior of p-block-element-doped α -CP monolayers, including B, N, Al, Si, and S dopants, using first-principles Density Functional Theory (DFT) calculations as implemented in the Quantum ESPRESSO package^[2]. According to the HER model proposed by Jens K. Nørskov and guided by the Sabatier principle, an efficient HER catalyst should exhibit hydrogen adsorption that is neither too strong nor too weak^[3]. The calculated hydrogen adsorption energy for B-doped α -CP is -0.91 eV, indicating excessively strong hydrogen binding, while the remaining doped systems exhibit endothermic adsorption behavior. Among them, S-doped α -CP shows the most favorable adsorption characteristics, with a hydrogen adsorption energy of 0.23 eV, suggesting a relatively balanced adsorption–desorption profile despite its mildly endothermic nature. These findings indicate that S-doped α -CP is a promising candidate in the screening of HER catalysts, providing useful insights for the design of efficient 2D catalytic materials.

Keywords: α -CP Monolayer; Density Functional Theory; p-Block Doping; Hydrogen Evolution Reaction.

References:

1. B. Özdamar, G. Özbal, M. N. Çınar, K. Sevim, G. Kurt, B. Kaya, and H. Sevinçli, *Physical Review B*, 98, (2018), 045431.
2. P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, A. Dal Corso, S. Fabris, G. Fratesi, S. de Gironcoli, R. Gebauer, U. Gerstmann, C. Gougoussis, A. Kokalj, M. Lazzeri, L. Martin-Samos, N. Marzari, F. Mauri, R. Mazzarello, S. Paolini, A. Pasquarello, L. Paulatto, C. Sbraccia, S. Scandolo, G. Sclauzero, A. P. Seitsonen, A. Smogunov, P. Umari, and R. M. Wentzcovitch, *Journal of Physics: Condensed Matter*, 21, (2009), 395502.
3. J. K. Nørskov, T. Bligaard, A. Logadottir, J. R. Kitchin, J. G. Chen, S. Pandalov, and U. Stimming, *Journal of The Electrochemical Society*, 152, (2005), J23–J26.

Atomic-Scale Interfacial Design of $Zn_2SnO_4/2D$ g- C_3N_4 Heterojunction for Enhanced Pseudocapacitive Energy Storage

Haribhau Gholap^{1*}, Krushna Zoting¹, Chaitanya Khade¹, Om G. Jadhav², Amol Tambe³, Balaji Ghule⁴

¹*Department of Physics, Fergusson Collage (Autonomous), Savitribai Phule Pune university, Pune-411004, Maharashtra, INDIA*

²*High-Performance Computing (HPC) Technologies Group, Center for Development of Advance Computing(C-DAC), Panchawati, Pune - 411008, Maharashtra, INDIA.*

³*Department of Chemistry, Savitribai Phule Pune university, Pune-411007, Maharashtra, INDIA*

⁴*School of Chemistry, University College Cork, Cork-T12YT20, Republic of Ireland.*

*Corresponding email - haribhau.gholap@fergusson.edu

Atomic-scale interfacial engineering is an effective strategy to activate pseudocapacitive charge storage in hybrid electrode materials. Herein, we report the atomic-scale interfacial design

of a $\text{Zn}_2\text{SnO}_4/2\text{D g-C}_3\text{N}_4$ heterostructure, where controlled Sn, Zn, and O anchoring at carbon and nitrogen sites tailors the interfacial electronic structure and enables enhanced pseudocapacitive energy storage. Density functional theory calculations reveal that metal-oxygen coordination induces crystal-field splitting and strong hybridization between metal d states and O/N p states, generating redox-active states near the Fermi level. Charge density difference and electrostatic potential analyses demonstrate pronounced oxygen-centered charge accumulation and interfacial dipole formation, resulting in a significant enhancement of quantum capacitance near E_{F} . Electrochemically, the optimized $\text{Zn}_2\text{SnO}_4/2\text{D g-C}_3\text{N}_4$ electrode exhibits clear faradaic features with surface-controlled kinetics, delivering a high areal capacitance of $10\text{--}11 \mu\text{F cm}^{-2}$ at room temperature and maintaining stable pseudocapacitive behavior over a wide potential window. The electrode shows excellent rate capability with minimal capacitance loss at high scan rates and outstanding cycling stability, retaining 90–95% of its initial capacitance after 5000 cycles. A pouch-type asymmetric device based on $\text{Zn}_2\text{SnO}_4/2\text{D g-C}_3\text{N}_4$ achieves an energy density of $\sim 10 \text{ Wh kg}^{-1}$ at a power density of $\sim 900\text{--}1000 \text{ W kg}^{-1}$, along with near-unity coulombic efficiency. This work establishes atomic-scale interfacial design as a viable pathway to engineer oxygen-driven pseudo-capacitance in 2D heterostructure energy storage electrodes.

Keywords—Atomic-scale interfacial; ZTO/2D g-C₃N₄ heterostructures; pseudocapacitive energy storage, quantum capacitance; density functional theory.

References:

1. Z. Zhu, H. Xia, B. Ren, S. Han, H. Li, *Inorganic Chemistry Communications*, 149, (2023), 110432.
2. L. Gracia, A. Beltrán, J. Andrés, *J. Phys. Chem. C*, 115, (2011), 7740–7746.
3. Q. Zhang, M. Sun, C. Yuan, Q. Sun, B. Huang, H. Dong, Y. Zhang, *ACS Catal*, 13, 2023, 7055–7066

Hydrogen-Assisted Selective NO Reduction on Defective Graphene-Stabilized Ni₄ Single-Cluster Catalysts

Sourav Ghoshal¹, Subhash Kumar¹, Pranab Sarkar^{2,*}

¹*School of Applied Sciences, Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, Bhubaneswar-751024, Odisha, India*

²*Department of Chemistry, Visva-Bharati University, Santiniketan 731235, India*

*Corresponding Author E-mail: sourav.ghoshalfch@kiit.ac.in

The development of efficient and selective catalysts for nitric oxide (NO) conversion remains a significant challenge in heterogeneous catalysis. Nickel-based catalysts, particularly Ni nanoclusters, have been widely explored for NO-related reactions;¹⁻² however, investigations on Ni-based single-cluster catalysts (SCCs) for NO oxidation and reduction remain limited. In this work, a defective graphene-supported Ni₄ SCC (Ni₄@G) is rationally designed and investigated for both thermocatalytic and electrochemical pathways using DFT. The Ni₄ cluster exhibits strong

anchoring and favorable electronic interactions with the defect site, enabling efficient activation of NO and co-reactants. A comprehensive mechanistic analysis reveals that NO oxidation and reduction proceed via both Langmuir-Hinshelwood (LH) and Eley-Rideal (ER) pathways, with the ER mechanism being kinetically preferred due to lower free energy barriers and more facile product desorption. In addition to conventional routes, an alternative LH (ALH) pathway is explored, highlighting the role of N-N coupling intermediates. Importantly, H₂ assistance significantly enhances catalytic performance by facilitating the removal of surface oxygen species (*O) as H₂O, thereby improving catalyst regeneration and turnover. Microkinetic simulations demonstrate that H₂-assisted ER pathways dominate over a wide temperature range, leading to substantially higher reaction rates, with N₂O identified as the primary product under thermocatalytic conditions. Furthermore, electrochemical NO reduction (NOER) on Ni₄@G proceeds via a fully downhill pathway involving *HNOH intermediates, enabling efficient NH₃ formation while suppressing the competing hydrogen evolution reaction. Structural stability analyses, including ab initio molecular dynamics and Ni atom migration studies, confirm the strong resistance of the Ni₄ cluster against sintering under reaction conditions. Overall, this study establishes Ni₄@G as a promising single-cluster catalyst for selective NO conversion and provides mechanistic insights for the rational design of advanced catalytic systems.

Keywords: NO Reduction; Defective graphene; Nickel Single Cluster Catalyst; Density Functional Theory; Microkinetic

References:

1. S. H. Talib, S. Hussain, S. Baskaran, Z. Lu, K. Li, ACS Catalysis, 10, (2020), 11951-11961.
2. A. Mahata, K. S. Rawat, I. Choudhuri, B. Pathak, Scientific reports, 6, (2016), 25590.

Interfacial 2DEG in Oxide Heterostructures: Unlocking Spintronic and Spin-Caloritronic Functionalities

Payal Wadhwa^{1,2*} and Alessio Filippetti²

¹*Department of Applied Physics, Indian Institute of Information Technology Vadodara, Gandhinagar-382028, Gujarat, India*

²*Department of Physics, University of Cagliari, Monserrato, CA, 09042-I, Italy*

*email-id: payal_wadhwa@iiitvadodara.ac.in

The interfacial quasi two-dimensional electron gas (2DEG) formed between complex oxides such as LaAlO₃ (LAO) and SrTiO₃ (STO) has emerged as one of the most compelling platforms in oxide electronics, distinguished by a rich spectrum of functionalities including field-effect driven superconductivity, high electron mobility and magnetoresistance. Concurrently, the realization of a tightly confined 2DEG with high carrier mobility and large spin-polarization is of paramount importance for the development of next-generation spintronic and spin-caloritronic devices. In this work, we demonstrate that oxide heterostructures, namely LaAlO₃/EuTiO₃/SrTiO₃ (LAO/ETO/STO) along the (001) direction and EuTiO₃/SrTiO₃ (ETO/STO) heterostructure along the (111) direction, serve as prototype material systems for this purpose. The presence of Eu induces strong spin-polarization in the 2DEG spontaneously formed at the interface, along with ferromagnetic order at low temperature, underscoring the potential of LAO/ETO/STO and ETO/STO heterostructures for spintronic applications [1,2]. Furthermore, the tight 2D confinement and spin-polarization in LAO/ETO/STO can be significantly enhanced upon charge depletion, in turn generating large thermopower associated with the phonon-drag mechanism. Most importantly, the pronounced difference in the population of the two spin channels results in the giant spin-polarized Seebeck effect and in turn, giant spin voltages of mVK⁻¹ order at the two

ends of an applied thermal gradient. Overall, these findings highlight the strong potential of LAO/ETO/STO interface for low-temperature spin-caloritronic applications [1].

Keywords: Oxide heterostructures, Magnetic materials, Density Functional theory, Spintronics, Spin-Caloritronics

References:

1. P. Wadhwa et al., Mater. Horiz., 10, 3559 (2023).
2. P. Wadhwa et al., ACS Appl. Electron. Mater., 6, 3657–3665 (2024).

Electrocatalytic Detection of Epinephrine using an Engineered 2D Carbon / Metal-Oxide Disposable Sensor in a Point-of-Care setting

Ankita Dutta Chowdhury and Somenath Roy*

Functional Materials and Devices Division, CSIR-Central Glass and Ceramic Research Institute, 196 Raja S. C. Mullick Road, Kolkata 700032, India
sroy.cgcri@csir.res.in

Epinephrine is a biologically important catecholamine that functions as both a neurotransmitter and hormone, and its reliable determination is important for clinical diagnosis, stress-related physiology, and biochemical monitoring. In this work, an engineered 2D carbon/metal-oxide interface was developed as an electrochemical platform for the sensitive detection of epinephrine in simulated urine matrix. The modified electrode produced a distinct anodic peak for epinephrine oxidation, indicating favourable charge transfer at the electrode-electrolyte interface.

The cuprous spinel-reduced graphene oxide ($\text{CuM}_2\text{O}_4\text{-rGO}$, $\text{M}=\text{Co, Cr, Mn}$) nanocomposites were synthesized via one pot co precipitation technique and thoroughly characterized using sophisticated characterization techniques such as XRD, XPS, Transmission Electron Microscopy, Raman Spectroscopy etc. spectroscopy to establish structure–property–function correlations relevant to electrocatalytic sensing. An overall comparative study was also well established between the spinel compounds and integration of the most sensitive nanocomposite on indigenously fabricated screen-printed electrodes showed a broad detection range from 30 nM to 100 μM , with a linear calibration response within the studied concentration window. The lower limit of detection was estimated to be 26 nM, and the response was obtained within less than 1 min.

As a step towards translational deployment, the fabricated electrodes were integrated with an ultraportable, IoT enabled, miniaturized pocket friendly device that can selectively detect neurotransmitters from a drop of urine sample in less than a minute, without the help of trained clinicians and operators. The results appear on the patient's smartphone through a customized application software which can be easily shared with clinicians, thereby bridging functional material design to real-world point-of-care neurodiagnostic applications. The present study thus demonstrates that an interface-engineered 2D carbon/metal-oxide interface electrode can serve as a simple, rapid, and sensitive disposable platform for electrochemical detection of catecholamine biomarkers.

Keywords: electrocatalysis, catecholamines, 2D Carbon/Metal-Oxide, point-of-care

References:

1. Dutheil, F., Fournier, A., Perrier, C. et al. Impact of 24 h shifts on urinary catecholamine in emergency physicians: a cross-over randomized trial. *Sci Rep* 14, 7329 (2024).
2. Y. Fan, N. W. Duncan, M. de Greck, and G. Northoff, *Neuroscience & Biobehavioral Reviews*, 35, (2011), 903–911.

Integration of InAs@ZnSe nanocrystals with 2D Materials for Infrared Optoelectronics

Mukesh Kumar Thakur

Department of Applied Physics, Indian Institute of Information Technology Vadodara (IIITV ICD), Gandhinagar, Gujarat 382028.

*Corresponding Author: mukesh@iiitvadodara.ac.in

Harvesting infrared (IR) light is essential for enhancing the efficiency of photovoltaic and photoelectric applications, including imaging and communication. However, existing IR technologies primarily rely on toxic materials such as lead, cadmium, and mercury chalcogenides, raising significant environmental and health concerns.

Indium arsenide (InAs) has recently attracted attention due to its tunable IR absorption, high quantum efficiency, and compliance with the European Union's "Restriction of Hazardous Substances" (RoHS) directives. However, the long-chain organic ligands present in colloidal InAs nanocrystals hinder their direct integration into optoelectronic devices. To overcome this limitation, we explore the potential of combining InAs with 2D materials such as graphene, fluorographene, and MoS₂, where InAs serve as an IR-absorbing material while 2D materials act as efficient charge transporters.

In this study, we investigate the annealing effect on InAs nanocrystals to facilitate ligand removal. Our initial results indicate that InAs nanocrystals exhibit thermal stability up to 300°C. Furthermore, annealing enhances ligand removal, particularly in InAs@ZnSe core-shell structures, as confirmed by optical and electrical characterization. Photoluminescence and Raman spectroscopy further reveal the successful integration of InAs with 2D materials. These findings highlight a promising approach for optimizing InAs nanocrystals in next-generation infrared optoelectronic applications.

References:

1. Di Stasio et al., *Adv. Sci.* 2024, 11, 2400734
2. M. De Franco et al., *ACS Energy Lett.* 2022, 7, 3788
3. D. Zhu., *Adv. Mater.* 2023, 35, 2303621.
4. Thakur et al., *Adv. Funct. Mater.* 2024, 34, 2409951.

Tribological and Mechanical Behaviour of Spray-Coated MoO₃ and MoS₂–MoO₃ Coatings on SS321 Steel

Satyam Upadhyay¹, Rishi Sharma^{1*}, Anupam Roy^{1*}

¹ Department of Physics, Birla Institute of Technology Mesra, 835215

*Corresponding Author E-mail: rishisharmabit@hotmail.com, royanupam@bitmesra.ac.in

Transition metal dichalcogenides based solid lubricants are widely used due to their excellent tribological and mechanical behaviors, making them suitable for high-temperature, space, and marine applications [1]. In this study, the tribological and mechanical behaviors of MoO₃ and MoS₂-MoO₃ solid lubricants spray-coated on SS321 steel substrates are studied [2]. Different characterization techniques, e.g., field-emission scanning electron microscopy (FESEM) and X-ray diffraction (XRD) are used to evaluate the surface morphology and structural properties of the coatings. Raman spectroscopy measurements show different vibrational modes corresponding to MoO₃ and MoS₂, while energy dispersive spectroscopy (EDS) confirms the presence of Mo, S, and O elements in the coatings. The mechanical properties, including the microhardness, are evaluated using a microhardness tester. The tribological behaviour of the coatings is studied using a tribometer under different loads. Wear properties are further analysed using a profilometer under the same loading conditions. Results show that the wear rate increases with increasing applied load for both coatings.

Keywords: Tribology, solid lubricant, MoS₂, MoO₃, Spray-coating

References:

1. Bhushan, B., & Ko, P. L. (2003). Introduction to tribology. Appl. Mech. Rev., 56(1), B6-B7.
2. Upadhyay, S., Prakash, B., Kumar, V., & Sharma, R. (2025). Ti-Al-doped MoS₂ coatings on M50 steel for superior tribological applications. Surface Engineering, 41(6), 664-673.

Crystal field splittings in rare-earth honeycomb magnets RCl₃ (R= Yb, Er)

Shivam Parasar^{1*}, Rajyavardhan Ray^{1,†}

¹Department of Physics, Birla Institute of Technology Mesra, Ranchi, India - 835215

phdap10002.24@bitmesra.ac.in, r.ray@bitmesra.ac.in

Honeycomb magnets are well-known to host a variety of interesting magnetic phenomena including the Kitaev quantum spin liquid (QSL). α -RuCl₃ is a proximate Kitaev magnet with a long-range ordered ground state in the form of zigzag antiferromagnet, and shows a spin-liquid-like behavior under ~ 10 T magnetic fields.¹ The layered two-dimensional (2D) structure where $S = 1/2$ Ru atoms span a hexagonal network supported by face-sharing RuCl₆ octahedra is also shared by the rare-earth based honeycomb magnets YbCl₃ and ErCl₃. In YbCl₃, however, the ground state is found to be a Neel antiferromagnet.² The specific heat shows a broad dominant peak at $T_1 \sim 1.20$ K, followed by a small sharp kink at $T_2 = 0.60$ K, suggesting a short range order at 1.20 K and a long range order at 0.60 K. Here, we calculate the ligand field splitting of the rare-earth based 2D honeycomb magnets RCl₃ (R=Yb, Er) using density functional theory (DFT) and accounting for the hybridization between the R - $4f$ states and the valence states of the surrounding ligands — the only free parameter in our calculations. We find that both the compounds possess effective spin $S = 1/2$ at low-temperatures. For YbCl₃, our estimates for the crystal field splitting and magnetic anisotropy, in term of g -tensor, is in a very good agreement with the experimental reports.³ In comparison to YbCl₃, the splitting between the lowest $4f$ states

is ~ 80% smaller in ErCl_3 . Consequently, the effective $S = 1/2$ description in ErCl_3 is valid only at low temperatures. We further obtain and compare the ground states and magnetic properties of the two compounds.

Keywords: Honeycomb magnets, Rare-earth trichlorides, Density functional theory (DFT), Crystal field splitting.

References:

1. A. Banerjee, C. A. Bridges, J.-Q. Yan, et al., *Nature Materials*, 15, (2016), 733-740.
2. G. Sala, M. B. Stone, B. K. Rai, et al., *Nature Communications*, 12, (2021), 171.
3. G. Sala, M. B. Stone, B. K. Rai, et al., *Phys. Rev. B*, 100, (2019), 180406(R).

Scalable Fabrication and Device Architecture Optimization of MXene Gas Sensors for Wearable Applications

Varsha Biswas¹, Vinayak B Kamble¹.

*School of Physics, Indian Institute of Science Education and Research,
Thiruvananthapuram, Kerala, India¹*

*varshabiswas23@iisertvm.ac.in

2D transition metal carbides or MXenes, are emerging as promising candidates for gas sensing, with $\text{Ti}_3\text{C}_2\text{T}_x$ being the most studied system. While high conductivity, mechanical flexibility, room-temperature operation and scalability make them promising for gas sensing, practical devices require consistent sample-to-sample performance.[1]. Although several reports have elucidated the response of MXenes and their composites, the influence of device architecture on sensing performance remains to be examined [2,3]. In this work, the roles of electrode geometry and film morphology were studied using the scalable spray-coating and drop-casting method to deposit MXene films on Au-Inter-Digitated Electrodes (IDEs) with 7-finger (7F) and 2-finger (2F) geometries on flexible PET substrates. These devices were tested against multiple gases, including CO , CH_4 , and H_2 , as well as VOCs such as Ethanol, Acetone, Toluene, and Methanol, across a range of concentrations at parts per million (PPM) levels, and the electrical response was analysed. Devices with different sensor architectures showed distinct responses and stability. It was observed that the spray-coated 7-finger device had almost an order of magnitude higher responsivity than the drop-casted 7-finger device, whereas the drop-casted 2-finger device had the lowest signal-to-noise ratio among all films. Thus, the spray-coated MXene films offer the advantages of higher response and scalability. Overall, this study aims to demonstrate that the gas-sensing behaviour in MXene-based devices depends on both electrode configuration and film-deposition method. Therefore, this work provides further practical insights into the existing literature on MXenes for the optimisation of room-temperature flexible gas sensors.

Keywords: MXenes; Film morphology; Gas sensing;

References:

1. Halim, J., Kota, S., Lukatskaya, M. R., Naguib, M., Zhao, M. Q., Moon, E. J. and Barsoum, M. W. *Advanced Functional Materials*, 26(18), (2016), 3118-3127.

2. Wu, M., An, Y., Yang, R., Tao, Z., Xia, Q., Hu, Q., and Zhou, A. ACS Applied Nano Materials, 4(6), (2021), 6257-6268.
 3. Hermawan, A., Zhang, B., Taufik, A., Asakura, Y., Hasegawa, T., Zhu, J. and Yin, S. ACS Applied Nano Materials, 3(5), (2020), 4755-4766.
-

Endodontic file wear assessment using multi-sensor fusion & CRITIC

Apoorv Tripathi^a, Pavan Kumar Kankar^{a*}, Ankur Miglani^b, Hem C. Jha^c, Niharika Gupta^d,
Rohit Mishra^d

^a*System Dynamics Lab, Department of Mechanical Engineering, Indian Institute of Technology, Indore, Madhya Pradesh, India*

^b*Microfluidics and Droplet Dynamics Lab, Department of Mechanical Engineering, Indian Institute of Technology, Indore, Madhya Pradesh, India*

^c*Infection Bioengineering Group, Department of Biosciences and Biomedical Engineering, Indian Institute of Technology Indore, Simrol, Indore 453552, Madhya Pradesh, India*

^d*Hitkarini Dental College & Hospital Jabalpur*

Nickel titanium instruments are used in RCT (Root Canal Treatment) and are prone to unexpected failure. This study aims to develop a CRITIC (CRiteria Importance Through Intercriteria Correlation) optimised CWI (Composite Wear Index) using Force, acceleration and acoustic sensory input for real-time prediction of file failure. Statistical parameters like RMS (Root Mean Square), PTP (Peak-To Peak) etc. were computed using the raw data obtained from sensors during RCT. These parameters were then optimised using CRITIC framework and assigned weights based on contrast and conflict. Also, in conjunction to the sensor fusion data, physical measurements of files post experimentation were done to corroborate wear data with sensor prediction. The optimised weighted parameters resulted in a strong correlation between physical wear and computed CWI of sensors. Hence this framework will prove essential in predicting faults in NiTi instruments and preventing sudden failures.

Abbreviations

CRITIC - CRiteria Importance Through Intercriteria Correlation, CWI - Composite Weart Index, RMS - Root Mean Square, TKEO - Teager-Kaiser energy operator, RCT - Root Canal Treatment, PTP - Peak – to – peak, NiTi - Nickel Titanium, ZCR – Zero Crossing Rate

Bi-Doped g-C₃N₄ as a Sustainable Electrocatalyst for Ammonia Production

Aman Bharti¹, Pulkit Garg¹, Ankit Tyagi^{1*}

¹*Department of Chemical Engineering, Indian Institute of Technology Jammu, Jammu, Jammu and Kashmir, 18122, India*

*Corresponding Author E-mail: ankit.tyagi@iitjammu.ac.in

With the growing emphasis on sustainable chemical processes, the need for eco-friendly alternatives to the conventional Haber-Bosch process for ammonia synthesis has become increasingly important. Ammonia is a vital nitrogenous fertilizer for global food production and is also emerging as a promising hydrogen carrier in the green economy. However, the Haber-Bosch process is highly energy-intensive, consuming 1-2% of global energy and emitting significant CO₂, primarily because hydrogen is produced from fossil fuels. In this context, the electrochemical nitrogen reduction reaction (ENRR) offers a sustainable pathway for ammonia synthesis under ambient conditions [1]. Nevertheless, challenges such as low selectivity and the competing hydrogen evolution reaction (HER) hinder its practical application. In this study, a two-dimensional (2D) material, graphitic carbon nitride (g-C₃N₄), was employed as a support and modified via bismuth (Bi) doping to enhanced catalytic performance.

A series of Bi-doped g-C₃N₄ catalysts with varying Bi loadings (1, 2, 3, 4 and 5 wt.%) were synthesized, denoted as B1GCN, B2GCN, B3GCN, B4GCN and B5GCN, respectively. Structural and morphological analyses (XRD, FTIR, FESEM with EDS mapping and XPS) confirmed successful Bi incorporation while preserving the layered 2D structure of g-C₃N₄ [2].

Electrochemical measurements were conducted in an H-type Cell connected with acid traps for gas purification separated by Nafion membrane with acidic electrolyte. Ammonia production was quantified using Indophenol blue method and UV-Vis spectroscopy. The enhanced performance is attributed to the optimized electronic structure induced by Bi, which facilitates N₂ adsorption and activation while suppressing HER [3].

Additionally, the catalysts demonstrated good stability and reusability, indicating for long-term application. This work highlights Bi-doped g-C₃N₄ as a promising electrocatalyst for sustainable ammonia synthesis and provides insights into dopant-support interactions for improved catalytic activity.

Keywords: Nitrogen reduction reaction; Electrocatalysis; Ammonia Synthesis

References:

1. Qing G, Ghazfar R, Jackowski ST, et al (2020). Chem Rev 120:5437–5516.
2. Yuan C, Liang S, Lu Z, et al (2021). Ma Y, Lu Y, Li C, et al (2024). Catal Letters 154:5830–5837.

Biological Response of ZX50 Mg Alloy: The Role of Surface Microstructure

Sheetal Yamalakonda¹, Khandu Wadhonkar², Nilesh K. Kumbhar¹, Mirza S. Baig², Santosh S. Hosmani¹

¹ Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Indore 453552, India

² Mehta Family School of Biosciences and Biomedical Engineering, Indian Institute of Technology Indore, Indore 453552, India

* Corresponding author: Santosh S. Hosmani (sshosmani@iiti.ac.in)

The surface mechanical attrition treatment (SMAT) and subsequent controlled annealing were performed on a ZX50 Mg5Zn0.2Ca alloy to optimise its near-surface microstructure for biodegradable applications. Graded ultrafine grain structures were produced by systematically varying the annealing temperature (100 °C to 400 °C), and the resulting specimens were tested in DMEM-FBS to study L929 cell cytocompatibility and DMEM-Media to study their corrosion behaviour, the formation of a protective salt layer, and BSA (bovine serum albumin) solution tested for protein adsorption. The synergistic effect of SMAT-induced microstructural refinement and annealing at 100 °C and 250 °C resulted in a thicker, more uniform, and chemically more stable Ca and P-rich protective layer in DMEM media. In a cytocompatibility study, the non-

SMATed samples showed a considerably lower cell viability of ~16% in the 50% extract solution after 24 h; however, the SMATed samples showed significant protection, with ~50% viability. The SMATed samples annealed at 100°C and 250°C exhibited statistically better cell viability of ~77% and ~82%, respectively. Moreover, these samples exhibited the highest BSA protein adsorption (0.071 $\mu\text{g}/\text{cm}^2$ at 250°C). The improved behaviour of the alloy decreased as the annealing temperature increased to 400°C, due to an adverse effect of grain coarsening, which formed an inferior protective layer in biological media. In general, the integrated SMAT-annealing pathway outlines a viable processing regime that yields a stable, biologically favourable surface on the ZX50 alloy, enabling controlled resorption and enhancing the biological response of future biodegradable osteosynthesis devices. However, by using the 2D material like graphene oxide along with polymer on optimized substrate improved the cytocompatibility and degradation rate which may goes to 90 days until the complete tissue engineer and bone repairment.

Keywords: ZX50 Mg alloy; Surface Mechanical Attrition Treatment; Annealing; Biocorrosion; Cytocompatibility.

References:

1. Kumbhar, N. K.; Yamamoto, A.; Wadhonkar, K.; Baig, M. S.; Hosmani, S. S., J. Alloys Compd. 2025, 1010, 178259.
2. Dittrich, J.; Knappek, M.; Veselý, J.; Kubásek, J.; Bohlen, J.; Šašek, S.; Afonso, C. R. M.; Minarik, Journal of Materials Research and Technology 2025, 36, 6024–6038.
3. Aghamohammadi, H.; Rabiee, S. M.; Hosseinipour, S. J.; Jamaati, R., Journal of Materials Research and Technology 2023, 27, 5154–5164.

Architecture-Dependent Emergence of Self-Powered Gas Sensing in Junction Engineered WSe₂ Device

Dipanjana Mondal¹, Dhariya Sood¹, Vinayak B. Kamble^{1*}

¹*School of Physics, Indian Institute of Science Education and Research Thiruvananthapuram, Kerala, India, 695551.*

*Corresponding Author E-mail: kbvinayak@iisertvm.ac.in

Self-powered gas sensors capable of operating without an external energy supply represent a critical frontier for next-generation environmental monitoring and Internet-of-Things platforms. Here, we present a systematic investigation of Chemical Vapor Deposition (CVD) grown WSe₂-based devices across three distinct architectures: a monolayer single-flake channel, a controlled twisted-stacked few-layer single-flake system, and a multi-domain flake network. This study elucidate that how structural complexity governs charge transport, broadband photodetection and gas-sensing behaviour within a single material system. The multi-domain network, comprising a heterogenous assembly of monolayer, few-layer, and twisted WSe₂ domains, exhibits robust ultra-broadband (280-850 nm) photoresponse in self-driven condition with a detectivity of 1.5×10^9 Jones and response/recovery time of <200 msec. Furthermore, the device exhibits highest photocurrent compared to the other two devices. Strikingly, this same network architecture enables NO₂ detection at concentrations as low as 10 PPM at 0V in the dark. Moreover, at a low bias of few millivolts the limit of detection is ~1 PPM, and further improved to sub-ppm sensitivity under light illumination. This behaviour is attributed to a network of nanoscale junctions, where spatially distributed built-in electric fields enable carrier separation, while gas adsorption modulates interflake barrier heights and carrier pathways. By contrast, the stacked device exhibits a weak but measurable response for ~200 PPM of NO₂ at zero and dark bias condition, which enhanced under light illumination. While, the monolayer device is entirely

insensitive at 0 V in dark, exhibiting only marginal response under finite bias and phot-activation. These reveals a clear mechanistic hierarchy from bulk-dominated transport in the monolayer device to interface-mediated response in the stacked few layer device, and ultimately to junction-controlled transport in heterogenous network device. This architecture-dependent engineering of device in a single 2D material system establishes a compelling design paradigm for simultaneously self-powered, broadband- photoresponsive, and ultrasensitive gas sensors.

Keywords: WSe₂, Self-powered gas sensor, Chemical Vapour Deposition, Junction effects, Photo-assisted sensing.

References:

1. Patra, C., Guna, V.K., Chakraborty, S., Mondal, S. and Nandakishora, Y., 2025. ACS sensors, 10(9), pp.6347-6379.
2. Furchi, M.M., Pospischil, A., Libisch, F., Burgdorfer, J. and Mueller, T., 2014. Nano letters, 14(8), pp.4785-4791.
3. Late, D.J., Huang, Y.K., Liu, B., Acharya, J., Shirodkar, S.N., Luo, J., Yan, A., Charles, D., Waghmare, U.V., Dravid, V.P. and Rao, C.N.R., 2013. ACS nano, 7(6), pp.4879-4891.

WS₂ vertically aligned sheet like structure for humidity sensing at room temperature

Thameemul Ansari M*, Sornadurai D, Sourav Pan, Shanthini S, Arun K Prasad

Materials Science Group, Indira Gandhi Centre for Atomic Research, A CI of Homi Bhabha National Institute, Kalpakkam 603102, Tamil Nadu, India.

* Corresponding author: thameemulansarians@gmail.com*

The ability to rapidly detect humidity changes within a narrow range at room temperature is critically important for agriculture, pharmaceutical, and semiconductor applications [1]. Existing humidity sensors face stability challenges at room temperature and have a limited detection range. Recently, sensors based on transition metal dichalcogenides (TMDCs) have garnered significant attention for their exceptional ability to detect gases and moisture at room temperature, a capability attributed to their unique electrical properties arising from their atomically thin, two-dimensional structure. However, pristine TMDCs often exhibit poor sensing performance, with low sensitivity and long response times [2]. These undesirable characteristics can be substantially enhanced through the controlled introduction of defects into the TMDC structures. Current research has focused on synthesizing WS₂ nanosheet-like structures via the hydrothermal method while carefully controlling defect incorporation. These nano-sheets have been characterized using scanning electron microscopy (SEM), X-ray diffraction (XRD), Raman spectroscopy, and photoluminescence spectroscopy. It was noted that the conductivity of the sensing layer increases with humidity over a broad range, from 60% to 86% relative humidity (RH). The sensing mechanism is explained by the Grotthuss mechanism of proton conductivity, which posits that conductivity arises from proton tunneling between water molecules via hydrogen bonding. The developed humidity sensor demonstrates a high response (~90%), rapid response and recovery times (11 and 20 seconds, respectively), and excellent repeatability and stability.

Keywords: WS₂ nanosheets, Hydrothermal method, Humidity sensor,.

Reference:

1. Jha et al IEEE Transactions on Nanotechnology 17(3), (2018), 582-589.
2. Gupta et al Sensors and Actuators A: Physical 331, (2021), 113008.

Deterministic Switching of Graphitic Stacking Order via Superlubric Interfaces

Nirmal Roy*, Moshe Ben Shalom

School of Physics and Astronomy, Tel Aviv University, Tel Aviv, Israel

*Corresponding Author E-mail: nirmalroy1378@gmail.com

The stacking order in few-layer graphene serves as a powerful and tunable degree of freedom for engineering exotic electronic phases, including spontaneous electric polarization, orbital magnetism, and unconventional superconductivity. Despite this potential, achieving deterministic, reversible, and energy-efficient switching between stacking configurations at the nanoscale remains a significant experimental hurdle. Here, we introduce a novel device architecture—the Super-Lubricant Array of Polytypes (SLAP) that enables precise and programmable control of stacking boundaries. By inserting a rotationally misaligned spacer with pre-patterned nanoscale etched cavities between otherwise aligned graphene layers, we spatially confine distinct stacking orders (polytypes) while preserving superlubric, near-frictionless interfaces in the surrounding regions. This design enables controlled switching of polytypes with lateral dimensions down to ~ 30 nm, driven by sub-nanonewton shear forces and with energy consumption below 1 fJ per switching event. We show that the switching process is mediated by the nucleation and propagation of highly mobile, edge-driven solitons, reflecting weak pinning and efficient long-range elastic relaxation across the heterostructure. Moreover, by engineering the cavity geometry, we tune elastic coupling between adjacent stacking domains, thereby controlling switching thresholds and domain wall dynamics. These results establish a practical route for nanoscale control of stacking order and introduce a scalable platform for reconfigurable van der Waals devices, with potential applications in multiferroic and neuromorphic systems.

Keywords: Graphitic Polytypes; Super lubricity; non-volatile Switching.

References:

1. Nirmal Roy et al., Nat. Nanotechnol. 21, 530–537 (2026).
2. Nirmal Roy et al., Adv. Mater. 37, no. 47 (2025): e09947.
3. Atri, S.S. et al., Advanced Physics Research, 3(5), p.2300095 (2024).

Gate-Tunable Density-of-States Valve for Charge Injection at a Two-Dimensional Superconductor/Semiconductor Interface

Abhishek Chandrakant Patil, Ayeon Yun, Kyeong Min Lee, Myung Gwan Hahm^{1*}

¹⁾Inha University, Incheon, South Korea

E-mail : mghahm@inha.ac.kr

Two-dimensional semiconductor/superconductor interfaces have emerged as promising building blocks for cryogenic electronics; however, microscopic origin of charge injections across such heterointerfaces remains poorly resolved once the contact enters the superconducting state. Here, we demonstrate an all-two-dimensional field-effect transistor consisting of chemical-vapor-deposited WSe₂ channels and split NbSe₂ source/drain electrodes revealing that the

superconducting interface acts as a gate-tunable density-of-states valve for charge injection [1]. Above the superconducting transition, temperature-dependent transport scaling separates barrier-controlled and hopping-assisted regimes, allowing the interfacial injection landscape to be quantified as a function of gate bias. Below the transition, the device exhibits a counterintuitive suppression of low-bias current and conductance despite the emergence of superconductivity in the contact electrodes [2]. Differential conductance mapping reveals that this suppression is confined to a low-bias window and is progressively lifted by perpendicular magnetic field, consistent with field-induced filling of the superconducting density of states. By defining an injection-efficiency metric and statistically tracking its magnetic-field evolution across multiple devices, we show that gate bias and magnetic field act as two independent control knobs: the gate tunes the interfacial barrier, whereas the magnetic field tunes the superconducting density of states [3]. This two-knob control establishes a physically transparent mechanism for programmable charge injection at van der Waals superconductor/semiconductor interfaces and introduces a design principle for cryogenic logic circuits, adaptive contacts, and quantum-compatible two-dimensional electronic devices.

Keywords: Two-Dimensional Materials; Superconductor-Semiconductor Interface; NbSe₂ Superconductivity; WSe₂ Field-Effect Transistors; Charge Injection Mechanism.

References:

1. J. Guan, H.-J. Chuang, Z. Zhou, and D. Tománek, ACS Nano 11 (2017) 3904-3910.
2. Y. Kim, A. R. Kim, J. H. Yang, K. E. Chang, J. -D Kwon, S. Y. Choi, J. Park, K. E. Lee, D. -H Kim, S. M. Choi, K. H. Lee, B. H. Lee, M. G. Hahm, B. Cho, Nano Letters 16 (2016) 5928-5933
3. A. Inane, M. Gomez, G. O. Steffensen, T. Kanne, J. Nygard, A. Levy Yeyati, E. J. H. Lee, Nature Communications 14 (2023) 2873

Redox-Active 2D Layers Metal-Organic Framework as Ion Highways Towards Fast and Reversible Ammonium-Ion Supercapacitors

Nissar Hussain¹ Altaf Husain¹ and Shaikh M. Mobin^{*, 1}

¹ Department of Chemistry, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore 453552, India.

*Corresponding Author E-mail: xray@iiti.ac.in

The design of advanced materials for ammonium-ion (NH₄⁺) energy storage systems is gaining momentum due to their environmental benignity and fast ion mobility. In this work, we introduce a new two-dimensional cobalt metal-organic framework (2D Co-MOF) composed of redox-active organic linkers and cobalt centers, which adopts a crystallographically defined 2D layered architecture. The intrinsic planar structure offers extensive surface accessibility, uniform pore channels, and abundant redox-active sites conducive to efficient NH₄⁺ intercalation. It exhibits a high specific capacitance of 564 Fg⁻¹ at 1 Ag⁻¹, excellent rate capability, and a capacity retention of 91% over 10,000 charge-discharge cycles. Furthermore, the ammonium-ion supercapacitors (AISC) exhibited an outstanding energy density of 42.5 Whkg⁻¹, maintaining 88 % capacitive retention after 10,000 cycles. This study highlights the potential of crystal-engineered, redox-active 2D layered MOFs as high-performance NH₄⁺ ion storage materials. It offers a rational strategy for developing advanced coordination frameworks for next-generation green energy storage technologies.

Keywords: 2D layered MOF, Intercalation/deintercalation, Ammonium ion Supercapacitors

References:

1. Hussain. N. Husain, A. Mobin. M. S. *ACS Materials Lett.* (2026).
2. Gao M. Zhiyong Wang Z., Liu Z, Huang Y. Faxing Wang F, Mingchao Wang M, Yang S, Li J, Jinxin Liu, Haoyuan Qi H. L. J. Panpan Zhang P, Xing Lu, Xinliang Feng X. L. X. *Advance Materials*, 35, (2023), 2305575.

Capacitive Pressure Sensing Behaviour of Microporous Dielectric Elastomer Composites

Dhananjay Sahu¹, Raj Kumar Sahu^{1*}, Yadagiri Naik Banothu², Somnath Bhattacharya¹, Himanshu Pathak³

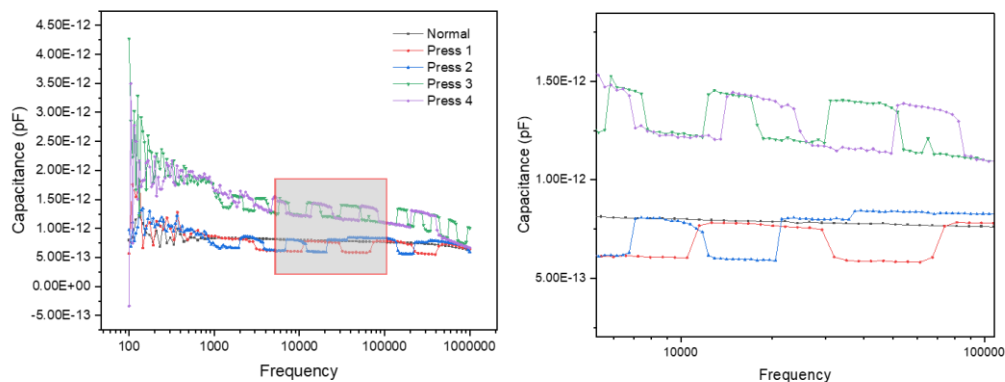
¹*Department of Mechanical Engineering, National Institute of Technology Raipur, Raipur, Chhattisgarh, 492010, India*

²*DRDO Young Scientist Laboratory for Smart Materials, Ministry of Defence, Kanchanbagh, Hyderabad, 500058, India*

³*School of Mechanical and Materials Engineering, Indian Institute of Technology Mandi, Himachal Pradesh, 835205, India*

*Corresponding Author E-mail: raj.mit.mech@gmail.com

The presence of microporosity in dielectric elastomers effectively improves their pressure-sensing performance by enhancing deformability and relative permittivity through reduced air volume. In this work, the capacitive pressure-sensing performance of in-lab-fabricated sensors, made from commercially available microporous dielectric elastomer (VHB 4910), is explored, focusing on the effects of microporosity on linearity, sensitivity, and response time. The porosity of the elastomer is tuned by incorporating filler-dispersed interpenetrated polymer network of trimethylolpropane trimethacrylate, followed by characterisation of the mechanical compliance and dielectric properties. The results obtained for conductive and ceramic fillers are compared to determine the best-performing combination correlating with the aspects of microporous structure. This work may guide future efforts in the design and development of flexible and stretchable pressure sensors for wearable electronics and related fields of application.

Initial Results:

Keywords: microporous; pressure sensor; dielectric; sensitivity; response time

References

1. Sahu D, Sahu RK, Bhattacharya S, et al (2026), J Phys D Appl Phys 59.
 2. Sahu D, Prabhakar OP, Sahu RK, et al (2025), Polym Eng Sci 1–14.
-

POSTER ABSTRACTS

True Strain Estimation and Energy Band Gap Modulation in Silicon Nanomembrane-based 2D Structures

Soumya Tiwari^{1,*}, and Anjali Chaudhary¹.

¹Department of Physics, Indian Institute of Technology Bhilai, Chhattisgarh, 491002, India

*soumyat@iitbhilai.ac.in

Strain engineering has emerged as a powerful and versatile approach to modulate the electronic and mechanical properties of two-dimensional (2D) materials, enabling tunable functionalities without altering their chemical composition.[1] Ultrathin silicon nanomembranes (Si NMs) represent a unique class of 2D architectures owing to their nanoscale thickness, high surface-to-volume ratio and mechanical flexibility. These attributes make Si NMs promising candidates for strain-engineered applications, facilitating precise modulation of their properties for advanced flexible electronic, photonic and optoelectronic devices.[2] Traditional strain engineering methods often fail to decouple substrate effects from intrinsic material behavior, leading to inaccuracies in strain characterization and limiting their applicability. To address this, we developed a versatile technique that employs a mechanical pressure cell to tensilely stress Si NMs transferred over a hole onto a flexible polyimide substrate, accurately measuring strain while eliminating substrate interactions, with broad applicability to other transferable 2D materials.[3]

We fabricated Si NMs of lateral dimensions 1mm × 1mm and 220 nm thickness, from commercially available SOI wafers, using maskless photolithography and wet chemical etching methods. The roughness of the Si NM is estimated before and after the transfer process using Atomic Force Microscopy. In-situ Raman spectroscopy is used to quantify the amount of strain in tensilely strained Si NM. We achieved a maximum strain of approximately 0.9% at 20 psi, resulting in a reduction of the bandgap from 1.1 eV to 0.8 eV, as predicted by deformation potential theory.[3] Our work establishes a robust approach to precisely control the amount of strain in Si NM based 2D structures, highlighting their significant potential for emerging applications in flexible electronics, optoelectronics, sensing platforms, and energy-efficient technologies.

Keywords: silicon nanomembrane; strain engineering; bandgap modulation; flexible electronics

References:

1. A. K. Katiyar and J.H. Ahn, *Small Methods*, 9, (2025), 2401404.
2. S. Tiwari, T. Kumar, S. K. Saxena, R. Prakash, and A. Chaudhary, *Advanced Optical Materials*, 13, (2025), 2402065.
3. S. Tiwari, T. Kumar, D. K. Rath, S. R. Gupta, R. Kumar, R. Prakash, and A. Chaudhary, *ACS Applied Nano Materials*, 8, (2025), 5694–5699.

Impact of Built-in E-field and Covalency on Excitonic Properties of PtXY (X,Y = S, Se, Te) Janus layers

Anjana E Sudheer¹, Fidha Saheer M^{1,2}, G. Tejaswini¹, Karthikeyan C¹, and D Murali1a

¹Department of Sciences, Indian Institute of Information Technology
Design and Manufacturing, Kurnool, India.

²Department of Physics, Aligarh Muslim University, Aligarh, Uttar
Pradesh, 202002, India.

^{a)} Corresponding author: dmurali@iiitk.ac.in

The discovery of two-dimensional (2D) Janus layers has attracted considerable research interest due to the intrinsic built-in electric field arising from their structural asymmetry [1]. In addition, their promising excitonic properties and pronounced dielectric responses further enhance their relevance for optoelectronic and catalytic applications [2-3]. It is interesting to study how the built-in electric field influences and modifies the excitonic properties of Janus layers in comparison with their parent structures. In this work, we design prototypical PtXY (X, Y = S, Se, Te) Janus monolayers along with their parent PtX₂ (X = S, Se, Te) structures and systematically investigate their excitonic properties using many-body perturbation theory within the GW-Bethe Salpeter Equation (GW-BSE) framework. The built-in electric field is quantitatively analyzed through electrostatic potential energy profiles. Our results reveal a significant decrease in excitonic binding energy with increasing built-in electric field. Furthermore, enhanced covalency is also found to reduce the excitonic binding energy. Interestingly, the Janus monolayers exhibit excitonic characteristics due to the combined effect of dual bond nature and the intrinsic electric field. The spatial distribution of excitonic states is analyzed in real space through simulations of the excitonic wave function, while the exciton coupling strength in reciprocal space is explored using fat-band analysis. Finally, we qualitatively analyse the life time of bright and dark excitons formed from the oscillator strength of the excitonic peaks.

Keywords: Density Functional Theory, Covalency, Built in E-field, Janus layers

References :

1. Wen-Jin Yin, Hua-Jian Tan, Pei-Jia Ding, Bo Wen, Xi-Bo Li , Gilberto Teobaldi and Li-Min Liu Mater. Adv., 2, (2021), 7543-7558/13.
2. Jan Kopaczek, Mohammed Y. Sayyad, Cheng-Lun Wu, Renee Sailus, Robert Kudrawiec, and Seth Ariel Tongay. Nano Lett. 24, (2024), 15700-15706/49
3. Blake Povilus, Renee Sailus, Jan Kopaczek, Shuai Feng, Mohammed Y. Sayyad, Seyed Tohid Rajaei Moosavy, Hayley Ruddick, Sui Yang , Seth Ariel Tongay Appl. Phys. Lett. 127 (2025), 163102/16

Magnetic and transport properties of a Gd/Co multilayer

Srijani Haldar^{1,2}, Neha Sharma³, P. Gupta^{2,4}, V. Malik³, M. Gupta⁵, Surendra Singh^{1,2,*}

¹ Solid State Physics Division, Bhabha Atomic Research Centre, Mumbai- 400085, India

² Homi Bhabha National Institute, Anushaktinagar, Mumbai-400094, India

³ Department of Physics, Indian Institute of Technology Roorkee, Roorkee, 247667, India

⁴ Engineering Applications Lab, Raja Ramanna Centre for Advanced Technology, Indore 452013, India

⁵UGC-DAE Consortium for Scientific Research, University Campus, Khandwa Road, Indore 452001, India

*surendra@barc.gov.in

Rare earth(RE)-transition metal (TM) heterostructures exhibit novel magnetic properties due to strong antiferromagnetic coupling at the interfaces. Interface structure and morphology tend to play an important role in determining depth dependent magnetism of these heterostructures [1-3]. Annealing of the heterostructures modify the interface morphology and magnetic properties. In this work, we have studied the effect of isothermal annealing on structure and magnetic properties of a Gd (RE)/Co (TM) multilayer grown by magnetron sputtering technique. Multilayer was vacuum annealed isothermally at 300 °C for different times. The evolution of in-

depth magnetic properties of Gd /Co multilayer induced by isothermal annealing at 300°C in vacuum for different times have been studied by polarized neutron reflectivity (PNR) technique. Annealing of the multilayers for a very short period of time (0.5 hours) resulted in significant large interdiffusion of atoms between the Gd and Co layers accompanied by formation of GdCo₂ alloy phase at the interfaces. In addition, we have also investigated the temperature-dependent longitudinal resistance (R_{xx}) and transverse (Hall) resistance (R_{xy}) measurements as a function of magnetic field applied perpendicular to the plane of the Gd/Co multilayer in pristine and annealed states to correlate the effect of annealing on the magnetoresistance and anomalous hall resistance of the multilayer. We found unconventional and complex MR and anomalous Hall effect for the multilayer.

Keywords: Heterostructures; Magnetron sputtering; Isothermal annealing; Polarized Neutron Reflectivity; Anomalous Hall Effect.

References:

1. M. A. Basha et al., Physical Chemistry Chemical Physics 20, 21580 (2018).
2. S. Singh et al., Physical Chemistry Chemical Physics 24, 6580 (2022).
3. J. Park et al., Physical Review B 103 , 014421 (2021)

Unravelling the lattice architecture of MnCo₂O₄

Chinmay Gavade, Vandana A Mooss *

Department of Chemistry, School of Science and Environmental Studies,

Dr. Vishwanath Karad MIT World Peace University, Pune, India

Email:vandana.mooss@mitwpu.edu.in

Among transition metal oxides, the spinel architecture represents a premier class of materials for catalytic applications, primarily due to the inherent synergistic effects of its dual-metal redox centers. The functional performance of these oxides is fundamentally dictated by structural parameters and morphology beyond their elemental composition. MnCo₂O₄ have gained considerable momentum as an advanced material for ORR(Oxygen Reduction reaction) and OER (Oxygen reduction and evaluation) in fuel cell and material air batteries (Wang et al., 2021). However, the durability and activity over many cycles is often limited mainly due to structural instability, relatively lower than noble metals. These drawbacks restrict their efficiency and long-term durability for real world applications. High susceptibility of Jahn-Teller distortions associated with Mn³⁺(d⁴) ions causing geometric distortions in octahedral sites trigger the phase transition between cubic to tetragonal phase (Ulpe et al., 2019). Therefore, the current study attempts to bridge the gap between synthesis and its influence on lattice parameters. Establishing such a precise correlation is vital to optimize catalytic efficacy, demonstrating that tailoring the synthesis is a fundamental requirement for engineering high-performance spinel catalysts.

Keywords: MnCo₂O₄, lattice parameter, microstrain

References:

1. Ulpe, A. C., Bauerfeind, K. C. L., & Bredow, T. (2019). ACS Omega, 4(2), 4138–4146.
2. Wang, H., Zhang, W., Meng, J., Pei, Y., Qiu, X., Meng, F., & Liu, X. (2021). ACS Applied Materials and Interfaces, 13(20), 24329–24340.

COUPLED STRUCTURAL, MAGNETIC, AND ELECTRONIC EVOLUTION IN OXYGEN-DEFICIENT $\text{BaFeO}_{3-\delta}$ THIN FILMS

Priti Roy¹, Subho Saha¹, Bharti Ojha¹, Rajan Mishra¹, Shivam Choudhary¹, Bharathi Paltiya¹, Ram Janay Choudhary^{1,*}

¹UGC-DAE Consortium for Scientific Research, Indore, India, 452001

*Corresponding Author: ram@csr.res.in

Perovskite transition metal oxides (ABO_3) have long remained a focal point of condensed matter physics due to their ability to host a vast array of emergent functionalities, including multiferroicity, colossal magnetoresistance, and metal-insulator transitions [1]. Among these, the class of iron-based perovskites featuring iron in the unusually high Fe^{4+} valence state (AFeO_3 ; $\text{A}=\text{Ca, Sr, Ba}$) has attracted significant attention for its versatile magnetic textures and potential in next-generation spintronic and magneto-optical devices [2]. Here we report structural, magnetic and electronic properties of $\text{BaFeO}_{3-\delta}$ epitaxial thin film grown over STO(001) via pulsed laser deposition technique, exploring substrate induced strain effect in epitaxial thin films. In situ post annealing of as grown the film at 1 atm oxygen partial pressure for 1.5 hour has been done to stabilize metastable cubic perovskite phase. Figure 1 a) shows the xrd pattern of post annealed film with and without oxygen. Figure 1 b) shows Temperature dependent magnetization of the film carried out at 0.1 T Field cooled (FC) protocol via 7 T SQUID-VSM for both the films. The one with post annealing in absence of oxygen doesn't show any magnetic phase transition down to 5 K, whereas the one with post annealing in presence of oxygen shows antiferromagnetic to ferromagnetic phase transition at around 150 K as we lower down the temperature. By combining Pulsed laser deposition method, lab source and synchrotron-based spectroscopic techniques, we elucidate how lattice distortions, oxygen-vacancy-induced instability is responsible for magnetic arrangement. These findings provide a strategic pathway for engineering iron-based perovskite derivatives for advanced magnetic-optic transparent spintronics.

Keywords: Oxygen Vacancies; Epitaxial Perovskite Oxides; Electronic Structure; Magnetoproperties; Synchrotron Photoemission

References:

1. Chakraverty, S., Matsuda, T., Ogawa, N., Wadati, H., Ikenaga, E., Kawasaki, M., Tokura, Y., & Hwang, H. Y. (2013). Applied Physics Letters, 103(14), 142416. <https://doi.org/10.1063/1.4824210>
2. Ou, J., Zhou, H., Huang, H., Rao, F., Zeng, X., Chen, L., Shao, R., Duan, M., & Huang, C. (2025). Applied Physics Reviews, 12(1), 011412. <https://doi.org/10.1063/5.0241360>

Unleashing the Potential of Layered Nb_2CT_x MXene Inks Towards the Flexible Electronic Applications

Calvin Prakash Raj Martin¹, Suguna P¹ and Ayyappan Sathya^{1,*}

¹ Advanced Nanostructured Materials Laboratory, Department of Physics, School of Electrical & Electronics Engineering (SEEE), SASTRA Deemed to be University, Thirumalaisamudram, Thanjavur 613401, Tamil Nadu, India

*Corresponding author: avyappan@phy.sastra.edu

In this generation, flexible electronic devices play a major role in the field of sensors, telecommunication and health monitoring. It has several advantages like wearability, compatibility, mechanical stability, and flexibility compared to the traditional electronic devices. These unique properties lead the flexible electronic devices towards the applications such as electro-magnetic interference (EMI) Shielding, flexible pressure sensor, electronic-skin, energy storage etc., [1]. MXenes are layered 2-dimensional (2D) material with unique properties like exceptional electrical conductivity, high hydrophilicity, flexibility and good mechanical stability [2]. MXenes have a layered morphology and tuneable surface functional groups, which ensures an efficient material for flexible EMI shielding and pressure sensor applications[3]. Here, Nb_2CT_x based MXene is synthesised through the LiF/HCl etching of lab prepared Nb_2AlC MAX phase at room temperature for 72h. The Nb_2AlC MAX phase is synthesised via a pressureless sintering of elemental powders at 1600°C for 1h at inert atmosphere. The phase formation of MAX and MXene is confirmed through the structural analysis (X-ray diffraction) and obtaining of colloidal solution. The MXene ink coated on a flexible substrate like polyethylene terephthalate sheet, cotton fabric etc., for the flexible pressure sensor and EMI shielding applications.

Keywords: MXene, Flexible, Sensor, Shielding

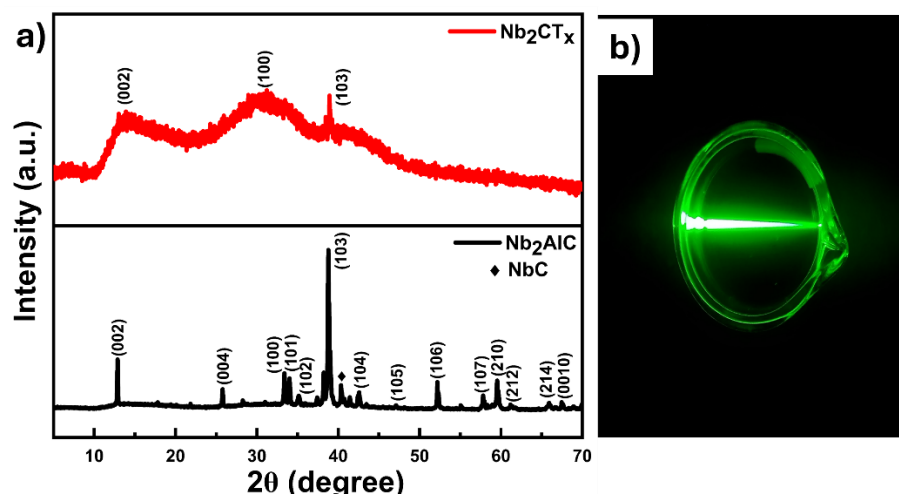


Figure 1:(a) XRD pattern of Nb_2AlC MAX and Nb_2CT_x MXene (b) Photographical image of Tyndall effect in Nb_2CT_x MXene Colloidal solution

References:

1. Hongyan Yin, Lizhi Chen, Fangfei Liu, Tursun Abdiryim, Jiaying Chen, Xinyu Jing, Yancai Li, Mengyao Su, Xiong Liu, Materials Today Physics 47 (2024) 101537
2. Lianjia Zhao, Kang Wang, Wei Wei, Lili Wang, Wei Han, Info Mat. (2019) 1:407–416.
3. Dianbo Zhang, Rui Yin, Yanjun Zheng, Qianming Li, Hu Liu, Chuntai Liu a, Changyu She, Chemical Engineering Journal 438 (2022) 135587.

Enhanced Valley Polarization and Valley Lifetime in Fe_3O_4/MoS_2 Heterostructure via Magnetic Proximity

Laxmi Saran¹, Nagendra S Kamath¹, Suman Kalyan Pal¹

Indian Institute of Technology Mandi, Mandi, Himachal Pradesh 175005, India

*Corresponding author: saranlaxmisaran@gmail.com

Two-dimensional transition metal dichalcogenides (TMDs) exhibit coupled spin and valley degrees of freedom, making them promising candidates for valleytronic and spintronic applications. However, rapid valley depolarization and limited control over valley lifetimes remain major challenges in pristine TMDs such as MoX_2 and WX_2 (where X = S, Se, Te). Several approaches, including strain engineering, electrostatic gating, defect engineering, and magnetic proximity coupling through magnetic material/TMD heterostructures, have been explored to manipulate valley dynamics [1,2]. In particular, magnetic proximity coupling has emerged as a powerful route to control spin-valley physics through interfacial exchange interactions [3]. In this work, we explore Fe_3O_4/MoS_2 heterostructure as a platform for tuning valley properties using helicity-resolved ultrafast spectroscopy. We demonstrate an enhanced valley degree of polarization and longer valley depolarization time due to magnetic material heterostructure with 2D TMDs. Magnetic proximity from Fe_3O_4 could induce exchange fields that may suppress valley depolarization through valley splitting, leading to longer depolarization time in the Fe_3O_4/MoS_2 heterostructure. This work highlights magnetic heterostructures as an effective strategy for controlling valley physics toward next-generation valleytronic devices.

Keywords: Valleytronic; Spintronic; Magnetic Proximity

Reference:

1. Nagendra S. Kamath, Ashish Soni and Suman Kalyan Pal, J. Phys. Chem. Lett., 16 (1), 25 (2025).
2. Nagendra S. Kamath, A. Soni, D. Maity and S. K. Pal, J. Phys. Chem. C, 130, 4504 (2026).
3. J. Li, Z. Chen and J. Kang, ACS Nano, 18, 10921 (2024).

Exciton-phonon coupling in few-layer 2D $CrCl_3$

Hien Giap¹, Carlo Sta. Maria¹, Chien-Chih Lai¹, Yuan-Ron Ma^{1,2,3*}

¹*Department of Physics, National Dong Hwa University, Hualien 974301, Taiwan*

²*Department of Applied Informatics, Fo Guang University, Yilan 262307, Taiwan*

³*Office of Postgraduate Studies, UCSI University, Kuala Lumpur 56000, Malaysia*

*Corresponding Author E-mail: ronma@gms.ndhu.edu.tw

Chromium chloride ($CrCl_3$) is a two-dimensional (2D) magnetic semiconductor with potential applications in nanoscale magnetic devices. Most works focus on the magnetic properties of $CrCl_3$, and there few studies on its layer-dependent optical properties [1]. In this work, cryogenic photoluminescence (PL) spectroscopy was used to investigate the temperature-dependent peak broadening of the few-layer $CrCl_3$ flakes. The PL spectra of the 2D $CrCl_3$ has one

peak that is centered at 1.45 eV, which exhibits peak broadening when heated from 80 K up to 300 K. All PL emission peak from 74L down to bilayer CrCl_3 exhibit this peak broadening, however, the bilayer flake shows a larger overall change in full width at half maximum (FWHM). When fitted using the exciton-phonon coupling model, a noticeable pattern is observed in which the exciton-phonon coupling strength is larger in the bilayer and then decreases with increasing thickness. In contrast, the temperature-independent inhomogeneous broadening parameter is proportional to the thickness. These results suggest that the lattice vibrations cause a large broadening in the bilayer CrCl_3 , and when suppressed at low temperatures, results in an overall sharper PL emission.

Keywords: CrCl_3 , exciton-phonon coupling, photoluminescence, 2D materials

References:

1. D.R. Klein, D. MacNeill, Q.L. Song, D. T. Fang, S.X. Fang, R.A. Ribeiro, P.C. Canfield, E. Kaxiras, R. Comin, Nat. Phys. 15, (2019) 1255-1260.
2. W. Jin, H.H. Kim, Z. Ye, G. Ye, L. Rojas, X. Luo, B. Yang, F. Yin, J.S.A. Horng, S. Tian, Y. Fu, G. Xu, H. Deng, H. Lei, A.W. Tsen, K. Sun, R. He, L. Zhao, Nat. Commun. 11, (2020), 4780.

Anisotropic light matter interaction in 2D $\alpha\text{-In}_2\text{Se}_3$

Divya Jangra¹, Binoy Krishna de^{1,2}, Pragati sharma¹, Mayanak gupta³, Ranjan mittal³, Pavan Nukala², Arvind k yogi¹, Praveen Velpula¹, Vasant G. Sathe^{1,*}

¹UGC-DAE Consortium for Scientific Research, University Campus, Khandwa Road, Indore-452001,

²Centre for Nano Science and Engineering, Indian Institute of Science, Bangalore-560012, India

³Solid State Physics Division, Bhabha Atomic Research Centre, Mumbai 400085, India

*vasant@csr.res.in

Anisotropic light-matter interactions in 2D materials have attracted significant attention due to their potential for polarisation-sensitive optoelectronic applications, including photodetectors and photo-actuators, etc. [1]. Due to their intrinsically anisotropic crystal structures, 2D materials are considered to be promising candidates for light-polarization-based device applications. It is generally believed that the crystal structure decides the optical anisotropy. In this work, we have studied the polarisation-dependent light-matter interaction in ferroelectric $\alpha\text{-In}_2\text{Se}_3$ with a 3R structure using Angle-Resolved Polarised Raman Spectroscopy (ARPRS) and show that the light-matter interaction depends not only on the anisotropic crystal structure but also on the excitation wavelength used. Polar plots showing the intensity of A_1 Raman mode as a function of the azimuthal angle, measured with two excitation lasers of 473nm and 633 nm are shown in Figure 1 (a) and (b), respectively. The experimental results are supported by Density Functional Theory calculations [2]. Scanning transmission electron microscopy (STEM) analysis revealed that the crystal structure is highly anisotropic along the c -direction. This structural anisotropy, along with the complex interplay of electron-photon and electron-phonon interactions, enforces the optical anisotropy, which is also reflected in complex Raman tensor elements. These anisotropic light matter interactions lead to an anisotropic electrical

response when illuminated by different wavelengths of light, 473 nm and 633nm (shown in Figures 1 (c) and (d)). Interestingly, the photo-response anisotropy can be controlled by light polarisation as well as the wavelength of the light, which opens a new dimension in its usage in polarisation-sensitive optoelectronic applications.

Keywords: Light matter interaction; Optical response; Polarised Raman spectroscopy; Photodetection;

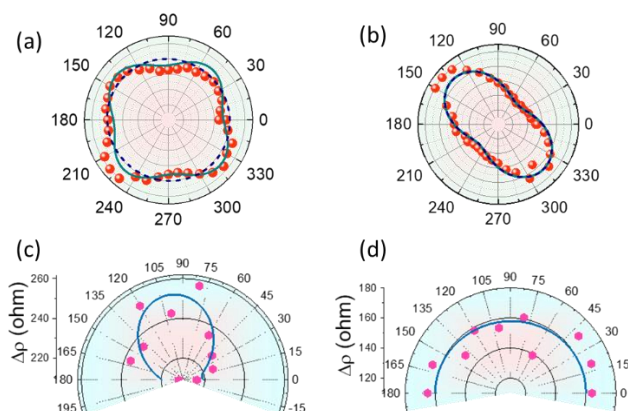


Figure 1: Polar plots of the intensity of an A_1 Raman mode (104 cm^{-1}) (a) and (b) and Photo-resistance (c) and (d) with respect to crystallographic orientation θ obtained using 473nm and 633nm excitations.

References:

1. L. Li, P. Gong, D. Sheng, S. Wang, W. Wang, X. Zhu, X. Shi, F. Wang, W. Han, S. Yang, K. Liu, H. Li, and T. Zhai, *Advanced Materials*, 30 (50), 2018, 1804541
2. D. Jangra, B. K. De, P. Sharma, K. Chakraborty, S. Parate, A. K. Yogi, R. Mittal, M.K. Gupta, P. Nukala, P. K. Velpula, and V. G. Sathe, *ACS Appl. Mater. Interfaces* 17,15, (2025), 22903–22911.

Machine Learning-Accelerated Design of Near-High-Entropy O3-type Low-Strain Cathodes for Sodium-Ion Batteries

Novotna Seal^{1,2} and Amreen Bano^{1,2*}.

¹Research Institute for Sustainable Energy (RISE), TCG Centres for Research and Education in Science and Technology (TCG-CREST), Kolkata 700091, India,

² Academy of Scientific and Innovative Research (AcSIR), Ghaziabad 201002, India

*Corresponding Author E-mail: amreen.bano@tcgcrest.org

Sodium-ion batteries (SIBs) are promising low-cost alternatives to lithium-ion batteries; however, the structural instability of Ni/Co-free sodium layered oxide (SLO) cathodes remains a major challenge. In this work, pristine CFMO ($\text{Na}_{0.95}\text{Mg}_{0.05}\text{Cu}_{0.20}\text{Fe}_{0.35}\text{Mn}_{0.40}\text{O}_2$) [1] was used as the parent composition to design near-high-entropy (NHE) cathodes through hierarchical Ti, Li, and F doping, using MACE [2] MatPES r2SCAN, Machine Learning Interatomic Potentials (MLIPs) [3]. The NHE strategy introduces controlled compositional complexity, enabling improved structural stability without excessive disorder associated with fully high-entropy

systems. The designed NHE compositions were systematically evaluated using an MLIP-assisted DFT framework to investigate structural stability, electronic properties, redox behaviour, voltage profiles, and Na^+ migration kinetics. Compared to pristine CFMO and the experimentally reported HECFMO cathode $\text{Na}_{0.95}\text{Li}_{0.05}\text{Mg}_{0.05}\text{Cu}_{0.20}\text{Fe}_{0.22}\text{Mn}_{0.35}\text{Ti}_{0.13}\text{O}_2$ (denoted here as HE_4), the NHE compositions exhibit enhanced structural stability during charging and partially suppress the detrimental O3-P3 phase transition. The NHE cathodes exhibit moderate interlayer spacing (5.14–5.17 Å), consistent with experimentally reported HE-CFMO system with moderate interlayer spacing ($d = 0.54$ nm) that demonstrated significantly improved cycling stability and enhanced Na^+ diffusion kinetics compared to pristine CFMO ($d = 0.55$ nm). Furthermore, the simulated (003) peak shift for NHE_1 (0.01°) was significantly lower than pristine CFMO (0.71°) and HE_4 (1.00°), indicating reduced structural distortion and improved phase stability, consistent with experimentally reported HE-CFMO behaviour. Li substitution within the transition metal layers further improves theoretical capacity while mitigating Jahn-Teller distortion, highlighting the effectiveness of the NHE design strategy for developing high-capacity, structurally robust, and sustainable cathodes for next-generation sodium-ion batteries.

Keywords: Na layered cathodes; High entropy cathodes; Machine learning; anion redox; O3 type cathodes;

References:

1. Z. Huang, S. Wang, X. Guo, Adv. Mater, 36, (2024), 2410857.
2. Batatia, D. P. Kovács, G. N. C. Simm, arXiv, (2023), 2206.07697.
3. R. Jacobs, D. Morgan, S. Attarian, Current Opinion in Solid State and Materials Science, 35, (2025), 101214.

Correlation Between Growth Parameters, Morphology, and Defect Formation in CVD-grown MoS_2

Rohit Kumar Mahto* and Anupam Roy*#

*Department of Physics, Birla Institute of Technology, Mesra, Ranchi, India, 835215

#E-mail: royanupam@bitmesra.ac.in

The broad range of remarkable properties of semiconducting two-dimensional (2D) transition metal dichalcogenides (TMDs) makes them attractive for next-generation electronics and optoelectronics applications. Despite several favorable properties, the practical application of TMDs faces several challenges, primarily due to limitations of synthesis methods. Among different methods, chemical vapor deposition (CVD) has emerged as the predominant technique for synthesizing large-area monolayer TMD films. In this work, the effect of CVD growth parameters on the morphology of 2D molybdenum disulfide (MoS_2) was studied. It is observed that even modest variations in material flux and temperature can significantly alter the resulting film morphology, affecting domain shape, size, nucleation, and growth modes. In addition, this work emphasizes the role of these growth parameters in the formation of various types of defects, including vacancies, grain boundaries, screw-dislocation-driven spirals, the nucleation of MoS_2 nanodots along the domain edges, and triangular and hexagonal pits. These findings provide valuable insights into the influence of growth parameters for achieving more uniform and homogeneous 2D-TMDs using CVD.

Keywords: Transition metal dichalcogenides; MoS_2 ; CVD; defects.

Modulating the optoelectronic properties of MoS₂-WS₂ verticle heterostructure through different strain mechanisms: a DFT study

Shakti Singh¹, Sunil Verma^{1,2}

¹Photonic Materials Technology Section, Raja Ramanna Centre for Advanced Technology, Indore 452013, India

²Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai 400094, India

*Corresponding Author E-mail: shaktis@rrcat.gov.in

Van der Waals heterostructures (vdWH) are a fascinating class of layered materials already known for exhibiting strain-tunable optoelectronic properties. This topic has been revisited for the most versatile heterobilayered MoS₂-WS₂ 2D material. We have employed the vdW-corrected DFT (GGA-level) with truncation of the Coulomb interaction in the z-direction [1] that mimics a true 2D material in the xy plane to study the effects of strain on the electronic and optical properties. The ground state structure was thoroughly confirmed via phonon band structure, upon which different types of strain were applied: biaxial strain, while the effects of uniaxial strain were studied along orthogonal directions aligned with the zigzag and armchair axes. Additionally, the uniaxially strained structures were volume-relaxed in the unstrained direction together with ionic relaxation. This selective relaxation of the simulation cell also yields the Poisson's ratio of the material when strained along these two directions, which is found to be 0.20 along zigzag axis and 0.22 along armchair axis, reported for the first time and agreeing well with indirectly arrived experimental value for the vdWH [2]. From the calculated band structures, 8% biaxial tensile strain was found to close the band gap, resulting in a semiconductor-to-metal transition, whereas above a 2% biaxial compressive strain, the heterostructure shows a direct bandgap, which is highly sought after for its application in optoelectronic devices. The modulation rate for biaxial tension is 214 meV/% strain at small strain (1-2%), while it drops to 137 meV/% strain at large strain (6-7%). For uniaxial tension along the armchair as well as zigzag directions, the rates are similar (a constant 85 meV/% strain for 1-5% range) in the two directions which is also observed experimentally [3]. We observed a difference in modulation rates for uniaxial compression along zigzag (~20 meV/% strain) and armchair (~35 meV/% strain) directions. Overall, the band gap and various optical properties are tunable through strain, with biaxial strain offering coarse control and uniaxial strain providing finer tuning.

Keywords: heterostructure; MoS₂-WS₂; strain engineering; optoelectronic.

References:

1. T. Sohler, M. Calandra, F. Mauri, Physical Review B, 96, (2017), 075448.
2. K. Liu, Q. Yan, M. Chen, W. Fan, Y. Sun, J. Suh, D. Fu, S. Lee, J. Zhou, S. Tongay, J. Ji, J. B. Neaton, J. Wu, Nano Letters, 14, (2014), 5097-5103.
3. X. He, H. Li, Z. Zhu, Z. Dai, Y. Yang, P. Yang, Q. Zhang, P. Li, U. Schwingenschlogl, X. Zhang, Applied Physics Letters, 109, (2016), 173105.

Self-Powered Photodetection in a 2D MoS₂/MoSe₂ p-p Heterojunction with Enhanced Current Density

Lenin O. L.¹, Gowri Sankar², E S Kannan², Saidi Reddy Parne^{1,*}

¹Department of Applied Sciences, National Institute of Technology Goa, Goa 403703, India

²*Department of Physics, Birla Institute of Technology and Science Pilani K. K. Birla Goa Campus, Goa 403726, India*

*Correspondence: psreddy@nitgoa.ac.in

The development of self-powered photodetectors is crucial for next-generation low-power and autonomous electronic systems. In such devices, the short-circuit current (I_{sc}) is a key parameter, as it determines the photocurrent generated under zero external bias and directly influences zero-bias responsivity. Low-dimensional transition metal dichalcogenides (TMDCs) have emerged as promising materials for optoelectronic applications due to their atomically sharp interfaces, tunable band structures, and strong light–matter interaction. Although TMDC-based p–n homo- and heterojunctions have been widely explored, their zero-bias performance can be limited by interfacial recombination and inefficient carrier extraction under self-powered operation.

Here, we demonstrate a two-dimensional p–p heterojunction composed of p-type MoS₂ and p-type MoSe₂ for self-powered photodetection. The heterointerface enables favorable band alignment driven by work-function differences and interfacial electronic interactions, facilitating efficient carrier transport and extraction under zero external bias. The proposed p–p device delivers a short-circuit current density of 46.8 mA/cm², compared to 3.3 mA/cm² reported for a representative 2D MoS₂ p–n junction under similar illumination conditions^[1], corresponding to a substantial enhancement in zero-bias current generation. These results establish p–p heterostructure engineering as an effective strategy for achieving high current density and improved self-powered operation in 2D material-based optoelectronic devices.

Keywords: Self-powered photodetectors; p–p heterojunction; Transition metal dichalcogenides (TMDCs); Short-circuit current density (I_{sc}); MoS₂/MoSe₂ heterostructure.

References:

1. Gate tunable photovoltaic effect in MoS₂ vertical p–n homostructures, S. A. Svatek, E. Antolin, D. Lin, R. Frisenda, C. Reuter, A. J. Molina-Mendoza, M. Muñoz, N. Agrait, T. Ko, D. P. de Lara and A. Castellanos-Gomez, *J. Mater. Chem. C*, 2017, 5, 854

Interplay of Ripples, Strain, and Grain Boundaries on Thermal Conductivity of Polycrystalline Graphene

Abhishek Kumar¹, Amit Singh¹

¹*Department of Mechanical Engineering, IIT Bombay, Mumbai 400076, India*

*Corresponding Author E-mail: amit.k.singh@iitb.ac.in

Thermal transport in polycrystalline graphene is strongly governed by out-of-plane fluctuations (ripples) and grain boundaries (GBs), both of which are critical to its performance in nanoelectronics and thermal dissipation applications. Although applied strains are known to modulate ripple morphology [1], the coupled influence of strain, ripples, and GB structure on the thermal conductivity (TC) of polycrystalline graphene remains insufficiently understood. The present study addresses this gap by investigating these effects systematically across a wide strain range using large-scale polycrystalline models.

Equilibrium molecular dynamics simulations employing the Green–Kubo method [2] were performed on large-scale polycrystalline graphene models incorporating realistic grain boundaries and out-of-plane ripples under varying tensile strains. To isolate the intrinsic phonon response to strain — independent of ripple and GB disorder — spectral energy density (SED) data from ripple-free pristine graphene systems reported in our prior work are used as a reference baseline [3].

Two mechanistically distinct strain regimes, (0-3)% and (3-10)%, are identified for not only pristine graphene but also for graphene with grain boundaries. Regardless of grain boundary angle, ripple amplitudes become negligible at around 3% tensile strains. Between (0-3)% tensile strains, TC increases with respect to strain, due to two conflicting trends of decreasing ripple amplitudes and increasing strain. The decrease in ripples increases TC whereas the increase in tensile strain decreases TC because of strain-induced phonon softening. In the low-strain regime (0–3%), where ripples and strain coexist prominently, we successfully decouple ripple contributions to TC from those of applied strain.

Using the spectral energy density approach, it can be shown that TC for samples without ripples decreases by around 36% between (0-3)% strain [3]. Our analysis reveals that ripples alone suppress TC by nearly (50-60)% in unstrained pristine or polycrystalline graphene, establishing ripple-induced phonon scattering as the dominant mechanism of heat transport reduction in these systems. At higher tensile strains (3–10%), when samples are ripple-free, TC decreases by approximately 30%, driven by strain-induced phonon softening phenomenon. We also compare the effects of ripples and strains for different grain boundaries.

Keywords: Ripples, Grain boundaries, Graphene, Green-kubo, Spectral energy density

References:

1. K.H.Park and U.Ravaoli. *Journal of Applied Physics* 122, 025115 (2017).
2. A.Singh and E.B.Tadmor, *J. Comput. Phys.* 299, 667 (2015).
3. Abhishek Kumar, Kunwar Abhikern, and Amit Singh. *Journal of Applied Physics* 137.13 (2025).

Impact of Different Metal Precursors on APCVD Growth of α -MoO₃

Vedant Chavan¹, Ayushi Tyagi¹, Tanushree Choudhury¹

¹ *Indian Institute of Technology Bombay*

Corresponding Author E-mail: tanuhc@iitb.ac.in

Molybdenum trioxide (α -MoO₃) is a layered transition metal oxide possessing a wide bandgap (~3.0–3.2 eV), high work function, and strong in-plane anisotropy, making it attractive for optoelectronic devices, sensing applications, electrochromic systems, and low-dimensional electronics. Its van der Waals layered structure also enables mechanical exfoliation and integration into thin-film device platforms. Despite these advantages, achieving localized and stoichiometric MoO₃ growth through vapor-phase methods remains challenging because of competing gas-phase nucleation, precursor decomposition behavior, and the strong influence of substrate surface properties and thermal gradients. Atmospheric pressure chemical vapor deposition (APCVD) offers a scalable route for crystalline oxide synthesis; however, understanding the influence of precursor chemistry on growth behavior remains essential for obtaining high-quality α -MoO₃ films.

Growth experiments were performed in a horizontal APCVD system at a furnace temperature of 700 °C under an oxygen flow rate of 100 sccm using ammonium molybdate tetrahydrate (AMT), MoO₃ powder, and molybdenum hexacarbonyl (Mo(CO)₆) as precursor sources. Identical growth conditions including precursor loading, precursor position, substrate placement, and gas flow rate were maintained for all precursors to comparatively study precursor-dependent growth behavior. Under these conditions, crystal formation was predominantly observed through homogeneous vapor-phase nucleation followed by deposition on the inner walls of the quartz reactor tube. Significant variations in crystal yield were observed, with AMT producing the highest yield followed by MoO₃ and Mo(CO)₆. Annealed c-plane sapphire substrates placed downstream exhibited finger-like crystal structures for AMT and MoO₃ precursors, indicating comparatively favorable growth behavior. Raman measurements further confirmed MoO_{3-x} formation on these

substrates, suggesting enhanced substrate-mediated nucleation and oxygen non-stoichiometry. In contrast, such structures were not observed for the Mo(CO)_6 precursor, likely due to earlier precursor decomposition and higher volatility during growth.

Raman spectroscopy confirmed the formation of $\alpha\text{-MoO}_3$ for all precursor systems, while notable differences in defect-related Raman features were observed across precursors. The intensity ratio of the peaks near 281.5 cm^{-1} and 335.4 cm^{-1} showed significant variation, indicating precursor-dependent defect density and crystalline quality. Among the investigated precursors, AMT exhibited the lowest defect-related intensity ratio and comparatively cleaner Raman features, whereas Mo(CO)_6 showed the highest defect-associated contribution. X-ray photoelectron spectroscopy (XPS) measurements revealed the presence of both Mo^{6+} and Mo^{5+} states in all samples, indicating oxygen vacancy-related sub-oxide formation. The Mo(CO)_6 -grown samples additionally showed broader carbon-related features and relatively stronger substrate-associated signals.

Based on the comparatively higher crystal yield obtained using AMT, a temperature-dependent growth study was further performed between $500\text{--}900\text{ }^\circ\text{C}$ to optimize crystal growth conditions. The largest lateral crystal dimensions were obtained near $700\text{ }^\circ\text{C}$, suggesting an optimum balance between precursor vaporization and surface diffusion kinetics. Atomic force microscopy measurements showed crystal thicknesses in the range of $\sim 100\text{--}500\text{ nm}$. Mechanically exfoliated flakes transferred onto Si/SiO_2 substrates were subsequently used for electrical characterization. Initial current–voltage measurements showed nonlinear transport behavior consistent with trap-assisted conduction. Further exfoliation to reduced flake thicknesses along with improved contact fabrication resulted in nearly linear $I\text{--}V$ characteristics with minimal hysteresis, indicating improved charge transport behavior.

Overall, the study demonstrates that precursor selection strongly influences the growth behavior, defect formation, stoichiometry, and electronic properties of APCVD-grown $\alpha\text{-MoO}_3$ thin films. The findings provide insight into precursor-dependent growth mechanisms and offer guidance for optimizing vapor-phase synthesis of layered MoO_3 for future electronic and optoelectronic applications.

Keywords: $\alpha\text{-MoO}_3$; layered transition metal oxide; APCVD; Different Precursors.

References:

1. M. Dhanasankar, K.K. Purushothaman, G. Muralidharan, *Solid State Sciences* 12 (2010) 246–251
2. L. Seguin, M. Figlarz, R. Cavagnat, J. C. Lassegues, *Spectrochimica Acta Part A* 51 (1995) 1323–1344
3. Reetu Sharma, Ranjana Jha, Anjana Sarkar, Amit Kumar Sharma, Darshan Sharmaa, Medha Bhushana, Rekha Bhardwaj, *Ceramics International* 46 (2020) 23084–23097

Synthesis of Layered Two-dimensional Vanadium Disulfide (VS_2) by Chemical Vapor Deposition for Optoelectronic Applications

Ramesh Rajarapu¹, and Tanushree H. Choudhury¹

¹*Metallurgical Engineering and Materials Science, IIT Bombay, Powai, India*

*Corresponding Author E-mail: (tanuhc@iitb.ac.in)

VS_2 is a layered two-dimensional (2D) transition metal dichalcogenide (TMDC) that exhibits excellent optical, electrical, catalytic, and magnetic properties and has emerged as a promising material for next generation nonlinear optoelectronic applications.^{1,2} However, phase-controlled and high-quality VS_2 remains challenging because of existence of sub stoichiometric phases in the vanadium-sulfur system such as V_2S_3 , V_3S_4 and V_5S_8 and the narrow thermodynamic

growth window.³ Here, we report a two step atmospheric pressure chemical vapor deposition (APCVD) process to investigate the synthesis mechanism of phase pure-VS₂ using V₂O₅ and sulfur precursors. In the first step, V₂O₅ was reacted under sulfur vapor in a forming gas (Ar/H₂: 95:5) ambient, resulting predominantly in the formation of VO₂ rather than direct conversion of VS₂. In order to develop phase-pure VS₂, in the subsequent step, the as-grown VO₂ based substrate was further sulfurized in an Ar ambient. During this process, the intermediate VO₂ film undergoes sulfur-assisted volatilization, producing volatile vanadium species that are transported downstream and reacted with the sulfur vapor, leading to the deposition of VS₂ on the c-sapphire substrate. This process results in the formation of triangular VS₂ domains with the sizes of ~ 1 μm. These results suggest that the intermediate VO₂ phase may play a vital role in facilitating the nucleation and growth of VS₂. Further, Raman measurement confirms the distinctive first order Raman peaks of VO₂ and VS₂, optical and scanning electron microscopy show the domain morphology, and AFM validates the VS₂ domains thickness. Therefore, this study shows a feasible route for the scalable synthesis of layered and phase pure VS₂ for the development of optoelectronic applications.

Keywords: 2D materials; VS₂; CVD; Optoelectronic

References:

1. Su, J.; Wang, M.; Li, Y.; Wang, F.; Chen, Q.; Luo, P.; Han, J.; Wang, S.; Li, H.; Zhai, T. *Adv. Funct. Mater.* 2020, 30 (17), 1–9.
2. Yuan, J.; Wu, J.; Hardy, W. J.; Loya, P.; Lou, M.; Yang, Y.; Najmaei, S.; Jiang, M.; Qin, F.; Keyshar, K.; Ji, H.; Gao, W.; Bao, J.; Kono, J.; Natelson, D.; Ajayan, P. M.; Lou, J. *Adv. Mater.* 2015, 27 (37), 5605–5609.
3. Pandey, A. K.; Mishra, A. K. *J. Cryst. Growth* 2026, 678 (December 2025), 128483.

Oxygen-Tuned MoS₂ Thin Film: From Surface Chemistry to Humidity Sensing Performance

Megha Mishra¹, Teny Theresa John^{1*}

¹*Department of Physics, Birla Institute of Technology and Science, Pilani K K Birla Goa Campus, Zuarinagar, Sancoale, Goa-403726*

*Corresponding Author E-mail: teny@goa.bits-pilani.ac.in

Engineering humidity-sensitive materials capable of maintaining reliable charge transport across widely varying adsorption regimes remains a central challenge in humidity sensing. In this work, oxygen-modified MoS₂ (MSO) thin films were synthesized by thermal evaporation. Structural, morphological, and chemical characterization was carried out using Raman spectroscopy, atomic force microscopy (AFM), scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), and X-ray photoelectron spectroscopy (XPS). The analyses confirmed retention of the layered β-2H-MoS₂ structure together with the formation of an oxygen-rich hydrophilic surface. The films exhibited a mixed morphology consisting of nanoparticles, stacked nanosheets, and flower-like structures, leading to increased surface roughness and a larger number of adsorption sites for water molecules. The MSO thin films exhibited stable and reproducible humidity sensing over a wide relative humidity (RH) range of 10%-95% RH. Figure 1 shows the variation in current response of the MSO thin film sensor at different RH levels. The inset shows the transient response of the sensor for 55% RH with 50% RH used as the baseline condition. A humidity response greater than 10⁴ was observed in the 50%-95% RH range, while the inferred conductance variation across the complete 10%-95% RH range approached 10⁶. The sensor also showed response and recovery times of 30 s and 20 s, respectively in the high humidity regime, along with stable sensing behaviour in the low-humidity

region as well. The enhanced sensing performance is attributed to oxygen-induced surface states and the mixed hierarchical morphology, which together improve water adsorption and charge transport. At lower humidity levels, proton hopping through adsorbed water molecules dominates the sensing process, whereas ionic conduction through continuous water layers becomes significant at higher humidity levels. The results demonstrate that oxygen incorporation and morphology tuning are effective approaches for improving the humidity sensing performance of thermally evaporated MoS₂ thin films.

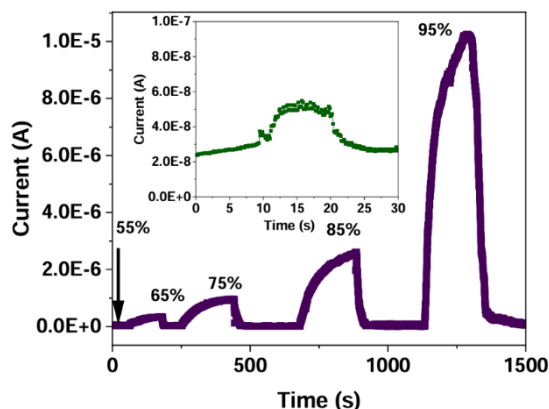


Figure 1: The variation in current response of the MSO thin film sensor at different RH levels. The inset shows the transient response of the sensor for 55% RH with 50% RH used as the baseline condition.

Keywords: Oxygen-modified MoS₂; Wide range detection; X-ray photoelectron spectroscopy.

Synthesis and characterization of PDMS based Ru-Si membrane by spin coating method for dissolved oxygen sensor application

Gangadhar Hattale ^{1,2}, Rutuja Kadam ¹, Onkar Dube ^{1,3}, Vaibhav Varpe ¹, Shubhankar Kane ¹, Ashish Yengantiwar ², Rajesh Kanawade ^{1,3*}.

¹Physical and Materials Chemistry Division, CSIR-National Chemical Laboratory, Pune 411008, India

²Department of Physics, Fergusson College (Autonomous), Pune, Affiliated to Savitribai Phule Pune University, Pune 411004, India

³Academy of Scientific and Innovative Research (AcSIR), Ghaziabad- 201002, India

*Corresponding Author E-mail: rv.kanawade.ncl@csir.res.in

With growing concern about climate change, the monitoring of dissolved oxygen (DO) parameters become one of the sensitive parameters. Deviation in oxygen concentration levels at different depths from its preferred range can have serious consequences for the functioning of marine ecosystems. Luminescent membranes are widely used in monitoring DO concentration for sensing, optimization, calibration, and validation. Therefore, we aim to prepare and characterize the luminescent Polydimethylsiloxane (PDMS) based Ru-Si membrane having DO sensing properties. In this work, DO measurement setup was developed, and the fabricated sensing membranes were evaluated over the full DO concentration range using the intensity-based detection method, calibration, repeatability, stability, effect of temperature on DO and effect of temperature-salinity on DO were measured. The effort of this study is to provide an alternative

approach to synthesize and characterize the PDMS based Ru-Si membrane by spin coating method which is used for DO sensor applications[1].

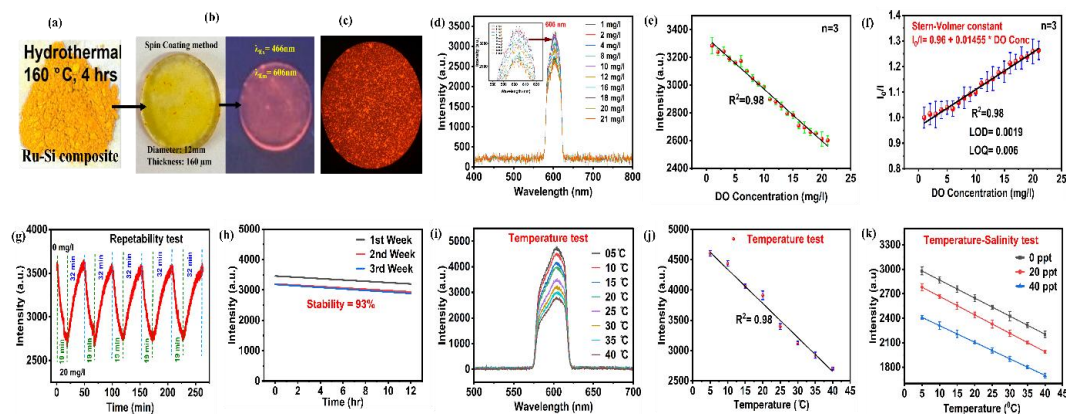


Figure 1. (a-d) Synthesis of Ru-Si composite by hydrothermal method and PDMS based Ru-Si membrane synthesis by spin-coating method on sapphire window and microscopy image; (d,e) full range DO testing by intensity method on developed in-house sensor and its analysis, respectively; (f) calibration curve with varying DO concentrations; (g) Repeatability test; (h) Stability test for three continuous weeks; (i,j) Effect of temperature on PDMS based Ru-Si membrane; (k) Effect of temperature and salinity on PDMS based Ru-Si membrane.

Keywords: Optical DO sensor; Spin coating; Intensity method; PDMS based Ru-Si membrane.

References:

1. Y. Zhao, X. Liu, Y. Jiang, Spectrochimica Acta Part A: Molecular and Biomolecular spectroscopy, 261 (2021) 120003.

Carbon Surface-Engineered CuO/Fe₂O₃ nano-composite as efficient photocatalyst

Purnima Patil,^{1,2} Aarti Gunjal,¹ Niladri Ballav,³ Suparna Saha,³ Vikram Pandit^{2*} and Satishchandra Ogale^{1,3*}

¹ Center for Energy Science, Indian Institute of Science Education and Research, Pune-411008, India

² Department of Chemistry, H. V. Desai College, Pune 411002, India

³ Research Institute for Sustainable Energy (TCG-CREST) Kolkata-700091, India

*Corresponding Author E-mail: satishogale@gmail.com

A composite comprised of earth abundant CuO and Fe₂O₃ is synthesized by a simple ball milling process followed by annealing at moderate temperature. The development of the interface phase as a function of ball milling time and high temperature annealing treatment is studied by using a number of techniques such as X-ray Diffraction, X-ray photoelectron spectroscopy, Raman Spectroscopy etc. The CuO, Fe₂O₃, and CuO/Fe₂O₃ composites are then subjected to ball milling in the presence of a small quantity of graphite, urea, or g-C₃N₄ for surface layer incorporation of carbon or carbon and nitrogen dopants, respectively. The samples are then oxidized for several hours to get rid of remaining carbon or g-C₃N₄ residues which interfere with

the photocatalytic process by strong dye adsorption. The photocatalytic properties of the materials are then examined by standard protocols using different dyes. It is found that these materials with surface carbonization exhibit impressive catalytic performance. DFT calculations are performed to understand the band alignments and charge transfer effects.

Keyword: Earth abundant, Carbon, doping, Photocatalysis, DFT

References:

1. A. Taravati, A. Poursattar Marjani, S. Bibak, and H. Sarreshtehdar Aslaheh, *Scientific Reports*, 15, (2025), 36817
2. R. Bhosale, S. Jian, C. P. Vinod, S. Kumar, and S. Ogale, *ACS Applied Materials & Interfaces* 11, (2019), 6174-6183

Valence Band Engineering in Mg-Doped α -GaCrO₃ for Efficient Electron-Blocking in Photovoltaic Devices

Rishav Sharma¹, V. Srihari², Ashok Kumar Bhakar³, Mangla Nand⁴, Dinesh Kumar Shukla⁵, Rupesh Devan^{1*}, and Ravindra Jangir^{3*}.

¹Indian Institute of Technology, Indore, Madhya Pradesh-453552, India. Accelerator Physics and ²Synchrotrons Utilization Division,

³High Pressure and Synchrotron Radiation Physics Division, Bhabha Atomic Research Center, Mumbai, India

⁴Indus Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Indore- 452013 (M.P), India.

⁵BARC Beamlines Section, Raja Ramanna Centre for Advanced Technology, Indore- 452013 (M.P), India.

UGC-DAE Consortium for Scientific Research, Indore 452001, India

*Corresponding author: ravindrajangir@rrcat.gov.in, rupesh@iiti.ac.in

The stability of organic and perovskite solar cells highly depends on the interfacial layer that enables selective charge transport while suppressing recombination losses. Commonly used organic hole-transport/electron-blocking layers suffer from thermal stability, moisture sensitivity, and complex doping processes. Here, we report valence band engineering in Mg-doped α -GaCrO₃ as a robust inorganic alternative for electron-blocking applications. Polycrystalline α -(Mg_xGa_{0.8-x})Cr_{1.2}O₃ ($0 \leq x \leq 0.02$) was synthesised using a solid-state synthesis route. The solubility of Mg in α -GaCrO₃ is investigated using synchrotron-based X-ray diffraction (SXRD) measurements and found to be prominent up to $x = 0.01$. Mg incorporation leads to a significant reduction in resistivity ($\approx 10^3$) and promotes enhanced hole delocalization, as revealed by impedance spectroscopy. Comparative photoelectron spectroscopy demonstrates that Ga incorporation induces strong Cr 3d-O 2p hybridization, resulting in modified valence band structure and improved charge transport relative to Mg-doped α -Cr₂O₃. Seebeck measurements confirm p-type conduction with a high Seebeck coefficient ($\sim 449 \mu\text{V K}^{-1}$). Ultraviolet photoelectron spectroscopy reveals an ionization energy of $\sim 5.2 \pm 0.2$ eV and a conduction band minima level establishes a large electron-blocking barrier of $\sim 3.0 \pm 0.2$ eV relative to typical perovskite and organic absorbers. Combined with a wide band gap (~ 3.95 eV), these results demonstrate Mg-doped α -GaCrO₃ as a promising wide-bandgap p-type material for efficient and stable electron-blocking layers for next-generation photovoltaic devices.

Harnessing electric-field-driven interfacial charge modulation to control exchange bias in complex oxides

Shivam Choudhary¹, Rajan Mishra¹, Sourav Chowdhury², Sumit Sarkar³, Ram Janay Choudhary^{1*}

UGC-DAE Consortium for Scientific Research, Indore 452001, India
Deutsches Elektron-Synchrotron Desy, Notkestrasse 85, 22607, Hamburg, Germany
Department of Physics and Astronomy University of Waterloo, Waterloo, Canada
 ramjanay@gmail.com

The realization and control of exchange bias are highly desirable for energy-efficient spintronic applications. Heterostructures composed of ferromagnetic (FM) and antiferromagnetic (AFM) oxides provide an ideal platform for studying exchange bias and its electric-field control. In this work, epitaxial bilayers of AFM GdBaCo₂O_{5.5} (GBCO) and FM La_{0.7}Sr_{0.3}MnO₃ (LSMO) were grown on conducting Nb-doped SrTiO₃ (001) substrates using pulsed laser deposition (PLD). Magnetometry reveals an exchange-bias effect in the LSMO/GBCO heterostructure, evidenced by the shift of the field-cooled M–H hysteresis loop with an exchange-bias field of about 225 Oe at 5 K. X-ray spectroscopic measurements further confirm strong FM/AFM interfacial charge coupling. GBCO is an oxygen sponge material, where oxygen vacancy modulation strongly influences its electronic and magnetic states, which can be controlled via electric bias. We observe a systematic tuning of the exchange-bias strength by applying an electric bias across the heterostructure, indicating electric-field control of the interfacial magnetic interaction. The observed modulation is attributed to bias-driven oxygen vacancy redistribution, interfacial charge transfer, and orbital reconstruction, which alter the exchange coupling at the interface, as evidenced by the in-operando bias-dependent synchrotron X-ray spectroscopy. These results demonstrate electrically controllable exchange bias in complex oxide heterostructures and provide a pathway toward low-power, low-cost spintronic devices.

Keywords: Exchange bias, Electric control of magnetism, Spintronics, Oxide heterostructures

References:

1. S. Chowdhury, et. Al. J. Phys. Chem. C 125, 17535–17547 (2021)
2. Tsukasa Katayama et. Al. Chem. Mater. 33, 5675–5680 (2021)
3. Tsukasa Katayama et. Al. Chem. Mater. 12, 1295–1300 (2023)

Isolation and Purification of Phycobiliprotein from Nostochopsis Lobatus HKAR-21

Ayushi Kumari¹, Yogita Rani¹, Km Mamata Patel¹, Saurabh Chaubey¹, Payel Rani², Prabhat Tripathi^{1*}

¹Department of Chemistry, Indian Institute of Technology (Banaras Hindu University), Varanasi 221005

²Department of Botany, Banaras Hindu University, Varanasi 221005

*Corresponding Author Email: prabhat.chy@itbhu.ac.in

The photosynthetic apparatus of cyanobacteria comprises a light-harvesting antenna complex called phycobilisomes, consisting of different types of phycobiliprotein (PC, APC and PE). A naturally occurring blue pigment-protein found in cyanobacteria as phycobiliprotein, called phycocyanin, has garnered significant attention due to its diverse biotechnological applications, including its reported anticancer and anti-inflammatory properties. The commercial viability of phycocyanin is heavily dependent on its purity and stability, necessitating efficient extraction and purification methods to maximise its utility. Our goals here are to extract the phycocyanin from cyanobacteria and reconstitute it in a liposome. The extraction process generally involves cell wall disruption (sonication and freezing - thawing), removal of water-soluble colouring protein, and concentration and purification steps. We are exploring various methods for phycocyanin purification, which often involves ammonium sulphate precipitation, dialysis and sucrose density gradient to achieve an initial purity index, followed by advanced chromatography techniques. We found a UV absorption peak at 620 nm in the purified phycocyanin and distinctive peaks of fluorescence emission at 645 nm and excitation at 620 nm. FTIR spectral analysis confirmed the presence of characteristic functional groups associated with phycocyanin. We will present the circular dichroism (CD) spectroscopy of the purified protein to understand if it possesses an α -helical secondary structure. Because the phycobilisome is very luminous, it can be encapsulated in liposomes along with phycocyanin, for further characterisation using UV, fluorescence spectra, gel electrophoresis, TEM and DLS.

Keywords: Cyanobacteria, Isolation and Purification, Phycocyanin, Characterisation

References:

1. Jaiswal, J.; Singh, A. P.; Sinha, R. P., *Anal. Biochem.* 2025, 706, 115953.
2. Kumar, D.; Dhar, D. W.; Pabbi, S.; Kumar, N.; Walia, S., *Indian J. Plant Physiol.* 2014, 19 (2), 184–188.
3. Kissoudi, M.; Sarakatsianos, I.; Samanidou, V., *J. Sep. Sci.* 2018, 41 (4), 975–981

Stabilization of Orthorhombic Ferroelectric Phase in HfO₂ thin film

Mangla Nand^{1,*}, Shilpa Tripathi^{1,2}, Yogesh Kumar¹, R. J chodhary³, S. K. Rai^{4,2}

¹*Beamline Development and Application Section, Bhabha Atomic Research Centre, Mumbai 400085,*

²*Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai 400094, India*

³*UGC-DAE Consortium for Scientific Research, Khandwa Road, Indore, Madhya Pradesh. 452017,*

⁴*Indus Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Madhya Pradesh, 452013*

*Corresponding Author E-mail: kappu001@gmail.com

The discovery of ferroelectricity in binary oxide HfO₂ thin films has renewed interest in further miniaturizing the non-volatile memory devices and advanced ferroelectric-based technologies beyond conventional complementary metal oxide semiconductor (CMOS) architectures [1]. The unique ferroelectric behaviour of HfO₂, combined with its exceptional scalability with CMOS technologies, facilitates the integration of ferroelectric functionality beyond conventional CMOS architectures. Although the switchable polar phase in HfO₂ has been demonstrated in ultrathin films (<10 nm) [2], stabilizing this polar phase remains a significant challenge due to its metastable nature. In this study we have successfully stabilized the ferroelectric orthorhombic phase in pulsed laser deposited HfO₂ thin films by varying the trivalent (Nd) doping concentration. The thickness of the film was determined using XRR and found to be

~13 nm. The structural and compositional investigation were carried out through the analysis of x-ray absorption near edge structure (XANES) spectra, x-ray diffraction (XRD) along with x-ray photoelectron spectroscopy (XPS). The structural analysis using XRD showed the film $\text{Hf}_{1-x}\text{Nd}_x\text{O}_2$ ($x = 0, 0.08, 0.12$) transform from monoclinic ($x=0$) to orthorhombic ($x=0.08$) to cubic ($x=0.12$) as Nd concentration increases in the film. The XRD analysis also revealed that the orthorhombic thin film $\text{Hf}_{0.92}\text{Nd}_{0.08}\text{O}_2$ exhibits a highly preferred orientation along the (010) direction in the out-of-plane geometry, as shown in Fig. 1. The XANES analysis of Hf L_3 edge shows that the Hf is in 2+ oxidation states in all the thin films. The XPS analysis also confirms the 2+ oxidation of Hf in all the thin films. The O 1s core level fitting shows that oxygen vacancy plays a major role in the stabilization of polar orthorhombic phase. This study provides significant insights into the stabilization of the ferroelectric phase in HfO_2 thin films, which is crucial for the development of next-generation non-volatile memory devices for advanced electronic applications.

Keywords: XANES; Polar Phase; Ferroelectric.

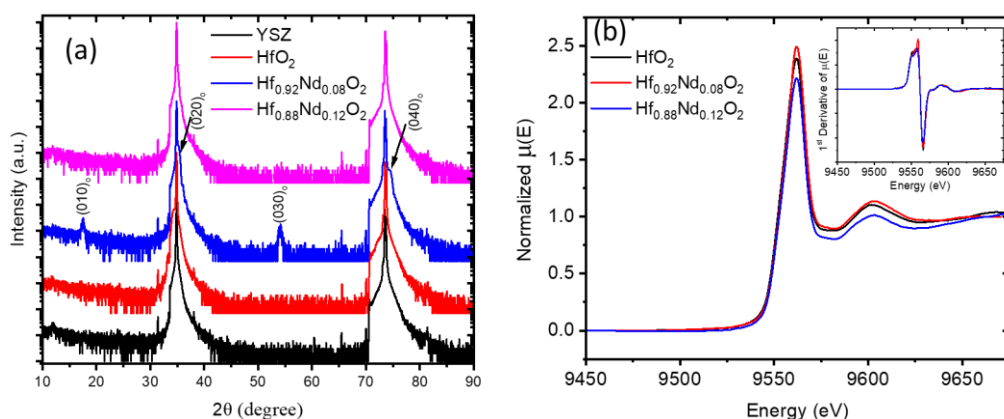


Fig. 1 (a) X-ray diffraction, (b) XANES spectra of Hf L_3 edge of $\text{Hf}_{1-x}\text{Nd}_x\text{O}_2$ ($x = 0, 0.08, 0.12$) films.

References:

1. K. Lee et al., ACS Nano, 18 (2024), 12707–12715.
2. J. Liao, S. Dai, R.-C. Peng et al., Fundamental Research 3 (2023) 332–345.

Ultrafast water flow in graphene oxide and its implications for environmental applications

Sangram Kishor Behera¹, Vikas Yadav¹, Anjan Das¹, Chitti Venkata Krishnamurthy^{1,*}, Manu Jaiswal^{1,*}

¹Department of Physics, Indian Institute of Technology Madras, Chennai-600036, India

*Corresponding Author E-mail: manu.jaiswal@iitm.ac.in

Graphene oxide (GO) membranes exhibit ultrafast water transport, with flow rates comparable to those through open apertures; however, the underlying mechanism remains unclear [1]. In this work, we combine transport and sorption studies across a wide humidity range and liquid water flow under pressure to develop a unified understanding of water dynamics in GO under confinement. The lamellar system of GO flakes shows strongly anisotropic diffusion, with in-plane transport is two orders of magnitude faster than out-of-plane transport. The role of

pinholes on the basal plane of GO in increasing the water transport rate is revealed. The diffusion coefficient of water inside GO was found to increase with interlayer spacing but remains lower than the self-diffusion coefficient of bulk water. Despite this, high permeation rates arise from the large sorption capacity of GO [2]. Sorption analysis further reveals coexistence of surface-bond and clustered water, with clustering dominating at high humidity. These results show that ultrafast transport arises from the interplay of sorption, confinement, and nanoscale pinhole structure in the GO membrane [3]. Our findings are useful towards environmental applications of GO in filtration and atmospheric water harvesting, both of which benefit from rapid kinetics of water in the respective regimes.

Keywords: Sorption; Permeation; Diffusion coefficient; Confinement; Pinholes.

References:

1. R. R. Nair, H. A. Wu, P.N. Jayaram, and A. K. Geim, *Science*, 335, (2012), 442-444.
2. V. Yadav, S. K. Behera, A. Das, C. V. Krishnamurthy, and M. Jaiswal, *Faraday Discussions*, Accepted manuscript, (2026), Article No. D5FD00121H.
3. V. Yadav, A. Das, S. K. Behera, C. V. Krishnamurthy, and M. Jaiswal, *Carbon*, 243, (2025), 120563.

Study of Dielectric Tunability of Polycrystalline Barium Strontium Titanate Thin Films.

Priyanshu Chandrakar¹, V. Raghavendra Reddy^{1*}

¹*UGC-DAE Consortium for Scientific Research, University Campus, Khandwa Road, Indore 452001, India*

*Corresponding Author E-mail: vrreddy@csr.res.in

Barium strontium titanate (BST) polycrystalline thin films have widely studied due to their excellent dielectric tunability and ferroelectric properties for advanced electronic device applications devices. In the present work, we explored the effects of DC bias field on polycrystalline BST thin films of different thickness deposited on p-Si substrates with pulsed laser deposition, as a function of temperature and frequency. The films are characterized using X-ray diffraction (XRD), Atomic Force Microscopy (AFM), Energy Dispersive X-ray (EDX) spectroscopy, Scanning electron microscopy (SEM) and temperature dependent dielectric measurement techniques. DC bias field-dependent dielectric measurements reveal field-induced response with considerable temperature dependent dielectric tunability, elucidating key information about domain wall motion and space charge contribution. The results provide important insights about the field-driven polarization dynamics in BST thin film systems. A thickness-dependent study was performed to understand the role of crystallite size on dielectric response and tunability behaviour. These results improve the understanding of polarization mechanisms in polycrystalline BST systems and their potential for tunable dielectric and ferroelectric device applications.

Keywords: Barium strontium titanate (BST); Dielectric tunability, Pulsed Laser Deposition

References:

1. S. P. Balmuchu, R. K. Gupta, and A. Kumar, *Journal of Alloys and Compounds*, 1040, (2025), 183574.
2. A. Selmi, A. E. Maayoufi, F. Jomni, and J.-C. Carru, *Journal of Materials Science: Materials in Electronics*, 36, (2025), 1796.

Structural & electrical studies of epitaxial LaNiO₃ thin film grown by RF magnetron sputtering

Dhananjay Moharana*, Hemant Singh, Balaganapathi T, D. K. Shukla

UGC-DAE Consortium for Scientific Research, Indore (452001), India

*Corresponding Author Email: djay@csr.res.in, djaym1610@gmail.com

Perovskite oxide thin films have attracted significant research interest owing to their remarkable structural tunability and intriguing charge, spin and lattice correlations, enabling diverse electronic and multifunctional device applications. Among these, LaNiO₃ (LNO) is a rare metallic perovskite oxide that exhibits paramagnetic metallic behaviour down to 500 mK and possesses a pseudocubic lattice parameter of approximately 3.838 Å, closely matching with many functional perovskite oxides [1]. Additionally, its high electrical conductivity and chemical stability, makes it an excellent choice of buffer layer and bottom electrode material. It also enhances epitaxial growth quality and improves ferroelectric fatigue endurance [2]. While pulsed laser deposition is commonly employed for epitaxial LNO growth, RF magnetron sputtering offers a cost-effective alternative.

In this work, we report the structural and electrical studies of epitaxial LaNiO₃ thin film on (001) oriented SrTiO₃ substrates using RF magnetron sputtering. Successful high-quality growth has been achieved through systematic optimization of deposition parameters including substrate temperature, working pressure, gas flow ratio, and target to substrate distance. Detailed structural characterization including phi-scans and high-quality reciprocal space mapping of specular and off-specular reflections were carried out using synchrotron X-ray diffraction at GIXS Beamline (BL-13), Indus-2, RRCAT. Reciprocal space mapping around the asymmetric (103) reflection reveals coherent epitaxial growth with the STO substrate, and in-plane lattice parameter of film layer being identical to substrate. Rocking curve measurements exhibit a narrow (0.0292 degree) full width at half maximum, signifying excellent crystalline quality and low mosaicity. Electrical transport measurements performed using a 14T QD PPMS demonstrate predominantly metallic behaviour over a broad temperature range, with resistivity decreasing upon cooling. A weak low-temperature resistivity upturn is observed below ~60K, suggesting the presence of weak localization or electron-electron interaction effects typical of correlated metallic oxides [3]. These results demonstrate that RF magnetron sputtering can yield device quality epitaxial LaNiO₃ thin films for using as bottom electrodes for complex oxide heterostructure devices for future applications.

Keywords: LaNiO₃, RF Magnetron Sputtering, Synchrotron XRD, RSM, Rocking Curve

References:

1. M. L. Medarde, J. Phys.: Condens. Matter 9, (1997), 1679.
2. K. M. Satyalakshmi, K. B. R. Varma, and M. S. Hegde, J. Appl. Phys. 78, (1995), 1160.
3. G. Herranz, F. Sanchez, J. Fontcuberta, V. Laukhin, J. Galibert, M. V. Garcia-Cuenca, C. Ferrater, and M. Varela, Phys. Rev. B 72, (2005), 014457.

Ni and Mg Doped α -AlCrO₃: Wide Band Gap P-type Transparent Conducting Oxides

Prabhat Kumar Singh^{1,2}, Dharavath Venkatesh⁵, Rishav Sharma³, Ashok Bhakhar^{1,2}, Mangla Nand², Rupesh Devan^{3,4*}, Ravindra Jangir^{1,2,*}

¹ *Indus Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Indore- 452013 (M.P), India.*

² *Homi Bhabha National Institute, Training school complex, Anushakti Nagar, Mumbai- 400094, India*

³ *Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Simrol, Indore 453552, India*

⁴ *Multidisciplinary Academic Research Centre, National Dong Hwa University, Hualien 974301, Taiwan*

⁵ *Centre For Nanoscience And Engineering, IISC Bangalore-560012, India.*

*Corresponding Author E-mail: ravindrajangir@rrcat.gov.in

P-type transparent conducting oxides (TCOs) are essential for the development of high-performance bipolar optoelectronic devices. Unlike their n-type counterparts, p-type TCOs typically exhibit significantly lower hole mobility and conductivity. This limitation stems from a "flat" valence band dominated by the highly electronegative oxygen 2p orbitals, which leads to heavy hole effective masses. To address this, Kawazoe proposed the chemical modulation of the valence band (CMVB)[1], a method that reduces valence band localization through the hybridization of metallic d-orbitals with O 2p orbitals. While CuAlO₂ was initially proposed as a p-type TCO based on this principle, it suffers from a narrow bandgap and strong d-d interactions that cause hole self-trapping, thereby limiting transparency and mobility. Consequently, chromium-based p-type TCOs with quasi-filled d-orbitals have emerged as a promising alternative [2-3]. In this work, we alloyed α -Cr₂O₃ with α -Al₂O₃ (α -Al_xCr_{1-x}O₃) to enhance the optical properties, achieving a tunable bandgap ranging from 3.5 to 5.1 eV. To improve electrical performance, the effects of Ni and Mg doping were investigated. Mg-doped α -AlCrO₃ demonstrated better conductivity compared to Ni-doped samples. The p-type nature of the conductivity was confirmed by positive Seebeck coefficients. Photoelectron spectroscopy revealed ionization energies of 5.7 eV for Ni-doping and 5.1 eV for Mg-doping. These energy levels align well with common active layers in perovskite solar cells, suggesting that these materials are highly suitable for use as hole transport layers (HTLs) in perovskite photovoltaic's and other bipolar optoelectronic applications.

Keywords: Wide-Band gap, P type TCO, Hole transport layer.

References:

1. Kawazoe, H.; Yasukawa, M.; Hyodo, H.; Kurita, M.; Yanagi, H.; Hosono, H. Nature 1997, 389 (6654), 939–942.
2. Nagarajan, R.; Duan, N.; Jayaraj, M.; Li, J.; Vanaja, K. A.; Yokochi, A.; Draeseke, A.; Tate, J.; Sleight, A. International Journal of Inorganic Materials 2001, 3, 265–270.
3. R. Jangir, V. Srihari, A.K. Bhakar, M. Nand, D.K. Shukla, S.N. Jha, T. Ganguli, ACS Appl. Energy Mater. 5 (2022) 8629–8638.

Reactive Glass–Metal Interaction as a Platform for Designing Hybrid Functional Interfaces

Sinorul Haque¹, Amarnath R. Allu^{1*}

¹CSIR-Central Glass and Ceramic Research Institute, Kolkata 700032, India.

*Corresponding Author E-mail: aareddy@cgcrici.res.in

Recent studies^[1] on reactive glass–metal systems have demonstrated that nominally inert glass substrates can actively participate in interfacial chemistry with deposited metals. H-glass (sodium-alumino-phospho-silicate glass) exhibits a unique ability to chemically and electronically interact with thermally dewetted Au nanoislands under ambient conditions at moderate temperatures (550 °C). This interaction leads to the incorporation and redistribution of glass-derived species such as Na, P, Al, Si, and O at the metal interface, resulting in multi-elemental hybrid nanoislands with modified surface energetics, lattice perturbations, and altered plasmonic response. Such reactive glass–metal coupling represents a departure from conventional inert-support behavior and establishes glass as an active participant in interfacial engineering. Building on this concept, the reactive glass surface can serve as a versatile platform for constructing hybrid functional systems. By introducing additional functional layers such as two-dimensional (2D) materials like molybdenum disulfide (MoS₂) or perovskites like materials like cesium bismuth iodide (CBI), the interfacial chemistry can be further tuned to regulate charge transfer, adsorption behavior, and carrier dynamics. This strategy enables rational design of multifunctional architectures for applications including sensing, catalysis, and optoelectronic devices.

Keywords: Glass-metal interaction; Hybrid nano-island;

References

1. S. Haque, S. R. Keshri, G. Gadudhula, K. Chatterjee, S. Majumdar, K. Biswas, S. Ganiseti, I. Mandal, D. A. Basha, P. Pal, P. K. Chowdhury, N. Joshi, S. Sappati, N. N. Gosvami, P. Nukala, E. Varrla, N. M. A. Krishnan and A. R. Allu, *Adv. Funct. Mater.*, 2025, e27669.

A study of quantum capacitance in defect engineered WS₂ monolayer

Snehal S Shinde¹, Appasaheb P Torane¹, Laxman D Kadam¹ and Bhilchandra S Pujari²

¹ Solid State Physics laboratory, Department of Physics, Yashwantrao Chavan Institute of Science, Karmaveer Bhaurao Patil University, Satara 415001, India

² Department of Scientific Computing Modelling and Simulation, Savitribai Phule Pune University, Pune – 411 007, INDIA

*Corresponding Author E-mail: bspujari@scms.unipune.ac.in

*Corresponding Author E-mail: kdilaxman@yahoo.com

Tungsten disulfide (WS₂) monolayers are promising materials for next-generation semiconductor and energy storage applications because of their atomic thickness, thermal stability, and favorable electronic properties. In this work, the effect of substitutional transition-metal defects in WS₂, created by replacing one W atom with a transition-metal atom, was investigated using first-principles calculations. The introduction of these defects generates localized electronic states near the Fermi level, which increases the density of states and enhances the intrinsic quantum capacitance. Defect-induced broadening of electronic states further influences the charge distribution and overall electronic behaviour of the system. Among, the investigated defect configurations, vanadium substitution shows the most stable and effective

performance. The V-substituted WS₂ system exhibits the highest quantum capacitance of 450 F/g at the Fermi level. Transport analysis also indicates a negative Seebeck coefficient of approximately 15 μ V/K confirming electron-dominated conduction. In addition, the V-substituted system demonstrates a favorable thermoelectric figure of merit. These results suggest that substitutional defect engineering in WS₂, particularly through vanadium replacement, is a promising strategy for advanced nanoelectronic and energy storage applications.

Keywords: Transition metals, Defect Engineering, Density Functional Theory, Quantum Capacitance

References:

1. P Giannozzi, O Andreussi, T Brumme, O Bunau, M Buongiorno Nardelli, M Calandra, R Car, C Cavazzoni, D Ceresoli, M Cococcioni, N Colonna, I Carnimeo, A Dal Corso, S De Gironcoli, P Delugas, R A DiStasio, A Ferretti, A Floris, G Fratesi, G Fugallo, R Gebauer, U Gerstmann, F Giustino, T Gorni, J Jia, M Kawamura, H-Y Ko, A Kokalj, E Küçükbenli, M Lazzeri, M Marsili, N Marzari, F Mauri, N L Nguyen, H-V Nguyen, A Otero-de-la Roza, L Paulatto, S Poncé, D Rocca, R Sabatini, B Santra, M Schlipf, A P Seitsonen, A Smogunov, I Timrov, T Thonhauser, P Umari, N Vast, X Wu, and S Baroni. Advanced capabilities for materials modelling with Quantum ESPRESSO. 29(46):465901.
2. Stefan Grimme, Journal of Computational Chemistry, 27(15):1787–1799, 2006.
3. Georg K. H. Madsen, Jesús Carrete, Matthieu J. Verstraete, Computer Physics Communications, Volume 231, 2018, Pages 140–145, ISSN 0010-4655.

Superexchange-Induced Positive Exchange Bias and Field-Cooling Reversal in Co /CoO Multilayer

Ambrish Dwivedi¹ Sharma Saurabh¹ and Dileep Kumar^{1*}

¹UGC-DAE CSR, University Campus, Khandwa Road, Indore 452017, INDIA

^{*}dkumar@csr.res.in

Exchange bias in magnetic multilayers plays a crucial role in stabilizing magnetic configurations through interfacial exchange coupling [1,3]. This interfacial interaction induces a unidirectional anisotropy that shifts the magnetic hysteresis loop, enabling controlled magnetization switching. Such behavior is important for applications in spintronic devices, magnetic sensors, and magnetic memory elements, where stable and controllable magnetic states are essential.

In the present work, exchange bias in Co/CoO/Co/CoO multilayers has been investigated. The film structure is characterized in situ during growth using Reflection High-Energy Electron Diffraction (RHEED), X-ray Reflectivity (XRR), and Magneto-optical Kerr Effect (MOKE) techniques. The independent magnetic contributions of the top and bottom Co layers were separated by fitting the hysteresis loops with an appropriate mathematical function [2]. The combined analysis reveals that both Co layers exhibit positive exchange bias at room temperature (RT), whereas a transition from positive to negative exchange bias occurs under field-cooled (FC) conditions. The unusual positive exchange bias observed at RT is attributed to antiferromagnetic superexchange coupling between Co³⁺–O–Co³⁺ at the interfaces. Under FC conditions, enhanced ordering of the antiferromagnetic (AFM) moments leads to the emergence of negative exchange bias. The increased AFM moment ordering strengthens the negative exchange bias, which eventually dominates over the interfacial superexchange coupling.

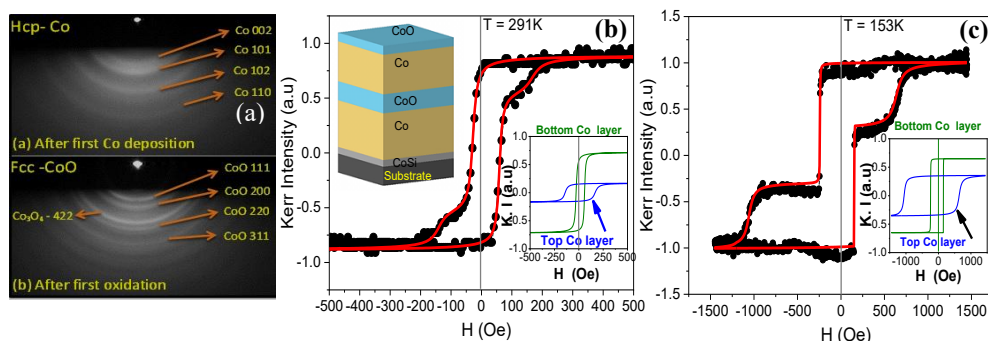


Figure : (a) RHEED images of the Co and CoO layers recorded during growth. Magnetic hysteresis loops measured by MOKE at (b) RT and (c) under FC conditions. The inset shows the two separated contributions from the top and bottom Co layers.

Keywords: Exchange bias, Magnetic thin film, in-situ MOKE, Superexchange coupling.

References:

1. S. Peng, D. Zhu, W. Li, H. Wu, A. J. Grutter, D. A. Gilbert, J. Lu, D. Xiong, W. Cai, P. Shafer, K. L. Wang, and W. Zhao, *Electronic Nature*, 3, (2020), 757-764.
2. S. Singh, P. Kumar, A. Gupta, and D. Kumar, *Journal of Magnetism and Magnetic Materials*, 513, (2020), 167186.
3. S. Singh, P. Kumar, A. Gupta, and D. Kumar, "Origin of exchange bias in orthogonal magnetic anisotropy system," *Physical Review B*, 108, (2023), 075414.

Spin-Photon Correlation in Few-Layer AgCrP₂S₆: Unraveling Antiferromagnetic Order Through Helicity Photoluminescence

Denny Pratama Hasibuan¹, Clara Sinta Saragih¹, Carlo C. Sta. Maria¹, Duy Van Pham², Yuan-Ron Ma^{1,3,4*}

¹Department of Physics, National Dong Hwa University, Hualien 974301, Taiwan

²Department of Physics, Chung Yuan Christian University, Taoyuan 320314, Taiwan

³Department of Applied Informatics, Fo Guang University, Yilan 262307, Taiwan

⁴Office of Postgraduate Studies, UCSI University, Kuala Lumpur 56000, Malaysia,

*Corresponding Author E-mail: ronma@gms.ndhu.edu.tw

The emergence of intrinsic magnetism in two-dimensional (2D) semiconductors offers new opportunities for exploring spin-dependent optical phenomena and their applications in spintronic and optoelectronic devices. Here, we investigate the spin photon correlation in few-layer AgCrP₂S₆ using helicity photoluminescence (PL) spectroscopy to probe its antiferromagnetic ordering. AgCrP₂S₆ is a recently emerging member of the layered metal thiophosphate (MPX₃) family [1], where it is an air-stable layered semiconductor with a direct band gap of approximately 1.31 eV, enabling efficient optical transitions strongly coupled to spin states [2]. Few-layer samples were mechanically exfoliated and characterized under polarization-selective optical measurements at temperatures below the Néel temperature (T_N ≈ 21 K) [3]. Distinct circular polarization responses were observed in the PL spectra, revealing strong coupling between magnetic ordering and excitonic recombination. We manipulate the photoluminescence polarization by applying an in-plane magnetic field. Our findings indicate that the magnetic

arrangement in thin-layered ACPS can be detected through the helicity of the emitted photons. That means helicity photoluminescence is an effective and non-invasive method for observing antiferromagnetic transitions in ultrathin van der Waals magnets, which provide insight into spin photon interactions in AgCrP₂S₆ for future quantum and spin-optoelectronic applications.

Keywords: AgCrP₂S₆; Spin photon couplings; helicity photoluminescence; antiferromagnetic;

References:

1. S. Selter, Y. Shemerliuk, B. Büchner, and S. Aswartham, *Crystals*, 11, (2021), 1-8/500.
2. Y. Sun, R. Zhang, J. Tan, S. Zeng, S. Li, Q. Wei, Z.-Y. Zhang, S. Zhao, X. Zou, B. Liu, and H.-M. Cheng, *Sci. Adv.*, 11, (2025), 1-7/eadr3105.
3. J. Xu, L. Li, N. Wang, X. Chang, R.-C. Xiao, X. Hu, H. Zhu, G. Wu, B. Li, X. Yang, and Z. Sheng, *Adv. Funct. Mater.*, 36, (2026), 1-9/e18545.

Circularly Polarized Photoluminescence Helicity in Stacked CrI₃ van der Waals Layered Crystals

Clara Sinta Saragih¹, Denny Pratama Hasibuan¹, Carlo C. Sta. Maria¹, Yuan-Ron Ma^{1,2,3,*}

¹*Department of Physics, National Dong Hwa University, Hualien 97401, Taiwan*

²*Department of Applied Informatics, Fo Guang University, Yilan 262307, Taiwan*

³*Office of Postgraduate Studies, UCSI University, Kuala Lumpur 56000, Malaysia*

*Corresponding author: ronma@gms.ndhu.edu.tw

CrI₃ is a layered van der Waals magnetic material that has garnered significant attention due to its unique layer-dependent magnetic properties and strong spin-orbit coupling. In this work, the CrI₃ single crystals were synthesized using the chemical vapor transport (CVT) technique [1]. Thin flakes with controlled layer thicknesses were obtained through mechanical exfoliation [2] and subsequently stacked at precisely controlled angles using the high-pressure atomic stacking (HPAS) method, which was investigated for the photoluminescence (PL) helicity of stacked few-layer CrI₃ crystals. In this study, we compare the photoluminescence (PL) helicity of monolayer (1L), bilayer (2L), and stacked layers. Magneto-optical measurements reveal a distinct layer-dependent magnetic behavior among the stacked CrI₃ samples. The monolayer (1L) CrI₃ exhibits ferromagnetic ordering, in contrast, the bilayer (2L) CrI₃ demonstrates antiferromagnetic ordering [3]. Notably, the trilayer (3L) stacked CrI₃ reverts to ferromagnetic ordering; that reflects a combination of the magnetic characteristics observed in both the monolayer and bilayer configurations. The stacking angle of the CrI₃ layers is found to modulating both the magnetic phase and the optical response of the system. These findings provide valuable insights into the magnetic and optical properties of few-layer CrI₃ and highlight its potential for applications in next-generation spintronic and magneto-optical devices.

Keywords: CrI₃; CVT methods; helicity photoluminescence; stacked layer; magneto-optical.

References:

1. C.S. Saragih, D.V. Pham, J.-X. Lin, W.-J. Chen, P.-H. Wu, C.-C. Yang, C.-C. Lai, C.-H. Tsai, H.-S. Hsu, and Y.-R. Ma, *Mater Today Adv*, 23, (2024), 100525.
2. P. V. Pham, T.-H. Mai, H.-B. Do, M. Vasundhara, V.-H. Nguyen, T. Nguyen, H. V. Bui, V.-D. Dao, R. K. Gupta, V. K. Ponnusamy, and J.-H. Park, *Chem. Rev*, 124, (2024), 10619.

3. J.L. Seyler, D. Zhong, D.R. Klein, S. Gao, X. Zhang, B. Huang, E. Navarro-Moratalla, L. Yang, D.H. Cobden, M.A. McGuire, W. Yao, D. Xiao, and P. J.-Herrero, X. Xu, *Nature Phys*, 14, (2018), 277.

Photo Luminescence features of Ce doped YAG synthesized by modified sol Gel method

Dr. Nanda Bhatia

Devi Ahilya Vishwavidhyalaya, Indore, Department of Chemistry, Bherulal Patidar Government PG College, Mhow
n.kaithwas305@gmail.com

Yttrium aluminum garnet (YAG) is an extensively used laser host material. Rare-earth doped YAG nanoparticles are promising phosphor materials for display devices. Ce-doped YAG nanoparticles were synthesized by modified sol–gel technique in an aqueous media using citric acid and ethylene glycol as chelating agents. Single phase nanocrystalline material was obtained at 850 °C. It is much lower temperature compared to other similar techniques used for preparing nanoparticles of YAG. The Ce-doping concentration was optimized and kept at 1 mol%. The prepared material was annealed at different temperatures and were characterized by XRD, SEM and TEM techniques to establish the particle sizes and structure. Average particle sizes were found ranging ~20–50 nm. The YAG: Ce nanophosphors showed intense, green-yellow emission, corresponding to $Ce^{3+} 5d1 \rightarrow 2F5/2, 2F7/2$ transitions and its photoluminescence excitation spectrum contains the two $Ce^{3+} 4f1 \rightarrow 5d1, 5d2$ bands. The crystal field splitting energy levels positions 5d1 and 5d2 and the emission transitions blue shift with annealing temperatures have been discussed. It was found that the $Ce^{3+} 4f1$ ground state position relative to valence band maximum of YAG host nanomaterial decreases with increasing the temperature.

Keywords: Nanophosphors, Nanomaterial, Photoluminescence, Nanoparticles, Ce:YAG

References:

1. K. S. Bartwal S. Kar, N. Kaithwas, M. Deshmukh, M. Dave, N. P. Lalla and H. Ryu, *Advanced material research* vols. 26-28 (2007) pp 667-670.
2. N. Kaithwas, M. Dave, S. Kar, K. S. Bartwal, Structural Feature of Ce doped YAG nanoparticles synthesized by modified sol-gel method, *Physica E*, 44 (2012) 1486-1489.
3. N. Kaithwas, M. Dave, S. Kar, K. S. Bartwal, Investigations on single phase formation by Dy Doped YAG Nanoparticles, *oper journal of modern physics*, vol. 1, No 1, February 2014, 30-34.

Anisotropic Phonon Modes in PLD-Grown α - MoO₃ Thin film Probed by Polarization-dependent Raman Spectroscopy

Manish Kumar¹, Krushna R. Mavani^{1*}

¹*Department of Physics, Indian Institute of Technology Indore (IIT Indore),
Indore, Madhya Pradesh, India – 453552*

*Corresponding Author E-mail: krushna@iiti.ac.in

In the last decade, orthorhombic molybdenum trioxide has gained attention as a layered 2D material due to its weak van der Waals interactions between adjacent layers and its ability to support highly anisotropic phonon polaritons in the mid-infrared up to the terahertz frequencies of light [1,2]. By rotating α -MoO₃ layers (twisting) at nanoscale, the anisotropic propagation of these phonons can be tuned. Understanding the phonon anisotropy of α -MoO₃ thin films is therefore essential for their potential applications in nanophotonics, optoelectronics, and anisotropic optical devices.

In this work, α -MoO₃ was deposited on a quartz substrate by pulsed laser deposition (PLD). X-ray diffraction (XRD) confirms a highly oriented (020) crystalline structure, indicating layered growth along the b -axis of crystallographic lattice. Polarization-dependent Raman spectroscopy was employed to investigate the anisotropic vibrational properties of this highly oriented film. The polarization-resolved Raman measurements demonstrated that only the A_g and B_{2g} vibrational modes exhibit pronounced four-fold symmetry, confirming strong in-plane phonon anisotropy in the film. Low-frequency modes corresponding to interlayer vibrations along the a , b , and c crystallographic directions were also studied. The modes associated with the a and b directions show pronounced four-fold symmetry. The polarization dependence confirms that these van der Waals-coupled interlayer vibrations are highly sensitive to the polarization angle and the crystallographic orientation of the film. Different anisotropic phonons related to stretching and bending within the layers of Mo–O bonds were previously observed at higher wavenumbers in α -MoO₃ films [3], while the present work reports the anisotropy in lower-frequency mode of PLD-grown α -MoO₃ film. These results provide important insight into the lattice dynamics and anisotropic phonon response of PLD-grown α -MoO₃ thin films, highlighting their suitability for future anisotropic photonic and polaritonic applications.

Keywords: 2D materials; Van der Waals interactions; Anisotropic phonon; Thin Film, Raman Spectroscopy; Pulsed laser deposition (PLD).

References:

1. Zhang, S., Ma, P., You, O. et al. *Nature Nanotechnology*, 21, 223–228 (2026).
2. Chen, M., Lin, X., Dinh, T.H. et al. *Nature Materials*, 19, 1307–1311 (2020).
3. Y.Gong, Y.Zhao, Z.Zhou, D.Li, H.Mao, Q.Bao, Y.Zhang, G. P.Wang, *Adv. Optical Mater.*, 2022, 10, 2200038.

Simultaneous Extraction of Correlated Fermi Surface Parameters in WTe₂ via Shubnikov-de Haas Oscillations and Profile Likelihood Analysis

Jaydeep Lohia^{1*}, Somnath C. Roy¹, Sanjeev Kumar²

¹*Department of Physics, IIT Madras, Chennai, Tamil Nadu, India*

²*Department of Electronic & Electrical Engineering, University College London, London, UK*

*Corresponding Author E-mail: ph22d001@smail.iitm.ac.in

Tungsten ditelluride (WTe₂) in the orthorhombic Td phase is a Type-II Weyl semimetal, exhibiting near-perfect electron-hole compensation and extreme, non-saturating magnetoresistance (XMR) [1]. Reliable extraction of Fermi surface parameters - effective masses, Dingle temperatures, quantum scattering lifetimes, and Berry phases - from Shubnikov-de Haas (SdH) oscillations is essential for understanding its transport physics, yet prior analyses rely on approximations and independent pocket fitting that introduce systematic biases and underestimate uncertainties.

We report SdH magneto-transport measurements on bulk WTe₂ single crystals grown by chemical vapour transport (Cl₂ transport agent, introduced by WCl₆ decomposition), performed at 0.3-10 K in magnetic fields up to 12 T with B || c. Fourier analysis of the oscillatory

magnetoresistance resolves four Fermi pockets at frequencies $F = 91.9, 127.7, 143.1$, and 164.2 T, attributed to two electron (e_1, e_2) and two hole (h_1, h_2) pockets via the Onsager relation [2,3]. The electron-hole compensation ratio of 0.946 ± 0.029 , defined as the ratio of summed hole to electron pocket frequencies, confirms near-perfect compensation.

The central methodological contribution is a coordinate descent χ^2 minimisation over the full (B, T) dataset, fitting all four pockets simultaneously. Parameter uncertainties are obtained via profile likelihood at a 1% χ^2 threshold [4], a statistically rigorous method capturing asymmetric errors that covariance-based methods underestimate, applied here to SdH analysis. The procedure yields effective masses $m^* = 0.214 \pm 0.015$ to $0.350 \pm 0.043 m_e$, Dingle temperatures $T_D = 1.5$ – 2.7 K, quantum scattering lifetimes $\tau_q = 0.45$ – 0.81 ps, and a mean quantum mobility $\mu_q \approx 3964 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. Berry phases extracted by three independent methods - Landau fan intercept, LK phase fit, and coordinate descent optimised phase- show mutual consistency, confirming the reliability of the simultaneous fitting procedure.

Keywords: WTe₂; Shubnikov-de Haas oscillations; Lifshitz–Kosevich analysis

References:

1. M. N. Ali, J. Xiong, S. Flynn, J. Tao, Q. D. Gibson, L. M. Schoop, T. Liang, N. Haldolaarachchige, N. P. Ong, and R. J. Cava, *Nature*, 514, (2014), 205–208.
2. I. M. Lifshitz and A. M. Kosevich, *Soviet Physics JETP*, 2, (1956), 636–645.
3. R. B. Dingle, *Proceedings of the Royal Society A*, 211, (1952), 517–525.
4. G. Cowan, K. Cranmer, E. Gross, and O. Vitells, *European Physical Journal C*, 71, (2011), 1554.

Probing Plasmons in MXenes using Surface Spectroscopy

Yugal Rathod^{1,2,†}, Or Messer^{3,†}, M. W. Barsoum⁴, Maxim Sokol³, Varun Natu^{1,2*}

¹*Physical and Materials Chemistry Division, National Chemical Laboratory, Pune, India, 411008*

²*Academy of Scientific and Innovative Research (AcSIR), Ghaziabad, Uttar Pradesh, India, 201002*

³*Department of Materials Science and Engineering, Tel Aviv University, P.O.B 39040, Ramat Aviv 6997801, Israel*

⁴*Department of Materials Engineering, Drexel University, Philadelphia, PA, USA*

[†]*Authors contributed equally*

*Corresponding Author E-mail: vr.natu@ncl.res.in

Plasmonic materials have recently garnered significant interest due to their applications in various fields, ranging from optoelectronics to photocatalysis. Traditionally, metal nanoparticles (NPs) have been explored for these applications, but plasmons are not limited to metal NPs; they are also observed in other materials with high free charge carrier density including 2D materials, such as graphene, h-BN, and phosphorene. The most recent addition to this list is 2D transition metal carbides/nitrides, known as MXenes. To date, plasmons in MXenes have primarily been investigated using high-resolution electron energy loss spectroscopy (EELS) inside a transmission electron microscope (TEM). While this technique is crucial for studying plasmons, it requires state-of-the-art TEMs and highly skilled operators. In this work, we demonstrate that simple X-ray photoelectron spectroscopy (XPS) measurements can be used to probe bulk plasmons in MXenes. Additionally, by using the reflected-EELS (REELS) attachment

inside an XPS, it is possible to probe low-energy interband transitions as well. We show that our data coincides with that obtained from TEM, thereby simplifying the analysis of plasmons. Based on the XPS data, we also calculate the electron density of different MXenes and corroborate these findings with electrical and optical measurements from the literature.

Keywords: MXenes; Plasmons; X-ray photoelectron spectroscopy; Reflected electron energy loss spectroscopy (REELS).

References:

1. V. Mauchamp, M. Bugnet, E. P. Bellido, G. A. Botton, P. Moreau, D. Magne, M. Naguib, T. Cabioch, and M. W. Barsoum, *Physical Review B*, 89 (2014), 235428.
2. V. Natu, M. Benchakar, C. Canaff, A. Habrioux, S. Célrier, and M. W. Barsoum, *Matter*, 4 (2021), 1224-1251.
3. J. K. El-Demellawi, S. Lopatin, J. Yin, O. F. Mohammed, and H. N. Alshareef, *ACS Nano*, 12 (2018), 8485-8493.

Electric field-driven protein translocation through a single-walled carbon nanotube embedded in a lipid bilayer membrane

Saurabh Chaubey,¹ Km Mamata Patel,¹ Yogita Rani,¹ Simaran Nasa,² Manoj Verma,² and Prabhat Tripathi^{1*}

¹*Department of Chemistry, Indian Institute of Technology (Banaras Hindu University), Varanasi, UP, India*

²*Center for Nano Science and Engineering, Indian Institute of Science, Bangalore, Karnataka, India.*

*Corresponding Author E-mail: prabhat.chy@itbhu.ac.in

Cell membranes primarily act as boundaries that create distinct compartments, with each membrane separating an internal and external environment. One of the essential components of the physical structure of life is the lipid membrane. Carbon nanotube porins (CNTP) are short, nanoscale channels derived from carbon nanotubes that can spontaneously insert into lipid bilayers and mimic biological membrane proteins such as aquaporins. These synthetic channels demonstrated the selective transport of ions and even the translocation of single-stranded DNA, a DNA hairpin for single-molecule sensing. However, the application of these artificial channels for protein translocation has been envisioned but never been experimentally demonstrated. We will present the demonstration of the protein translocation through single-walled carbon nanotubes (SWCNTs). We have scissored long carbon nanotubes of the μm range to ultra-short length to 10-20 nm to incorporate it into the lipid bilayer. We have observed characteristic ionic current jumps in the single-molecule electrophysiology measurements due to spontaneous insertions of 10-20 nm long CNTs into the lipid bilayer membrane. Here, we report the translocation behaviour of cytochrome c protein induced by electric fields through a single-walled carbon nanotube of inner diameter ~ 1.3 nm. The electric field at the CNTP was found to sufficiently unfold the cytochrome c protein to facilitate the translocation, as confirmed by a decrease in the average residence time of the protein at the CNTP upon increasing the transmembrane potential.

Keywords: carbon nanotube, cytochrome c, lipid bilayer, nanopore, translocation

References

1. Liu, H.; He, J.; Tang, J.; Liu, H.; Pang, P.; Cao, D.; Krstic, P.; Joseph, S.; Lindsay, S.; Nuckolls, C. *Science* 2010, 327 (5961), 64-67.
2. Kam, N. W. S.; Dai, H., *Journal of the American Chemical Society* 2005, 127 (16), 6021-6026.
3. Geng, J.; Kim, K.; Zhang, J.; Escalada, A.; Tunuguntla, R.; Comolli, L. R.; Allen, F. I.; Shnyrova, A. V.; Cho, K. R.; Munoz, D.; Wang, Y. M.; Grigoropoulos, C. P.; Ajo-Franklin, C. M.; Frolov, V. A.; Noy, A. *Nature* 2014, 514 (7524), 612-615.

Curcumin-Derived (3-Aminopropyl) trimethoxysilane-Functionalized Carbon Quantum Dots: A Colorimetric and Fluorometric Nanoprobe for Antioxidant Activity, Liposome Encapsulation, and Picric Acid Detection

Yogita Rani, Abhishek, Mamata Patel Km and Prabhat Tripathi*

Department of Chemistry, Indian Institute of Technology (BHU), Varanasi-221005, U.P., India

**Corresponding Author E-mail: prabhat.chy@itbhu.ac.in*

The development of a method to detect highly harmful picric acid (PA) is important for environmental safety and human health. In this study, we developed a simple and fast fluorescence-based sensing method using curcumin carbon quantum dots (CQDs) modified with 3-aminopropyl-trimethoxysilane (3-APTMS) for accurate detection of PA. The CQDs were prepared by solvothermal treatment of curcumin and then functionalized with 3-APTMS. Different techniques such as UV–visible spectroscopy, FTIR, XRD, zeta potential, TEM, and XPS were used to study their properties. The synthesized CQDs had an average size of about 3.4 nm and showed excitation-dependent fluorescence behavior. They also exhibited a fluorescence yield of 63.85% and a quantum yield of 1.59%. When picric acid was added, the fluorescence intensity of the CQDs decreased significantly, along with a noticeable color change under UV light. The sensor showed good linear detection of PA in the concentration range of 0.1–2.5 μM , with a very low detection limit of 88.96 nm. At low concentrations (<0.07 mg/mL), the CQDs also demonstrated antioxidant activity, confirmed through DPPH colorimetric and electrochemical assays. Furthermore, the CQDs were encapsulated inside liposomes to improve their biocompatibility for future cell imaging studies. Overall, these CQDs are effective and selective probes for detecting picric acid. They also retain the antioxidant properties of natural curcumin while offering better water solubility and bioavailability. In addition, they have potential applications in cell imaging and semiconductor-based photocatalytic systems.

References:

1. Yogita Rani, Mamata Patel Km, and Prabhat Tripathi, *ACS Applied Materials & Interfaces*, 2024, 16(50), 68936–68949.

Direct Observation of Electric Field-Driven Protein Translocation Through the Single-Walled Carbon Nanotube Embedded in a Lipid Bilayer Membrane Revealed a High Dipole Moment of the Transition State

Km Mamata Patel¹, Saurabh Chaubey¹, Ayushi Kumari¹, Prabhat Tripathi^{1*}

¹*Department of Chemistry, Indian Institute of Technology, (Banaras Hindu University), Varanasi, 221005*

*prabhat.chy@itbhu.ac.in

Lipid bilayer is a pivotal constituent of the cell membrane that facilitate the translocation of the selective molecules and ions through it when subject to external electric field, called electroporation. Here, we have studied the statistical defects in lipid bilayer, which caused the short-lived pore formation in μs -ms time scale, and $\text{\AA}$ -nm range in size and breakdown behaviour in the lipid bilayer membrane for four varieties of phospholipids DPhPC, POPC, DOPC and DOPG. Further, we have studied the sucrose effect on the POPC lipid bilayer integrity, as sucrose acts as a cryoprotectant and osmotic agent that influences membrane mechanical and electrical properties. In a typical experiment, we have used the Montal-Muller method to form a vertical lipid bilayer membrane on a 100 μm Teflon aperture and applied a linear stepwise voltage ranging from 0 to 1 V with a 10 mV voltage step with a 20 ms step duration, while monitoring the ionic-current with pA current and μs time resolution in 1M KCl buffer solution. We found that the measured distribution of membrane breakdown voltage $V_{\text{breakdown}}$ follows the Weibull distribution, and its shape changes with the identity of the lipid molecule, and concentration of sucrose. Our analysis of the observed short-lived pores both during and prior to the membrane breakdown suggests that the pores during the instantaneous breakdown at $V_{\text{breakdown}}$ and the pores prior to the breakdown occur independently and have different population of hydrophilic and hydrophobic pores. The effect of sucrose molecules on the population and stabilization of hydrophilic/hydrophobic pores in POPC lipid bilayer will be presented.

Keywords: Lipid bilayer membrane, pores, metastable pores, breakdown, and the breakdown rate.

References:

1. Patel, K. M.; Chaubey, S.; Rani, Y.; Tripathi, P. ACS Nano 2026, 20 (4), 3582–3592.
2. Lafarge, E. J.; Muller, P.; Schroder, A. P.; Zaitseva, E.; Behrends, J. C.; Marques, C. M. Proc. Natl. Acad. Sci. 2023, 120 (11), e2213112120.

Probing Strain-Driven Modifications in Electronic Structure of Epitaxial GdVO₃ Thin Films

Bharathi Palthya¹, Subho Saha¹, Bharti Ojha¹, Priti Roy¹, Rajan Mishra¹, Shivam Choudhary¹, Ram Janay Choudhary^{1*}

UGC-DAE Consortium for Scientific Research, University Campus, Khandwa Road, Indore (M.P.) 452001.

*Corresponding Author E-mail: ram@csr.res.in

Rare-earth Ortho vanadate's RVO_3 exhibit strong coupling between spin, orbital, and lattice degrees of freedom. Structural distortions of the VO_6 octahedra, including Jahn–Teller distortions and octahedral rotations, lift the degeneracy of the V^{3+} t_{2g} orbitals, leading to remarkable electronic, magnetic, and electrical properties. In the present study, we report the structural and electronic properties of epitaxial GdVO_3 thin films grown on LaAlO_3 (001) substrates via pulsed laser deposition (PLD), focusing on the effects of substrate-induced strain and film thickness variation. X-ray diffraction (XRD) confirms phase purity and epitaxial growth of the GdVO_3 thin films on the LAO substrate. X-ray photoemission spectroscopy (XPS) and soft X-ray absorption spectroscopy reflect that V ion is dominantly in 3+ charge state with minor fraction of 4+ charge state. The hetero-valency of V ions arise from oxygen vacancy present in the films, as confirmed from XPS data. Resonant photoemission spectroscopy shows that the band at the valence band maxima site near the Fermi level is dominated by V 3d state and shows Mott insulating nature of the grown thin film.

Keywords: Jahn–Teller distortions; orbital coupling; Resonant photoemission spectroscopy; Mott insulator.

Resonant Photoemission Study of PLD-Grown Textured Ti_3AlC_2 Thin Films

Apurva Karahe^a, Subho Saha^b, Ekta Choudhary^{a,c}, Manopriya Samtham^a, Ravindra Jangir^{d,e,f},
Ram J. Choudhary^{b*}, and Rupesh Devan^{a,d*}

^a Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.

^b UGC-DAE Consortium for Scientific Research, Khandwa Road, Indore, 452001, India.

^c Department of Physics, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India

^d Centre for Electric Vehicle & Intelligent Transport System, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.

^e Accelerator Physics and Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Indore 452013, India

^f Homi Bhabha National Institute, Anushakti Nagar, Mumbai, 400094, India

*Corresponding Author E-mail: rupesh@iiti.ac.in, ram@csr.res.in

Thin Film growth of layered ternary carbides, popularly known as MAX phases, is crucial owing to their distinctive combination of metallic and ceramic properties, suitability as a precursor for two-dimensional MXene production, and compatibility with device applications. Among various MAX phases, Ti_3AlC_2 has gained significant attention due to its facile synthesis and excellent electrical and thermal properties. Herein, we present a study of structure and electronic properties of Ti_3AlC_2 MAX thin films deposited by pulsed laser deposition on glass substrate. X-ray diffraction studies reveal the formation of Ti_3AlC_2 phase with (004) reflection, which suggests the textured growth. Film morphology observed from FESEM consists of triangular platelet-like features with out-of-plane ordering across the surface. The electronic structure has been characterized using x-ray photoelectron spectroscopy (XPS) and resonant photoemission spectroscopy (RPES). XPS analysis reflects the presence of Ti–Al–C metallic bonds, Ti (III) and Ti (IV) oxide states in the thin film. Valence band spectra disclose the metallic nature of the grown thin film. RPES studies reveal the dominance of a characteristic feature of the Ti 3d state at the valence band maxima site, near the Fermi level.

Keywords: Ti_3AlC_2 ; MAX phases; pulsed laser deposition; RPES

In-situ characterization of temperature-dependent thermal conductivity in MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) using Raman spectroscopy

Priyam Srivastava¹, Rani Rohini¹

¹Department, Department of Materials Engineering, Indian Institute of Technology Jammu, Jammu-181221, Jammu and Kashmir, India.

*Corresponding Author E-mail: 2022rmt0022@iitjammu.ac.in

MXene is a group of 2D material which is made up of early transition carbides, carbonitrides, and nitrides. The first MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) was synthesized by Prof. Yury Gogotsi et. al. in 2011. It has also shown the highest electrical conductivity among all 30 types of MXene synthesized till date. Because of tunable and metallic properties of MXene, it has been used in many applications such as thermoelectric, sensor, EMI shielding, composites and many more.

In 2D materials like MXene, Raman spectroscopy is one of the best approaches for nanoscale thermal conductivity measurement since it overcomes numerous limitations posed by traditional methods. It is appropriate for thin films, few-layer flakes, and nanostructures when large, defect-free samples are difficult to prepare. It is non-contact, non-destructive, and extremely localized, unlike bulk measuring methods like laser flash analyzer. Herein, we have used Raman spectroscopy to calculate thermal conductivity of $\text{Ti}_3\text{C}_2\text{T}_x$ using intensity and temperature variation. The approach measures heat dissipation by measuring the Raman peak shift of phonon modes with temperature. The MXene was synthesized using modified-MILD method. The results were also verified with laser flash analyzer.

Keywords: MXene ($\text{Ti}_3\text{C}_2\text{T}_x$); Raman spectroscopy; thermal conductivity; modified-MILD method.

References:

1. Srivastava, P.; Rohini, R. The Journal of Physical Chemistry C 2026.

Evidence of Spin-phonon Coupling in Magnetic Two-dimensional Van Der Waals Single-crystalline $\text{Fe}_{3-x}\text{Co}_x\text{GaTe}_2$

Rosni Roy¹, Sandip Paul¹, Gourab Kanti Jana¹, Rajib Mondal^{1*}

¹UGC-DAE Consortium for Scientific Research, Kolkata Centre, Sector III, LB-8, Salt Lake, Kolkata 700 106, India

*Corresponding Author E-mail: mondal.rajib1988@gmail.com, rmondal@csr.res.in

Investigations of long-range magnetic order in low dimensional materials are growing rapidly in the field of condensed matter physics owing to their novel physical properties and advanced technological applications (1). Recently, single-crystalline Fe_3GaTe_2 and its doped compounds are of tremendous interest due to several intriguing properties such as skyrmionic behaviour, large magnetocrystalline anisotropy etc. (2). The material under this study $\text{Fe}_{3-x}\text{Co}_x\text{GaTe}_2$ ($x \sim 0.25$) has been grown by self-flux method. Laue diffraction and X-ray diffraction performed on a cleaved crystal confirms single-crystalline nature and hexagonal crystal structure ($P6_3/mmc$), respectively. From temperature (T) dependent dc magnetization (M) data, the ferromagnetic to paramagnetic phase transition temperature (T_C) is ~ 250 K, at 1 kOe applied field.

In T -dependent field cooled cooling and field cooled warming M curves for $H//[c]$, there is a thermomagnetic irreversibility in the T -regime 140–220 K, indicating occurrence of any structural transition and/or presence of additional magnetic phase. Crystallographic c -axis is the easy axis of M . Isothermal magnetic entropy change (ΔS_m) for $\Delta H = 20$ kOe is -0.3 J/kg·K and -0.95 J/kg·K along $H//[ab]$ and $H//[c]$ respectively. Along $H//[ab]$, inverse magnetocaloric effect has been observed below T_C . Longitudinal resistivity increases with increasing T , indicating metallic nature. When H applied along c -direction and current along ab -plane, anomalous Hall behaviour was observed similar to M - H curves. Raman spectroscopic study in the T -range 153 K–323 K was performed to investigate lattice dynamics of the studied material. Room temperature pattern reveals two well resolved vibrational peaks at ~ 117 cm^{-1} and 138 cm^{-1} , assigned to E_{2g} and A_{1g} modes, respectively (3). The Raman shift decreases with increasing T for both the modes and follows anharmonic model upto T_C , and then deviates below T_C , indicating spin-phonon coupling present. These findings on magnetic, magnetocaloric, magnetotransport and Raman spectroscopic are very crucial for several spin-based applications.

Keywords: Ferromagnetism; Magnetocaloric effect; Skyrmion; Anomalous Hall effect; Raman spectroscopy.

References

1. K. S. Burch, D. Mandrus, and J.-G. Park, *Nature*, 563, (2018), 47–52.
2. K. Zhu, M. Wang, Y. Deng, M. Tian, B. Lei, and X. Chen, *Phys. Rev. B*, 109, (2024), 104402.
3. X. Chen, X. Zhang, W. He, Y. Li, J. Lu, D. Yang, D. Li, L. Lei, Y. Peng, and G. Xiang, *Nano Lett.*, 25, (2025), 4353–4360.

Multiplexed Au/Pt/Pd Nanorattles Enable Electronic Tongue Detection of Breast Cancer Biomarkers in Urine

Khushal Singh¹, Parastoo Vahdatiyekta², Tan-Phat Huynh² and Amit Jaiswal¹.

¹ School of Biosciences and Bioengineering, Indian Institute of Technology, India

² Chemistry and Chemical Technology, Åbo Akademi University, Finland

*Corresponding Author: j.amit@iitmandi.ac.in

Global statistics reveal that cancer is one of the most prevalent and lethal diseases globally, accounting for ~ 9.7 million deaths in 2022, with forecasts predicting a rise to 35.3 million deaths by 2050. Lung cancer is the predominant cancer among males globally, while breast cancer is the most commonly diagnosed cancer among females. Timely detection of breast cancer significantly improves the survival chances. Mammography, although a standard screening method, exhibits limited sensitivity and specificity, particularly for early-stage, and further diagnostic procedures, such as biopsy are invasive ones. Non-invasive procedures are particularly beneficial, as they simplify sample collection and enabling continuous monitoring of disease prognosis and diagnosis process. This study, an electronic tongue (e-tongue) for breast cancer detection in artificial urine samples, utilizing six unique plasmonic nanorattles with cubic and tetrahedral morphologies, each having solid Au-octahedral and tetrahedral cores encased in porous Au, Pt, or Pd shells. These materials were applied to glassy carbon electrodes (GCEs) along with graphite electrodes, yielding consistent signals throughout a duration of seven days and 300 cyclic voltametric (CV) cycles. They demonstrated negligible leaching (< 1.5 ppb) and no alteration in chemical state before and after 300 CV cycles, as confirmed by energy dispersive X-ray spectroscopy (EDX), inductively coupled plasma mass spectrometry (ICP-MS), and X-ray photoelectron spectroscopy (XPS) analysis. The larger Au-tetrahedron nanorattles exhibit superior electrochemical performance compared to cubic nanorattles due to the compositional and morphological factors. Differential pulse voltammetry (DPV) was selected as transduction

method of the e-tongue, with its data then processed and analyzed using machine learning and principal component analysis (PCA). The random forest model attained a precision of 0.99, sensitivity of 1.00, and specificity of 0.98. The results indicate the promise of a nanorattle-based e-tongue as a non-invasive diagnostic instrument for breast cancer identification.

Keywords: Plasmonic nanorattles; Breast cancer; E-tongue; sensing.

Layer-Controlled Growth of WS₂ Films on Al₂O₃ and Interface Band Alignment via HXPES Analysis

U. K. Goutam^{1#}, Jagannath^{1#}, Jyoti Prakash^{2#}, S. Khan^{3#} and D. Bhattacharyya^{4#}

¹Technical Physics Division, Bhabha Atomic Research Centre, Mumbai-400085, India

²Glass and Advance Materials Division, Bhabha Atomic Research Centre, Mumbai-400085, India

³Nano Science Lab., Semiconductor Lasers & Devices Division, Raja Ramanna Centre for Advanced Technology, Indore-452013, India

⁴Atomic and Molecular Physics Division, Bhabha Atomic Research Centre, Mumbai-400085, India

[#]Homi Bhabha National Institute, Anushaktinagar, Mumbai-400094, India

*Corresponding Author E-mail: ukgoutam@rrcat.gov.in

Tungsten disulfide (WS₂), a prominent member of the transition metal dichalcogenides (TMDs) family, has garnered significant attention due to its remarkable electronic, optical, and mechanical properties. These properties exhibit pronounced variation between the bulk (indirect band gap ~1.3 eV) and monolayer (direct bandgap ~2.0 eV) forms, which opens up extensive possibilities for novel applications in nanoelectronics and optoelectronics technologies. Also, understanding the band alignment at WS₂/Al₂O₃ interfaces is critical for achieving optimal device performance as band alignment directly affects charge carrier injection, transport, and recombination. In view of the above, WS₂ thin films, spanning from a few layers to monolayer in thickness, were effectively grown over large areas on sapphire (Al₂O₃) substrates through electron-beam (e-beam) deposition followed by a sulfurization process. Structural analysis confirmed that the films adopt the 2H crystalline phase, as evidenced by Grazing Incidence X-Ray Diffraction (GIXRD). The formation of monolayer and few-layer WS₂ films on Al₂O₃ was verified using Raman spectroscopy, with further confirmation provided by Photoluminescence (PL) spectroscopy. The electronic band alignment at the WS₂/Al₂O₃ interface was investigated non-destructively via Hard X-ray Photo-Electron Spectroscopy (HXPES). Using core level positions of W 4f and Al 2p in different samples and their valence band maxima (VBM) positions, the Valence Band Offsets (VBO) and also utilizing band gap values, the Conduction Band Offsets (CBO) were determined for different samples. Both VBO and CBO values were observed to increase with additional WS₂ layers. These findings enhance the understanding of WS₂-based nanoelectronics and optoelectronics, underlining the significance of accurate control over layer thickness and interface characteristics.

Keywords: TMD; WS₂; Band offset; X-ray photoelectron spectroscopy; monolayer; thin film interface.

References:

2DMTG 2026

1. Amritanand Sebastian, Rahul Pendurthi, Tanushree H. Choudhury, Joan M. Redwing & Saptarshi Das; Nat. Commun., 12 (2021), 693.
2. Chunxiao Cong, Jingzhi Shang, Yanlong Wang, and Ting Yu, Adv. Optical Mater., 6 (2018), 1700767.
3. U. K. Goutam, Jagannath, Jyoti Prakash, S. Khan, Pabitra Sahu, D. Bhattacharyya and L. M. Pant; J. Appl. Phys.;137 (2025), 144303.

Scan-Number Scaling and Structural Convergence in Fluorescence EXAFS: A Reverse Monte Carlo Study of O3-NaFeO₂

Harshini V. Annadata¹, Chiranjeev Rao¹, Biplab Ghosh^{1*}

¹*Beamline Development & Application Section, Physics Group, Bhabha Atomic Research Centre, Trombay, Mumbai 400085.*

*Corresponding Author E-mail: biplab@rrcat.gov.in

Determining the number of scans required for reliable fluorescence EXAFS analysis is a key practical challenge in synchrotron spectroscopy. Although scan averaging improves the signal-to-noise ratio following expected counting-statistical behavior, whether continued spectral improvement translates into improved structural reliability remains unclear. Here, scan-number scaling in Fe K-edge fluorescence EXAFS of layered O3-NaFeO₂ is systematically investigated using ensemble reverse Monte Carlo (RMC) refinement in wavelet-transform space.

A total of 64 fluorescence scans were collected and reorganized into subsets defined by $N=n^2$ ($n=1-8$). Three independent RMC refinements were performed at each averaging level, yielding 24 calculations in total. The evolution of $k^2\chi(k)$, Fourier transforms, partial radial distribution functions, and RMC-derived structural parameters was tracked as a function of scan averaging. Spectral noise was quantified from the high-k region (8–12 Å⁻¹) using root-mean-square (RMS) analysis.

Spectral noise decreases continuously with increasing scan averaging, consistent with counting-statistical expectations. However, structural quantities stabilize significantly earlier. The Fe–O bond distance remains stable even at moderate averaging levels, while the Fe–O mean-square relative displacement (MSRD) converges near $n=5$ ($N=25$ scans). Beyond this threshold, additional averaging yields only marginal improvement in structural refinement despite continued noise reduction. Wavelet-space residuals and radial distribution functions corroborate that structural convergence occurs well before complete spectral noise minimization.

These findings demonstrate that spectral improvement and structural convergence are not equivalent, and provide a practical framework for defining sufficient scan averaging in fluorescence EXAFS experiments.

Keywords: Fluorescence EXAFS; Reverse Monte Carlo; Wavelet transform; Structural convergence; Noise analysis.

References:

1. J. Timoshenko, A. Kuzmin, and J. Purāns, J. Phys.: Condens. Matter, 26, (2014), 055401.
2. A. Kuzmin, J. Timoshenko, A. Kalinko, I. Jonane, and A. Anspoks, Radiat. Phys. Chem., 175, (2020), 108112.

PEDOT: PSS induced Supramolecular G-Quartet Hydrogel for Diffusion-Controlled High-Performance Flexible Supercapacitor

Yoshita Chakravarty[#], Suryakamal Sarma[#], Anjali Singh, Dr. Tridib K Sarma*

Department of Chemistry, Indian Institute of Technology Indore, Simrol, Indore-453552, India

E-mail: tridib@iiti.ac.in

Soft and wearable electronics require conductive hydrogels that seamlessly integrate efficient charge storage, biocompatibility, and structural adaptability. Herein, we present a minimalistic, metal-free, and facile strategy in which PEDOT:PSS acts as a structure-directing agent to induce the self-assembly of the small biomolecule guanosine 5'-monophosphate (5'-GMP) into a fibrillar supramolecular network. This process transforms aqueous PEDOT:PSS dispersions into a free-standing, conductive hydrogel without the need for external crosslinkers or harsh conditions. The assembly is driven by PEDOT:PSS-induced spontaneous condensation of 5'-GMP into hierarchically organized G-quartet fibrils, further stabilized by Na⁺-mediated ion channels. In this hybrid system, PEDOT:PSS plays a dual role as both an electroactive component and a supramolecular structure-directing agent, thereby enhancing mechanical robustness and structural integrity while overcoming the limited charge-storage capacity of pristine PEDOT:PSS. The intimate coupling between ion transport through G-quadruplex channels and reversible polaronic-bipolaronic transitions along the PEDOT backbone enables efficient, diffusion-controlled, and reversible charge-storage behaviour. Notably, the hydrogel was successfully integrated into an all-solid-state flexible supercapacitor, delivering a high specific capacitance of 1436 F g⁻¹ and an energy density of 71.8 Wh kg⁻¹, along with excellent cycling stability and mechanical flexibility, highlighting its promise for next-generation soft energy-storage devices.

Keywords: supramolecular self-assembly, G-quadruplex, hydrogel, ion-diffusion, flexible supercapacitor

References:

1. Lu, B.; Yuk, H.; Lin, S.; Jian, N.; Qu, K.; Xu, J.; Zhao, X., *Nat Commun*, 10, (2019), 1043.
2. Sarma, S.; Jain, N.; Bansal, L.; Vishwakarma, R.; Prasun, A.; Sahu, T. K.; Kumar, R.; Sarma, T. K., *ACS Appl. Mater. Interfaces*, 17, (2025), 48532–48546.
3. Horii, T.; Li, Y.; Mori, Y.; Okuzaki, H., *Polym J*, 47, (2015), 695–699.

Two-Dimensional Molybdenum carbide as an efficient electrocatalyst for Hydrogen Evolution Reaction

Shubham Singh Kushwaha, Rohit Kumar Gupta and Ashish Kumar Mishra*

Nano Photonics & Energy Materials Laboratory (NPEML)

School of Materials Science and Technology

Indian Institute of Technology (BHU), Varanasi, India

Email: shubhamskushwaha.rs.mst24@itbhu.ac.in, akmishra.mst@itbhu.ac.in*

Hydrogen gas is considered as promising fuel source to meet growing energy demands due to its high energy density, clean combustion, and renewability. It can be used for powering

vehicles, heating homes, and generating electricity. Among various hydrogen production methods, electrochemical water electrolysis is a promising approach for producing green hydrogen using Pt-, Ir-, and Ru-based electrocatalysts [1]. However, the high cost of these noble metal catalysts has driven the scientific community to search for new, cost-effective electrocatalysts. MXenes (2D transition metal carbides, nitrides such as Mo_2C , $\text{Ti}_3\text{C}_2\text{T}_x$, Ti_2N , etc.) can be used as highly efficient electrocatalysts for hydrogen production due to their large surface area, high electrical conductivity, variable surface chemistry and good stability in acidic/basic medium [2,3]. Here, we synthesized 2D Molybdenum carbide (Mo_2C) and characterized it using different microscopy and spectroscopy techniques. We have further demonstrated its excellent electrocatalytic activity for hydrogen evolution reaction (HER) using electrochemical techniques like LSV and EIS studies.

Keywords: MXenes, Molybdenum carbide, Hydrogen evolution reaction

References:

1. P. K. Maurya and A. K. Mishra, ACS Applied Energy Materials, 7, (2024), 487-498.
2. M. P. Bilibana, Advanced Sensor and Energy Materials, 2, (2023), 100080.
3. C. Tang, A. Sun, Y. Xu, Z. Wu, and D. Wang, Journal of Power Sources, 296, (2015), 18-22.

Superior 3D Conductive $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}/\text{rGO}$ Composite Cathode for Advanced Sodium-Ion Batteries

V. M. Khetmalis¹, Mohammad Zaid², Surendra K. Martha², R. J. Deokate^{1*}

¹Department of Physics, Vidya Pratishthan's Arts Science and Commerce College, Baramati 413 133, (MS), India. *idya Pratishthan's Arts Science and Commerce College, Baramati 413 133, (MS), India.*

²Department of Chemistry, Indian Institute of Technology Hyderabad, Kandi, Sangareddy, Telangana 502284, India

*Corresponding Author E-mail: deokate2000@yahoo.co.in

$\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}$ (NVOPF) is considered one of the most promising cathode material for sodium ion batteries due to its high operating potential and theoretical capacity. However, its practical performance is deeply affected due to its poor electronic conductivity. In this study focused on preparation of polyanionic $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}/\text{rGO}$ (NVOPF/rGO) composite cathode material by solgel method for sodium-ion batteries (SIBs). The structural, morphological, and electrochemical properties of the NVOPF/rGO composite were systematically investigated using X-ray diffraction (XRD), scanning electron microscopy (SEM), transmission electron microscopy (TEM), X-ray photoelectron spectroscopy (XPS), and electrochemical impedance spectroscopy (EIS). Electrochemical results demonstrated that the NVOPF/rGO exhibits enhanced performance, including a high discharge capacity, excellent cycling stability, and superior rate capability. The material delivered initial specific discharge capacity of approximately 120 mAh/g at a current density 0.1C and retention up to 91% after 200 cycles at 0.1C in the voltage range 2.5 - 4.3V. The half coin cells were made in 70:20:10 ratio for active material: active carbon: PVDF binder in NMP solvent. These results indicate that NVOPF/rGO is a promising candidate for use in high-performance sodium-ion batteries, offering a balance of high capacity, good cycling stability, and fast sodium-ion diffusion.

Keywords: Sodium ion, NVOPF, cathode, batteries, energy storage

References:

1. Yameng Yin, et al, Nano Energy, Volume 41, (2017), Pages 452-459/152270.
2. Hongyun Jin et. al., Journal of Materials Chemistry A, Volume 3, (2015) 17563-17568/17563.
3. Caiyun Ma et. al., Journal of Power Sources, Volume 482 (2021), 228906.

Osmotic Energy Conversion using MoSe₂/MoS₂ Heterostructure Nanopores in Multi-Ion Environments

Mukesh Kumar¹, Simran Nasa², Samriddhi Kushwaha¹, Sumanth Sekhar Samal¹, Manoj Varma²
and Pramoda K. Nayak^{1,3,*}

1Centre for Nano and Material Sciences, Jain (Deemed-to-be University), Jain Global Campus, Kanakapura, Bangalore 562112, Karnataka, India

2Centre for Nano Science and Engineering, Indian Institute of Science, Bangalore-560012, India

3Department of Physics, 2D Materials Research and Innovation Group, Indian Institute of Technology Madras, Chennai 600036, India

*Email: pramoda.nayak@jainuniversity.ac.in

Efficient harvesting of salinity gradient energy demands membrane systems capable of operating under realistic multi-ion environments [1]. Although two-dimensional (2D) materials have demonstrated excellent performance in single-electrolyte systems [2,3], their behavior in complex ionic conditions remains insufficiently understood. In this work, we report osmotic energy generation using MoSe₂/MoS₂ heterostructure nanopores and systematically investigate ion transport in both monovalent and divalent electrolyte systems. The MoSe₂/MoS₂ heterostructure enables interfacial charge modulation, leading to asymmetric ion transport. Nanopores with diameters of ~5 nm are fabricated and characterized via current–voltage measurements under a wide range of concentration gradients ($\Delta C = 10\text{--}100000$). Pronounced ion-dependent transport behavior is observed for potassium chloride (KCl), sodium chloride (NaCl), and magnesium chloride (MgCl₂) electrolytes. At $\Delta C = 1000$, the generated power reaches ~127 nW for KCl, compared to ~66 nW for NaCl and ~88 nW for MgCl₂, highlighting strong ion selectivity. At higher concentration gradients ($\Delta C = 100000$), MgCl₂ exhibits the highest power output (~188 nW), indicating enhanced transport governed by multivalent ion interactions and charge screening effects.

Furthermore, experiments using mixed electrolytes mimicking artificial seawater show intermediate performance (~78 nW at $\Delta C = 1000$), underscoring the influence of competing ionic species on transport characteristics. The osmotic power increases consistently with concentration gradient, demonstrating scalable energy generation. Overall, these findings provide direct insight into multi-ion transport mechanisms in heterostructure nanopores and establish MoSe₂/MoS₂ systems as promising platforms for practical and scalable blue energy harvesting applications.

Keywords: MoSe₂/MoS₂ heterostructure, Multi-ion transport, Artificial seawater, Nanopores, Osmotic energy.

References:

1. PATTLE, R. E., Nature, 174, (1954), 660–660.
2. Manikandan, D.; Karishma, S.; Kumar, M.; Nayak, P. K., npj 2D Mater. Appl. 8, (2024), 1–19.

3. Feng, J.; Graf, M.; Liu, K.; Ovchinnikov, D.; Dumcenco, D.; Heiranian, M.; Nandigana, V.; Aluru, N. R.; Kis, A.; Radenovic, Nature, 536, (2016), 197–200.

Thermal Stability of 2D PtSSe-SnSSe Janus Lateral Heterostructures – A DFT Study

Karthikeyan C¹, G. Tejaswini¹, Anjana E Sudheer¹, Amrendra Kumar^{2,3}, M Vallinayagam^{4,5}, M. Zschornak^{4,5}, C. Kamal^{2,3}, D. Murali^{1, a}.

¹Department of Sciences, Indian Institute of Information Technology Design and Manufacturing Kurnool, Andhra Pradesh 518008, India

²Theory and Simulations Laboratory, Theoretical and Computational Physics Section, Raja Ramanna Centre for Advanced Technology, Indore 452013, India

³Homi Bhabha National Institute, Training School Complex, Mumbai 400094, India

⁴Institute of Experimental Physics, TU Bergakademie Freiberg, Leipziger Str. 23, 09596 Freiberg, Germany

⁵Technical Physics, University of Applied Sciences, Friedrich-List-Platz 1, 01069 Dresden, Germany und Wirtschaft, Friedrich-List-Platz 1, 01069 Dresden, Germany

^aCorresponding Author: dmurali@iiitk.ac.in

In the present work, we design the Janus lateral heterostructure of (PtSSe)_n – (SnSSe)_n in a horizontal(in-plane) direction, where n = 2,3,4 represents the number of unit cells in each monolayer. The successful fabrication of graphene-hexagonal boron nitride (h-BN) lateral heterostructure has opened a new area of research across a wide range of materials [1,2]. Herein, we explore the structural and electronic properties of these designed PtSSe-SnSSe lateral heterostructures. The lattice mismatch between the monolayers PtSSe and SnSSe is found 3.2%, which is well within the limit to build lateral heterostructures. The observed formation energy of these lateral heterostructures is negative, indicating that these lateral heterostructures are energetically stable. Ab initio molecular dynamics reveals that these 1T-phase Janus lateral heterostructures exhibit thermal stability at 300 K [3]. From electronic band structures, it is found that the (PtSSe)₂ – (SnSSe)₂ heterostructure exhibits an indirect band gap of 1.25 eV. After increasing the number of unit cells corresponding to each of the individual layers, it makes a transition from an indirect to a direct band gap semiconductor with a band gap of 1.08 eV for (PtSSe)₃ – (SnSSe)₃ and 0.94 eV for (PtSSe)₄ – (SnSSe)₄, respectively. This observation highlights an interesting and significant phenomenon inherent to lateral heterostructures. The Layered Resolved Density of States (LDOS) reveals that these Janus lateral heterostructures exhibit type-1 (straddling) band alignment, with the conduction and valence bands dominated by the PtSSe monolayer, indicating that these structures hold promise for light-emitting diodes, solar cells and catalytic applications. These results suggest that the proposed work on Janus lateral heterostructures holds promise for applications in thermoelectric and photocatalytic materials, as well as in optoelectronic devices.

Keywords: Lateral heterostructure, thermal Stability, electronic properties, type -1 band alignment and Density Functional theory.

References:

1. Zheng Liu, Xuebei Yang, Wu Zhou, Yongii Gong, Sidong Lei, Xuebei Yang, Jiangnan Zhang, Jingjiang Y, Ken P. Hackenberg, Aydin Babakhani, Juan-Carlos Idrobo, Robert Vajtai, JunLOu and Pulickel M. Ajayan, Nature Technology, 8, 2, 119-124 (2013).
2. Zhongliang Shen, Kairen, Ruxing Zheng, Zhaoming Huang, Zhen Cui, Zijun Zheng, Li Wang, Frontiers in materials, 9, 838648 (2022).

3. M. Vallinayagam, Anjana E Sudheer, A. Kumar, G. Tejaswini, M. Posselt, C. Kamal, D. Murali and M. Zschornak, arXiv preprint arXiv:2512.12626 (2025).

A-Site Cu Engineering in $\text{Cs}_3\text{Bi}_2\text{I}_9$ -Derived Perovskite-Inspired Materials for Enhanced Charge Transport and Magneto-Optoelectronic Performance

Gopal Sa¹, K. D. M Rao*

¹ School of Applied & Interdisciplinary Sciences, Indian Association for the Cultivation of Science, Kolkata, Jadavpur, 700032, India

*Corresponding Author E-mail: saiskdmrao@iacs.res.in

Bismuth-based halide perovskite-inspired materials (PIMs) are emerging as promising, environmentally friendly alternatives to lead-based perovskites due to their superior stability and low toxicity.^[1] However, their practical application in optoelectronics is often limited by wide band gaps (≥ 2 eV) and low electronic dimensionality. In this work, we have synthesized $\text{Cs}_{2.86}\text{Cu}_{0.14}\text{Bi}_2\text{I}_9$, a two-dimensional perovskite-inspired material obtained via partial substitution of Cs^+ with Cu^+ at the A-site of $\text{Cs}_3\text{Bi}_2\text{I}_9$ via low-cost solution process. Single-crystal X-ray diffraction confirms that Cu atoms preferentially occupy edge positions within the hexagonal lattice, resulting in reduced lattice parameters compared to the parent compound. This A-site cation engineering enhances orbital hybridization between Ag 5s and I 6p states,^[2] leading to a reduced band gap of 1.73 eV and a more delocalized electronic structure. As a result, $\text{Cs}_{2.86}\text{Cu}_{0.14}\text{Bi}_2\text{I}_9$ exhibits electrical resistivity reduced by six orders of magnitude. X-ray photoelectron spectroscopy (XPS) analysis confirms that Cu exists in the +2-oxidation state, corresponding to a d^9 electronic configuration. This configuration contains an unpaired electron, rendering the system magnetically active. The magnetic nature is further supported by the presence of a pronounced signal in electron paramagnetic resonance (EPR) measurements at both room and low temperatures. Additionally, magnetization versus magnetic field (M-H) measurements at low temperature exhibit magnetic hysteresis loops reveal weak ferromagnetic behaviour^[3] with saturation magnetization 150 memu/g. Leveraging these properties, photodetectors based on $\text{Cs}_{2.86}\text{Cu}_{0.14}\text{Bi}_2\text{I}_9$ demonstrate a Narrow spectral response with a photosensitivity of 150 which is 15 order higher than pristine. This study highlights the effectiveness of A-site cation engineering in designing high-performance, low-toxicity PIMs for optoelectronic applications.

Keywords: Perovskite; Solution process; Ferromagnetic; Narrowband; Photodetector

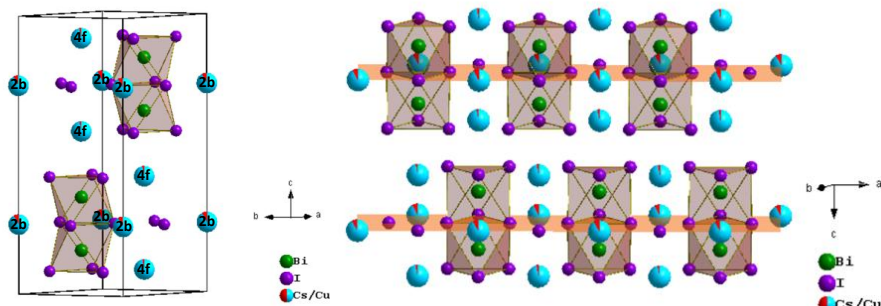


Figure 1. Unit cell from solved SCXRD of as synthesized solution process $\text{Cs}_{2.86}\text{Cu}_{0.14}\text{Bi}_2\text{I}_9$ crystal with $R_{\text{int}} = 9\%$

References:

1. M. Hossain, K. Singh, A. Narwal, M. S. Sheikh, S. K. Reddy, K. Vankayala, A. Singh, S. Khan, S. Khan, P. K. Velpula, M. Chirumamilla, S. S. R. K. C. Yamijala, G. K. Grandhi, P. Vivo, and K. D. M. Rao, *Chem. Mater.*, **36**, (2024), 7781–7791
2. S. Li, S. Song, P. Lv, S. Wang, J. Hong, and G. Tang, *Phys. Chem. Chem. Phys.*, **27**, (2025), 14948.
3. J. Zhao, J. Zhao, C. Shang, K. Yang, H. Wang, C. Yan, Z. Wang, C. Fang, and J. Luo, *Adv. Funct. Mater.*, **0**, (2026), e75359.

Screening of Energetically Stable High Entropy Hailed Perovskites Using Cluster Expansion Method for Optoelectronic Devices

Vinay V. Bhatt^{1*}, Kishan H. Mali¹, Alpa Dashora¹,

¹*Department of Physics, Faculty of Science, Maharaja Sayajirao University of Baroda, Vadodara, 390020*

*E-mail: bhattvinay139@gmail.com

As the increasing demand of renewable energy and limited amount of energy sources like fossil fuels, a new class of materials with better properties are required for optoelectronic applications. Multi component alloys specifically halide perovskites are promising candidates for optoelectronic applications with large configuration space which allows fine tuning desired property by changing local environmental chemistry of elements [1]. First-principles methods like Density Functional Theory are computationally expensive and not suitable for such large number of structures. This study shows a methodology for screening suitable configurations among large number of candidates on basis of heat of formation. Primary step of this process is to generate all possible configuration and obtaining energies of all structures using Machine Learning Force Fields [2]. After generating sufficient data set (Structure-Energy pairs), it can be used to train the Cluster Expansion (CE) model. Integrated Cluster Expansion Toolkit [3] (ICET)-a python library is used to efficiently handle structure generation and CE model training part. Once the model is trained and tested over sufficient structures outside training dataset with acceptable error margins, it can be used with Monte Carlo (MC) method to simulate over large configuration space to obtain desired parameters with balanced computational efficiency and accuracy. Using entropy with heat of formation $\Delta G_{\text{formation}}$ can be calculated for each structure which can be used as a parameter to comment on stability of structures. At the end Python data analysis tools are used to plot and visualize configurationally stable candidates. Screened materials now can be used for further property analysis which reduces the time and cost required to select from all configurations. This approach opens the door for further research on high entropy materials with desired properties like high power conversion efficiency material for absorber layer of high entropy halide perovskite solar cell, narrow emission range for LEDs and LASER.

Keywords: High Entropy Alloys; Halide Perovskites; Cluster Expansion; Optoelectronics.

References:

1. Jun-Zhong Xie et. al, Journal of Chemical Theory and Computation, 20, (2024), 6207-6217.
2. Ångqvist, M., Muñoz, W.A., Rahm, J.M., Fransson, E., Durniak, C., Rozyczko, P., Rod, T.H. and Erhart, P. (2019), Adv. Theory Simul., 2: 1900015.
3. Søren Smidstrup, Kurt Stokbro, Anders Blom, Troels Markussen, Jess Wellendorff, Julian Schneider, Tue Gunst, Brecht Verstichel, Petr A Khomyakov, Ulrik G Vej-Hansen, Mads Brandbyge et. al, J. Phys: Condens. Matter (APS), 32, (2020), 015901.

Fluorophosphate: A Promising Cathode for Sodium-Ion Battery

Arshiya Pathan^a, Ranjit Kate,^b Vandana Mooss^{a*}

^a Department of Chemistry, Dr. Vishwanath Karad, MIT World Peace University, Pune, India

^b R & D Department, Dr. Vishwanath Karad, MIT World Peace University, Pune,

*Corresponding Author Email: Vandana.moss@mitwpu.edu.in

Sodium-ion batteries (SIBs) have emerged as a sustainable and cost-effective alternative to lithium-ion batteries due to the low cost and the abundant availability of sodium. Among various polyanionic compounds, NaVPO₄F stands out as a high-voltage cathode material due to its excellent structural stability, high operating voltage, and favorable sodium-ion transport characteristics. In this study, NaVPO₄F was successfully synthesized using a sol-gel method to achieve uniform cathode material morphology, crystallinity, and high phase purity. The synthesized material was characterized using X-ray diffraction (XRD) and scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM-EDX) to confirm its crystal structure, elemental identification, and surface morphology. Electrochemical performance was evaluated through the cyclic voltammetry (CV), galvanostatic charge-discharge tests, and electrochemical impedance spectroscopy (EIS). The synthesized material demonstrated promising capacity, good rate capability, and stable cycling behavior. These results confirmed the potential of sol-gel-derived NaVPO₄F as an efficient cathode material for sodium-ion batteries, offering valuable insights into the development of advanced energy storage systems.

Keywords: Energy storage, cathode material, sol-gel synthesis, Na-ion battery, cost-effective.

Reference:

1. Y. Xu, L. Yin, C. Yang, Y. Lei, and H. Zhang, ACS Sustain. Chem. Eng., vol. 13, no. 35, pp. 14336–14345, Sep. 2025, doi: 10.1021/acssuschemeng.5c03685.
2. Y. Wang, D. Bin, C. Wang, Y. Cao, and Y. Xia, Batter. Supercaps, vol. 6, no. 10, Oct. 2023, doi: 10.1002/batt.202300275.

Top-Down Synthesis of Ti₂CT_x MXene from Ti₂AlC Precursor for Advanced Electrochemical Energy Storage

Suguna Palanimuthu¹, Calvin Prakashraj Martin², Ayyappan Sathya^{1*}

¹Advanced Nanostructured Materials Laboratory, Department of Physics, School of Electrical & Electronics Engineering (SEEE), SASTRA Deemed to be University, Thanjavur-613 401, Tamil Nadu, India.

*Email: ayyappan@phy.sastra.edu

MAX phase, a class of layered ternary carbides and nitrides, has a unique metallic and ceramic nature, with properties such as high thermal conductivity, damage tolerance, and oxidation resistance [1]. MXenes, two-dimensional materials derived from the parent MAX phase, have unique properties such as higher electrical conductivity, mechanical strength, and surface area [2]. Here, Ti_2CT_x MXene is synthesised via 72 hrs of LiF/HCl etching of Ti_2AlC MAX phase synthesised at 1400°C for 1 hour at an inert atmosphere. Fig. 1 shows the XRD pattern of pristine Ti_2AlC MAX phase, multilayer and single layers of Ti_2CT_x MXene inks after the etching process. Ti_2CT_x MXene possesses higher capacitance due to its surface area, charge transportation and diffusion nature. Due to its good conducting nature, it will be further taken towards the electrochemical applications [3]. The layered surface morphology of Ti_2CT_x , along with large interlayer spacing, facilitates ion intercalation and diffusion, for better charge storage mechanisms towards batteries and supercapacitors.

Keywords: MAX phase, Milling, Pressure-less sintering, Etching, MXenes

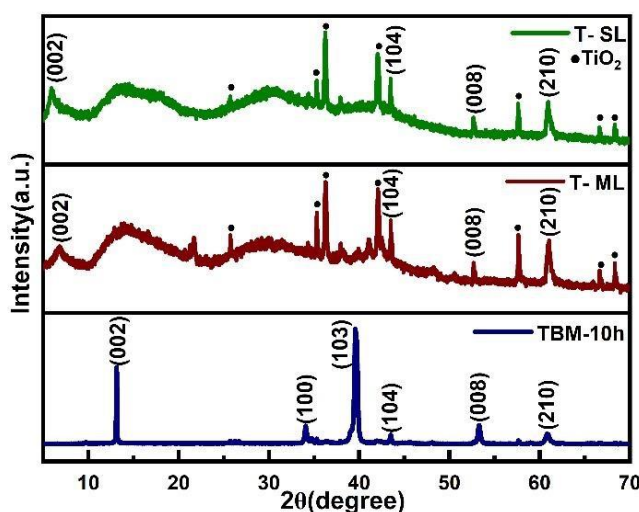


Fig 1: XRD Pattern for the structural confirmation of Ti_2AlC MAX phase with MXene

References:

1. Edrisi, A., Aghajani, H., Seyedein, S. H., & Tabrizi, A. T. *Ceramics International*, 50(23), (2024), 50846–50854
2. Li, J., Yuan, X., Lin, C., Yang, Y & Xu, L., Du, X., *Advanced Energy Materials*, 7(15), (2017)
3. Zhu, K., Jin, Y., Du, F., Gao, S., & Gao, Z., Meng, X., *Journal of Energy Chemistry*, 31, 11–18, (2018)

From Jackfruit Waste to Supercapacitor Electrode

Neha Joshi ¹, Vaishnavi Hangarge ², Shobha Birajdar ² & Satishchandra Ogale ^{1,3*}

¹ *Centre for Energy Science, Indian Institute of Science Education and Research, Dr. Homi Bhabha Road, Pashan, Pune 411008, India*

² *The Department of Chemistry at Nowrosjee Wadia College, Pune 411001, India*

³ *Research Institute for Sustainable Energy, TCG-CREST, Kolkata 700091, India*

*Corresponding Author E-mail: satishogale@gmail.com

Amongst various fruits, jackfruit stands out for a very large quantity of waste biomass vis a vis the edible fruit content, including peels, seeds, and fibrous residues. These present a rich bioresource of carbonaceous material that can be transformed to high quality carbon for energy applications. Thus, in this work, the collected jackfruit biomass was subjected to controlled pyrolysis to produce carbon, for use in batteries and supercapacitors. To improve the material performance, the carbon obtained by high temperature pyrolysis was further treated by acid washing, which enhanced its surface properties and porosity. Among the prepared samples, the acid-treated carbon showed the best characteristics, including higher surface area 290 m²/gm and improved purity. This material was used to fabricate a symmetric supercapacitor using carbon cloth as the current collector. In addition to multiple materials characterizations such as XRD, RAMAN, FESEM, XPS, charge–discharge measurements. These demonstrated a specific capacitance of 3.43 F g⁻¹ and impressive cycling stability. Overall, this study highlights a simple and sustainable pathway to transform jackfruit waste into value-added materials, contributing to greener and more cost-effective energy storage technologies.

Keywords: Jackfruit biomass, Biomass-derived carbon, Electrochemical energy storage

References:

1. Y.L. Bai, C.C. Zhang, F. Rong, Z.X. Guo, and K.X. Wang, Chemistry A European Journal, 30 (2024), e202304157.
2. M. Sevilla and A. B. Fuertes, ACS Nano, 8 (2014), 5069–5078.

Towards defect-free laser-induced graphene coating on copper foil for anode-free Li metal batteries

Aarti Gunjal¹, Swati Jadhav² and Satishchandra Ogale^{1,2*}

¹ *Centre for Energy Science, Department of Physics, Indian Institute of Science Education and Research, Pune 411008, India*

² *Research Institute for Sustainable Energy, TCG-CREST, Salt Lake, Kolkata-700091, India*

Corresponding Author Email: satishogale@gmail.com

Laser-induced graphene (LIG) is a very well-established process for realizing functional carbon coatings on different substrates. Depending on the laser processing parameters it yields structurally, morphologically and chemically interesting forms which are also adherent because of the thermal energy input. In most cases, however, a defect band (D band) is invariably observed in the Raman spectrum of such coatings, in addition to the so-called G-band corresponding to desired graphitic carbon. In this work we have applied CO₂ laser processing to carbonize the coatings of the dried powder of aromatic eucalyptus (EU) leaves on metals. The effects of laser power density, scanning speed, substrate choice, and precursor coating thickness on the quality of

carbon are carefully examined and optimized to obtain a unique almost defect-free few layers graphene coating, as reflected by a miniscule D-band and a significant 2D signature in the Raman spectra of few-layer graphene. Comparative studies on other aromatic leaves sample (e.g. lemon grass) and non-aromatic leaves suggest that aromaticity is important for realizing an enhanced 2D band, which could be due to internal turbulent cavitation upon laser-induced transient heating. *In situ* nitrogen doping is also achieved by applying the LIG process to a mixture of EU powder loaded with urea. Remarkably, the nitrogen incorporated coatings on copper and aluminium current collectors render far superior performance in anode-free Li metal batteries, respectively, as against the defect-free few layers graphene coating. In fact, half cells with alkali metals show an impressive cycling stability of over 450 cycles for Li ion battery. Through thorough characterizations employing multiple techniques, an attempt is made to develop a mechanistic understanding of the issues involved.

Keywords: Laser induced graphene, Composites, Earth Abundant Materials, Electrode, Green Hydrogen, Li metal battery.

References:

1. M.Wable, M. Furquan, M. Paygude, A. Shetti, S. Ogale, * and A. Banerjee*, ACS Appl. Energy Mater, 5 (9), 2022, 10940-10951.
2. S. Saha, S. Hiwase, S. Mondal, A. Deshmukh, and S. Ogale, Carbon, 228 (2024), 119275.

Structural and Thermoelectric Properties of High-Entropy Spinel Oxide

Jitendra Kumar*, Ashutosh Kumar

Functional Materials Laboratory, Department of Materials Science and Metallurgical Engineering, Indian Institute of Technology Bhilai, Chhattisgarh 491002, India

*Email: jitendrakum@iitbhilai.ac.in

High entropy oxides (HEOs), consisting of five or more cations in equimolar ratio, have attracted attention as functional materials due to their high configurational entropy, lattice disorder, and tunable transport properties. In the present study, a novel high-entropy spinel oxide ($\text{Co}^*\text{3O}_4$) was synthesized using a hydrothermal method, and its structural and thermoelectric properties were systematically investigated. Phase formation and crystal structure were analyzed using X-ray diffraction followed by Rietveld refinement, confirming the formation of a spinel structure. The homogeneous distribution of constituent elements was verified by energy-dispersive X-ray spectroscopy, depicting successful incorporation of multiple cations in the system. X-ray photoelectron spectroscopy (XPS) was used to determine the oxidation states of the elements. Thermoelectric properties viz. Seebeck coefficient, electrical conductivity, and thermal conductivity, were measured across a wide temperature range from 300K-1100K. The positive Seebeck coefficient observed for all samples confirms hole-dominated conduction, while the increase in electrical conductivity with temperature suggests non-degenerate semiconducting transport behavior. A low thermal conductivity of $\sim 1.37 \text{ W m}^{-1} \text{ K}^{-1}$ was obtained at 300 K, which is significantly lower than that of conventional spinel oxides. This remarkable reduction is attributed to enhanced phonon scattering arising from severe lattice disorder and compositional complexity in the high-entropy framework. This study highlights entropy-engineering as an efficient strategy to develop promising p-type material at higher temperatures for power generation application.

Keywords: High-entropy oxides, Thermal conductivity, Electrical conductivity, Seebeck coefficient, Configurational entropy, Figure of merit

Impact of Different Solvents on the Structural, Optical, and Photocatalytic Performance of Pristine Bi₂S₃

Priyanka¹, Chandni Devi^{1,2,*}, Gaurav Kumar Yogesh³

¹*Department of Physics and Astrophysics, School of Basic Sciences, Central University of Haryana, Mahendergarh, 123031, Haryana*

²*Department of Applied Sciences and Humanities, School of Engineering & Technology, Central University of Haryana, Mahendergarh, 123031, India*

³*Materials and Production Engineering, The Sirindhorn International Thai-German Graduate School of Engineering, King Mongkut's University of Technology North Bangkok, Bangkok-10800, Thailand*

*Corresponding Author Email: chandni@cuh.ac.in

In this study, pristine Bi₂S₃ was synthesized via a facile hydrothermal method using different solvents such as double-distilled water (DDW)[1], ethylene glycol (EG)[2], and ethanol (Eth)[3]. The structural, functional, and morphological impacts of these solvents were analyzed by X-ray diffraction (XRD), Fourier-transform infrared spectroscopy (FTIR), and field-emission scanning electron microscopy (FESEM). The XRD data have been analyzed by the Fullproof programme through the process of Rietveld refinement, and visualization of the lattice cell structure has been produced using VESTA software. The electronic charge transfer resistance was measured by electrochemical impedance spectroscopy (EIS), and the band gap by UV-DRS spectroscopy. Finally, the practical impact of different solvents was demonstrated by photocatalytic dye degradation under natural sunlight irradiation.

Keywords: Bi₂S₃, Hydrothermal, Rietveld Refinement, Photocatalysis, MB dye.

References:

1. P. Jangra, S.K. Sharma, C. Devi, et.al. Mater. Sci. Semicond. Process. 197 (2025) 109710.
2. M. Iqbal, A. Ibrar, A. Ali, et.al. J. Mol. Struct. 1267 (2022) 133598.
3. M.K. Verma, A. Gupta, R. Kumar, et.al. Mater. Today Proc. (2023).

Tailoring defects in solution-processed SnO₂ for enhanced interfacial charge dynamics in perovskite solar cells

Yash Taneja^{1,2}, Ranbir Singh^{1,2*}

¹*School of Mechanical and Materials Engineering, Indian Institute of Technology Mandi, Kamand, Mandi, Himachal Pradesh, India*

²*Advanced Energy Conversion Laboratory (AECL), Indian Institute of Technology (IIT) Mandi, Mandi, Himachal Pradesh, 175005, India*

*Corresponding Author E-mail: ranbir@iitmandi.ac.in

Aqueous synthesis of SnO₂ provides a scalable approach for electron transport layers in perovskite solar cells (PSCs), though precise control of defect chemistry remains difficult. While oxygen vacancies are necessary for n-type conductivity, excessive vacancies introduce deep trap states that promote non-radiative recombination. Here, we report a simple room-temperature chemical method to regulate defect density in SnO₂ by tuning the ratio of hydrogen peroxide (H₂O₂) to a fixed tin oxalate precursor, enabling controlled oxidation. Structural disorder and

chemical bonding were examined using FTIR and UV-Vis spectroscopy, while film morphology was analyzed via AFM and SEM. Urbach energy analysis was employed to quantify sub-bandgap disorder and identify the optimal condition that minimizes trap states without compromising film quality. The optimized SnO₂ films are expected to improve device stability and efficiency in PSCs, offering a low-cost alternative to complex and energy-intensive fabrication processes for electron transport layers in optoelectronic devices.

Keywords: Defect engineering, Oxygen Vacancies, Tin Oxide, Perovskite Solar Cells.

References:

1. Taneja, Yash, Dheeraj Dube, and Ranbir Singh. Journal of Materials Chemistry C 12, no. 37 (2024): 14774-14808.
2. Taneja, Y., Chandila, M., Kumar, P., Naidu, B. S., Sahoo, M. R., Dube, D., & Singh, R. (2026). Small, e12808.
3. Yash Taneja and Singh, Ranbir. EES Solar (2026).

Surface-Modified Pt/Al₂O₃ Nanocatalysts for Hydrogen Recombination application in Humid Environments

Sushmita Bhartiya^{1,2*}, Rashmi Singh¹, Ashish Singh^{1,2}, Gurvinderjit Singh^{1,2}, Sunil Verma^{1,2}

¹Photonic Materials Technology Section, Raja Ramanna Centre for Advanced Technology, Indore 452 013, India

²Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai, India

*Corresponding Author E-mail: sushmitab@rrcat.gov.in

One of the most effective and reliable routes for hydrogen control in the event of an accidental release is the catalytic H₂–O₂ recombination. Platinum (Pt) based catalysts are widely employed for this reaction, however, their high cost and performance degradation under humid conditions remain significant challenges [1,2]. In the present work, a hydrophobic alumina-supported Pt nanoparticle based catalyst was synthesized, aiming to achieve both platinum economy and enhanced stability in wet conditions. High surface area γ -Alumina (~195 m²/g) was used as a support to ensure uniform metal dispersion and strong metal-surface interaction. Pt/Al₂O₃ powders with varying Pt loadings (2%, 5% and 7.5 wt.%) on the powder surface were prepared by polyol method [3]. Among these 5 wt% Pt loading was found to be optimum, with an active metallic surface area of ~5.4 m²/g and average particle size 2.1 nm. It also exhibits good thermal stability under thermal exposure for 4 hrs at 450°C for 20 cycles. A hydrophobic catalyst was prepared by coating 5 wt% Pt-loaded Al₂O₃ powder using polytetrafluoroethylene (PTFE) binder on SS mesh. Catalytic activity was evaluated in closed reactor system, where change in H₂ concentration with time was recorded by the TCD detector. The H₂ recombination activity of the synthesized catalyst, expressed in terms of half-life (T_{1/2}), was approximately 63s at an initial hydrogen concentration of 4.5% in air. Further testing for humidity tolerance, demonstrated no loss in catalytic activity after the catalyst was exposed to steam for two hours. This stability is attributed to the hydrophobic PTFE coating, which likely prevents water from deactivating the active sites. The combined effect of efficient Pt utilization in nanoscale form and enhanced

moisture resistance makes this catalyst a promising candidate for hydrogen mitigation applications at ambient conditions.

Keywords: H₂-O₂ recombination; Catalyst; Al₂O₃ support; Pt nanoparticles; Hydrophobic.

References:

1. C. Xiao, Y. Yang, X. Zhou, Y. Du and Z. Tan, *Microporous and Mesoporous Materials*, 279, (2019) 395-399 .
2. D. Wang, Z. Wang, J. Xiong, Z. Yu, and X. Lu, *Chem Cat Chem*, 17, (2025), <https://doi.org/10.1002/cctc.202500182>.
3. R. Singh, M.K. Singh, S. Bhartiya, A. Singh, D.K. Kohli, P. C. Ghosh, S. Meenakshi and P.K. Gupta, *International Journal of Hydrogen Energy*, 42, (2017), 11110-11117.

Enhancing Mechanical Energy Harvesting with Robust Sponge TENG Parallel Arrays

Shubham Thakur^{1,2}, Bablesh Gupta^{1,2}, Sakshi Dhiman^{1,2}, Ranbir Singh^{1,2*}

¹ *School of Mechanical and Materials Engineering, Indian Institute of Technology Mandi, Kamand, Mandi, Himachal Pradesh, 175005, India*

² *Advanced Energy Conversion Laboratory (AECL), Indian Institute of Technology Mandi, Kamand, Himachal Pradesh, 175005, India*

Corresponding Email ID: ranbir@iitmandi.ac.in

The escalating demand for self-powered portable electronics has driven the development of efficient ambient mechanical energy harvesting technologies. However, triboelectric nanogenerators (TENGs) struggle with direct battery charging due to low output current and irregular voltage. To overcome this, a sponge-based TENG integrated power management circuit was developed for efficient Li-ion battery charging. Where triboelectric porous and elastic sponge structure was used to enhances surface charge generation, producing high output current under tapping frequency from 1 to 4 Hz. Further a circuit designed for current rectification and regulation that converts the irregular AC into stable DC output suitable for charging and powering the low-energy devices. We also explored the parallel configurations of multiple TENG units, increasing output current from 1.48 to 1.94 μ A. This parallel configuration improves the charging of Li-ion battery from 0.99 V for a single TENG device to 1.005 V for three interconnected devices, indicating the superior performance of parallel sponge-based TENGs for battery applications.

Keywords: Triboelectric; Parallel interconnections; Power management circuit; Sponge-based TENG; Energy storage, Li-ion battery.

References:

1. X. Pu, L. Mengmeng, L. Linxuan, Z. Chi, *Advanced Science*, 3, (2016), 1500255/1.
2. S. Dhiman, B. Gupta, A. Dabrey, *Sustainable Energy Fuels*, 10, (2026), 2028-2039
3. Fan, Feng-Ru, T. Zhong-Qun, and W.L Zhong, *Nano energy*, 1, (2012), 328-334/2.

Multimodal Computational Insights Into Hydrogen Evolution Reaction Mitigation in Zn Electrolytes: The Role of Anion-Surface Coupling

Soumya U K¹, Amreen Bano^{1,2*}

¹Research Institute for Sustainable Energy (RISE), TCG Centres for Research and Education in Science and Technology (TCG CREST), Sector V, Salt Lake, Kolkata 700091, India,

²Academy of Scientific and Innovative Research (AcSIR), Ghaziabad, 201002 India

*Corresponding Author E-mail: amreen.bano@tcgcrest.org

Aqueous zinc-based batteries are promising candidates for large-scale energy storage because of their low cost, high safety, and environmental compatibility. However, their practical performance is often hindered by parasitic side reactions, particularly the hydrogen evolution reaction (HER), which is strongly governed by the Zn^{2+} solvation environment and interfacial electrolyte chemistry.¹ In this work, molecular dynamics simulations were performed to investigate the role of anion chemistry in tuning Zn^{2+} solvation structure and its influence on HER activity in five representative electrolytes: ZnCl_2 , $\text{Zn}(\text{ClO}_4)_2$, ZnSO_4 , $\text{Zn}(\text{OTf})_2$, and $\text{Zn}(\text{TFSI})_2$. Radial distribution function and coordination analyses revealed that Zn^{2+} retains a water-rich first solvation shell in ZnCl_2 , $\text{Zn}(\text{ClO}_4)_2$ and ZnSO_4 systems, exhibiting Zn-H₂O coordination numbers of 5, 4, and 5, respectively. In contrast, bulky fluorinated anions in $\text{Zn}(\text{OTf})_2$ and $\text{Zn}(\text{TFSI})_2$ partially replaced coordinated water molecules², showing Zn-anion coordination numbers of 2 and 3. This substitution increased the free-water population in $\text{Zn}(\text{OTf})_2$ and $\text{Zn}(\text{TFSI})_2$ electrolytes to 841.29 and 849.89 molecules, respectively, indicating a greater tendency toward HER due to the enhanced reducibility of uncoordinated water molecules. To investigate interfacial effects, representative solvation structures were further examined on Cu, Sn, and Zn surfaces. Adsorption studies revealed comparatively stronger interfacial binding and suppressed HER activity on Sn surfaces. Experimental observations reported in previous studies showed similar trends, where anions preserved stronger hydrogen-bonded water networks, reduced electrolyte degradation, and improved Zn deposition stability and cycling reversibility compared to fluorinated electrolyte systems.³ Overall, this study provides atomistic insight into the interplay between anion chemistry, solvation structure, and surface reactivity, offering useful design principles for developing stable and efficient aqueous zinc-based energy storage systems.

Keywords: Electrolyte engineering; Aqueous zinc batteries; Hydrogen evolution reaction; Solvation structure; Molecular dynamics simulation.

References

1. J. Cao, D. Zhang, X. Zhang, Z. Zeng, J. Qin, and Y. Huang, *Energy & Environmental Science*, 15, (2022), 499-528.
2. L. Cao, D. Li, E. Hu, J. Xu, T. Deng, L. Ma, Y. Wang, X.-Q. Yang, and C. Wang, *Journal of the American Chemical Society*, 142, (2020), 21404-21409.
3. Y. Zhu, S. Thomas, T. Wang, X. Guo, Y. Wang, C. Liu, S. M. Sarathy, X. Zhang, O. M. Bakr, O. F. Mohammed, and H. N. Alshareef, *Science Advances*, 11, (2025), eadx8413.

Machine Learning-Driven Multiscale Computational and Experimental Investigation of Graphite-Based Anodes for Lithium-Ion Batteries

Antra Mohini ^{a,b}, Divyanshi Soni^c, Binson Babu^c and Amreen Bano^{*a,b}

- (a) Research Institute for Sustainable Energy (RISE), TCG Centres for Research and Education in Science and Technology (TCG-CREST), Kolkata 700091, India
- (b) Academy of Scientific and Innovative Research (AcSIR), Ghaziabad 201002, India
- (c) Department of Physics, School of Natural Science, Shiv Nadar Institution of Eminence, Uttar Pradesh-201314, India
- *Corresponding author Email: amreen.bano@tcgcrest.org

The development of high-performance graphite anodes is essential for improving the efficiency and durability of lithium-ion batteries (LIBs). In this study, multi-dopant engineering of three-layered graphite was investigated through the incorporation of Nitrogen (N), Phosphorus (P), Oxygen (O), and Boron (B) in different compositional configurations to enhance electrochemical performance. Density Functional Theory (DFT)-assisted MACE-MP0 machine learning interatomic potential (MLIP) calculations integrated with structural optimization in the Atomic Simulation Environment (ASE) were employed to design and evaluate doped graphite systems. The dopants were strategically selected to improve electrical conductivity, Li-ion intercalation behaviour, oxidation-state flexibility, and structural stability during lithiation-delithiation processes. Computational analysis identified the most energetically favourable configuration with enhanced Li-ion adsorption characteristics, reduced structural distortion, and improved structural stability. To validate the computational findings, experimental synthesis and electrochemical characterization of the optimized doped graphite compositions were carried out using conventional electrode fabrication and battery testing techniques. Electrochemical measurements performed in a two-electrode configuration using Li metal and LP30 electrolyte demonstrated enhanced Li-ion storage capacity, improved cycling stability, and favourable charge-transfer characteristics for the optimized doped graphite compared to pristine graphite. The improved electrochemical behaviour was attributed to the synergistic effect of N and O dopants, which create strong polar sites for Li-ion adsorption and stabilization, in good agreement with the computational predictions. These findings provide valuable insights into multi-dopant engineering strategies for designing advanced graphite-based anode materials for next-generation energy storage applications.

Keywords:

Lithium-ion batteries; Graphite anode; Multi-dopant engineering; Density Functional Theory (DFT); Energy storage materials

References:

1. Liu, J. et al. Polyaniline-induced nanocarbon with multi-doped heteroatoms for high-rate lithium-ion battery anodes. *Nanotechnology* 36, (2025), 365401.
2. Burke, K. & Wagner, L. O. DFT in a nutshell. *Int. J. Quantum Chem.* 113, (2013), 96-101.

Nitrogen Ion Irradiation-Induced Defect Engineering of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene for Supercapacitor Applications

Gauri J. Salunkhe¹, Amol Sonawane¹, Rajesh Jaiswar², Pravin S. Walke², Sanjay D. Dhole¹, Shailendra S. Dahiwal^{1*}

¹Department of Physics, Savitribai Phule Pune University (SPPU), Ganeshkhind Road, Ganeshkhind, Pune, Maharashtra 411007, India.

²National Centre for Nanosciences and Nanotechnology (NCNNUM), University of Mumbai, Kalina, Santacruz East, Mumbai, Maharashtra 400098, India

*Corresponding Author E-mail: ssd@physics.unipune.ac.in

Defect engineering is an effective strategy for tailoring the properties of two-dimensional materials for various applications. In the present work, nitrogen ion irradiation was employed to introduce controlled defects in $\text{Ti}_3\text{C}_2\text{T}_x$ MXene films for supercapacitor applications. The films were irradiated with low-energy nitrogen ions at fluences ranging from 10^{12} to 10^{16} ions/cm² to investigate the influence of irradiation-induced defects on their structural, surface, and electrochemical properties. X-ray diffraction analysis revealed changes in the (002) peak position, peak broadening, and reduced crystallinity after irradiation, indicating lattice distortion and defect formation. Raman spectroscopy showed peak shifts, intensity variations, increased FWHM, and changes in the D/G band intensity ratio, confirming enhanced structural disorder and defect density. Fourier transform infrared spectroscopy revealed modifications in surface functional groups and bonding environments. X-ray photoelectron spectroscopy confirmed changes in elemental composition and oxidation states, while the appearance of the N 1s peak indicated successful nitrogen incorporation and modifications to the electronic structure. Scanning electron microscopy demonstrated surface roughening, exfoliation, and irradiation-induced microstructural defects in the MXene layers. The electrochemical properties were investigated using cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS) in a three-electrode configuration using 0.5 M H_2SO_4 as the electrolyte. The CV curves exhibited a nearly rectangular shape, indicating dominant electric-double-layer capacitive behaviour, with slight deviations due to defect-induced resistance and ion-transport limitations. GCD measurements revealed enhanced capacitive performance for the irradiated samples. Compared with pristine $\text{Ti}_3\text{C}_2\text{T}_x$ MXene, the specific capacitance increased from 233 F g^{−1} to 426 F g^{−1} after nitrogen ion irradiation due to irradiation-induced active sites and improved ion accessibility. EIS analysis further revealed defect-induced variations in charge-transfer resistance and ion-transport kinetics. The study demonstrates that controlled nitrogen ion irradiation effectively tailors the defect density and electrochemical characteristics of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene for next-generation supercapacitor applications.

Keywords: $\text{Ti}_3\text{C}_2\text{T}_x$ MXene, Nitrogen Ion Irradiation, Low energy ion, Defect Engineering, Supercapacitors.

References:

1. Yang, Q., Wang, W., Li, X. et al. J Solid State Electrochem 27, 1955–1967 (2023).
2. Ikhiya, I.L., Nkele, A.C., Ahmad, I. et al. Indian J Phys (2025).

Graphene Oxide Integrated P (VDF–TrFE) Nanocomposite Films: Studies on Structural, Electrical and Dielectric properties for Energy Harvesting Applications

Sampa Sutradhar^{1,2,*}, Srinibas Satapathy^{1,2,*}, Raj Mohan S.¹, Bhumika Sharma¹, Pratik Deshmukh^{1,2}, Shovan Kumar Majumder^{1,2}

¹ *Laser Biomedical Applications Division, Raja Ramanna Centre for Advanced Technology, Indore-452013*

² *Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai-400094, India.*

*Corresponding Author: sampasutradhar@rrcat.gov.in, srinu73@rrcat.gov.in

Graphene Oxide (GO) has emerged as promising two-dimensional nanofillers for reinforcing flexible piezoelectric polymer nanocomposites, including Poly(vinylidene fluoride) and its copolymers, for various applications such as wearable health monitoring devices, energy harvesting etc. The performance of these materials is strongly governed by β -phase content and key dielectric properties. In this study, GO was synthesized using a modified Hummer's method and then incorporated into the poly (vinylidene fluoride–trifluoroethylene) (P(VDF–TrFE)), a lead-free piezoelectric polymer, to fabricate nano-composite films with varying GO concentrations (0.25, 0.5, 0.8, and 1 vol. %). Uniform thick nano-composite films ($\sim 40 \mu\text{m}$) were fabricated using doctor blade technique. X-ray diffraction studies confirmed the formation of the electro active β -phase of P(VDF–TrFE) ($2\theta \sim 19.98^\circ$) in both pristine and composite films. FTIR analysis further established the formation of pure β -phase only, which ensures superior piezoelectric properties. Electrical measurements unambiguously shows the decrease in leakage current with increase in GO loading, which indicates improved electrical insulation behavior. Frequency dependent dielectric measurements of these films, carried out in frequency range 10^2 – 10^6 Hz, showed a significant increase in permittivity (from 8.1 to 35.6) upon increasing GO concentration along with typical frequency-dependent dispersion, where higher values were observed at lower frequencies. Conductivity ($\sigma \sim 1.6 \times 10^{-11}$ – 2.65×10^{-6}) and dielectric loss ($\tan\delta \sim 0.019$ – 0.025) are enhanced with GO content. Overall, our studies confirms that the incorporation of GO effectively tailor the structural and dielectric properties of P(VDF–TrFE) films. Hence, this work highlights the potential of GO-based polymer nanocomposites in energy harvesting applications.

Keywords: Ferroelectric polymers; Graphene oxide reinforcement; β -phase crystallinity; Charge transport; piezoelectricity

Type II Based $\text{TiSe}_2/\text{TiO}_2$ Heterostructures for Hydrogen Evolution Reactions (HER)

Laveena Pursharthi¹, Somnath Chanda Roy^{2,*}, Abhishek Misra^{2,*}

^{1,2} *Department of Physics, Indian Institute of Technology Madras, Chennai, India, 600036*

*somnath@iitm.ac.in

The hydrogen evolution reaction (HER) confers a sustainable method for producing hydrogen. The main goal of this work is to create a type-II heterostructure based on titanium diselenide/titanium dioxide ($\text{TiSe}_2/\text{TiO}_2$). For TiSe_2 , selenium and titanium oxide powder were used via Chemical vapour deposition (CVD) on silicon and for TiO_2 , hydrothermal synthesis in order to increase HER control. TiO_2 nanorods are combined with transition metal dichalcogenides, such as TiSe_2 , which are renowned for their exceptional qualities, to create a potent photocatalytic system. Field-emission scanning electron microscopy (FE-SEM), Raman spectroscopy, and X-ray diffraction (XRD) were among the characterization methods used. Both ultraviolet photoelectron spectroscopy (UPS) and X-ray photoelectron spectroscopy (XPS) provide a broad understanding of charge transport kinetics and electronic interactions. The UPS experiments

demonstrate that an improvement in catalytic active sites is defined by a shift in peak positions in XPS spectra. It was confirmed that the heterostructure's enhanced energy environment to start the charge transfer strengthened the creation of hydrogen. According to the UPS research, the heterostructure minimum work function (TiSe_2) calls for improvement in charge transfer. $\text{TiSe}_2/\text{TiO}_2$ heterostructure is therefore deemed a likely material for sustainable hydrogen generation.

Keywords: $\text{TiSe}_2/\text{TiO}_2$ heterostructure; Chemical vapour deposition; Hydrothermal synthesis.

References:

1. Arellano, Lucia G., et al. International Journal of Hydrogen Energy 48.96 (2023): 37500-37509.
2. Nguyen, Cong Thanh, D. Navanitha, and Arunadevi Natarajan. International Journal of Hydrogen Energy 165 (2025): 150927.
3. Dincer, Ibrahim, and Canan Acar. International journal of hydrogen energy 40.34 (2015): 11094-11111.

Effect of Epitaxial Strain on the Structural and Magnetic Properties of BiCrO_3 Thin Films

Bharti Ojha¹, Subho Saha¹, Ram Janay Choudhary^{1*}

¹UGC-DAE Consortium for Scientific Research, University Campus, Khandwa Road, Indore 452001, India

* bharti.ojha99@gmail.com

$\text{BiCrO}_3(\text{BCO})$ is a promising multiferroic material which has gained interest owing to its wide range of structural, electronic and magnetic properties. It can be used to realize a vast number of devices such as spintronic devices, optoelectronics, sensors, data storage memories, integrated circuits, catalysts, etc. BCO crystalizes in monoclinic structure with C2/c space group and antiferromagnetic below 120K and ferroelectric below 410K. Since BCO is metastable in bulk, stabilization through substrate induced strain provides an effective route to tailor its multifunctional properties. The tuning of strain in films leads to significant modifications bond lengths, bond angles, crystal-field splitting, and orbital hybridization, thereby influencing charge transfer, electronic bandwidth, and magnetic exchange interactions. Although BCO thin films have attracted considerable attention as multiferroic materials, the influence of strain on the correlation between structural distortion, chemical states, and magnetic behaviour remains insufficiently understood. In the present work, we aim to investigate the effects of substrate-induced epitaxial strain on the electronic and magnetic properties of thin films of BCO grown on SrTiO_3 and LaAlO_3 substrates using Pulsed Laser Deposition. XRD analysis of grown films confirms single phase growth of BCO film along 001 plane with tensile strain of 1% in out of plane. XPS spectra confirms 3+ charge state of Cr and Bi. Magnetization data shows multiple magnetic transitions. The BCO/LAO film exhibits a sharp magnetic transition at 155 K which can be associated with the ferromagnetic moments parasitic to the antiferromagnetic ordering of Cr^{3+} ions (140K) suggesting enhanced ordering. BCO/STO film shows persistent FC–ZFC bifurcation up to room temperature, suggesting magnetic frustration in addition to above transitions which is previously unreported. These observations demonstrate that substrate-induced strain critically governs the balance between antiferromagnetic exchange and weak ferromagnetism in epitaxial BiCrO_3 thin films. **Keywords:** multiferroic material; thin films; Pulsed Laser Deposition

References

1. Kanungo, Sudipta, and Tanusri Saha-Dasgupta, Physical Review B 83.10 (2011)
-

Thermo-Mechanical Hybrid Piezo-Pyroelectric Nanogenerator Based on Graphene-Reinforced BCZT-PVDF Composite for Enhanced Energy Harvesting

Abhijit Das¹, Sumit Chahal¹, Saket Asthana^{1*}

¹Advanced Functional Material Laboratory, Department of Physics, Indian Institute of Technology, Hyderabad 502285, India

* Corresponding author: asthanas@phy.iith.ac.in

Self-powered systems harnessing mechanical and thermal energies based on nanogenerators (NGs) that are obtained using low-cost materials and simple preparation methods have garnered significant attention. Herein, we report a lightweight, flexible hybrid NG based on the piezoelectric and pyroelectric effects of a Graphene nanosheets (GNS)-reinforced in barium calcium zirconate titanate (BCZT), embedded in polyvinylidene fluoride (PVDF), which can be directly used as an active layer without any post-poling treatment. The GNS were synthesized by a simple liquid-phase exfoliation method and mixed with BCZT ceramic powder. This mixture was combined with PVDF polymer at varying GNS concentrations from 10 mg to 40 mg to fabricate composite films for energy harvesting using solution casting. With increasing graphene concentrations the electroactive β -phase content $F(\beta)$ of PVDF increase, which is confirmed by Fourier transform infrared spectroscopy (FTIR) and we observed maximum $F(\beta)$ for 30 mg GNS (Gr-30) which is 79.46 %. The dielectric constant and conductivity also increases with increase the GNS up to the percolation threshold concentration $f_c = 0.0375 \pm 0.003$. For all the composite mechanical harvesting demonstrate with frequency from 1 Hz to 5 Hz and we observed maximum output voltage is 144.6 V for Gr-30 at 5 Hz under 20 N force and power density $89.6 \mu\text{W}/\text{cm}^2$. Moreover, a hybrid piezoelectric-pyroelectric energy harvesting was demonstrated from constant temperatures ranging from 50 °C to 90 °C, with an increment of 10 °C and we observed maximum output voltage 272 V at 90 °C for Gr-30 with power density $359.64 \mu\text{W}/\text{cm}^2$. The device powered an LED under mechanical impact and charged capacitors, demonstrating durability and practicality. This novel design of the GNS reinforced hybrid nanogenerator demonstrated here holds promise for self-powered wearable electronic textiles and to power generation on a large scale.

Keywords: GNS-BCZT hybrid system, Piezoelectric-Pyroelectric hybrid energy harvester, Percolation threshold, ‘Web-like’ or ‘bumpy’ texture, Agglomeration.

References:

1. Missaoui, Sarra, et al. ACS Applied Electronic Materials 6.8 (2024): 5608-5625.
 2. Jena, Lalit Kumar, et al. ACS Applied Electronic Materials 7.3 (2025): 1314-1328.
 3. Jiang, Z. Y., et al. Journal of Applied Physics 115.20 (2014).
-

Layer-Engineered MoS₂-Integrated Photo-Triboelectric Nanogenerator for Dual Energy Harvesting and Detection

Rudresh Kundugolmath¹, Sumit Chahal¹, Navneet Soin², Nisa Salim², Saket Asthana^{1*}

¹ *Department of Physics, Indian Institute of Technology Hyderabad, 502285, India*

² *Swinburne University of Technology, Hawthorn, VIC 3122, Australia*

*Corresponding Author E-mail: asthanas@phy.iith.ac.in

Dielectrics and fluoropolymers remain the primary choice for active layers in triboelectric nanogenerators (TENGs). However, their low surface-to-volume ratio, inherent polarization, charge density and limited functionalities limit the overall power density of the TENG devices^[1]. Whereas the 2D transition metal dichalcogenides, such as Molybdenum disulfide (MoS₂) address these challenges and also its broad photodetection range provides an opportunity to increase surface charge density when illuminated with UV and NIR light sources.^[2,3] This study demonstrates the potential integration of 2D-MoS₂ and graphite in TENG devices to simultaneously harness optical and mechanical energy while addressing the high in-plane resistance of the MoS₂ deposited substrates. 2D-MoS₂ obtained from liquid phase exfoliation technique was deposited on the already graphite coated cellulose paper. 2D-MoS₂/Graphite/Paper element was used as a tribonegative layer and kapton was employed as tribopositive layer. Under periodic contact and separation, an electrical signal was generated, which is demonstrated to be photo-dependent owing to the presence of a photo-active MoS₂ layer. To further investigate, the electric output performance of the TENG was compared in the dark, UV-illumination and NIR illumination conditions. The peak-to-peak voltage exhibited a sharp increase from ~430V in dark to ~490V and ~550V in UV and NIR light illuminations respectively. The higher output in NIR conditions in comparison to the UV illumination is backed with the high intensity NIR source used, leading to excess photo and thermally generated charge carriers due to thermal effects from the NIR source. The reported TENG exhibited a maximum output power density of 10.76 Wm⁻² thereby addressing the issue of low power density observed in the existing photosensitive TENG devices.

Keywords: Triboelectric nanogenerators; 2D MoS₂; Cellulose paper; Photoactive tribolayers.

References:

1. Zhao, P., Huang, Y., Li, P., Sharma, N., Murdoch, B. J., Rogers, A., ... & Soin, N. Nano Energy, 141, (2025), 111046.
2. Khan, S. A., Saqib, Q. M., Bae, J., & Cho, S. W. Chemical Engineering Journal, 516, (2025), 163854.
3. Chahal, S., Kundugolmath, R., Shinde, S. L., & Asthana, S. ACS Applied Energy Materials, 8(23), (2025), 17431-17441.

Role of Fe doping on OER activity of Ni₃N Thin Films

Shivani Sahu¹, Mukul Gupta^{1,*}

¹ *UGC-DAE Consortium for Scientific Research, University Campus, Khandwa Road, Indore 452001, India*

Corresponding author: mgupta@csr.res.in

Clean energy production is an important requirement for future generations and the electrochemical water splitting is an efficient route for green hydrogen production. However, oxygen evolution reaction (OER) is limited by sluggish reaction kinetics and high overpotential which reduce overall efficiency [1]. To overcome these reaction barriers and improve electrocatalytic performance by lowering overpotential, nickel nitride electrocatalysts are being widely investigated as cost-effective alternatives of noble metal based electrocatalyst like RuO_2 , IrO_2 etc. [2]. Nickel nitrides are formed in different phases e.g. Ni_2N , Ni_3N , Ni_4N , but in particularly Ni_3N exhibits promising OER activity [2]. In this work, we deposited Ni_3N thin films on an indium tin oxide (ITO) substrate via reactive sputtering process. Recently, it was observed that Fe doping in Ni-N creates active sites and boost the OER activity [3], therefore, we performed by co-sputtering of Fe during the growth of Ni_3N films, nominally at 5, 10, and 25 at.% Fe. The crystal structure of the films was investigated using x-ray diffraction as shown in Fig. 1(a), while XANES measurements were carried out at BL-01 beamline of Indus-2 synchrotron radiation source as shown in Fig.1(a, b). The Ni L-edge spectra exhibit a systematic shift toward lower energy with increasing Fe concentration indicating the incorporation of Fe atoms into the hcp- Ni_3N structure. This shift suggests a reduction in the Ni oxidation state due to charge transfer between Fe and N atoms. In contrast, the N K-edge (fig. 1c) shows a shift in the opposite direction compared to the Ni edge [3]. Electrochemical measurements for the OER along with stability analysis and reaction kinetics of Fe-doped Ni_3N , will be presented in the manuscript.

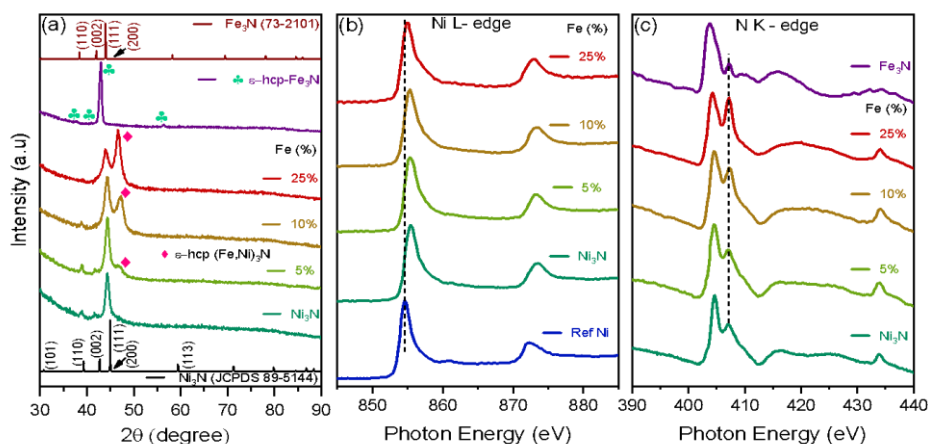


Figure 1(a) x-ray diffraction pattern of Fe-doped Ni_3N thin films and (b) Ni L and N K-edge XANES spectra for Fe doping of 0, 5, 10 and 25% at $\text{RN}_2 = 50\%$

Keywords: Thin films, Nickel nitride, electrocatalyst, oxygen evolution reaction.

References:

1. A. T. Khan, G. S. Priyanga, K. Khan, E. Pervaiz, T. Thomas, and M. Yang. *ChemSusChem* 12, no. 17 (2019): 3941-3954.
2. S. Sahu, P. Gupta, P. Rajput, A. Gloskovskii, and M. Gupta. *physica status solidi (RRL)–Rapid Research Letters* 20, no. 1 (2026): e202500327.
3. S. Zhao, M. Li, Han, D. Xu, J. Yang, Y Lin & Bao, J. *Advanced Functional Materials*, 28 (2018): 1706018.

Development and evaluation of transition metal compound-carbon based composites as electrodes for supercapacitor applications

Shwetha K P^{1,2*}

¹Centre for Nanomaterials and Devices, RV college of Engineering, Bengaluru 560056, India

²Department of Physics, RV College of Engineering, Bengaluru, 560059, India.

*Corresponding Author E-mail: shwethaprabhakarao@gmail.com

Climate change and depleting conventional energy sources require sustainable, high-performance energy storage. Supercapacitors have high power density and long lifespans, but low energy densities hinder commercialization. Hybrid electrode materials that combine the high theoretical capacitance of transition metal sulphide's (TMS) with the superior electrical conductivity of functionalized multi-walled carbon nanotubes (f-MWCNTs) address this limitation. The three TMS systems studied were monometallic nickel sulfide (NiS), cobalt disulfide (CoS₂), and bimetallic Nickel cobalt sulfide (NiCo₂S₄). A novel technique for the accurate regulation of nanoparticle growth employs a one-step hydrothermal synthesis at 180°C for 2 hours, utilizing cetyltrimethylammonium bromide (CTAB) as a surfactant. The carboxylic (-COOH) functionalization of multi-walled carbon nanotubes (MWCNTs) enhances surface energy, inhibits TMS aggregation, and increases chemical stability. This method creates flake- and rod-like architectures for fast ion intercalation and electron transport. Electrochemical studies showed that NiS@f-MWCNT had a peak specific capacitance of 1966 Fg⁻¹ at 1 Ag⁻¹ and 1400 Fg⁻¹ for pristine NiS. The CoS₂@f-MWCNT composite had 623 Fg⁻¹ and 97.4% stability over 10,000 cycles. A superior energy density of 78.3 Wh kg⁻¹ was achieved by the bimetallic NiCo₂S₄@f-MWCNT system. Kinetic modelling confirmed that capacitive and diffusion-controlled processes synergize across all tested TMS systems. Asymmetric supercapacitor (ASC) coin cells proved viability. The high-power densities of the NiS@f-MWCNT//AC and CoS₂@f-MWCNT//AC devices powered an Arduino-driven temperature sensor (accurate to 1°C), digital stopwatches, and LEDs. This study connects laboratory synthesis to practical electronic applications by providing a scalable, cost-effective framework for designing next-generation hybrid energy storage systems. The exceptional performance and practical feasibility of these hybrid materials, as thoroughly documented in peer-reviewed articles in the Journal of Energy Storage, provide a scalable and sustainable foundation for high-performance energy storage, facilitating the global transition to a carbon-neutral society.

Keywords: Transition metal sulfide; Carbon nanotubes; energy density; Supercapacitor; asymmetric coin cell device.

References:

1. K. P. Shwetha, et al., Journal of Energy Storage, 97, (2024), 112699.
2. K. P. Shwetha, et al., Journal of Energy Storage, 75, (2024), 109404.
3. K. P. Shwetha, et al., Journal of Energy Storage, 72, (2023), 108657

Machine Learning-Guided Electrochemical Behaviour of NiV₃O₈@MXene Asymmetric Supercapacitors

Nagendra Babu A P¹, Bahumanya R G², Shwetha K P^{3,4,†}, Sudha Kamath M K^{3,4,†}.

¹Department of Physics, Jnana Bharathi Campus, Bangalore University, Bengaluru, 560056, India

²*Department of Computer Science and Engineering, RV College of Engineering, Bengaluru-560059, India.*

³*Centre for Nanomaterials and Devices, RV college of Engineering, Bengaluru 560056, India*

⁴*Department of Physics, RV College of Engineering, Bengaluru, 560059, India.*

[†]Corresponding authors: sudhakamath@rvce.edu.in, shwethaprabhakarao@gmail.com

This paper introduces a synergistic experimental-computational framework for $\text{NiV}_3\text{O}_8/\text{MXene}$ (NIV-MX) heterostructures designed for advanced asymmetric supercapacitor (ASC) applications. The hydrothermal technique yielded phase-pure NiV_3O_8 , which was subsequently hybridized with conductive 2D MXene nanosheets to create a high-affinity heterointerface. Physicochemical analysis indicates that robust interfacial coupling induces localized alterations in electronic structure, facilitating accelerated ion diffusion and enhanced redox activity. The NIV-MX electrode exhibits a specific capacitance of 724 F g^{-1} at 1 A g^{-1} and maintains 87.44% of its capacity after 10,000 charge-discharge cycles, as validated by kinetic analyses. A systematic evaluation of negative electrode candidates (MXene, rGO, and AC) revealed that activated carbon is the best option for charge equilibrium and improved full-cell performance. The NIV-MX//AC asymmetric device is highly stable within a 1.8 V operating range, with an energy density of 62 Wh kg^{-1} at 978 W kg^{-1} and a cycling durability of 85.42%. This paper introduces a new AI-driven informatics layer, Quantum-Kernel Latent Representations, which condense high-dimensional datasets while preserving nonlinear interfacial relationships, as well as an Ising-type statistical framework that addresses scan-rate-dependent kinetic heterogeneity. Furthermore, Physics-Informed Machine Learning provides accurate performance predictions across a wide range of operational scenarios. This comprehensive approach creates a scalable framework for advanced energy storage systems by combining nanoscale material engineering and data-driven predictive analytics for next-generation power electronics.

Keywords: NiV_3O_8 , MXene, asymmetric supercapacitor, Quantum Kernel, Machine Learning.

References:

1. V. Soundharrajan, V. Vaiyapuri, et al., *Ceram. Int.*, 43 (2017) 13224–13232.
2. B. Sambandam, V. Soundharrajan, J. Song, S. Kim, J. Jo, D. T. Pham, et al., *J. Electroanal. Chem.*, 810 (2018) 34–40.

Study of Silver-Based Catalyst for Selective Catalytic Reduction of NO_x Using Hydrogen: Water Tolerance

Vinod Lohar,^{1,2} Vedang Banpurkar ¹ Shubhangi Umbarkar ^{2,3} Rajesh Kanawade^{*1,2}

¹*Physical and Material chemistry Division, CSIR –National Chemical Laboratory, Dr. Homi Bhabha Road, Pune, 411008, India*

²*Academy Of Scientific and Innovative Research (AcSIR), Ghaziabad 201002, India*

³*Catalyst and Inorganic chemistry Division, CSIR –National Chemical Laboratory, Dr. Homi Bhabha Road, Pune, 411008, India*

^{*}Corresponding author's email: rv.kanawade.ncl@csir.res.in

Rapid industrialization and transportation engines have significantly increased air pollution by emitting harmful gases like NO_x, SO_x, hydrocarbons, and carbon monoxide [1]. Among these pollutants, NO_x is a major contributor to environmental degradation and adverse

health effects, creating a strong need for efficient emission control technologies. Selective Catalytic Reduction (SCR) is one of the most effective methods for NO_x reduction. Though widely studied, conventional ammonia and hydrocarbon SCR systems are limited by ammonia slip, catalyst poisoning, fouling, and high operating costs. In recent years, hydrogen-assisted SCR (H₂-SCR) has emerged as a promising alternative due to its ability to reduce NO_x efficiently at low temperatures below 200 °C. Hydrogen is also an environmentally friendly reducing agent, as its combustion produces only water without CO₂ emissions. Despite these advantages, catalyst stability and water tolerance remain major challenges in H₂-SCR systems. Precious metal catalysts such as platinum (Pt) and palladium (Pd) exhibit high catalytic activity; however, their high cost, limited water tolerance, and sulphur poisoning restrict their practical applications [2]. In this study, a novel silver-based catalyst was synthesized using the Deposition–Precipitation method and evaluated under H₂-SCR conditions for NO_x reduction. A dedicated experimental setup was developed, consisting of controlled gas flow systems for H₂, O₂, and NO gases, a gas mixing unit, and a furnace-equipped reactor operating in the temperature range of 100–500 °C. Water humidity was precisely controlled using a peristaltic pump to evaluate catalyst water tolerance. The catalyst was deposited onto a honeycomb-structured substrate, commonly used in catalytic converters, to enhance surface area and improve the conversion of NO_x to N₂. Catalytic activity and NO_x conversion efficiency were analysed using a NO_x analyser. The results demonstrate the strong potential of silver-based catalysts as efficient, economical, and environmentally sustainable materials for catalytic converters in hydrogen internal combustion engines.

Keywords: H₂-SCR, Silver-based catalyst, Catalytic Converter

References:

1. Arsie, I., Battistoni, M., Brancaloni, P., Cipollone, R., Corti, E., Millo, F., Occhicone, A., & Zembi, J. (2024). A New Generation of Hydrogen-Fueled Hybrid Propulsion Systems for the Urban Mobility of the Future. *Energies*, 17(1), 34.
2. Sheikh Muhammad Farhan, Wang Pan, Chen Zhijian, Yin JianJun. Innovative catalysts for the selective catalytic reduction of NO_x with H₂: A systematic review. *Fuel* 355(2024)129364.

From Pre-Catalyst to Active Phase: DFT Insights into Surface Evolution of 2D TMPs toward β-NiOOH Interfaces for OER

Rohan Arya^{1*}, Alpa Dashora¹.

¹*Department of Physics, Faculty of science, The Maharaja Sayajirao University of Baroda, Vadodara, Gujarat, India - 390002*

*Corresponding Author E-mail: rohanarya97@gmail.com

Transition metal phosphides (TMPs) have emerged as a leading class of pre-catalysts for the Oxygen Evolution Reaction (OER); however, their true catalytic nature defined by the dynamic surface transformations they performed under harsh oxidative potentials. This study utilizes Density Functional Theory (DFT)¹ to rigorously model the surface reconstruction and phase evolution of 2D metal-rich phosphides², elucidating the atomistic mechanism by which they transform into catalytically active species. Addressing the complexity of the active interface, we successfully modelled the thermodynamic evolution of the phosphide pre-catalyst into the stable

β -nickel oxyhydroxide (β -NiOOH) phase. A rigorous four-step relaxation protocol was developed to determine the ground-state structure of the reconstructed surface, ensuring a true local minimum consistent with the magnetic and structural properties expected under OER conditions. Thermodynamic mapping of the reaction pathway indicates a theoretical overpotential of 0.499 V, with the formation of the hydroperoxyl intermediate identified as the rate-determining step. Crucially, we demonstrate that the enhanced catalytic performance is driven by the specific electronic structure of the evolved active phase. Our analysis reveals that the surface-reconstructed β -NiOOH exhibits a d-band centre (ϵ_d)³ at -2.467 eV. This optimized electronic configuration moderates the anti-bonding character of metal-adsorbate states, ensuring ideal intermediate binding strengths that facilitate product desorption and prevent catalyst poisoning. These findings highlight the critical role of surface reconstruction and electronic structure engineering in 2D TMP-derived interfaces, providing a mechanistic foundation for designing efficient electrocatalysts for sustainable oxygen evolution and green hydrogen production.

Keywords: 2D TMPs; Surface Evolution; OER; DFT; Green Hydrogen.

References:

1. Kohn, W. and Sham, L.J., Physical review, 1965, 140(4A), p.A1133.
2. Xu, J., Li, J., Xiong, D., Zhang, B., Liu, Y., Wu, K.H., Amorim, I., Li, W. and Liu, L., Chemical science, 2018, 9(14), pp.3470-3476.
3. Greeley, J., Nørskov, J.K. and Mavrikakis, M., Annual review of physical chemistry, 2002, 53(1), pp.319-348.

Ligand-Tuned Co-MOF Thin Films Electrode for Efficient and Scalable Hydrogen Production

Reshma Khandekar^a, Susmita Patil^a, Rohit Sutar^a, Rahul Patil^b, Ashok Chougale^c, Jyotiprakash Yadav^{**}

^a Energy Research and Thin Film Laboratory, USIC, Shivaji University, Kolhapur, 416004, Maharashtra, India.

Department of Physics, Shri Yashwantrao Patil Science College, Solankur, Shivaji University, Kolhapur, 416212 Maharashtra, India.

^c Department of Chemistry, The New College, Kolhapur, Shivaji University, Kolhapur, 416012, Maharashtra, India.

*Corresponding author: Dr. Jyotiprakash B. Yadav
E-mail: jbyadavphy@gmail.com

The design of a cost-effective electrocatalyst for Hydrogen Evolution Reaction (HER) is urgently required to reduce the reliance on expensive noble metals and to enable sustainable, low-cost hydrogen production[1]. In this study, we report a cost-effective and binder-free synthesis of Co-MOF thin films on a stainless-steel substrate via a chemical method. The concentration ratio of metal to ligand is systematically varied (1:3, 1:4, 1:5, and 1:6) to optimize the catalytic performance. Among the synthesized samples, the optimized Co-MOF electrode, denoted as (1:5) CoM-3, exhibits superior HER performance, delivering an overpotential of 258 mV at 10 mA cm⁻² and a Tafel slope of 76 mV dec⁻¹. In addition, the electrode demonstrates a high electrochemically active surface area (ESCA) of 11.97 cm². When evaluated in a prototype water electrolysis setup, the optimized catalyst achieves the highest hydrogen production rate of 625.33 mL h⁻¹ among all samples. The binder-free synthesis of the Co-MOF thin film directly on the substrate further enhances the catalytic performance by improving electrical conductivity and ensuring efficient charge transfer. Moreover, the electrode exhibits excellent durability, maintaining stable performance over 100 hours of continuous operation. These results highlight

that Co-MOF thin films are promising, economical, scalable electrocatalysts for HER, offering significant potential for sustainable hydrogen production and clean energy applications.

Keywords: Co-MOF; HER; hydrogen production.

References:

1. A. Dong, Z. Li, Y. Ma, W. Liao, F. Zhao, X. Zhang, H. Gao, *Nanomaterial* (2025) 1–35.
-

Photo-Response of Pristine and In-doped ZnO Thin Films prepared via a Cost-effective Spray Pyrolysis using a Medical Nebulizer

Mohd Vasim¹, Niketa Dey², Anjali Kumrawat³, Poonam Singh¹, Prof. Somaditya Sen^{1*}

¹*Department of Physics, Indian Institute of Technology, Indore*

²*Department of Physics, Manipal University, Jaipur*

³*School of Electrical and Electronics Engineering, VIT Bhopal University, Bhopal*

^{1*} Corresponding Author: sens@iiti.ac.in

Wurtzite-structured pure and In-doped ZnO (IZO; $\text{Zn}_{1-x}\text{In}_x\text{O}$; $x = 0, 0.0078, 0.015, 0.03125$) thin films were synthesized on quartz substrates at 450 °C using a cost-effective spray pyrolysis technique utilizing a medical jet nebulizer. XRD studies revealed polycrystalline wurtzite $P6_3mc$ structure for all the IZO films. UV-Vis studies revealed an unchanged unconventionally high bandgap of 3.88 eV for all the films, associated with the presence of probable defect states in the IR range ~ 0.25 eV (corresponding to shallow donor states like O vacancies (V_O), Zn interstitials (Zn_i)), Orange range ~ 2.1 eV (like singly ionized oxygen vacancies (V_O^+), oxygen interstitials (O_i), or zinc vacancies (V_{Zn})), blue range ~ 2.75 eV (Zinc interstitials (Zn_i), zinc vacancies (V_{Zn})), and violet range ~ 3.2 eV representing near band edge states (excitonic recombination or transitions involving shallow defects). The light–matter interaction of these films was investigated under UVB (~ 4.13 eV). Conductivity increases with illumination differently than the dark conditions. The current increases exponentially with time under different lights at a particular bias voltage of 5V. The photogenerated electron–hole pair formation and recombination were studied under such conditions along with the processes involved with the adsorption–desorption dynamics of atmospheric molecules on the film surface. Photo-response to the UVB light in pure ZnO is solely due to generation of electron-hole pairs ($t \sim 120$ s). However, this process becomes slower with In-doping ($t \sim 150$ –200 s). The lifetime of adsorption-desorption process is much faster than the e-h pair generation-recombination.

Keywords: In-doped ZnO; spray pyrolysis technique; photo-response; transport properties; electron-hole pair generation;

References:

1. M. Caglar, Y. Caglar, and S. Ilcan, *Physica Status Solidi (C)*, 4, (2007), 1337–1340.
 2. W. Song, R. Yu, M. Chen, R. Liu, R. Zhang, T. Li, and L. Zhang, *Nature Publishing Group Scientific Reports*, 15, (2025), 37216.
 3. Z. Ge, S. Yang, Z. Zhang, M. Hong, M. Liu, A. Zia, Y. Jiang, B. Zou, and L. Tang, *Small*, 21, (2025), 2411433.
-

Electrocatalytic OER Activity of Cr Oxyhydroxides, from Molecular Clusters to Surfaces: a DFT Study

Shaonsikta Chatterjee, Arunendu Das, and Bidisa Das*

Research Institute for Sustainable Energy (RISE),
TCG-CREST (DTBU)
Salt Lake, Sector V, Kolkata 700091, India
E-mail: bidisa.das@tcgcrest.org

Earth-abundant oxyhydroxide minerals represent a promising yet underexplored class of catalysts for electrochemical water splitting. Motivated by recent advances¹ in computational catalysis, we systematically investigate homogeneous² Cr oxyhydroxide $[\text{Cr}_x(\text{OH})_y(\text{H}_2\text{O})_z]^{m+/-}$ clusters and heterogeneous CrOOH surfaces including dopants ($\text{Fe}^{2+}/\text{S}^{2-}$), defect (OH/O) sites for oxygen evolution reaction (OER) activity. Particular attention is devoted to the γ -CrOOH (010) surface, whose catalytic determinants remain poorly understood. Density functional theory (DFT) calculations reveal that the γ -CrOOH (010) surface exhibits a low OER overpotential of 0.46 V, with a free energy change of 1.69 eV at the potential-determining step (PDS). The origin of the enhanced activity is elucidated through detailed structural, magnetic moment, and electronic structure analyses. Moreover, γ -CrOOH displays excellent hydrogen evolution reaction (HER) performance due to favorable adsorption energies of $^*\text{H}$ intermediates. These findings establish γ -CrOOH as a promising bifunctional catalyst for fuel cell technologies.

Keywords: Cr- oxyhydroxides clusters, DFT, OER and HER

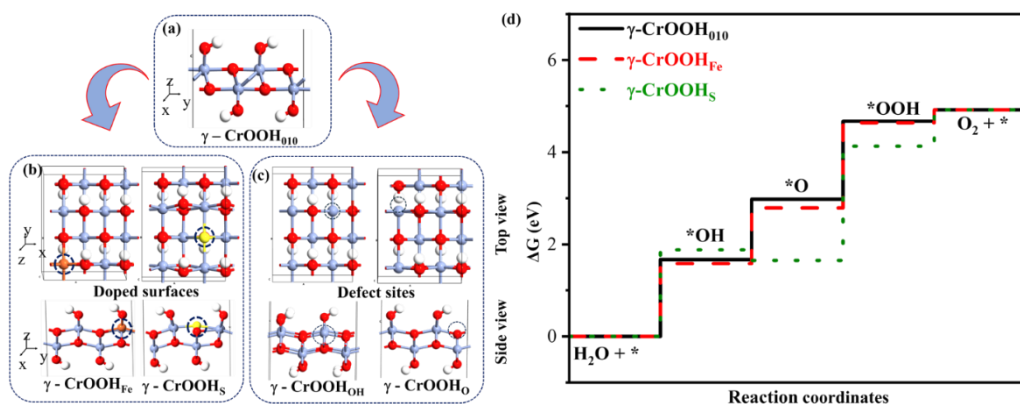


Figure 1: Surface monolayer (a) (010) g-CrOOH, (b) top and side views of Fe and S doped g-CrOOH ($\text{g-CrOOH}_{\text{Fe/S}}$), (c) top and side views of OH and O vacancy sites on g-CrOOH ($\text{g-CrOOH}_{\text{OH/O}}$), (d) Cumulative Gibbs's free energy for pristine g-CrOOH₀₁₀, and g-CrOOH_{Fe}, g-CrOOH_S doped surfaces at $U=0\text{V}$. Blue balls: Cr atoms; orange ball: Fe atom; yellow ball: S atom, red balls: O atoms and white balls: H atoms.

References:

1. N. T. Suen, S. F. Hung, Q. Quan, N. Zhang, Y. Xu, H. M. Chen, Chem. Soc. Rev., 46, (2017), 337–365.
2. De, S., and Das, B. ACS Appl Energy Mater, 9(1), (2026), 629–644.

Bidirectional Catalytic Functionality of $\text{Ti}_3\text{C}_2\text{-F}$ MXene to Hydrogen Evolution and Storage: Theoretical and Experimental Study

E. Choudhary,^{a,b} S. Yadav,^a A. Karahe,^a A. Patil,^a R. Jangir,^{c,d,*} R. S. Devan^{a,e,f,*}

^a*Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India*

^b*Department of Physics, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India*

^c*Accelerator Physics and Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Indore 452013, India*

^d*Homi Bhabha National Institute, Anushakti Nagar, Mumbai, 400094, India*

^e*Centre for Electric Vehicle & Intelligent Transport System, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India*

^f*Multidisciplinary Academic Research Centre, National Dong Hwa University, Hualien 974301, Taiwan*

*Corresponding author: email: rjangir.cat@gmail.com & rupesh@iti.ac.in

Hydrogen has been thought of as a clean energy carrier of the future in the field of sustainable technologies, but efficient hydrogen production and safe storage is a major issue. In this work, we demonstrate that the fluorine-terminated $\text{Ti}_3\text{C}_2\text{-F}$ MXene is a multifunctional 2D material that can serve as a bi-functional material in photocatalytic hydrogen evolution as well as solid-state hydrogen storage. DFT calculations show that $\text{Ti}_3\text{C}_2\text{-F}$ MXene has a quasi-metallic electronic structure with most electrons in 3d states, along with moderate hydrogen adsorption energies (-0.09 to -0.11 eV), which are ideally suited to charge transport and dissociative hydrogen activation. These properties aid an effective spillover-assisted catalytic mechanism. Under visible-light irradiation, $\text{Ti}_3\text{C}_2\text{-F}$ MXene has a hydrogen evolution rate of 3751 $\mu\text{mol/gh}$ in the presence of the sacrificial agent, methanol. Addition of Pt as a co-catalyst promotes charge separation and interfacial electron transfer and doubles the rate of hydrogen evolution to 4228 $\mu\text{mol/gh}$ with excellent photostability of 12 hours of continuous operation. Besides photocatalysis, $\text{Ti}_3\text{C}_2\text{-F}$ MXene can greatly improve the hydrogen storage of Mg-based systems. Mg + 2 wt% $\text{Ti}_3\text{C}_2\text{-F}$ composites have a hydrogen storage capacity of 7.02 wt% at 280 °C, as opposed to 6.13 wt% with pure Mg at 320 °C, indicating lower operating temperature and better sorption kinetics. The enhanced behavior has been credited to spillover-mediated hydrogen dissociation and enhanced diffusion routes at the Mg/MXene interface. Our work presents the potential of $\text{Ti}_3\text{C}_2\text{-F}$ MXene as an exceptional 2D multifunctional catalyst to enable combined hydrogen production and storage, which will provide new possibilities to the MXene-based material in the next-generation clean energy.

Keywords: $\text{Ti}_3\text{C}_2\text{-F}$ MXene; Hydrogen evolution; Sacrificial agent; Hydrogen storage; spillover mechanism.

References:

1. A. Patil, M. Samtham, E. Choudhary, A. Yadav, A. Miglani, S. Bimli, H. Jadhav, R. S. Devan, *Int. J. Hydrogen Energy* 2025, 112, 408-417.
2. E. Choudhary, V. Manjunath, R. Kalubarme, R. Jangir, R. S. Devan, *Nanoscale* 2024, 16 (37), 17599-17615.
3. S. Nishioka, F. E. Osterloh, X. Wang, T. E. Mallouk, K. Maeda, *Nature Reviews Methods Primers* 2023, 3 (1), 42.

Sustainable Hydrogen Generation Through Ammonia-Assisted Water Electrolysis Using Layered Double Hydroxide (LDH) Catalysts

Sanjivani Papal^{1,2}, Rajesh Kanawade^{1*}

¹*Physical & Materials Chemistry Division, CSIR - National Chemical Laboratory, Pune 411008.*

²*Academy of Scientific and Innovative Research (AcSIR), Ghaziabad 201002, India.
E-mail: rv.kanawade.ncl@csir.res.in*

Developing sustainable and clean energy alternatives is critical to address the current energy and environmental crisis. Hydrogen is one of the clean energy carriers, but challenges in storage and transport limit its practical use. Among hydrogen production methods, water electrolysis is a clean and widely studied approach, but it suffers from high energy requirements due to the sluggish oxygen evolution reaction (OER), which demands a high thermodynamic potential (~ 1.23 V) and significant overpotential[1]. To overcome these limitations, ammonia has gained attention as an alternative energy vector due to its high hydrogen content (17.7 wt%), low operating potential (~ 0.7 V), lower energy consumption, and ease of liquefaction[2]. Ammonia-assisted electrolysis, based on the ammonia oxidation reaction (AOR), enables energy-efficient hydrogen production but is limited by sluggish kinetics arising from complex multi-electron pathways described by the Gerischer–Maurer mechanism and Oswin–Salmon mechanism[3]. The development of sustainable, low cost, and efficient catalysts for AOR plays an important role in accelerating the reaction kinetics and boosting the performance of the AOR process. In this work, we explored the non-precious group metal catalysts, particularly layered double hydroxides (LDHs), as cost-effective and efficient alternatives to noble metal catalysts due to their tunable composition, high surface area, and abundant active sites. The catalyst is synthesized via controlled chemical method to achieve optimized nanostructures and enhanced catalytic performance. Structural and surface properties are characterized using XRD, SEM, TEM, and XPS. Electrochemical performance toward AOR is evaluated using cyclic voltammetry, linear sweep voltammetry, and electrochemical impedance spectroscopy. At 50 °C, ammonia-assisted electrolysis delivered a higher current density of 280 mA cm⁻² at only 0.52 V, compared to 244 mA cm⁻² for conventional water electrolysis at 2 V. The significantly reduced operating potential demonstrates the enhanced kinetics and superior energy efficiency of ammonia electrolysis for sustainable hydrogen production.

Keywords: Hydrogen; Ammonia oxidation; Electrocatalysts; Layered Double Hydroxides (LDH).

References:

1. Nadia Mohd Adli, Hao Zhang, Shreya Mukherjee, and Gang Wu, Journal of The Electrochemical Society, 165 (15) J3130-J3147 (2018).
2. Chenyu Zhu, Bin Guan, Zhongqi Zhuang, Junyan Chen, Zeren Ma, Xuehan Hu, Sikai Zhao, Kaiyou Shu, Hongtao Dang, Junjie Gao, Tiankui Zhu, Zhen Huang, Energy Conversion and Management, Volume 327, 1 March 2025, 119610.
3. Sol A Lee; Mi Gyoung Lee; Ho Won Jang, Sci China Mater, 2022, 65(12): 3334–3352.

Hierarchical Ni-MOF@Ti₃C₂T_x MXene composite for efficient bifunctional electrocatalytic water splitting

Ajay Patil,^a Ankit Yadav,^a Rishav Sharma,^a Ekta Choudhary,^{a,b} Ravindra Jangir,^{c,d} and Rupesh S. Devan^{a,e*}

^a Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.

^b Department of Physics, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.

^c Accelerator Physics and Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Indore 452013, India.

^d Homi Bhabha National Institute, Anushakti Nagar, Mumbai 400094, India.

^e Centre for Electric Vehicle & Intelligent Transport System, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.

*Corresponding Author E-mail: rupesh@iiti.ac.in

Developing cost-effective and efficient bifunctional electrocatalysts is essential for sustainable hydrogen (H₂) production via water electrolysis. Herein, a Ni-MOF@Ti₃C₂T_x MXene composite was rationally engineered via a solvothermal approach by integrating porous Ni-MOF microspheres with MXene nanosheets to enhance charge transport and active site accessibility. Ni-MOF@Ti₃C₂T_x catalyst delivering a specific surface area of 60.353 m²/g and an increased electrochemically active surface area (2C_{dl} = 20630 mF/cm²) facilitates improved electrolyte interaction and greater exposure of catalytic sites. Consequently, it achieves low overpotentials of 59 mV for the hydrogen evolution reaction and 149 mV for the oxygen evolution reaction at 10 mA/cm², and enables overall water splitting at a low cell voltage of 1.55 V. Its turnover frequency of 1.49 /s indicates favorable intrinsic reaction kinetics. The enhanced performance arises from strong interfacial coupling between Ni-MOF and Ti₃C₂T_x MXene, which promotes rapid charge transport and efficient utilization of redox-active sites. Overall, these results highlight the potentials of MXene-integrated MOF architectures as promising non-noble electrocatalysts for efficient water splitting.

Keywords: Ni-MOF@Ti₃C₂T_x MXene; Bifunctional electrocatalysis; Water splitting; Green hydrogen generation;

References:

1. A. J. Patil, M. Samtham, E. Choudhary, A. Yadav, A. Miglani, S. Bimli, H. Jadhav, and R. S. Devan, International Journal of Hydrogen Energy, 112, (2025), 408–417.
2. E. Choudhary, V. Manjunath, R. Kalubarme, R. Jangir, and R. S. Devan, Nanoscale, 16, (2024), 17599–17615.
3. H. Yang, G. Zhang, H. Zhou, Y. Sun, and H. Pang, Energy Material Advances, 4, (2023), Article 0033.

Mechanistic Insights into CoZn(VO₄)₂ Anodes Using In-Situ and Post-Cycling Analyses

Mohd Saqib^{1*}, Tejveer Singh Anand¹, Akshita Sharma², Gobinda Bag³, Harsh Yadav⁴, Amit Gupta⁵, Madhusudan Singh^{1*}

¹Department of Electrical Engineering, Indian Institute of Technology Delhi, New Delhi, India

²Department of Chemistry, Indian Institute of Technology Delhi, New Delhi, India

³Department of Chemistry, National Institute of Technology Manipur, Imphal, Manipur, India

⁴School of Engineering and Technology, National Forensic Science University, Gujarat, India

⁵Department of Mechanical Engineering, Indian Institute of Technology Delhi, New Delhi, India

*Corresponding Author E-mail: eez248654@ee.iitd.ac.in

The remarkable rise in demand for rechargeable batteries has led to significant exploration of novel electrode materials that exhibit improved coulombic efficiency, energy density, and stability [1]. Transition-metal vanadate oxides with a variety of cations and multiple valence states have recently exhibited exceptional electrochemical performance. This can be attributed to the complexity of their chemical composition, the complementary behaviour of various types of metal, the structural flexibility, and their interfacial properties [2]. In this study, the green synthesis of cobalt zinc orthovanadate ($\text{CoZn}(\text{VO}_4)_2$) nanosheets (CZVO) was achieved by a simple coprecipitation method followed by annealing at 300 °C.

In a synthesis, $\text{Co}(\text{CH}_3\text{CO}_2)_2 \cdot 4\text{H}_2\text{O}$, $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, and NH_4VO_3 were reacted at 120 °C in aqueous solution under vigorous stirring for 24 h, maintaining a pH of approximately 7. The obtained yellowish powder was washed with ethanol and deionised water, then annealed at 300 °C. Further material characterisation confirms the structure and elements of CZVO. Electrochemical evaluation is performed on the fabricated CR2016-type half- and full-coin cells. Li-ion half-cells were constructed using Li foil as the counter electrode, while full cells were assembled with LiFePO_4 (LFP) cathodes. Na-ion half-cells were created employing Na foil as the counter electrode.

Electrochemical tests conducted in lithium half-cells (0.01-3.0 V) demonstrated that CZVO delivered discharge capacities of 579.37, 297.59, 176.4584, 116.2976, and 34.2891 mAh g^{-1} at 1 C, 2.5 C, 5 C, 10 C, and 20 C, respectively. CZVO||LFP full-cell (1.0 V-3.62 V) exhibited a discharge capacity of 263.4299 mAh g^{-1} at a 2.5 C rate, equating to an energy density of ~608 Wh Kg^{-1} of cathode active material (7.03 mg cm^{-2}). Comparative Na-ion half-cell tests of the CZVO operated at a 2.5 C rate demonstrated a discharge capacity of 184.43 mAh g^{-1} compared to 297.59 mAh g^{-1} in Li-ion cells with equivalent current and mass of active material, emphasising the improved lithium storage capacity of this system. Ex-situ analyses, such as time-of-flight secondary ion mass spectrometry (ToF-SIMS), are being conducted to investigate lithium distribution and the formation of the solid electrolyte interphase (SEI). Moreover, techniques such as in-situ Raman spectroscopy, loop electrochemical impedance spectroscopy (loop-EIS), galvanostatic intermittent titration technique (GITT), and post-cycle field emission scanning electron microscopy (FESEM) have been used to better understand kinetics, charge transfer, diffusion processes, and structural changes during cycling.

Keywords: Zinc orthovanadate; Metal-ion battery; Layered morphology; In-situ

References

1. Choi, Woosung, et al. Journal of Materials Chemistry A 12.12 (2024): 7122-7131.
2. Shen, Laifa, et al. Advanced Functional Materials 24.18 (2014): 2630-2637

Supercapacitive Applications of Cobalt and Nickel Derived 3D Graphene Synthesized by Thermal Chemical Vapor Deposition

Anshuman Verma^{1,2}, Shubham Kumar^{1,2}, Rajath Alexander^{1,2}, Kinshuk Dasgupta^{1,2*}

¹Bhabha Atomic Research Centre, Mumbai, India- 400085

²Homi Bhabha National Institute, Mumbai, India- 400094

*kdg@barc.gov.in

Three-dimensional graphene (3D graphene) has emerged as a promising electrode material for high-performance supercapacitors because of its interconnected porous architecture, high electrical conductivity, and large accessible surface area. In the present work, pristine 3D graphene sheets were synthesized on cobalt and nickel catalysts via thermal chemical vapor deposition (CVD) using methane as the carbon precursor. A comparative investigation was carried out to understand the influence of catalyst-assisted growth on the structural characteristics and electrochemical performance of the synthesized graphene architectures.

Raman spectroscopic analysis revealed differences in graphitization behavior and graphene layer thickness. Nickel-derived graphene exhibited a high I_{2D}/I_G ratio of 4.05, indicating formation of predominantly single-layered graphene structures, whereas cobalt-derived graphene showed an I_{2D}/I_G ratio of 1.24 corresponding to multilayer interconnected graphene structures. FE-SEM and HRTEM analyses confirmed formation of porous graphene frameworks.

Electrochemical performance was evaluated using cyclic voltammetry (CV), galvanostatic charge-discharge (GCD), and electrochemical impedance spectroscopy (EIS) in 1 M KOH electrolyte using a three-electrode configuration. Nickel-derived graphene exhibited a high specific capacitance of 486 F/g at 0.5 A/g because of its large accessible surface area. However, cobalt-derived graphene demonstrated superior rate capability, lower impedance, improved capacitance retention, and excellent cycling stability over approximately 14000 cycles. Catalyst selection dictates graphene growth, morphology, and performance, with cobalt-assisted synthesis yielding robust and stable 3D graphene electrodes for superior supercapacitors.

Keywords: 3D graphene, supercapacitor, chemical vapor deposition, cobalt, nickel

References:

1. Verma, A., Alexander, R., Kaushal, A., Kar, S., and Dasgupta, K. (2024). A simple approach to synthesize high-quality 3D graphene sheet by chemical vapor deposition using cobalt catalyst template. *Diamond and Related Materials*, 148, 111433.
2. Chen, Z., Ren, W., Gao, L., Liu, B., Ma, L. P., and Cheng, H. M. (2011). Three-dimensional flexible and conductive interconnected graphene networks grown by chemical vapour deposition. *Nature Materials*, 10, 424–428.

Enhancing Cycle Life with solution-processed Laminar Li_3VO_4 Anodes for Li, Na, and aqueous Zn-Ion Batteries

Tejveer Singh Anand^{#1}, Aashish Joshi², Akshita Sharma³, Mohd Saqib¹, Shubham Aggarwal¹, Amit Gupta⁴, Madhusudan Singh^{*1}

¹*Functional Materials & Devices Laboratory, Department of Electrical Engineering, IIT Delhi, New Delhi, India, 110016.*

²*School of Interdisciplinary Research (SIRE), IIT Delhi, New Delhi, India, 110016.*

³*Department of Chemistry, IIT Delhi, New Delhi, India, 110016.*

⁴*Thermo-fluids and Energy Systems Laboratory, Department of Mechanical Engineering, IIT Delhi, New Delhi, India, 110016.*

[#]Presenting Author Email: tejveer.singh.anand@ee.iitd.ac.in

^{*}Corresponding Author Email: msingh@iitd.ac.in

Li_3VO_4 (LVO) anodes offer promising alternatives to graphite in lithium-ion batteries. This study showcases a surfactant-free sol-gel synthesis [1,2] method that enhances cycling stability and C-rate performance, confirmed by PXRD as an orthorhombic phase (Pmn21) [3]. This approach paves the way for improved energy storage solutions. Rietveld refinement provided

lattice constants of $a=5.448 \text{ \AA}$, $b=6.327 \text{ \AA}$, and $c=4.949 \text{ \AA}$. HRTEM revealed high crystallinity with laminar flake formation, evidenced by non-ring-like diffraction patterns. The resulting LVO||Li half-cell achieved a capacity of 764 mAh/g at a current density of 0.7 A/g, reaching a deep discharge voltage of 0.02 V. Rate performance tests demonstrated a 10.26 % reduction in intercalation potential when current density increased from 0.7 A/g to 3.5 A/g. Long-term testing at approximately 13.5 A/g for over 30,000 cycles revealed that the cell could still deliver a specific discharge capacity of around 155 mAh/g at the 15000th cycle (asymptotic point), exhibiting a coulombic efficiency (CE) exceeding 95 %. Following 30000 cycles of persistent deep charging ($3 \rightarrow 0.02\text{V}$), the cell maintained a 100-160 mAh/g capacity, with an estimated discharge capacity fading of approximately 0.002 mAh/g. Interestingly, the cell's initial capacity increased until it reached an asymptotic point. EIS measurements on aged cells showed reduced charge-transfer resistance compared to fresh cells, indicating effective lithium ion diffusion through the SEI layer. Further LVO||Li cell was combined with a commercially available LFP cathode to develop a full-cell configuration using a similar assembly method. The full-cell was cycled for over 1000 cycles, with a capacity of 165 mAh/g at 0.15 A/g within a voltage range of 1-3.5 V. Post-cycling TEM confirmed that the laminar structure remained intact following 30,000 cycles of high-rate continuous operation. For the LVO||Zn battery, electrodes were punched and assembled into CR2016 cells with an aqueous electrolyte containing 2 M ZnSO_4 & 0.5 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ in distilled water. The LVO||Zn half-cell exhibited an initial capacity of 54.71 mAh/g with a CE of approximately 99.68%, retaining 36.81 mAh/g (CE $\sim$ 99.87 %) after 6000 cycles at 0.5 A/g between 0.15 V and 1.6 V. We are conducting additional ex-situ analyses, including ToF-SIMS, to investigate further the distribution of lithium and operational byproducts aiming to gain deeper insights into the underlying mechanisms.

References:

1. M. Singh et al., pat. 202211074423 (US No. 18/393,750) (India) (IIT Delhi) Filed.
2. M. Singh et al., pat. 202311051106 (US No. 18/786,397) (India) (IIT Delhi) Filed.
3. Song, et al, J. Mater. Sci. Technol., 140, 142-152 (2023).

Unraveling growth dynamics of $\text{Sn}_3\text{O}_4/\text{FTO}$ photoanode for photoelectrochemical water oxidation under directional illumination

Aayushi Miglani¹, Kushal Dale¹, Shyam Lal Gurjar¹, Manopriya Samtham², Himanshu Srivastava³, Ravindra Jangir³, Ram Janay Choudhary⁴, Bharati Debanath⁵, Rupesh S. Devan^{2**}, Onkar S. Game^{1*}

¹Department of Physics, Indian Institute of Technology Indore, Khandwa Road, Simrol, Indore, Madhya Pradesh 453552, India

²Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Indore, Khandwa Road, Simrol, Indore, Madhya Pradesh 453552, India

³Accelerator Physics and Synchrotrons Utilization Division, Raja Ramanna Centre for Advanced Technology, Indore 452013, India

⁴UGC-DAE Consortium for Scientific Research (CSR), Indore 452001, India

⁵Research Institute for Sustainable Energy (RISE), TCG-CREST, Salt Lake, Kolkata 700091, India

** rupesh@iiti.ac.in *ogame@iiti.ac.in

Tin oxide-based materials have attracted significant attention in energy conversion and opto-electronic applications due to their tunable electronic structure and chemical stability. Among them, Sn_3O_4 , a mixed-valence two-dimensional layered oxide, is of particular interest, owing to its electronic structure due to the co-existence of Sn^{2+} and Sn^{4+} .^[1]

In this work, we demonstrate synthesis of phase-pure n-type Sn_3O_4 , a mixed valence Tin Oxide on FTO (fluorine-doped tin oxide) substrate through a solvothermal route by varying the growth duration. The study elucidates the morphological evolution of the films, revealing a transition in nano-structuring from compact architectures to porous, hierarchical structures. Prolonged growth hours led to enhanced crystallinity and an increase in crystallite size. The characteristics, including bandgap, energy band edges, and work function, were found to vary systematically with growth duration. A clear correlation was established between the structural configuration ranging from compact thin films to porous hierarchical thick films and their photoelectrochemical (PEC) water-splitting performance under different illumination directions. The investigation provides mechanistic insight into the photoelectrochemical water oxidation behavior of $\text{Sn}_3\text{O}_4/\text{FTO}$ photoanodes under both front and back illumination, emphasizing the interplay between growth dynamics, morphology, and light-matter interaction.

Keywords: mixed-valence tin oxide; Sn_3O_4 ; photoelectrochemical water oxidation; photo-active material; directional illumination

References:

1. X. Yu, C. Li, J. Zhang, L. Zhao, J. Pang, L. Ding, International Journal of Minerals, Metallurgy and Materials 2024, 31, 231.

Elucidating Lithium and Sodium-ion Storage Mechanisms in Layered $\text{Na}_2\text{Ti}_3\text{O}_7$

Akshita Sharma¹, Tejveer S. Anand², Antara sarkar³, Madhusudan Singh², Amit Gupta⁴, Ashok Kumar Ganguli^{1,5}

¹Department of Chemistry, IIT-Delhi, New Delhi, India.

²Department of Electrical Engineering, IIT Delhi, New Delhi, India

³European XFEL (X-ray free electron laser), Holzkoppel 4, 22869 Schenefeld, Germany

⁴Department of Mechanical Engineering, IIT Delhi, New Delhi, India

⁵Department of Chemical Sciences, Indian Institute of Science Education and Research Berhampur, Odisha, India*

*Corresponding E-mail: ashok@chemistry.iitd.ac.in

Layered titanates are widely recognized for their electrochemical intercalation properties due to their open layered structures, making them promising candidates for energy storage applications. However, their practical performance is often limited by low energy density arising from restricted vacant sites. In this study, $\text{Na}_2\text{Ti}_3\text{O}_7$ (NTO) is investigated as an anode material for both Li and Na-ion batteries to systematically understand the effect of ionic size on electrochemical behavior and structural stability. The results reveal distinct differences in structural and surface evolution between the NTO||Li and NTO||Na systems. The NTO||Li cell exhibits capacity retention of 70 % and 17 % at 0.2 C and 2 C, respectively, whereas the NTO||Na system retains only 50% and 10% capacity at the corresponding C-rates. This severe capacity fading in the sodium system is further investigated using in-situ Raman spectroscopy and in-situ EIS measurements. ToF-SIMS analysis at both low and high C-rates, along with in-situ and ex-situ FESEM imaging, reveals significant surface cracking in the NTO||Na electrode after galvanostatic charge–discharge cycling. This is accompanied by the presence of COOH^- , PO^- , and CHO^- species on the surface. These observations are further corroborated by ex-situ EXAFS

studies, highlighting structural distortion induced by the insertion of larger sodium ions. Overall, this work emphasizes the importance of detailed investigations into lithium and sodium intercalation mechanisms to optimize electrochemical performance and long-term stability of NTO-based anodes for advanced energy storage applications.

Keywords: In-situ Raman spectroscopy; XANES, NTO||Na.

References

1. J. Phys. Chem. C 2024, 128, 46, 19499–19509.
2. Energy Fuels 2024, 38, 13, 12150–12161.
3. ACS Appl. Energy Mater. 2024, 7, 1, 312–325.

Atomic-Level Engineering of Ni-MOF via Cobalt Incorporation: Structural insights and Enhanced supercapacitor electrode.

Ankit Yadav ^a, Manopriya Samtham^a, Shama Parveen ^a, Suman Yadav^a, Rupesh S. Devan ^{a,b}

^a *Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology*

Indore, Simrol, Khandwa Road, Indore, 453552, India.

^b *Centre for Electric Vehicle & Intelligent Transport System, Indian Institute of Technology Indore, Simrol, Khandwa Road, Indore, 453552, India.*

*Corresponding author: email: rupesh@iiti.ac.in

Bimetallic NiCo–MOFs is widely explored candidates for advanced energy storage; however, the structural evolution of Ni–MOF frameworks upon Co incorporation remains inadequately understood. To deep dive into the synergistic effect of Ni and Co, a series of NiCo–MOF with varying Co contents (NC-1, NC-2, and NC-3) were systematically synthesized to elucidate the composition–structure–property and electrochemical relationship. Among the synthesized samples, the optimally tuned NC-2 exhibits enhanced atomic ordering and improved crystallinity, which facilitate efficient ion transport pathways and improved electrical conductivity. Furthermore, Co incorporation induces a distinct morphological transformation from nanoflower-like architectures to interconnected wrinkled nanosheets, providing abundant electroactive sites and defect-rich surfaces that accelerate redox kinetics. As a result, the NC-2 electrode delivers a high specific capacitance of 630.99 F g⁻¹ at 1 A g⁻¹, along with an energy density of 16.96 Wh/kg and a power density of 1866.66 W/kg. This work highlights the critical role of controlled compositional engineering in optimizing the structural and electrochemical properties of bimetallic MOFs. The findings offer new insights into defect modulation and morphology control, paving the way for the rational design of next-generation MOF-based electrodes, with future prospects in operando mechanistic studies.

Keywords: NiCo-MOF, advance energy storage, wrinkled nanosheets, Synergistic effect.

References:

1. S.Y. Lee, H. J. An, J. Moon, D.H. Kim, K. W. Park, J. T. Park, Electrochim. Acta, 2023, vol. 451 142291.
2. Raissa, N. L.W. Septiani, S. Wustoni, F. Failamani, N. Wehbe, M. Eguchi, H. Nara, S. Inal, V. Suendo, B. Yulianto, J. Power Sources, 603(2024), 234423.

3D printed 0.937(Bi 0.5 Na 0.5)TiO 3 –0.063BaTiO 3 (BNT-BT) for Piezo/photo-catalysis

Abhishek Shukla, Vishal Singh Chauhan*, Rahul Vaish*

*School of Mechanical and Materials Engineering, Indian Institute of Technology Mandi
Himachal Pradesh 175005, India*

*Corresponding Author E-mail: rahul@iitmandi.ac.in

Using direct ink writing (DIW) 3D printing method, 0.937(Bi 0.5 Na 0.5)TiO 3 – 0.063BaTiO 3 (BNT-BT) was fabricated. Raw precursor of BNT-BT and Atlas binder paste were mixed into 60:40 (by wt%) and 3D printed in various shapes. The printing process was carried out under a pneumatic pressure of 0.6–0.8 bar using a 0.61 mm nozzle. Additionally, the layer height was maintained at 0.61 mm, while the printing speed was fixed at 15 mm/s throughout the fabrication process. After air drying, calcination and sintering was done and BNT-BT was successfully synthesised. To examine the prepared paste, rheological studies on viscosity and shear stress were performed using Herschel-Bulkley model. Value of shear thinning parameter was ~ 0.62 , which confirms shear thinning behaviour of prepared paste[1]. Furthermore, TGA-DSC (Thermogravimetric Analysis – Differential Scanning Calorimetry) examination of atlas binder was done to find out its decomposition range and temperature. TGA analysis showed that the binder lost $\sim 87\%$ of its mass between 25–80 °C and an additional $\sim 9\%$ between 80–200 °C. Further gradual decomposition occurred over a wide temperature range, with $\sim 3.8\%$ additional mass loss observed between 400–500 °C. X-ray diffractometer (XRD) of sintered 3D printed sample was performed to confirm the phase[2]. To evaluate sintering state and morphology, field-emission scanning electron microscope (FE-SEM) was done. Elemental area mapping of the prepared sample was also carried out. Using Diffuse reflectance spectra (DRS) the bandgap of 3D printed material was evaluated around ~ 3.08 eV. Synthesized 3D printed BNT-BT was examined for piezocatalysis, photocatalysis and synergistic piezo-photocatalysis for methylene blue (MB) dye (5ppm/10 mL) dye degradation. 3D printed sample showed $\sim 44\%$ (180 min), $\sim 52\%$ (210 min) and $\sim 47\%$ (150 min) degradation efficacy in piezocatalysis, photocatalysis and piezo-photocatalysis, respectively.

Keywords: Direct ink writing, 3D printing, 3D printed catalyst, BNT-BT, multicatalysis

References:

1. Xu, H.; Liu, Y.; Che, Y.; Chen, Z. Combination of Direct Ink Writing and Reaction Bonded for Rapid Fabrication of SiCw/SiC Composites. *Ceram. Int.* 2023, 49 (1), 392–402.
2. He, X.; Boda, M. A.; Chen, C.; Dun, R.; Wang, L.; Bao, Y.; Pang, D.; Guo, L.; Zeng, H.; Li, Y. Ultra-Large Electromechanical Deformation in Lead-Free Piezoceramics at Reduced Thickness. *Mater. Horizons* 2024, 11 (4), 1079–1087.

MoS₂@Ti₃C₂@CNT Ternary Heterostructure as a High-Performance Anode for Sodium-Ion Batteries

Trupti Aher^{1, 2}, Ajay Yadav², Milind Kulkarni³ and Ashish Yengantiwar^{1*}

1 Department of Physics, Fergusson College (Autonomous), Pune affiliated to Savitribai Phule Pune University, 411004, Maharashtra, India

2 Modern College of Arts, Science and Commerce, Shivajinagar, Pune 411005, India

3 Centre for Materials for Electronics Technology (C-MET), Government of India, Panchawati, Off Pashan Road, Pune 411008, India

*Corresponding Author E-mail: ashish.yengantiwar@fergusson.edu

The practical implementation of sodium-ion batteries (SIBs) is hindered by the slow diffusion kinetics of Na^+ ions, low intrinsic electrical conductivity, and significant volume fluctuations of anode materials during cycling. To address these challenges, a multidimensional $\text{MoS}_2@\text{Ti}_3\text{C}_2@\text{CNT}$ ternary heterostructure was successfully designed and synthesized as a high-performance anode material for SIBs. The hybrid architecture combines two-dimensional MoS_2 nanosheets, conductive Ti_3C_2 MXene layers, and one-dimensional carbon nanotubes (CNTs), forming an interconnected conductive framework with abundant active sites for sodium storage. The presence of Ti_3C_2 and CNTs effectively inhibits the restacking of MoS_2 layers, facilitates rapid electron transport, and accelerates Na^+ diffusion kinetics. Furthermore, the heterointerfaces generated within the ternary composite enhance charge-transfer processes and improve structural integrity during repeated sodiation/desodiation cycles. As a result, the $\text{MoS}_2@\text{Ti}_3\text{C}_2@\text{CNT}$ electrode delivers a reversible specific capacity of 510 mAh g^{-1} at 200 mA g^{-1} after 100 cycles, demonstrating excellent cycling durability and rate capability. The superior electrochemical performance is mainly attributed to the synergistic interaction among the three components, enhanced electrical conductivity, increased pseudocapacitive contribution, and efficient ion/electron transport pathways. This study presents a promising approach for developing advanced MXene-based heterostructured anodes and offers valuable insights into the design of high-performance sodium-ion energy storage systems.

Keywords: Sodium-ion batteries; $\text{MoS}_2@\text{Ti}_3\text{C}_2@\text{CNT}$; MXene heterostructure; Anode materials; Energy storage.

References:

1. C. Wu, X. Wang, Y. Zhang, and J. Liu, *Journal of Materials Science & Technology*, 100, (2022), 101–109.
2. B. Wen, H. Li, X. Zhao, and Y. Chen, *Energy & Fuels*, 36, (2022), 11290–11299.

Investigation of water retention capacity & swelling kinetics of biocompatible electrospun nanofibers for enhanced wound healing

Nisha Kumari¹, Dr. Varun Vyas¹

¹*Department of Biotechnology, Bennett University, Greater Noida, UP, India*

*Corresponding author email: varun.vyas@bennett.edu.in

This study aims to investigate water retention capacity & swelling behaviour of biocompatible electrospun nanofibers for & rapid regeneration of injured skin tissue, by using polyvinyl alcohol (PVA) integrated with plant-based materials. This research addresses the urgent demand for scaffolds that balance high fluid uptake/retention while sustaining structural integrity

in fresh wounds and promoting cellular recruitment. PVA is a well-known polymer with excellent film-forming properties, hydrophilicity, and biocompatibility, making it perfect for its biomedical applications [1]. However, its high-water solubility limits its long-term stability. To overcome this limitation, cross-linking agents were incorporated into the nanofibers to increase their mechanical strength, stability, and water resistance. The resultant crosslinked PVA nanofibers demonstrated durability in aqueous environments for over 24 hours in both water & PBS, with absorption of 350% and swelling behaviour, enabling the material to maintain adequate moisture at the wound site while maintaining its structural integrity. This regulated absorption will remove excess wound exudate, allowing therapeutic bioactive compounds to be released into the wound [2]. Mechanical testing shows enhanced stiffness & structural strength of the nanofiber sheet, also FTIR confirmed crosslinking and functional interactions within the nanofiber matrix, making them suitable for skin tissue applications.

Overall, the developed crosslinked electrospun nanofibers serve as a stable & swelling responsive scaffold. Future objectives include giving the nanofiber matrix regenerative, antioxidant, & antibacterial properties by including plant-based bioactive ingredients [3]. By lowering inflammation, accelerating tissue regeneration, and preventing microbial infections, these organic materials will enhance healing process.

Keywords: Electrospun nanofibers, Polyvinyl alcohol (PVA), Crosslinking, Water retention capacity, Wound healing

References:

1. Palani N, Vijayakumar P, Monisha P, Ayyadurai S, Rajadesingu S. Electrospun nanofibers synthesized from polymers incorporated with bioactive compounds for wound healing. *J Nanobiotechnology* 2024;22.
2. Li K, Zhu Z, Zhai Y, Chen S. Recent Advances in Electrospun Nanofiber-Based Strategies for Diabetic Wound Healing Application. *Pharmaceutics* 2023;15.
3. Tripathi D, Rajinikanth PS, Pandey P. Electrospun chitosan nanofibers for regenerative wound healing: from molecular design to functional scaffolds. *J Mater Chem B* 2025;13:10743–79.

Theoretical Investigation on the Magnetic Properties of 2D CrSSe Janus Material Under Tensile Strain

Abhinav Prashant Gotmare,¹ Cyrille Barreteau² and Suranjan Shil¹

¹*Manipal Centre for Natural Sciences (MCNS), Manipal Academy of Higher Education (MAHE), Madhav Nagar, Manipal, Karnataka, India 576104*

²*SPEC, CEA, CNRS, Université Paris-Saclay, CEA Saclay, Gif-sur Yvette F-91191, France*

Corresponding Author E-mail: suranjan.shil@manipal.edu

In this work, we have theoretically studied the magnetic behavior of the 2D CrSSe Janus material in its 1T and 1T' [1] phases. Previous studies have shown that the 1T phase CrSSe Janus structure exhibits a ferromagnetic behavior[2] whereas our results reveal that a structural distortion to the 1T' phase can lead to anti-ferromagnetic behavior. Under low tensile strain, the AFM configuration is energetically favored, whereas FM states become dominant at high tensile strain, indicating a strain-tunable magnetic and structural phase transition between 1T' and 1T

phases. This arises because the 1T' phase shows a Peierls-like distortion with zigzag chains and asymmetric bonding, while the 1T phase has a symmetrical octahedral geometry. Tensile strain reduces these structural distortions, stabilizing the 1T phase over the 1T', leading to ferromagnetic exchange interactions. The stabilized 1T phase also showed a strong Stoner-like ferromagnetism. These findings can be useful for developing the next-generation straintronic[3] and spintronic devices, including memory and logic devices.

Keywords: 2D Janus materials; Magnetism; Spintronics; Straintronics

References:

1. M. R. Habib et al., 'Nanoscale, vol. 11, no. 42, pp. 20123–20132, 2019,
2. Y. Wu, Q. Liu, P. Shi, J. Su, Y. Zhang, and B. Wang, ' , Physical Chemistry Chemical Physics, vol. 25, no. 14, pp. 9958–9964, 2023,.
3. S. Bandyopadhyay, J. Atulasimha, and A. Barman, ' , Appl. Phys. Rev., vol. 8, no. 4, Dec. 2021,

Novel 2D LiInP₂Se₆/WSe₂ Heterostructure for Efficient Photocatalytic Water Splitting: First-Principle insights

Divyanshu Mishra and Divya Somvanshi

Department of Physics, Harcourt Butler Technical University (HBTU), Kanpur, 208002, India

*dsomvanshi@hbtu.ac.in

Photocatalytic water splitting offers a clean and sustainable pathways for producing hydrogen (H₂) fuel from solar energy. In this work, we have investigated the electronic and optical properties of a layered LiInP₂Se₆/WSe₂ van der Waals heterostructure (vdWH) using density functional theory (DFT) calculations. Our results demonstrate that LiInP₂Se₆/WSe₂ forms a type-II staggered band alignment with excellent optical absorption across visible spectrum as compared to their individual monolayers. Due to type-II bandgap, this heterostructure configuration promotes spatial separation of photo generated electrons and holes driven by interfacial charge transfer which further improves photocatalytic efficiency. The band edge positions of the LiInP₂Se₆/WSe₂ heterostructure straddle the water redox potentials over wide pH range of 0 to 12, confirming its suitability for overall water splitting in real world applications. Analysis of photogenerated carrier migration behavior confirms that the as designed heterostructure follows a direct Z-scheme photocatalytic mechanism, which maintains strong oxidation as well as reduction capabilities simultaneously [1, 2]. These findings suggest that the LiInP₂Se₆/WSe₂ vdWH is a highly promising photo catalyst for solar-driven hydrogen production, paving the way for efficient, scalable clean energy technologies.

Keywords: Z-scheme photocatalyst, photocatalytic water splitting, metal thiophosphates, 2D van der Waals heterostructure, solar energy conversion

References:

1. X. Li, C. Garlisi, Q. Guan, S. Anwer, K. Al-Ali, G. Palmisano, et al., Materials Today, vol. 47, pp. 75-107, 2021/07/01/ 2021.
2. P. Bai, W. Xu, L. Shi, H. Wei, Y. Duan, R. Wang, et al., The Journal of Physical Chemistry C, vol. 129, pp. 11698-11706, 2025/06/26 2025.

Molten Salt-Assisted Low-Temperature Synthesis and Characterization of MoAlB MAB Phase and 2D MoB MBene Nanosheets

Muhammed Shabeer K V¹, Shreeya Paresh Kamat¹, Chinnam Durga Jaya Teja¹, Prabu K²,
Saumen Mandal^{1*}

¹*Department of Metallurgical and Materials Engineering, National Institute of Technology
Karnataka, Mangalore – 575025, Karnataka, India*

²*Department of Electronics and Communication Engineering, National Institute of Technology
Karnataka, Mangalore – 575025, Karnataka, India*

*Corresponding Author E-mail: smandal@nitk.edu.in

This study explored the synthesis and characterization of 2D molybdenum boride (MoB) MBene nanosheets derived from MoAlB MAB phase precursor. The MoAlB MAB phase was synthesized by mechanically mixing Mo, Al, and B elemental powders in a molar ratio of 1:1.3:1 along with NaCl flux, using a molten salt-assisted low-temperature approach, followed by heat treatment in a tube furnace under flowing argon atmosphere. The use of molten NaCl as a liquid reaction medium significantly enhanced atomic diffusion among the precursor powders, enabling phase-pure MoAlB formation at a lower sintering temperature compared to conventional dry sintering routes. The obtained MAB phase powder was subsequently subjected to fluorine-free hydrothermal etching using NaOH solution in a Teflon-lined autoclave, followed by ultrasonication to promote layer separation and obtain few-layer MoB MBene nanosheets. The synthesized MoAlB and MBene were systematically characterized using XRD to confirm phase formation and progressive Al removal upon etching, FE-SEM to study the morphological evolution from layered MAB grains to accordion-like delaminated MBene sheets, EDS to verify elemental composition and confirm the elimination of Al after etching, FTIR to identify surface terminations (–OH, –O) introduced during alkaline etching, DSC to investigate thermal behavior and phase formation events, and BET surface area analysis to evaluate the specific surface area enhancement upon exfoliation. The obtained MoB MBene exhibited a fluorine-free surface with abundant active sites and –OH/–O terminations, which are promising attributes for potential applications in room-temperature gas sensing, particularly for ammonia detection.

Keywords: MBene; MoAlB MAB phase; molten salt synthesis; hydrothermal etching; 2D molybdenum boride

References:

1. M. Ramezanzadeh, R. Ghamarpoor, A. Dashan, A. Fallah, S. Soleimani, and B. Ramezanzadeh, *Advanced Composites and Hybrid Materials*, 8, (2025), 269.
2. G. Ding, T. Li, J. Li, and H. Du, *Applied Surface Science*, 720, (2026), 165387.

CVD-Grown 2D SnS₂ Memristors for Low-Power Non-Volatile Memory Applications

Swaraj Mukherjee¹, Mubashir M. Ganaie^{2,3}, Arpita Biswas², Ayan Chatterjee¹, Mahesh Kumar^{2*}

¹*Department of Material Engineering, Indian Institute of Technology Jodhpur, Jodhpur 342030, India*

²*Department of Electrical Engineering, Indian Institute of Technology Jodhpur, Jodhpur 342030, India*

³*Department of Physics, Indian Institute of Technology Jodhpur, Jodhpur 342030, India*
Corresponding Author E-mail: *[mkumar@iitj.ac.in]

Two-dimensional (2D) van der Waals materials provide defect-tunable switching platforms wherein the inter-layer spacing, surface termination, and vacancies have a direct impact on filamentary transport mechanisms. As such, among van der Waals 2D materials, tin disulfide (SnS₂) an indirect-bandgap (~2.2 eV) semiconductor that possesses an n-type semiconducting nature becomes quite promising due to its high sulphur vacancy content, tuning ability of electrical characteristics through doping, and suitability for large-scale growth by sulphur-assisted CVD. Here, we show an Ag/SnS₂/Pt-Si memristor where few-layer SnS₂ was deposited directly on Pt/Si substrates using sulphur-assisted chemical vapor deposition (CVD). Such an approach enables us to avoid any transfer-related interface contamination, resulting in a pristine switching interface. BRS operations were performed under ultralow switching voltages, wherein the SET voltage and the RESET voltage were 0.15 V and -0.21 V, respectively. ON/OFF resistance ratios ranging from 10² to 10³ are observed across devices, with a stabilized ratio exceeding 10² consistently maintained beyond 1000 endurance cycles, reflecting stochastic Ag conductive filament formation within the SnS₂ matrix as the origin of device-to-device variability. Retention tests were conducted by applying a low read bias voltage (0.01 V), and a stable separation of HRS and LRS states could be observed for more than 25,000 s. The electrochemical impedance spectroscopy (EIS) analysis, conducted by Nyquist plot evaluation, corroborates our results by showing that the LRS impedance region, limited to $Z' \approx 43.3\text{--}43.5\ \Omega$, belongs to a metallic filament, while the HRS impedance of $\sim 3 \times 10^5\ \Omega$ corresponds to filament rupture after RESET operation. The extremely low switching voltages of our memristors imply switching energies of picojoule level, which is significantly lower than those of conventional flash memory and comparable to some state-of-the-art 2D material-based RRAM devices using MoS₂ and h-BN as switching layers.

Keywords: SnS₂; Memristor; Resistive Switching; 2D Materials; CVD; RRAM

References:

1. Z. Wang et al., Nature Electronics, 1, (2018), 130–136.
2. C. Tan et al., Chemical Reviews, 117, (2017), 6225–6331.

Synaptic Performance of 2D Bismuth Oxyiodide (BiOI) Based Memristors for Neuromorphic Computing

Saiprasad Sawant, Kalyanee Patil, Shivani Tiwari and Tejashree Bhavé*

Department of Applied Physics, Defence Institute of Advanced Technology, Pune

*Corresponding Author E-mail: tejashreebhavé@diat.ac.in

The development of efficient neuromorphic computing systems requires the realization of electronic analogs to biological synapses. This study investigates the potential of Bismuth Oxyiodide (BiOI) as a functional material for memristive synaptic devices. We report the synthesis of BiOI and its subsequent structural characterization using X-ray diffraction (XRD), Raman Spectroscopy and FESEM confirming the high crystallinity and phase purity of the synthesized material. We constructed a vertical memristor structure to study the typical resistive switching characteristics and biomimetic synaptic functions.

Preliminary electrical measurements reveal stable bipolar RS behaviors with a notable ON/OFF ratio up to 10^2 . The device demonstrates high-performance RS characteristics, including long retention time and excellent endurance. Furthermore, the adjustable resistance states will be utilized to emulate synaptic plasticity, specifically long-term potentiation (LTP) and long-term depression (LTD) cycles. These results suggest that BiOI-based synaptic memristors possess promising potential for integration into neuromorphic chip systems and the simulation of artificial neural networks for advanced image recognition tasks.

Keywords: Bismuth oxyiodide; 2D materials; Memristor; Resistive Switching, Synaptic plasticity

References:

1. P. Lei, H. Duan, L. Qin, X. Wei, R. Tao, Z. Wang, F. Guo, M. Song, W. Jie, and J. Hao, *Adv. Funct. Mater.*, 32, (2022), 2201276.
2. W. Xie, Y. Zhong, D. Wang, L. Zhong, L. Han, Q. Yang, and W. Jie, *Mater. Des.*, 234, (2023), 112367.
3. G. Zhang, Z. Zeng, X. Zhang, Y. Hu, J. Tang, H. Wang, C. Su, X. Yi, Y. Yan, and H. Chen, *Adv. Funct. Mater.*, 36, (2026), e31113.

Ultrafast Nonlinear Optical Response of Plasmonic-Au Nanoparticles Decorated 2D-V₂O₅ Nanoflakes at Near-IR Wavelengths

Shivani Vijay Tiwari, Bhavya Doshi, RNV Krishna, Rashmi Patil, Saiprasad Sawant, A.V.R Murthy, Shyamal Mondal, Tejashree Bhavé*

Defence Institute of Advanced Technology (DU), Girinagar, Pune 411025

Corresponding Author email: tejashreebhavé@diat.ac.in

We report comparative study of third-order nonlinear optical response of colloidal 2D-V₂O₅ nanoflakes and Au-decorated V₂O₅ nanoflakes composites investigated using single beam Z-scan under femtosecond excitation at 1550nm. V₂O₅ nanoflakes were synthesized via a

hydrothermal method and subsequently decorated with Au nanoparticles varying the concentration of Au-NPs to form plasmonic heterostructures. Structural and morphological analysis done using XRD, UV-Visible Spectroscopy, FESEM-EDAX and FTIR, confirms successful formation of nanoflake morphology and uniform Au-NPs decoration, enabling metal-semiconductor interaction.

Open aperture Z-scan measurements reveal that pristine V_2O_5 , Au-NPs and their compositional variation in Au- V_2O_5 exhibit reverse saturable absorption RSA under femtosecond excitation, indicating dominant excited-state absorption and multiphoton absorption processes at 1550nm.

The observed RSA behaviour in all samples suggest strong optical limiting capability in the near-IR region. The non-linear absorption coefficient (β) is highest for pristine V_2O_5 nanoflakes (3.27×10^{-3} m/W), while Au-decorated V_2O_5 nanoflakes exhibit slightly reduced values ($3.13\text{--}3.07 \times 10^{-3}$ m/W), yet significantly higher than standalone Au-NPs (2.97×10^{-3} m/W). This trend indicates intrinsic multiphoton absorption in V_2O_5 dominates the overall magnitude, incorporation of Au nanoparticles introduces plasmon-mediated local field effects and interfacial carrier interaction that enhance the nonlinear response relative to Au-NPs alone.

The marginal reduction compared to pristine V_2O_5 nanoflakes is attributed to partial screening and redistribution of optical pathways at the metal-semiconductor interface. Overall, Au decorated V_2O_5 nanoflakes demonstrate balanced NLO response combining RSA with plasmonic tunability, making it promising candidate for optical limiting and ultrafast photonic applications operating at near-IR wavelengths.

Keywords: Nonlinear Optics; 2D-Nanoflakes; Au-NPs; Optical Limiting Materials, RSA

References:

1. Molli, M., Thomas, S. R., Philip, R. Applied Physics A, 122, 217, (2016)
2. Shi, Q., Zhu, B., Wang, L., Lan, Y., Wang, J., Cheng, G. Nanomaterials, 11, 2078, (2021)
3. Chu, H., Wang, H., Huang, Y., Dai, H., Lv, M., Zhang, Z., Cheng, J., Nanomaterials, 13(10), 2662, (2023)

Zinc Aluminium Layer double hydroxide- A 2D material for the electrochemical detection of *Bacillus Anthracis*

Diksha Singh^a, Manish Gupta^a, Ajay Kumar Goel^a and Mukesh Kumar Sharma^a

^a Defence Research and Development Establishment, Jhansi Road, Gwalior, Madhya Pradesh 474002, India

Corresponding authors: E-mail address: dikshasinghraghuvanshi0@gmail.com

It's pretty challenging to develop a potential nanozyme which not only mimic the conventional enzyme but also potentially equivalent to it. Herein, we have harvested the superior peroxidase activity of Palladium as single atom catalyst. So, we have synthesized atomically dispersed Palladium on layer double hydroxide (Pd/LDH). The synthesized material (Pd/LDH) was characterized by XAS, XPS, EPMA, HRTEM, XRD, UV-Vis, TGA, IR, CV and EIS. As, the material have shown higher electrical conductivity, high stability, abundant surface area, unique catalytic activity and biological friendly, the material shows excellent peroxidase activity and

several radical scavenging activities. As nanozyme have found use as a means to amplify signals in immunosensor-based platforms and there by improve sensors detection limits. Therefore, Pd/LDH were explored for the practical application of *Bacillus Anthracis* (*B. anthracis*) detection electrochemically using the electro active dye 3,3',5,5'-tetramethylbenzidine (TMB) as substrate. Under optimized conditions, Pd/LDH exhibits admirable sensing performance for *B. anthracis* detection, with a linear range of 20 pg/mL to 100 ng/mL and a lower detection limit of 10 pg/mL. Therefore, this work can be exploit a feasible route to design nanozymes with high potential activity and that meets the practical use as an alternative to natural enzymes.

Exploring Family of 2D Transition Metal Phosphides Towards Spintronics Application: A DFT Study

Hardik Chavadiya^{1*} and Alpa Dashora¹

¹*Department of Physics, Faculty of Science, The Maharaja Sayajirao University of Baroda, Vadodara, Gujarat, India.*

*Email: hardikchavadiya7@gmail.com

Two-Dimensional (2D) materials have garnered attention of many researchers due to their unique properties and versatile applications. Among them, MXenes (a family of materials derived from MAX phases) have been widely investigated due to their high electrical conductivity, large surface area, and layered 2D structures that enable surface engineering for applications in energy conversion and storage, as well as sensing and electronic devices [1]. MXenes derived from MAX phases with the general configuration of $M_{n+1}X_n$ mostly with Carbon and Nitrogen as non-metal atoms. It is interesting to explore the possibility of new class of materials with configuration of MXenes with other non-metal atoms systematically. Phosphorous is a potential candidate as it belongs to same group as N in periodic table. Phosphorous has larger atomic radius compared to N, which enables possibility of M-P-M structure to be stable due to larger anion atoms. While prior studies mainly explored the catalytic properties and energy storage application of transition metal phosphides [2-3], our work employs DFT+U to accurately capture the electronic correlations relevant to magnetism. Here, we have explored family of MXene based transition metal phosphides described as *MPhene* (M_2P ; M= Cr, Mn, Fe, Co) using DFT+U towards structural, electronic and magnetic properties. Here, we analyzed stable structure for the family of material among two possible configuration named Alpha and Gamma stacking. Further, we have optimized the Hubbard Parameter U self consistently for each of the material. For the possibility of experimental synthesis, we have calculated the formation energy. Among these Cr_2P is found to be of half metallic in nature, making it an eligible candidate for Spintronics Applications [1], whereas Mn_2P and Co_2P are found to be having AFM-C type magnetic ground state, making them eligible candidate for Antiferromagnetic Spintronics Applications [1].

Keywords: DFT, 2D Transition metal phosphide, Magnetic materials, Spintronics.

References:

1. Hu, G., Xiang, B.; et. al., Nanoscale Res. Lett., 15, (2020), 15, 226.
2. Shao, Y., Shi, X.; et al., Chem. Mater., 29, (2017), 20, 8892–8900.

3. Wang, S., Meng, W.; et al., ACS Applied Materials & Interfaces, 16 (50), (2024), 69733-69743.

IR-Reflective CoAl_2O_4 Blue Pigments: Role of pH and Mineralizers

Samruddhi Mulay, Raj Chaudhari, Vandana A Mooss *

Department of Chemistry, School of Science and Environmental Studies,

Dr. Vishwanath Karad MIT World Peace University pune

Email: vandana.mooss@mitwpu.edu.in

High intensity of the urban heat island (UHI) effect due to global warming, climate change, and rapid urbanization has driven increasing demand for energy-efficient thermal management materials. Hence, there is growing interest in the development of cool and infrared (IR)-reflective pigments for better thermal management. The current attempt explores the synthesis and performance of blue CoAl_2O_4 , inorganic pigment for thermal management applications. The study unravels the systematic investigation of pH and mineralizer effects on structural and optical properties of the pigment. Various mineralizers like ammonia, sodium carbonate, sodium bicarbonate, sodium hydroxide, potassium hydroxide, and sodium fluoride were utilized under varying pH conditions. Among the mineralizers, sodium fluoride afforded the highest NIR reflectance, whereas ammonia offered advantages in terms of scalability. The synthesized pigments were characterized using X-ray diffraction (XRD), particle size analysis, UV–Vis–NIR spectroscopy, SEM EDAX and CIE $L^*a^*b^*$ color measurements, and their properties were benchmarked against a commercial blue pigment. The optimized pigment demonstrated a high NIR reflectance of approximately 65%. These findings demonstrate that precise control of pH and mineralizer selection, critical for tailoring the properties of CoAl_2O_4 providing an effective pathway toward high-performance IR-reflective materials for cool-roof and energy-efficient coating applications. The achieved high NIR reflectance and comparison with commercial pigments demonstrate practical relevance, making the work significant for cool-roof coatings and energy-efficient urban infrastructure applications.

Keywords: Cobalt aluminate, Near-infrared reflectance, Cool pigments, Mineralizer.

References:

1. M. Taguchi et al., Dalt. Trans., vol. 42, no. 19, pp. 7167–7176, 2013.
2. M. Taguchi et al., Dalt. Trans., vol. 42, no. 19, pp. 7167–7176, 2013.

Fabrication of 2D TMD-based Electrode for Multianalyte Detection of Neurotransmitters

Amit Rana^{1*}, Tikendra Kumar¹, Anjali Chaudhary¹

¹ *Department of Physics, Indian Institute of Technology Bhilai, Bhilai, Chhattisgarh 491002, India*

*amitrana@iitbhlai.ac.in

Neurotransmitters such as dopamine, serotonin, and epinephrine play an important role in maintaining both physical and mental health. Any imbalance in their optimal levels can lead to stress, anxiety, and depression [1,2]. Therefore, continuous and accurate monitoring of these biomolecules is essential for the early detection and prevention of mental health disorders. Electrochemical sensors have emerged as promising tools for such applications due to their rapid response, high sensitivity, and low detection limits. In this context, two-dimensional transition metal dichalcogenides (2D TMDs) have gained significant attention owing to their unique electronic properties, high surface-to-volume ratio, and excellent catalytic activity.

In this work, tungsten diselenide (WSe₂) was synthesized via a hydrothermal method [2] and utilized to fabricate a modified glassy carbon electrode (GCE) for the electrochemical detection of multiple neurotransmitters. The synthesized WSe₂ exhibited a nanoflake-like morphology, as confirmed by field-emission scanning electron microscopy (FESEM). Structural analysis revealed a polycrystalline nature with the coexistence of both semiconducting 2H and metallic 1T phases. The WSe₂-modified GCE was employed for multi-analyte detection in phosphate-buffered saline (PBS) solution. Differential pulse voltammetry (DPV) studies demonstrated that the fabricated sensor exhibits excellent electrochemical performance, with limits of detection (LOD) of 12.49 μ M, 2.97 μ M, and 19 μ M for dopamine, serotonin, and epinephrine, respectively.

These results highlight the potential of WSe₂-based electrodes as efficient and reliable platforms for the simultaneous detection of neurotransmitters, paving the way for advanced electrochemical sensing applications in biomedical diagnostics.

Keywords: 2D TMD; Electrochemical sensor; Neurotransmitters; Mental Health.

References:

1. A. Rana et al., Recent Research Development in 2D Transition Metal Dichalcogenides for Dopamine Sensing, 2026, Invited Review, under review in Journal of Materials Chemistry C.
2. A. Rana et al., Fabrication of 2D TMD-based Electrode for the multianalyte detection of neurotransmitters, 2026, in preparation.

A highly sensitive and efficient room-temperature sensing material for NH₃ gas: 2D Terraced Microplate MoO₃

Trupti Patel¹, Ren-Jang Wu², Yu-Cheng Shih², Hsiang-Chin Shen², Ting-Yi Yeh², Murthy Chavali^{3,4,*}, Parineeta Das^{1,*}

¹ Department of Chemistry, Dr. Vishwanath Karad MIT World Peace University, Pune.

² Department of Applied Chemistry, Providence University, Taichung, 43301, Taiwan.

³ NTRC, MCETRC, Guntur, AP 522201, India.

⁴ Department of Chemistry, Kakraparti Bhavanarayana College, Kothapeta, Vijayawada, Andhra Pradesh 520001 India (<https://orcid.org/0000-0003-3910-2322>)

*Corresponding Author E-mail: chavalim@gmail.com, pari.das2111@gmail.com

High-performance gas sensors have been developed to meet the growing need for accurate ammonia (NH_3) detection in industrial, agricultural, and medical applications. In this work, a cost-effective citrate gel technique is used to create molybdenum trioxide (MoO_3) with layered step structures that resemble a terraced microplate. X-ray Diffraction (XRD), Ultraviolet-Visible Spectroscopy (UV-Vis), Fourier Transform Infrared Spectroscopy (FTIR), Raman spectroscopy, and Field Emission Scanning Electron Microscopy (FESEM) are used to methodically characterize the structural, spectroscopic, and morphological properties of the prepared material. The synthesized MoO_3 is applied via the dip-coating method on gold-coated interdigitated electrodes in order to fabricate the device. The sensor exhibits a high response [$(R_g - R_a)/R_a$] of 18.4 toward 5 ppm NH_3 , with a response time of 284 s to achieve 90% of its maximum response, along with a very rapid recovery time of 1 s to return to 90% of the initial baseline. The sensor performs exceptionally well for trace-level NH_3 detection, as seen by its low detection limit of 50 ppb and sensitivity of 3.7. The distinctive terraced microplate morphology, which offers a large surface area and promotes effective gas adsorption and charge transfer, is responsible for the improved sensing properties. These results demonstrate the promising features of MoO_3 -based nanostructures as viable options for room-temperature and sensitive NH_3 sensing applications.

Keywords: 2D, Terraced Microplate, MoO_3 , NH_3 Sensor

References:

1. A. Mešćeriakovas, K. Murashko, S. Alatalo, T. Karhunen, J. Leskinen, J. Jokiniemi and A. Lähde, *Nanotechnology* 31, 2020, 335709
2. D. Kwak, M. Wang, K. J. Koski, L. Zhang, H. Sokol, R. Maric and Y. Lei, *ACS Appl. Mater. Interfaces*, 11, 2019, 40, 37379

Defect Engineering in Few-Layer h -BN and MoO_3 via Oxygen Ion Irradiation for Single-Photon Emission

Samriddhi Kushwaha¹, Mukesh Kumar¹, Fouran Singh², Mandeep Cahal², Pramoda K. Nayak^{1,3*}

¹*Centre for Nano and Material Sciences, Jain (Deemed-to-be University), Jain Global Campus, Kanakapura, Bangalore 562112, Karnataka, India*

²*Material Science Group, Inter University Accelerator Center, New Delhi-110067, India*

³*Department of Physics, 2D Materials Research and Innovation Group, Indian Institute of Technology Madras, Chennai 600036, India*

*E-mail: pramoda.nayak@jainuniversity.ac.in

Quantum light sources are central to emerging quantum technologies, including quantum computing, secure communication, and advanced sensing. Rapid progress in these areas has intensified the search for robust materials capable of hosting stable single-photon emitters (SPEs). Two-dimensional van der Waals materials, particularly hexagonal boron nitride (h -BN) and molybdenum oxide (MoO_3), have attracted considerable attention due to their wide band gaps, structural stability, and ability to support defect-based quantum emitters at room temperature.

Among various defect-engineering approaches, ion irradiation provides a precise and controllable route for tailoring defect states [1].

In this work, mechanically exfoliated h-BN and MoO₃ flakes are subjected to oxygen ion (O⁶⁺) irradiation to induce optically active defects. Raman spectroscopy and photoluminescence (PL) measurements are employed to investigate structural modifications and defect-induced optical signatures similar to earlier works [2, 3]. The irradiated h-BN exhibits very sharp PL emission peaks in the order of few meV at 2.16 eV and 2.21 eV, while MoO₃ shows a distinct emission peak at 2.3 eV, indicating the formation of quantum emitter-like defect states. To unequivocally establish single-photon emission, future measurements of the second-order correlation function, $g^2(\tau)$, are necessary. These findings demonstrate that defect-engineered h-BN and MoO₃ are promising platforms for quantum light sources, with potential applications in quantum sensing, entangled photon generation, and low-power integrated quantum photonic devices.

Keywords: O⁶⁺ ion-irradiation, defect engineering, h-BN, MoO₃, single-photon emitter

References:

1. Rui Gu et.al. ACS Photonics, 2021,8, 2912-2922.
2. Sarkar et al. Nano Letters 24 (2024), 43-50.
3. Vikash Mishra et. al. Appl. Phys. Lett. 123(2023), 151104.

Selective Detection of Isopropanol Using CuO-Doped SnO₂ Nanocomposites at Ambient Temperature

K. Sreebalan¹, Bhuvanika Ramesh², Sridharan Madanagurusamy^{1*}

¹ Functional Nanomaterials & Devices Lab, Centre for Nanotechnology & Advanced Biomaterials and School of Arts, Sciences, Humanities & Education, SASTRA Deemed to be University, Thanjavur, 613401, India

² Functional Nanomaterials & Devices Lab, Centre for Nanotechnology & Advanced Biomaterials and School of Electrical & Electronics Engineering, SASTRA Deemed to be University, Thanjavur, 613 401, India

Email id: m.sridharan@ece.sastra.edu*

Volatile organic compound (VOC) detection at room temperature is critical for environmental monitoring, industrial safety, and healthcare diagnostics [1]. In this work, CuO-doped SnO₂ nanocomposites with varying CuO concentrations (1, 3, and 5 wt%) were synthesized via a facile one-step hydrothermal method. Structural and morphological properties were investigated using X-ray diffraction (XRD), scanning electron microscopy (SEM), and energy dispersive X-ray spectroscopy (EDX). XRD confirmed the tetragonal rutile SnO₂ phase with systematic lattice parameter variations upon CuO incorporation, confirming successful doping [2]. Chemiresistive thick films fabricated by the doctor blade technique were evaluated at room temperature against ethanol, methanol, acetone, acetaldehyde, formaldehyde, isopropanol (IPA), dimethylamine, ammonia, and triethylamine at 25 ppm. Among the tested samples, the 5 wt% CuO-SnO₂ sensor exhibited excellent selectivity towards IPA with well-defined saturated response and rapid response-recovery characteristics. The enhanced performance is attributed to

p-n heterojunction formation at the CuO-SnO₂ interface, modulating the depletion layer and increasing reactive oxygen species density. These results establish CuO/SnO₂ as a promising room temperature VOC sensor.

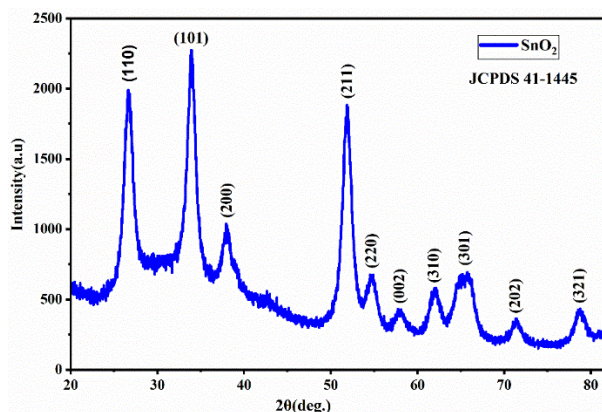


Fig 1. XRD pattern of Pristine SnO₂

Keywords: CuO/SnO₂, p-n heterojunction, hydrothermal synthesis, isopropanol, room temperature, VOC sensing

References

1. Zhang, J., Ma, S., Wang, B., & Pei, S. (2021). Journal of Alloys and Compounds, 886, 161299.
2. Liu, B., Zhang, L., Zhao, H., Chen, Y., & Yang, H. (2012). Sensors and Actuators B: Chemical, 173, 643–651.

Te-Rich MoTe₂/MWCNT Heterojunctions with Interface-Induced Carrier Inversion for Ultra-Low-Level Bipolar Detection of NO₂ and NH₃

Virendra Singh Choudhary¹, Surender Kumar Sharma¹

¹Department of Physics, Central University of Punjab, Bathinda

*Corresponding Author Email: surender.sharma@cup.edu.in

This work presents the development of a highly sensitive gas sensor based on a hybrid 2D/1D MoTe₂/MWCNT nanostructure with enhanced tellurium content, engineered to mitigate the noise issues typically associated with pristine transition metal dichalcogenides. The addition of 5 wt% multi-walled carbon nanotubes (MWCNTs) facilitates the formation of an effective percolation network, leading to nearly 90% noise reduction and stable electrical resistance in the kilo-ohm range. A distinctive self-induced transition from n-type to p-type conductivity is observed in the composite system. This phenomenon is validated through Mott–Schottky analysis and ultraviolet photoelectron spectroscopy (UPS), indicating charge redistribution driven by a work function difference ($\Delta\phi = 0.69$ eV), which promotes the formation of a hole accumulation layer (HAL).

The fabricated sensor demonstrates excellent room-temperature performance, achieving ultra-low theoretical limits of detection (LOD) of 0.43 ppb for NO₂ and 3.15 ppm for NH₃. It exhibits bipolar sensing behavior, characterized by a 2% decrease in resistance upon NO₂ exposure and a 4% increase when exposed to NH₃. At 1 ppb NO₂, the sensor shows response and recovery times of 76 s and 313 s, respectively, whereas for 8 ppm NH₃, the corresponding values are 80 s and 52 s. It is further observed that higher gas concentrations lead to faster response dynamics, although recovery tends to slow down at elevated concentrations. Additionally, the sensor maintains consistent performance over a period of 10 weeks and operates reliably under high humidity conditions (up to 90% RH), highlighting its robustness for practical applications.

Keywords: Gas sensor; Room temperature; Response time; ammonia

References:

1. Wu, E.; Xie, Y.; Yuan, B.; Zhang, H.; Hu, X.; Liu, J.; Zhang, D. ACS Publications, 3, (Year), 1719-1726.
2. Singh, S.; Saggi, I. S.; Singh, S.; Kumar, N.; Chen, K.; Xuan, Z.; Gupta, R.; Swihart, M. T.; Sharma, S. ACS Appl. Nano Mater., 6, (2023), 10698-10712.
3. Singh, S.; Deb, J.; Sarkar, U.; Sharma, S., Journal of hazardous materials, 435, (2022), 128821-128835.

MoS₂-Driven Conductive Networks in 3D Porous PDMS for High-Performance Flexible Piezoresistive Detection over a wide pressure range

Kiran W. Kumbhar¹, Revathi B.S¹, Dhanashri Sabale¹, Sangeeta N. Kale^{1*}

¹*Department of Applied Physics, Defence Institute of Advanced Technology, Pune*

*Corresponding Author E-mail: sangeetakale2004@gmail.com

Flexible piezoresistive sensors capable of simultaneously achieving high sensitivity and a broad pressure detection range remain a critical challenge due to the intrinsic trade-off between these two parameters. In this work, we report a synthesis method to fabricate a high-performance flexible sensor based on a three-dimensional porous polydimethylsiloxane (PDMS) architecture integrated with a MoS₂-driven conductive network. Unlike conventional hybrid systems that rely on multiple conductive fillers, this approach utilises uniformly distributed MoS₂ nanosheets to construct a stable and continuous percolation network within the porous elastomer framework via a simple dip-coating process. The porous microstructure, coupled with the pressure-responsive evolution of MoS₂ conductive pathways, enables effective modulation of contact resistance across a wide stress regime. The sensor exhibits a reliable piezoresistive performance over a broad pressure range from 0 to 700 kPa with moderate sensitivities of 0.0029 kPa⁻¹ in the low-pressure region (30–160 kPa) and 0.00054 kPa⁻¹ in the high-pressure region (160–700 kPa). The device also exhibits low hysteresis (~4.5%), fast response and recovery times of 20 ms and 41 ms, respectively, and stable performance during repeated mechanical loading over 1000 cycles. Importantly, this work shows the ability of a single-material MoS₂ conductive network to achieve stable sensing performance over a broad pressure range without complex fabrication or hybridization strategies. The proposed sensor demonstrates excellent mechanical compliance and

durability, suggesting its potential for practical applications in wearable electronics, human motion monitoring, and intelligent tactile systems.

Keywords: Piezoresistive Sensors; 3D porous structure; Transition metal dichalcogenides; human-machine interface; Tactile sensing

References:

1. He, Y., Lu, X., Wu, D., Zhou, M., He, G., Zhang, J., Zhang, L., Liu, H., and Liu, C, *Sensors and Actuators A: Physical*, 358(2023), p.114408.
2. Lu, H., Chen, B., Lai, X., Li, H., and Zeng, X, *Materials Today Nano*, 28(2024), p.100512.
3. Luo, Feng, Artur Ciesielski, and Paolo Samori, *Advanced Functional Materials* (2026): e75473.

Performance Benchmarking of CNT and Graphene Nanoplatelet Reinforced PVA Conducting Gel for Self-Powered Artificial Skin

Bablesh Gupta^{1,2}, Narendra Kumar Dhar³, Sachin Kumar³, Satinder K Sharma³, Ranbir Singh^{1,2*}

¹*School of Mechanical and Materials Engineering (SMME), Indian Institute of Technology (IIT) Mandi, Mandi, Himachal Pradesh, 175005, India*

²*Advanced Energy Conversion Laboratory (AECL), Indian Institute of Technology (IIT) Mandi, Mandi, Himachal Pradesh, 175005, India*

³*School of Computing and Electrical Engineering (SCEE), Indian Institute of Technology (IIT) Mandi, Mandi, Himachal Pradesh, 175005, India*

*Corresponding Author E-mail: ranbir@iitmandi.ac.in

The development of artificial electronic skin (E-skin) that replicates the tactile sensing capabilities of human skin offers transformative potential for intelligent robotics and next-generation wearable technologies. Nevertheless, designing conducting gels that simultaneously exhibit high flexibility and temperature sensitivity for multimodal sensing applications remains a major challenge. Nanomaterials like graphene nanoplatelet and carbon nanotubes have emerged as promising additives owing to their high surface-area-to-volume ratio, which enhances the sensitivity of conducting gels required for advanced E-skin systems. In this study, a comparative investigation of the nanomaterial-infused conducting gels comprising polyvinyl alcohol/sodium nitrate/multi-walled carbon nanotubes (PVA/NaNO₃/MWCNTs) and graphene nanoplatelets (PVA/NaNO₃/GNPs) on the basis of their mechanical, electrical, and thermal properties is presented. Although both nanomaterials improved the gel characteristics, the GNP-based conducting gel exhibited superior mechanical strength (3.0 MPa), enhanced electrical response (5.8 μ A), and improved sensing performance compared with the MWCNT-based counterpart. The optimized PVA/NaNO₃/GNPs gel was further utilized to develop a self-powered microchanneled triboelectric E-skin capable of generating an output voltage of 38.5 V and a power output of 0.42 mW, sufficient for powering low-energy electronic devices and monitoring human motion. The fabricated E-skin demonstrated remarkable sensitivities toward pressure, temperature, and stiffness, with values of approximately 1.81 V/kPa, 42.7 mV/K, and 7.43×10^{-6} V/Nm⁻¹, respectively. Furthermore, integration of the E-skin with a robotic arm enabled advanced

functionalities including object recognition, texture discrimination, and thermal sensing. The proposed scalable, cost-effective, and self-sustaining PVA/NaNO₃/GNPs nanocomposite gel-based E-skin presents a promising platform for future interactive robotics and human-machine interface applications.

Keywords: Nanomaterial; MWCNTs; GNPs; Microchannel; Artificial E-skin.

References

1. B. Gupta, S. Bano, and R. Singh, *Journal of Power Sources*, 632 (2025), 236-271.
 2. B. Gupta, S. K. Samanta, and R. Singh, *Materials Today*, 80 (2024), 681-709.
 3. S. Bano, et al., *ACS Applied Electronic Materials*, 7 (2025), 6967-6979.
-

All-Carbon Nanostructured Membranes Incorporating Carbon Quantum Dots for Antimicrobial Filtration Technologies

Kanchan Mone*, Ragini Ahiwale^a, Satishchandra Ogale*

^a *Center of energy science, Indian Institute of Science Education and Research, Pune
Dr. Homi Bhabha Road, Pashan, Pune, Maharashtra, 411008, India*

*satishogale@gmail.com

Carbon Quantum Dots (CQDs) based smart self-cleaning all-carbon membranes and hybrid all-carbon nanocomposite filtration systems have emerged as promising sustainable technologies for advanced air and water purification applications. In this work, CQDs were synthesized by citric acid and urea using hydrothermal method at 180°C for 9 h. The carbon nanodots were highly fluorescent and functionalized with excellent photocatalytic and antibacterial properties. These metal-free all-carbon systems employ the excellent photocatalytic activity, high surface area, conductivity, and antibacterial properties of carbon nanostructures to effectively remove contaminants and sterilize microbes. Under light irradiation, CQDs produce reactive oxygen species (ROS) that can continuously degrade organic pollutants, disrupt bacterial cell membranes, and inhibit biofilm formation, thus providing effective self-cleaning and antifouling performance., the interconnected all-carbon conductive network improves the electron transport, membrane permeability, mechanical stability and long-term filtration performance. The CQD-based all-carbon nanocomposite membranes exhibit tremendous potential for next generation antimicrobial filtration, healthcare protection, wastewater treatment and smart environmental remediation technologies owing to their lightweight nature, environmental-friendliness, low toxicity and sustainable synthesis route.

Keywords: Carbon Quantum Dots (CQDs), Reactive oxygen species (ROS), Antimicrobial, Nanocomposite, Filtration

References:

1. Dong, Y., et al. *Carbon*, 2012, 50, 2810-2815.
 2. Das, R., et al. *Advanced Materials*, 2019, 31, 1807831.
 3. Perreault, F., de Faria, A. F., & Elimelech, M. *Chemical Society Reviews*, 2015, 44, 5861-5896.
-

AI-Driven CNT based Sensor Arrays for Selective VOC Detection in Wearable Health Monitoring Platforms

Rohan Rohilla^{1,2}, Kinshuk Dasgupta^{1,2}, Jyoti Prakash^{1,2*}

*Affiliation:*¹Bhabha Atomic Research Centre, - Mumbai - 400085

²Homi Bhabha National Institute – Mumbai- 400094

*jprakash@barc.gov.in

Accurate and selective detection of volatile organic compounds (VOCs) in complex, real-world environments remain a fundamental challenge for wearable chemical sensing. Single-material sensors inherently suffer from cross-sensitivity and poor discrimination in multi-component gas mixtures, limiting their utility in clinical and environmental applications. Here, we present an AI-integrated chemiresistive sensing platform built on freestanding, additive-free carbon nanotube (CNT) fibers and films synthesized via floating catalyst chemical vapor deposition. The resulting networks combine high porosity, excellent electrical conductivity, and outstanding mechanical robustness with a Young's modulus of ~ 1.4 GPa and stable performance under repeated bending making them well-suited for flexible and wearable device integration.

Deliberate surface engineering of CNT films with CuO, V₂O₅, TiO₂, WO₃, and g-C₃N₄ yielded a six-element heterogeneous sensor array with intentionally complementary response characteristics. The array achieved ppb-level sensitivity, with a 2.3 ppb acetone detection limit and a 440% response to 10 ppm ethanol. A dual-branch deep neural network, trained on 18-dimensional feature vectors combining response magnitude and kinetic parameters, delivered 98% VOC classification accuracy and concentration prediction with $R^2 > 0.98$ across six structurally diverse VOCs and physiologically realistic diabetic breath-simulating mixtures. SHAP-based interpretability analysis confirmed that high performance originated from genuine synergistic multi-sensor interactions rather than single-channel dominance.

These results demonstrate that engineered sensing heterogeneity, combined with interpretable deep learning, can overcome the fundamental selectivity limitations of conventional chemiresistive sensors, providing a scalable, low-power framework for wearable diagnostics, environmental monitoring, and industrial safety applications.

Keywords: Carbon nanotubes, chemiresistive sensor array, volatile organic compounds, metal oxides, deep neural network, electronic nose, breath biomarkers, wearable sensors, SHAP interpretability, flexible sensing platform.

References:

1. R. Rohilla, J. Prakash, P. T. Rao, A. Pathak, and K. Dasgupta, *Journal of Materials Chemistry C*, 2025, 13, 18350 – 18364.
2. R. Rohilla, Jyoti Prakash, Kinshuk Dasgupta, *ACS Appl. Nano Mater.* 2025, 8, 49, 23414–23465.
3. R. Rohilla, J. Prakash, P. T. Rao, R. Alexander, and K. Dasgupta, *IEEE Sensors J.*, vol. 23, no. 24, pp. 30102-30111, 2023.

Au/ZnO Nanofilm E-Textile Sensor for Respiratory Humidity Monitoring

Anupam Kumari^{1,3,*}, Angan Sengupta^{2,3}, Ajay Agarwal^{1,3}

¹Department of Electronics Engineering, IIT Jodhpur, Rajasthan 342030.

²Department of Chemical Engineering, IIT Jodhpur, Rajasthan 342030.

³*Interdisciplinary Research Program, IIT Jodhpur, Rajasthan 342030.*

anupam.1@iitj.ac.in

Wearable respiratory monitoring demands highly sensitive, flexible, and conformable sensing systems capable of transducing subtle humidity variations associated with breathing dynamics ^[1]. This work reports a ZnO sputtered e-textile-based smart mask for real-time respiration monitoring. Interdigitated electrodes (IDEs) were embroidered on the porous mask using silver yarns to form mechanically robust, low-resistance electrodes ^[2]. A conformal ZnO thin film (~300nm) was deposited onto the pattern using RF magnetron sputtering, facilitating breathability while enabling a high surface-to-volume ratio. The sensor performance was further improved by Au decoration (~50s), resulting in improved sensitivity. The sensing mechanism is governed by quasi-2D surface conduction pathways, in which humidity-induced adsorption/desorption modulates charge transport, leading to measurable changes in the sensor's electrical resistance. Furthermore, incorporating Au nanoparticles introduces localized electronic modulation and catalytic activity, accelerating adsorption kinetics and improving response-recovery characteristics. The device's performance was characterized by resistance measurements over a relative humidity (RH) range of 10-80%, exhibiting a sensitivity of ~64 kΩ/%RH, minimal hysteresis, and good repeatability. The enhanced response is attributed to adsorption-driven charge modulation and the formation of continuous H₂O layers that facilitate Grotthuss-type proton hopping along the ZnO surface. The sensor was integrated with an IoT-enabled electronic unit for real-time signal acquisition, processing, and dissemination. Distinct inhalation-exhalation cycles were reliably captured through dynamic resistance changes, enabling stable, continuous monitoring of the respiratory pattern. The Au/ZnO-based e-textile smart mask provides a scalable, low-cost, and mechanically compliant solution for next-generation wearable healthcare systems, with strong potential in non-invasive respiratory diagnostics and continuous physiological monitoring.

Keywords: ZnO nanofilm, Au decoration, E-textile, Humidity sensor, Respiratory monitoring.

References

1. X. Wang, Y. Deng, X. Chen, P. Jiang, Y. K. Cheung, H. Yu, *Microsystems & Nanoengineering*, 2021, 7, DOI 10.1038/s41378-021-00324-4.
2. A. Kumari, A. Sengupta, A. Agarwal, *ACS Applied Materials & Interfaces*, 2025, 17, 42541.

Two-Dimensional Halide Perovskite-Based Flexible Memristor Devices for Neuromorphic Applications

Koushik Gayen¹, Sayandeep Guha¹, Suman Kalyan Pal^{1,*}

¹*School of Physical Sciences, Indian Institute of Technology Mandi
Kamand, Himachal Pradesh, 175005, India*

*suman@iitmandi.ac.in

The big data revolution has significantly increased the demand for efficient storage and processing, revealing the limitations of traditional computing, particularly the von Neumann bottleneck. Emerging technologies such as resistive random-access memory (ReRAM) and

neuromorphic computing offer promising alternatives by integrating memory and processing units.[1] To enable such systems, recent research has focused on novel materials including transition metal oxides, dichalcogenides, and halide perovskites due to their inherent memristive properties, which are crucial for synaptic emulation in neuromorphic devices.[2] In this study, we demonstrate a two-dimensional (2D) perovskite-based ReRAM device fabricated on a flexible substrate with the structure Ag/(4-FBA)₂PbBr₄/ITO/PET. The device exhibits endurance (>340 cycles), a high on/off ratio ($\sim 10^3$), and stable retention exceeding 10^4 seconds. Additionally, it successfully mimics neuromorphic functionalities, including long-term potentiation and depression, replicating biological synaptic behavior.[3] These results highlight the potential of 2D perovskite-based systems for next-generation flexible neuromorphic applications.

Keywords: 2D perovskites; Flexible, ReRAM; Neuromorphic computing.

References:

1. R. Khan, N. U. Rehman, S. Kalluri, S. Elumalai, A. Saritha, M. Fakhar-e-alam, M. Ikram, S. Abdullaev, N. Rahman, and S. Sangaraju, *Nanoscale*, 17, (2025), 13174–13206.
2. K. A. Nirmal, D. D. Kumbhar, A. V. Kesavan, T. D. Dongale, and T. G. Kim, *NPJ 2D Mater. Appl.*, 8, (2024), 83.
3. K. Gayen, S. Guha, and S. K. Pal, *ACS Appl. Electron. Mater.*, 7, (2025), 10341–10351.

Development of Novel Cost-efficient Environment Friendly High Performing Sulfide Based Thermoelectric Material

Pritam Deyl*, Tanmoy Maiti2

Department of Materials Science and Engineering, IIT Kanpur, Kanpur-208016

*pritamdey23@iitk.ac.in

The growing global energy demand and environmental concerns have intensified the search for sustainable energy solutions. Thermoelectric materials offer a promising pathway for directly converting waste heat into electricity, with no carbon footprint or environmental noise. The conversion efficiency of a thermoelectric generator is generally expressed by the thermoelectric figure of merit (zT). It is tricky to improve zT as all the inherent components are strongly correlated. Chalcogenide-based materials, such as Pb-based tellurides and selenides, are popular candidates for thermoelectric generators owing to their high zT and low electrical band gap. However, inherent environmental hazards, toxicity, and scarcity of those materials compelled the research community to shift its focus to sulfides. Although sulfur's environmentally benign nature and Earth's abundance are beneficial, low electrical conductivity is a drawback. Among all the available environmentally benign sulfides, Bismuth sulfide stands out for its low thermal conductivity ($\sim 0.4 \text{ Wm}^{-1}\text{K}^{-1}$) and good Seebeck coefficient ($\sim 450 \mu\text{VK}^{-1}$). However, Bi₂S₃ suffers from low zT (~ 0.11 at 573–723K) when compared to other Bi-based chalcogenides. The low zT is attributed to its higher band gap ($\sim 1.5 \text{ eV}$ at 298K) and higher electrical resistivity.

In the current work to enhance the thermoelectric efficiency of Bi₂S₃, we have developed advanced solid solutions incorporating 2D materials such as r-GO and Mxene, as well as transition-metal chalcogenides. These 2D materials necessarily improve mechanical strength, electrical conductivity and carrier mobility while suppressing thermal conductivity through enhanced phonon scattering, leading to higher zT values. FE-SEM and TEM studies confirm the 2D structure, whereas XRD revealed a single-phase crystal structure and UV-Vis-NIR spectra states low optical band gap. The hall effect measurement revealed electrical conductivity in the

order of $\sim 10^4$ S/m. Our thermoelectric study demonstrated a nearly 5-fold increase in zT w.r.t. the pristine materials.

Keywords: Sustainable energy, Thermoelectric materials, Bismuth sulfide, TMDC

Electrochemical Properties of Samarium-Substituted Mn Ferrites Prepared by Sol-Gel Auto-Combustion Technique for Supercapacitor applications

Balbhim S Maharnavar^{a,b,c}, Ashish Yengantiwar^{a*}, Ramesh J Deokate^{c*}

^aDepartment of Physics, Fergusson College (Autonomous), Savitribai Phule Pune University, Pune (MS)-411004, India

^bDepartment of Physics, Dada Patil Mahavidyalaya, Karjat, Dist-Ahilyanagar (MS)-414402 India

^cDepartment of Physics, Vidya Pratishthan's Arts Science and Commerce College Baramati (MS)-413102, India.

*Corresponding email: deokate2000@yahoo.co.in, ashish.yengantiwar@fergusson.edu

Samarium doped Mn ferrites ($\text{MnSm}_x\text{Fe}_{2-x}$) were synthesized using sol-gel auto-combustion technique with Sm^{3+} doping concentrations varying $x = 0.0, 0.03, 0.06, 0.09, 0.12$. The prepared nanomaterials were examined using X-ray diffraction analysis (XRD), Fourier transform infrared spectroscopy (FTIR), Field Emission Scanning electron microscopy (FE-SEM), and vibrating sample magnetometer (VSM). The XRD examination verified the pure FCC spinel phase of all samples with no impurity peak. The FTIR analysis for spinel molecules identified two frequency bands at the octa and tetrahedral sites. The FESEM analysis confirmed the shape, size and distribution of prepared nanoparticles. More specifically, the study will focus on the techniques which have been employed to raise the specific capacitance of the metal oxides. Electrochemical properties such as cyclic voltammetry (CV), potentiostatic and impedance were used to analyse the super capacitive performance of Sm-substitute MnFe_2O_4 electrodes. Based on the outcomes, it is determined that 0.06 Sm- MnFe_2O_4 has higher super capacitive behaviour than different samples. Precise capacitance for this nanocomposite from GCD was determined to be 556.39 F/g at a current density of 1 A/g. Furthermore, a solid-state symmetric supercapacitor device (SSD) demonstrates an energy density and power density of 9.23 Wh/kg and 1857.02 W/kg. The results indicate that including this synergy of metal ferrite can enhance the performance of supercapacitors.

Keywords: Nanomaterial, ferrite, supercapacitor, electrochemical study etc.

References:

1. Mehwish Sattar, Safia Anjum, M. Yasin Raja, Zeeshan Mustafa, Anam Mansoor, Rabia Khurram, Tafruj Ilayas a,b, *Ceramics International* 50 (2024) 16747–16764.
 2. Syed Khasim, Apsar Pasha, B.N. Ramakrishna, Prathibha B.S, Koushalya P. R, *Materials Chemistry and Physics* 325, (2024), 129675.
 3. E. Vivek, A. Arulraj, Mohammad Khalid, I. Vetha Potheher *Journal of Alloys and Compounds* 908, (2022), 164541.
-

Ultrasensitive Detection of Mercury Ions Using BaZnV₂O₇@ Ti₃C₂T_x MXene Functionalized AlGa_N/Ga_N HEMT Sensor

Pooja Kumari¹, Kirti Sankhala², Mahesh Kumar^{3*}

¹ Interdisciplinary Research Program of Internet of Things, Indian Institute of Technology Jodhpur, Jodhpur, Rajasthan, India

² Department of Center for Emerging Technologies for sustainable Development, Indian Institute of Technology Jodhpur, Rajasthan, India

³ Department of Electrical Engineering, Indian Institute of Technology Jodhpur, Rajasthan, India

*Corresponding Author E-mail: mkumar@iitj.ac.in

Rapid and ultrasensitive detection of mercury ions (Hg²⁺) is of paramount importance due to highly toxic nature of Hg²⁺ ions along with its persistence in the environment. In this work we report a mercury ions sensor based on a AlGa_N/Ga_N high electron mobility transistor (HEMT) functionalized with BaZnV₂O₇@ Ti₃C₂T_x MXene nanocomposite. This method exploits the 2D electron gas (2DEG) sensitivity of the AlGa_N/Ga_N heterostructure in response to BaZnV₂O₇@ Ti₃C₂T_x MXene nanocomposite interaction with Hg²⁺ ions. Due to its high chemical reactivity with vanadium species, Hg²⁺ ions induce redox reactions of V⁴⁺/V⁵⁺ ions, leading to surface potential accumulation and thus modulating the 2DEG. High electrical conductivity and large specific surface area of Ti₃C₂T_x MXene facilitates the rapid electron transfer further improving sensing performance.

This mercury ion sensor demonstrates excellent detection properties, with high sensitivity (~135 $\mu\text{A ppm}^{-1} \text{cm}^{-2}$), broad linear detection range (1 ppt–100 ppm), and low limit of detection (LOD, ~0.3 ppt). Besides, this mercury ion sensor shows a fast response time (<4 s), good repeatability, and stability during repeated measurements. The sensor exhibits better selectivity to Hg²⁺ ions compared to other heavy metals like Pb²⁺, Cd²⁺, Zn²⁺, Cu²⁺, and Ni²⁺ ions with minimal interference (<4% deviation in signal output). Selectivity arises from the higher binding capability of Hg²⁺ ions to the BaZnV₂O₇@ Ti₃C₂T_x MXene surface, leading to efficient control of the 2DEG channel.

Furthermore, the device maintains stable performance over extended periods, indicating good reliability for practical applications. The integration of BaZnV₂O₇@ Ti₃C₂T_x MXene with AlGa_N/Ga_N HEMT technology provides a promising platform for portable, low-power, and highly sensitive detection of mercury ions in environmental monitoring. This study highlights the potential of hybrid nanocomposite-functionalized HEMT sensors for next-generation water quality assessment systems.

Keywords: AlGa_N/Ga_N HEMT high-electron mobility transistor (HEMT), Hg²⁺ ions, Sensitivity, BaZnV₂O₇, Ti₃C₂T_x MXene

References:

1. Nigam, A., Goel, N., Bhat, T. N., Rahman, M. T., Dolmanan, S. B., Qiao, Q., & Kumar, M. (2020) Sensors and Actuators B: Chemical, 309, 127832.
2. Sharma, N., Nigam, A., Lobanov, D., Gupta, A., Novikov, A., & Kumar, M. (2021) IEEE Internet of Things Journal, 9(16), 14317-14324.

Comprehensive Assessment of GAA, DG, and GAP Vertically Stacked 2-D Nanosheet FETs for Advanced Technology Nodes

Aruna Patel¹, Laxmi Kumre²

Electronics and Communication Department, MANIT, Bhopal MP India

E-mail: arunapatel9685@gmail.com

This work presents a comprehensive simulation-based comparative study of vertically stacked two-dimensional (2D) nanosheet field-effect transistors (NSFETs), focusing on gate-all-around (GAA), dual-gate (DG), and gate-all-parallel (GAP) architectures as potential successors to conventional FinFETs for advanced technology nodes. Special emphasis is placed on multilayer stacked GAA MoS₂ FETs at ultra-scaled nanosheet dimensions. Key design parameters, including the number of nanosheet layers, spacer thickness, and doping profiles, are systematically analyzed to identify optimal device configurations. Additionally, practical non-idealities such as contact resistance and line-edge roughness (LER) are incorporated to ensure realistic performance evaluation.

The comparative analysis, performed using DG-FET as the reference, reveals that GAP 3L-FET exhibits a slight degradation in subthreshold swing (SS) of approximately 3.33%, while GAA 3L-FET maintains identical SS, indicating excellent electrostatic control. In terms of ON-state current (I_{ON}), significant enhancements are observed, with GAP and GAA architectures achieving improvements of approximately 36.75% and 241.88%, respectively. Furthermore, the impact of source/drain contact resistance (R_{SD}) is investigated in accordance with IRDS guidelines. Results show that increasing R_{SD} leads to noticeable degradation in I_{ON} , particularly in single-layer devices due to inefficient carrier injection. However, multilayer GAA structures effectively mitigate this limitation, maintaining high drive current even at elevated resistance levels. Notably, a three-layer GAA-FET achieves an I_{ON} of 1.8 mA/ μ m at three times the IRDS target resistance, significantly exceeding the projected requirement of 0.753 mA/ μ m.

Overall, the findings demonstrate that GAA NSFETs offer superior scalability, enhanced electrostatic integrity, and robust current performance, establishing them as promising candidates for future nanoscale transistor technologies.

Keywords: 2-D gate-all-around (GAA) nanosheet FET, 2-D materials (2DMs), vertical stacked MoS₂ FETs.

Reference:

1. Prabhat Kumar Dubey, Damiano Marian, Alejandro Toral-Lopez, Theresia Knobloch, "Simulations of Vertically Stacked 2-D Nanosheet FETs," IEEE Trans. Electron Devices, vol. 72, no. 3, Mar. 2025.
2. D. Marian, E. G. Marin, M. Perucchini, G. Iannaccone, and G. Fiori, "Multi-scale simulations of two dimensional material based devices: The NanoTCAD ViDES suite," J. Comput. Electron., vol. 22, no. 5, pp.1327–1337, Oct. 2023, doi: 10.1007/s10825-023-02048-2.

Observation of Resonant Tunneling Phenomena through the intrinsic valleys of Monolayer MoS₂

Abir Mukherjee, Kajal Sharma, Akash Kumar Sahu, and Samaresh Das

Centre for Applied Research in Electronics, Indian Institute of Technology Delhi, Delhi, 110016, India

Corresponding Author: crz258442@care.iitd.ac.in

This study demonstrates the successful fabrication of CVD-grown monolayer MoS₂ ultra-thin quantum well double-barrier resonant tunneling devices (RTDs), designed to be

compatible with standard CMOS processing. The quantized electronic states arising from multiple valleys in the momentum space of the 2D sheet, confined between Al_2O_3 barriers, produce distinct resonant tunneling peaks that broaden the negative differential resistance (NDR) region, as confirmed by experimental I–V measurements.

The electrical performance of these RTDs, evaluated from cryogenic to room temperature, establishes a significant benchmark, with peak-to-valley ratios (PVRs) reaching 170 at 4 K and 28 at room temperature. These results highlight the potential for further optimization toward practical room-temperature quantum devices. Moreover, both momentum-conserved and non-conserved tunneling from highly n-doped silicon into the multiple valleys of monolayer MoS_2 open new opportunities for gate-controlled spin-valley qubit technologies, particularly at millikelvin operating conditions^{1–3}.

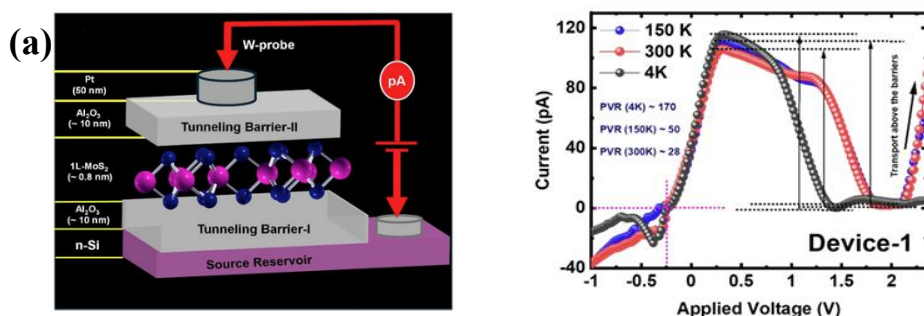


Figure: (a) Schematic of fabricated tunnel device and (b) I–V response of the device to determine the PVR at 4 K, 150 K and 300 K

Keywords: Quantum Transport, CVD, Monolayer MoS_2 , High PVR, and Room Temperature Quantum Application

References:

1. Mukherjee, A.; Sharma, K.; Bhatt, K.; Kandar, S.; Singh, R.; Das, S. APL Quantum 2025, 2.
2. Mukherjee, A.; Sharma, K.; Katiyar, A.; Das, S.; Ray, S.; Das, S; 2026.
3. Khan, B.; Mukherjee, A.; Georgiev, Y.; Colinge, J.-P.; Ghosh, S.; Das, S. 2025, 45423.

Frequency-Selective EMI Shielding with Near-Zero Transmission via Tunable Attenuation in Magnetic-Conductive Layered Structures

Priteesh Kumar Samal¹, Santosh Dasila², Akash Paramanik², V. Subramanian², Swetapadma Praharaj^{1**}, Dibyaranjan Rout^{1*}

¹Functional Materials Lab, Department of Physics, School of Applied Sciences, Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, Bhubaneswar-751024, Odisha, India

²Microwave laboratory, Department of Physics, Indian Institute of Technology Madras, Chennai-600036, Tamilnadu, India

*Corresponding Author E-mail: *droutfpy@kiit.ac.in, **spraharjfy@kiit.ac.in

The rapid growth of wireless electronics and modern communication technologies is driving an urgent demand of next generation Electromagnetic interference (EMI) shielding materials that are lightweight, ultrathin, flexible, and pattern-free with frequency-selective characteristics. To attain these features, a layered structure (L-CF) comprising Cobalt ferrite (CF)-

coated polyvinylidene fluoride (PVDF) membrane, a double-sided tape interlayer, and graphene nanoplatelets (GNP)-coated PVDF, is fabricated for frequency selective EMI shielding. The design exploits tuning of attenuation constant (α) to achieve selective X-band performance, the single-layer CF-coated PVDF layer (s-CF) exhibits minimal reflection, absorption, and transmission. Upon stacking the full layer structure, the total shielding mechanism enables via initial absorption in the magnetic s-CF magnetic layer, reflection at the graphene coated PVDF conductive layer, and re-absorption in the magnetic layer, with near-zero transmission. The frequency-selective shielding effectiveness of ~ 41 dB (43% absorption) @ 8.7 GHz and ~ 34 dB (30% absorption) @ 11.2 GHz is observed in X-Band frequency. To further validate the efficacy of the suggested layered structure, simulations are performed using COMSOL Multiphysics software. The simulation illustrates the power dissipation across each layer, along with the electric field and magnetic field distributions for a representative sample L-ZF20. The corresponding dual frequency bands were evaluated by radar cross-section analysis, demonstrating the potential of the material for stealth technology.

Keywords: Frequency-selective; Magnetic-conductive; Attenuation constant; COMSOL multiphysics; Radra cross-section

References

1. Nan, Ze, Wei Wei, Zhenhua Lin, Jianyong Ouyang, Jingjing Chang, and Yue Hao. Materials Science and Engineering: R: Reports 160 (2024): 100823.
2. Isari, Ali Akbar, Ahmadrza Ghaffarkhah, Seyyed Alireza Hashemi, Stefan Wuttke, and Mohammad Arjmand. Advanced Materials 36, no. 24 (2024): 2310683.
3. Durairaj, Lalitha, and Malathi Murugesan. Synthetic Metals 312 (2025): 117874.

Performance Evaluation and Comparative Analysis of Ternary Circuit based on CNTFET and CMOS

Vaibhav P. Nagrale¹, Laxmi Kumre²

Electronics and Communication Department, MANIT, Bhopal, MP, India
E-mail: nagrale111@gmail.com

Nanotechnology is defined as an engineering branch where materials are at a nanoscale near about 10 to 100 nm. At this nanoscale, material properties are different than properties that are at normal scale. Power efficiency comes up, a common constraint in ternary circuits at nanoscale. Complementary Metal-Oxide Semiconductor (CMOS) technology has an emerging concern in ternary circuits like power loss, scalability, and performance parameters. Carbon Nanotube Field Effect Transistor(CNTFET) can overcome CMOS performance parameters due to its enhanced electrical properties, low-threshold voltage and its remarkable scalability. Ternary logic operating voltages operate over three Logic states $\{0, 1, 2\}$ and encodes $\log_2(3) \approx 1.585$ bits per trit and gets a 37% reduction in digits count as compared with binary representations and enabling significant improve-ments in computational efficiency. On the basis of performance parameters, power consumption, propagation delay, power delay product and area efficiency are to be evaluated. Key parameter is that CNTFET-based Ternary circuits achieve superior PDP than CMOS-based ternary circuits, particularly at nanoscale geometries.

In ternary circuits like Standard Ternary Inverter, Ternary NAND, Ternary NOR, Ternary Multi-plexer, and Ternary Full Adder circuits, waveforms and their logic equations are mentioned. Our CNTFET Ternary Full Adder saves PDP up to 29% as compared with the CMOS Ternary Full Adder. Also, parameters of CNTFET Ternary Full Adder, like delay and power, are improved as compared with CMOS Ternary Full Adder. By varying the temperature

at particular values, its PDP is calculated by knowing its power and delay values. By varying temperature, its internal parameters change, and due to this, PDP changes.

Keywords: CNTFET; CMOS; Power; Delay; PDP;

References:

1. S. Kim, S.-Y. Lee, S. Park, K. R. Kim, and S. Kang, IEEE Trans. Circuits Syst. I, Reg. Papers, vol. 67, no.9, pp. 3138–3151, Sep. 2020
2. S. Heo et al., IEEE Electron Device Lett., vol. 39, no. 12, pp. 1948–1951, Dec. 2018
3. S. Nemati, M. Haghi Kashani, and R. F. Mirzaee, IET Circuits, Devices Systems, vol. 17, no. 3, pp. 111–134, 2023.

Sulfur-driven stoichiometric control of tungsten disulfide thin films grown via sulfurization of nanoscale tungsten layers

Md Samim Reza^{1*}, Henam Sylvia Devi^{1,2}, Tejveer Singh Anand¹, Madhusudan Singh¹

¹Functional Materials & Devices Laboratory, Department of Electrical Engineering, IIT Delhi, New Delhi, NCT, India, 110 016.

²Department of Chemistry, Dhanamanjuri University, Imphal, Manipur, India, 795001

*Corresponding Author E-mail: eez178182@ee.iitd.ac.in

A scalable two-step sulfurization approach for the growth of tungsten disulfide (WS₂) thin films was demonstrated using sputter-deposited nanoscale tungsten films and elemental sulfur precursors. The sulfurization process was carried out in a custom-built 12-zone horizontal split furnace (Quazar Technologies) under an argon atmosphere, without hydrogen, halides, or toxic gaseous sulfur sources, enabling a contamination-free, environmentally compatible growth route for transition-metal dichalcogenide (TMD) thin films. The effect of sulfur precursor amount on the stoichiometry and morphology of sulfurized tungsten films was systematically investigated by varying the sulfur quantity from 100 to 900 mg while keeping all other growth parameters unchanged.

Distinct sulfur-deficient, near-stoichiometric, and sulfur-rich WS_x phases were obtained by varying the sulfur quantity during growth. Energy-dispersive X-ray spectroscopy (EDX, Oxford Instruments) categorized the films into three compositional regions based on the measured S:W ratio: S:W < 2, S:W ~ 2, and S:W > 2. Under optimized sulfurization conditions (250–450 mg sulfur), near-stoichiometric WS₂ films (W:S ~ 1:2) were achieved with near-complete surface coverage. Field-emission scanning electron microscopy (FESEM, TESCAN Magna LMU) revealed diverse surface morphologies, including horizontally aligned flakes, randomly oriented nanosheet-like features, and micro-dome formations distributed across the substrate. X-ray diffraction (XRD, Rigaku Ultima IV, Cu K α = 1.54 Å) confirmed the formation of hexagonal 2H-WS₂ (ICDD: 08-0237) with dominant (002) orientation. Raman spectroscopy (Renishaw inVia, 514 nm, 1.0 mW) exhibited characteristic in-plane (E_{2g}¹) and out-of-plane (A_{1g}) vibrational modes consistent with multilayer WS₂ growth, while Raman mapping indicated spatially consistent sulfurization. X-ray photoelectron spectroscopy (XPS, Kratos Axis Supra, Al K α ~ 1486.6 eV) further confirmed successful sulfurization and the stoichiometric transition from sulfur-deficient to sulfur-rich growth regimes.

The study demonstrates that sulfur content is a critical parameter governing the compositional and structural evolution of sulfurized tungsten thin-films, a factor that has remained largely unexplored in sulfurization-based WS_2 growth. The presented hydrogen-free and halide-free synthesis strategy offers a simple, scalable, and versatile platform for the controlled growth of stoichiometric TMD thin films and can be extended to other layered chalcogenide systems.

Keywords: Tungsten disulfide, WS_2 , thin-films, sulfurization, stoichiometry

SPONSERS

thermo scientific



Anusandhan
National
Research
Foundation



WILEY



TROKUT
SOLUTIONS



सीएसआईआर
CSIR
भारत का नवाचार इंजन
The Innovation Engine of India



नाभिकीय विज्ञान अनुसंधान बोर्ड
Board of Research in Nuclear Sciences



SciAps



DYNOTECH
INSTRUMENTS
PVT LTD



ants



SAMYAK
COMPUTER PVT. LTD.



ISO Certified 9001:2015



A Joint Collaboration of Kapaku Analyte AB Sweden & Thermal Hazard Technology UK



About Malvern Panalytical:

Malvern Panalytical and Micromeritics provide market-leading material characterization technologies to the world's most innovative companies, academic institutions, and government laboratories. Our solutions help customers understand and optimize the physical and chemical properties of materials, enabling breakthroughs in pharmaceuticals, chemicals, advanced materials, batteries, mining, metals, energy, and manufacturing.

We support our customers through their biggest challenges—from developing life-saving vaccines to advancing sustainable fuels and critical materials. Through innovation, high-quality analytical solutions, and exceptional customer support, we help organizations accelerate research, improve product quality, and drive operational excellence.

Our portfolio includes particle size, shape, concentration, and zeta potential analysis; X-ray diffraction (XRD) for phase and structural characterization; X-ray fluorescence (XRF) for elemental analysis; molecular spectroscopy solutions; and technologies for surface area, porosity, density, adsorption, and catalyst characterization.

We also offer laboratory automation and integrated workflows to improve productivity and data quality.

We win customers with performance and keep them with service and insights. Our vision is to remain a recognized leader in material characterization, delivering trusted results and actionable insights. Driven by our people—experts in their fields—we foster a high-performance culture built on agility, transparency, collaboration, and customer focus, guided by our values: **Own It. Aim High. Be True.**

Learn More

Visit: <https://www.malvernpanalytical.com/>

Email: sangeetha.raj@malvernpanalytical.com

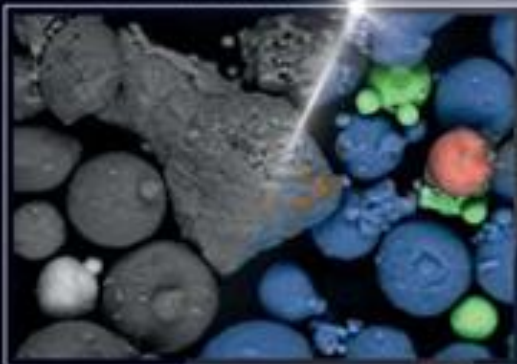
OXFORD
INSTRUMENTS

Making imaging elementary

Introducing Unity, the world's first combined Backscattered Electron and X-ray (BEX) imaging detector.

Accelerate your journey to scientific discovery with instant microstructural and chemical images, acquired simultaneously with the Unity detector.

Without the Unity detector



With the Unity detector



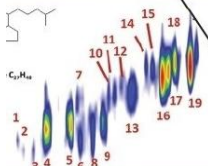
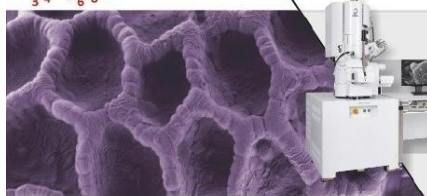
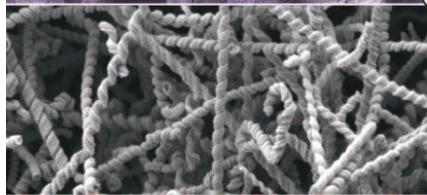
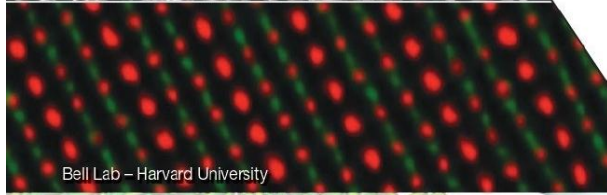
Find out how we're making sophisticated sample analyses simpler and faster than ever before.

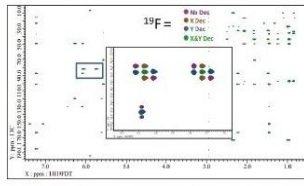
Visit us at nano.oxinst.com



Product Lineup

SMART • FLEXIBLE • POWERFUL

STEP INTO THE WORLD OF JEOL

- SEM
- TEM
- SAMPLE PREP
- NMR
- MASS SPEC
- EPMA
- LITHOGRAPHY








JEOL INDIA PVT LTD | www.jeol.com/in/



About the Conference

2D materials are far more than a scientific curiosity; they represent a transformative frontier in materials science, poised to redefine performance boundaries across diverse technological domains. With their atomically thin, layered structures, these materials exhibit unprecedented quantum, optical, and electrochemical behaviors that are unattainable in bulk or other morphological forms.

From the groundbreaking discovery of graphene to the emergence of advanced materials such as transition metal dichalcogenides, phosphorene, and MXenes, the 2D materials landscape is reshaping the future of energy, healthcare, defense, electronics, photonics, flexible devices, aerospace, and quantum technologies.

To capture the recent advancements in the field and dwell upon its future prospects, we are pleased to inform you that an International Conference on **"2D MatTech Global: Fundamentals to Applications"** is being organized by **Indian Institute of Technology Indore (IIT Indore)** in association with **UGC-DAE Consortium for Scientific Research (UGC-DAE CSR), Indore** and **Raja Ramanna Centre for Advanced Technology (RRCAT), Indore**.

This conference aims to serve as a dynamic, interdisciplinary platform uniting scientists, engineers, industry leaders, and policymakers around the world. It offers a vital space for sharing pioneering research, fostering interdisciplinary collaboration, and catalyzing the next wave of innovation.

By bringing together prominent figures and emerging experts in the field, the event seeks to accelerate the real-world integration of 2D materials and unlock their vast potential for global impact.

Venue



Indian Institute of Technology Indore, INDIA